Karlsruher Institut für Technologie

Module Handbook
M.Sc. Electrical Engineering and Information Technology 2025 (Master of Science)

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KIT DEPARTMENT OF ELECTRICAL ENGINEERING AND INFORMATION TECHNOLOGY



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1 Preamble

1.1 Overview of the degree program

The Master's degree program in Electrical Engineering and Information Technology at the Karlsruhe Institute of Technology (KIT) is divided into the four subjects:

- Field of Specialization (60 CP)
  - Fundamentals (24 CP)
  - Focus Area (dependent on Lab: 27-30 CP)
  - Lab course (exactly 1)
- Electives (24 CR)
- Interdisciplinary Qualification Area (6 CR)
- Master’s Thesis (30 CR)
  The Master's Thesis is written in English. At the student's request, the examiner may authorize the Master's Thesis to be written in German (SPO Section §14(4)).

Fields of Specialization

One of the four “Fields of Specialization” can be chosen.

- Automation, Robotics, and Systems Engineering
- Electrical Power Systems and Electromobility
- Information and Communication Technology
- Microelectronics, Photonics, and Quantum Technologies

Within each “Field of Specialization” four modules of the subject “Fundamentals” has to be chosen.

Furthermore, there are several “Focus Areas” that provide the students an orientation and recommendation for possible study courses and thus a guideline of reasonable module combinations. They are shown in the following chapter as a kind of profile within each “Field of Specialization”. The modules for each “Focus Area” are marked with a cross and are recommendations. Students are free to make their choice.

At last, exactly one lab or practical course has to be chosen within each “Field of Specialization”.

Electives

The elective modules provide a wide range of modules of further interest. Here, up to one additional lab or practical course may be chosen.

2 Introduction to the Module Handbook

2.1 General

The legal basis for the degree program and the conduct of examinations are the valid study and examination regulations (**Studien- und Prüfungsordnung** – SPO).

The degree program is divided into subjects. Each subject in turn is divided into modules. Each module consists of one or more module components ("Teilleistungen"), which are concluded by a competence certificate. The scope of each module is indicated by credit points (CP), which are recorded in the curriculum after successful completion of the module.

The study and examination regulations define the subjects (and their scope) that are assigned to the compulsory modules and/or the elective modules in the degree program.

The **compulsory modules** include the part of the program that constitutes the program-specific subject profile.

The **elective modules** serve to sharpen or expand the profile and allow interdisciplinary combinations or application-oriented supplements.

The module **Interdisciplinary Qualifications** contains courses with a predominantly non-technical content; these must be passed with evaluated proof of credit points. The courses have to be selected from the range of courses offered by the House of Competence (HOC), the FORUM (formerly ZAK), and the Language Center (SPZ), as well as from courses offered by the KIT-Department of Electrical Engineering and Information Technology or other KIT departments.

Achievements can be booked into the module "Interdisciplinary Qualifications" by the students themselves. Students can access the module via the menu item "Exam Registration and Unregistration", which can also be used to access the study schedule. Here you will find a new tab "ÜQ/SQ-Leistungen", which displays the list of your unassigned achievements.

Subsequently, these have to be assigned to the module components with the title 'Self Assignment-...' according to the grading scale, graded or ungraded. Title and credits of the achievement are then adopted automatically.

The **Module Handbook** describes the modules that are part of the degree program. It contains:

- the composition of the modules
- the size of the modules (in CP)
- the average workload (in hours)
- the interdependencies of the modules
- the competence goals of the modules
- the type of competence certificate
- the calculation of the module grade

The Module Handbook thus provides the necessary orientation in the degree program. Detailed information about the various lectures, exercises and seminars can be found in the **course catalog**.

You will find all information concerning the legal framework of your studies in the respective **study and examination regulations** of your degree program.

2.2 Notes on modules and module components

Level indication for the modules

Level 1 = 1st + 2nd semester Bachelor
Level 2 = 3rd + 4th semester Bachelor
Level 3 = 5th + 6th semester Bachelor
Level 4 = Master

Workload and credit points

Each credit point corresponds to an average workload of approx. 30 h. This effort is necessary for the students to achieve an average performance.

Versions of modules and module components

This specification provides information about the currently valid version of the module or the component. A new version is generated, for example, if an adjustment of the CP was carried out in the module or component. You will automatically receive the valid version in your curriculum. If you have already started a module, you can complete the module in the version you have started (grandfathering).

Type

Describes the type of competence certificate according to the study and examination regulations § 4. Competence certificates are subdivided into course works or examinations.

Examinations are graded

1. written examinations,
2. oral examinations, or
3. examinations of another type

Course works are ungraded written, oral, or practical achievements that students usually complete during the course.

Courses (lectures, exercises, tutorials, seminars)

In the chapter “Module Components” the corresponding courses of the current and the previous semester are shown in tabular form. For modules that are not offered every semester, you will thus receive complete information on the associated courses.

2.3 Registration and admission to module examinations

In order to take module examinations, students must register for the examination online in the **Campus Management Portal**.

In exceptional cases, registration may be made in written form at the *Studierendenservice* (Student Services) or at another facility authorized by the *Studierendenservice*. Registration deadlines for the competence certificates may be set by the examiners.

Where elective options exist, students make a binding declaration of module choice when registering for the examination. Upon application of the student to the examination board, the choice or assignment may be changed subsequently.

Each module and competence certificate may be assessed only once in the same degree program.

An examination will be passed, if the grade is at least “sufficient” (4.0). A module will be passed if all required module components are passed.

3 General Information

3.1 Study program details

KIT-Department	**KIT Department of Electrical Engineering and Information Technology**
Academic Degree	**Master of Science (M.Sc.)**
Examination Regulations Version	**2025**
Regular semesters	**4 semesters**
Maximum semesters	**8 semesters**
Credits	**120**
Language	**English**
Grade calculation	**Weighted average by credits**
Additional Information	**Link to study program** www.etit.kit.edu **Department** https://www.etit.kit.edu/english/master_etit.php **Business unit Studium und Lehre** https://www.sle.kit.edu/english/vorstudium/master-electrical-engineering-information-technology.php

3.2 Content

The Master's degree program serves to further enhance or complement the scientific qualifications acquired during Bachelor's studies. The graduates are able to independently apply scientific findings and methods and to assess their significance and scope for the solution of complex scientific and social problems.

Students choose the focus of their studies by choosing a Field of Specialization in the amount of 60 credits.

Each Field of Specialization covers defined, coordinated modules in the compulsory elective subjects "Fundamentals", "Focus Area", and "Lab Course". In the compulsory elective subject "Electives" (24 CP) students can choose modules freely. 6 credits of Interdisciplinary Qualifications complete the degree program.

3.3 Qualification Goals

The competence goals of the Master's degree program Electrical Engineering and Information Technology are divided into the following four main competence profiles:

1. Expertise: Students get to know the fundamentals of the discipline, as well as current research topics, processes, and results.
2. Research and problem-solving skills: Students learn the skills and techniques to meet challenges in research and industry.
3. Assessment and planning skills: Students participate in professional and research discourse and apply acquired knowledge, as well as learned techniques.
4. Personal and social skills: Students work on (their own) research projects, are integrated into a scientific team, are capable of independent and sustained professional and scientific development, and assess the social and societal impact of their activities.

For points 1 and 2 the focus is on lecturer activity, for points 3 and 4 correspondingly on student activity.

For the Master's degree program, these competence requirements can be further described in the following objectives:

Expertise

The graduates of the Master's degree program Electrical Engineering and Information Technology

- have an in-depth knowledge of mathematics and physics and an advanced expert knowledge of electrical engineering and information technology.
- are able to recognize and evaluate demanding technical and scientific tasks and problems in electrical engineering and information technology and to formulate approaches to solve them.
- master demanding scientific methods of their discipline and have learned to use them to analyze identified problems or subject-related issues according to the state of their knowledge.
- possess in-depth knowledge in a combination of the core competences of electrical engineering and information technology (e.g. automation and control technology, electrical energy systems, high-voltage technology, electrical drives, power electronics, digital technology, information technology, digital signal processing, communications engineering, high-frequency technology, measurement technology, imaging techniques, lighting technology, optoelectronics, circuitry, microelectronics, optical communication systems).

Research and problem-solving skills

The graduates of the Master's degree program Electrical Engineering and Information Technology

- are qualified to work as engineers and scientists in one of the main application fields of electrical engineering and information technology (e.g. electric mobility, medical engineering, microelectronic systems, communication technology, aerospace systems, photonics and optical technologies, renewable energies and smart grid, intelligent car).
- are familiar with the procedures for the analysis and design of components, circuits, systems, and equipment in electrical engineering.
- are familiar with advanced methods of presenting and processing information, programming, algorithmic formulation of processes, and the use of programming tools.
- possess an in-depth understanding of the methods of electrical engineering and information technology.
- are capable of further qualification through a doctorate program (PhD).

Assessment and planning skills

The graduates of the Master's degree program Electrical Engineering and Information Technology

- can evaluate designs of electrical and information technology, as well as various solution alternatives.
- recognize limits of the validity of theories and solutions in a wide variety of applications and new developments.
- can critically question results and transfer solutions to other areas of application.

Personal and social skills

The graduates of the Master's degree program Electrical Engineering and Information Technology

- are familiar with independent project work as well as interdisciplinary teamwork
- are able to grasp the results of others, and are able to communicate their own and team results in writing and orally.
- are able to familiarize themselves independently with new and complex subject areas in technical sciences and their methods.
- are able to work scientifically on research-related problems and develop complex assemblies or systems, possess a deeper understanding of applications of electrical engineering and information technology in various fields of work.

- know the limits and dangers involved and apply their knowledge responsibly and for the benefit of society, taking safety and ecological requirements into account. They actively contribute to the opinion-forming process in society with regard to scientific and technical issues and are able to communicate and cooperate with specialists in an interdisciplinary manner.

3.4 Studies and Examination Regulations

https://www.sle.kit.edu/english/vorstudium/master-electrical-engineering-information-technology.php

Master Electrical Engineering and Information Technology (SPO 2025)

<table>
<tr><th>Credits</th><th>1st term</th><th>2nd term</th><th>3rd term</th><th>4th term</th></tr>
<tr><td rowspan="4">10 CP

20 CP

30 CP</td><td>Fundamentals Module 1
(6 LP, 4 SWS)</td><td>Fundamentals Module 3
(6 LP, 4 SWS)</td><td rowspan="3">Electives
(24 CP)</td><td rowspan="4">Master's Thesis
(30 CP)</td></tr>
<tr><td>Fundamentals Module 2
(6 LP, 4 SWS)</td><td>Fundamentals Module 4
(6 LP, 4 SWS)</td></tr>
<tr><td>Fundamentals/ Focus Area Module
(6 CP)</td><td>Lab Course*
(6 CP)</td></tr>
<tr><td>Focus Area Modules
(12 CP)</td><td>Focus Area Modules
(12 CP)</td><td>Interdisciplinary Qualifications
(6 CP)</td></tr>
</table>

*** Lab Course: the credits may vary and are compensated in the Focus Area**

Subjects

Fundamentals	*24 or 30 CP*	*Field of Specialization: in sum 60 CP*
Focus Area	*min. 24 CP*	
Lab Course	*1 course*	
Electives	*24 CP*	
Interdisciplinary Qualifications	*6 CP*	
Master's Thesis	*30 CP*	

Field of Specialization: Automation, Robotics, and Systems Engineering

Fundamentals (24 CP)	CP winter	CP summer	Automation, Control & Robotics (Barth, Hohmann)	Measurement, Sensing & Signal-Processing (Heizmann, Wahls, Zwick)	Systems Engineering (Sax)
English modules					
Cyber-Physical Modeling		6	x	x	x
Optical Engineering and Machine Vision	6			x	
Optimimal Control	6		x		x
Robotic Intelligence: Navigation and Mobile Manipulation		6	x		
Signal Processing Methods	6			x	x
Systems and Software Engineering	6				x
Focus Area (depending on Lab: 21-30 CP)	**CP winter**	**CP summer**			
English modules					
Communication Systems and Protocols		5	x		x
Digital Twin Engineering	4		x		x
Hardware Modeling and Simulation	4				x
Hardware/Software Co-Design	6				x
Hardware Synthesis and Optimization		6			x
Human-Machine-Systems (*expected to start in SoSe 27*)		6	x		
IT/OT-Security Seminar	4		x	x	x
Laser Metrology		3		x	
Multivariable Control Systems	6		x		
Navigation and Localization Techniques		3	x	x	
Nonlinear Control Systems		6	x		
Practical Tools for Control Engineers	4		x		
Radar Systems Engineering	6		x	x	
Research Seminar Cooperative and Haptic Human-Machine Interactions		6	x		
Robotic Intelligence: Perception and Scene Understanding	6		x	x	
Seminar Embedded Systems		4			x
Seminar Industrial Process and Plant Engineering		4	x		x
Signal processing using nonlinear Fourier transforms and Koopman operators		6		x	
Software Engineering		3			x
Spaceborne Radar Remote Sensing		6		x	
Systems Engineering for Automotive Electronics		4			x
German modules					
Informationsfusion	4		x	x	x
Informationstechnik in der Industriellen Automation		3	x		
Praktisches Machine Learning		6		x	
Projektmanagement in der Entwicklung von Produkten für sicherheitskritische Anwendungen	4				x
Prozessanalyse: Modellierung, Data Mining, Machine Learning		3	x	x	
Quantum Machine Learning	3			x	
Sensoren		3		x	
Test eingebetteter Systeme im industriellen Umfeld	4				x
Verteilte ereignisdiskrete Systeme	4		x	x	
Lab Course (exactly 1)	**CP winter**	**CP summer**			
English modules					
Mechatronic Measurement Systems Lab	6			x	
Robotics, Automation, and Systems Control Lab		6	x		
Seamless Engineering	9				x
Signal Processing Lab		6		x	
Student Innovation Lab	9	6	x	x	x
System-on-Chip Lab	6				
German modules					
Praktikum Entwurf Digitaler Systeme		6			x
Praktikum Software Engineering		6			x
Praktisches Machine Learning		6		x	

Field of Specialization: Electrical Power Systems and Electromobility

Fundamentals (24 CP)	CP winter	CP summer	Electro-mobility	Electric Drives	Power Electronic Systems	Renewables	Electro-chemical Systems	Power Systems Engineering & Economics	Super-conductor Engineering
English modules									
Batteries, Fuel Cells, and Electrolysis	6		x			x	x		
Electric Power Transmission & Grid Control		6	x		x	x		x	x
Numerical Methods with Programming Practice		6	x	x	x	x	x	x	x
Optimal Control	6		x	x	x	x	x	x	x
Power Electronics		6	x	x	x	x	x	x	x
Superconductivity for Engineers	6								x
Focus Area (30 CP)	**CP winter**	**CP summer**							
English modules									
Battery and Fuel Cell Systems		3	x		x	x	x		
Communication Systems and Protocols		5	x	x	x	x	x	x	
Components of Power Systems		3						x	
Digital Real Time Simulations for Energy Technologies		3	x	x	x	x	x		
Electric Drives for E-Mobility		4	x	x	x		x		
Electrocatalysis		5					x		
Electrolysis and Electrosynthesis	6					x	x		
Electrolyzers and their Electrodes	3	3	x		x	x	x	x	
Energy Storage and Network Integration	4				x	x	x		
Hardware/Software Co-Design	6		x	x					
Liberalised Power Markets	6					x		x	
Nano- and Quantum Electronics		6							x
Power Electronic Systems	6			x	x	x	x	x	
Pulsed Power Technology and Applications	3								x
Quantum Detectors and Sensors	6								x
Radio-Frequency Electronics	6								x
Renewable Energy-Resources, Technologies and Economics	3					x		x	
Seminar New Power Electronic Systems and Technologies	4	4	x	x	x	x		x	
Seminar on Applied Superconductivity		3							x
Solar Energy (winter term) or Photovoltaik (summer term)	6	6				x	x		
Solar Thermal Energy Systems	4					x			
Superconducting Magnet Technology		4							x
Superconducting Materials (2-term module)	3	3							x
Superconducting Power Systems	4								x
Systems and Software Engineering	6		x	x					
Workshop Finite Element Method in Electromagnetics		3		x					
German modules									
Aufbau- und Verbindungstechnik für leistungselektronische Systeme	3			x	x				
Echtzeitregelung elektrischer Antriebe	6		x	x	x				
Einführung in die Energiewirtschaft		5						x	
Elektronische Systeme und EMV		3		x					
Energiewirtschaft	3							x	
Entwurf Elektrischer Maschinen	5		x	x					x
Grundlagen der Fahrzeugtechnik I	8		x						
Hochspannungsprüftechnik	4				x			x	
Hochspannungstechnik	6				x			x	
Praxis elektrischer Antriebe	4		x	x					
Regelung leistungselektronischer Systeme		6	x	x	x	x	x	x	
Schaltungstechnik in der Industrieelektronik	3			x	x				
Schutz- und Leittechnik in elektrischen Netzen	3				x			x	
Seminar Elektrokatalyse	3	3					x		x
Stromrichtersteuerungstechnik	3			x	x	x			
Lab Course (exactly 1)	**CP winter**	**CP summer**							
English modules									
Electric Drives and Power Electronics Lab		6	x	x	x				
Information Systems in Power Engineering Lab		6						x	
Practical Course: Smart Energy System	6	6				x		x	
Praktikum Batterien und Brennstoffzellen/ Batteries and Fuel Cells Laboratory	6		x			x	x		
Praktikum Nanoelektronik/ Lab Course on Nanoelectronics	6	6							x
Praktikum Solarenergie/ Laboratory Solar Energy	6	6				x			
Praktikum Supraleitende Materialien/ Lab Course on Superconducting Materials	6	6							x
Praktikum Supraleitende Quantenelektronik/ Lab Course on Superconducting Quantum Electronics	6	6							x
Robotic Winding Technology for Superconducting Wires Lab	6								x
German modules									
Energietechnisches Praktikum	6			x	x				

Field of Specialization: Information and Communication Technology

Fundamentals (24 CP)	CP winter	CP summer	Communication Systems (Schmalen, Rost, Zwick, Randel)	Communication Algorithms and Theory (Schmalen, Rost)	Signal and Information Processing (Wahls, Heizmann, Becker)	Microwave Systems (Zwick, Ulusoy, Younis)	Photonic Systems (Randel, Koos, Eichhorn)	Embedded System Integration (Becker)
English modules								
Advanced Communications Engineering	6		x	x	x	x	x	x
Hardware/Software Co-Design	6				x			x
Modern Radio Systems Engineering	6	6	x	x		x		
Numerical Methods with Programming Practice		6	x	x	x	x	x	x
Optical Networks and Systems	6		x	x	(x)		x	
Signal Processing Methods	6		x	x	x	x	x	x
Focus Area (30 CP)	**CP winter**	**CP summer**						
English modules								
Antennas and Beamforming	4		x			x		
Applied Information Theory	6		x	x	x			
Electromagnetics and Numerical Calculation of Fields	4					x	x	
Channel Coding: Algebraic Methods for Communications and Storage		3		x				
Channel Coding: Graph-Based Codes	6		x	x				
Coding of Audiovisual Signals		3		x				
Communication Systems and Protocols		5	x			x		x
Cyber-Physical Modeling		6						
Digital Signal Processing in Optical Communications		6	x		x		x	
Digital Twin Engineering	4							
Field Propagation and Coherence	4						x	
Hardware Modeling and Simulation	4							x
Integrated Photonics	6					x	x	
IT/OT-Security Seminar	4		x					
Machine Learning and Optimization in Communications		4	x	x	x			
Microwaves Measurement Techniques		4				x	x	
Mobile Communications	4		x	x				
Mobile Communications II		3						
Nonlinear Optics		6					x	
Optical Engineering and Machine Vision	6				x			
Optical Transmitters and Receivers	6						x	
Optoelectronic Components		4	x			x	x	
Photonic Integrated Circuits: Design and Applications		6					x	
Radar Systems Engineering	6					x		
Radio Frequency Integrated Circuits and Systems		6						
Radio-Frequency Electronics	6		x			x	x	

Robotic Intelligence: Perception and Scene Understanding	6							
Satellite Communications		3						
Semiconductor Quantum Dots	6						x	
Seminar Embedded Systems	4	4						x
Seminar Industrial Process and Plant Engineering		4						
Signal Processing for Communication Receivers		4	x	x	x	x	x	
Signal Processing with Nonlinear Fourier Transforms and Koopman Operators		6	x	x	x		x	
Spaceborne Radar Remote Sensing		6						
German modules								
Funkempfänger	3			x				
Hardware-Synthese und -Optimierung		6						x
Informationsfusion	4		x	x	x	x		x
Integrierte Intelligente Sensoren		3						x
Mikrosystemtechnik	3							
Praktisches Machine Learning		6			x			
Quantum Machine Learning	3				x			
Seminar: Ausgewählte Kapitel der Nachrichtentechnik	4	4						
Verteilte ereignisdiskrete Systeme	4							
Lab Course (exactly 1)	**CP winter**	**CP summer**						
English modules								
Communications Engineering Lab	6	6	x	x			x	
Digital Hardware Design Lab		6	x				x	x
Mechatronic Measurement Systems Lab	6				x			
Microwave Engineering Lab	6	6	x			x	x	
MMIC Design Laboratory	6	6				x	x	
Photonics and Communications Lab		6	x				x	x
Signal Processing Lab		6	x		x		x	
System-on-Chip Lab	6							x
German modules								
Praktisches Machine Learning		6			x			

Field of Specialization: Microelectronics, Photonics, and Quantum Technologies

Fundamentals (24 CP)	CP winter	CP summer	Microelectronics (Ulusoy, Becker)	Radio Frequency Electronics (Ulusoy, Zwick)	Quantum Technologies (Kempf)	Optics and Photonics (Koos, Lemmer)
English modules						
Hardware Synthesis and Optimization		6	x	x	x	x
Integrated Photonics	6		x	x	x	x
Numerical Methods with Programming Practice		6	x	x	x	x
Quantum Engineering		6	x	x	x	x
Radio-Frequency Electronics	6		x	x	x	x
Focus Area (30 CP)	**CP winter**	**CP summer**				
English modules						
Analog Circuit Design	4		x			
Antennas and Beamforming	4			x	x	
Digital Circuit Design		4	x			
Electromagnetics and Numerical Calculation of Fields	4			x	x	x
Hardware Modeling and Simulation	4		x			
High-Power Microwave Technology	3			x		
Microwaves Measurement Techniques		4		x	x	
Mixed-Signal IC Design		3	x			
Modern Radio Systems Engineering	6	6		x		
Modern VLSI Technologies		5	x		x	
Nano- and Quantum Electronics		6	x		x	x
Nonlinear Optics		6				x
Optical Transmitters and Receivers	6					x
Optical Waveguides and Fibers	4			x		x
Physics, Technology and Applications of Thin Films	4		x		x	x
Practical Amplifier Design		3	x	x		
Quantum Detectors and Sensors	6				x	x
Radio Frequency Integrated Circuits and Systems		6	x	x		x
Semiconductor Quantum Dots	6				x	x
Superconductivity for Engineers	6				x	x
German modules						
Optoelektronische Messtechnik		3				x
Quantum Machine Learning	3				x	
Lab Course (exactly 1)	**CP winter**	**CP summer**				
English modules						
Microwave Engineering Lab	6	6		x		x
MMIC Design Laboratory	6	6	x	x		
Photonics and Communications Lab		6				x
Praktikum Nanoelektronik/ Lab Course on Nanoelectronics	6	6	x		x	
Praktikum Nanotechnologie/ Laboratory Nanotechnology	6	6	x		x	
Praktikum Supraleitende Quantenelektronik/ Lab Course on Superconducting Quantum Electronics	6	6			x	
System-on-Chip Lab	6		x			
German modules						
Praktikum Schaltungsdesign mit FPGA	6	6	x		x	

Exemplary study plan:
MSc Electrical Engineering and Information Technology (SPO 2023)
(Field of Specialization “Automation, Robotics, and Systems Engineering”)

	Module	SWS	CR	Exam	Subject
1. Sem. (winter term)	Optimal Control	2+2	6	written	Fun
	Signal Processing Methods	2+2	6	written	Fun
	IT/OT-Security Seminar	2	4	oral	Foc
	Radar Systems Engineering	3+1	6	written	Foc
	Digital Twin Engineering	2	4	other type	Foc
	Coding of Audiovisual Signals	2	3	oral	Elec
	Sum CR		**29**		
2. Sem. (summer term)	Cyber-Physical Modeling	3+1	6	written	Fun
	Navigation and Localization Techniques	2	3	oral	Foc
	Laser Metrology	2	3	oral	Foc
	Robotics, Automation, and Systems Control Lab	4	6	other type	Lab
	Communication Systems and Protocols	2+1	5	written	Elec
	Deep Learning and Neural Networks	4	6	written	Elec
	Sum CR		**29**		
3. Sem. (winter term)	Systems and Software Engineering	2+2	6	written	Fun
	Digital Twin Engineering	2	4	other type	Foc
	Multivariable Control Systems	3+1	6	written	Foc
	Mechatronic Measurement Systems Lab	4	6	oral	Elec
	Practical Tools for Control Engineers	2	4	other type	Elec
	Interdisciplinary Qualifications		6	ungraded	IQ
	Sum CR		**32**		
4. Sem. (summer term)	Master’s Thesis		30		Ma
	Sum CR		**30**		
	In total		**120**		

Subjects of the Master ETIT	Abbreviation	Extent
Field of Specialization		60 CR
Fundamentals	Fun	min. 24 CR
Focus Area	Foc	min. 24 CR
Lab Course	Lab	Number: 1
Electives (incl. max. 1 Lab)	Elec	24 CR
Interdisciplinary Qualifications	IQ	6 CR
Master’s Thesis	Ma	30 CR

7 Study Program Structure

Mandatory	
Master's Thesis	30 CP
Field of Specialization	60 CP
Electives	24 CP
Interdisciplinary Qualifications	6 CP
Voluntary	
Additional Examinations *This field will not influence the calculated grade of its parent.*	

7.1 Master's Thesis

Credits
30

ID	Name	Language	Term offered	Credits
Mandatory				
M-ETIT-107191	Master's Thesis	**EN**	**WT+ST**	**30 CP**

7.2 Field of Specialization

Credits
60

Elective Area - Field of Specialization (Elective: 1 item)	
Electrical Power Systems and Electromobility	**60 CP**
Information and Communication Technology	**60 CP**
Automation, Robotics, and Systems Engineering	**60 CP**
Microelectronics, Photonics, and Quantum Technologies	**60 CP**

7.2.1 Electrical Power Systems and Electromobility

Part of: Field of Specialization

Credits
60

Columns: module code, title, language, semester, credits

Election notes
All modules are listed with English titles, regardless of the course language.

To see English modules with English titles and German modules with German titles (and descriptions) visit the German website.

ID	Name	Language	Term offered	Credits
Fundamentals (Elective: at least 24 credits)				
M-ETIT-107005	Batteries, Fuel Cells, and Electrolysis	EN	WT	6 CP
M-ETIT-105394	Electric Power Transmission & Grid Control	EN	ST	6 CP
M-MATH-106972	Numerical Methods with Programming Practice	EN	ST	6 CP
M-ETIT-107497	Optimal Control	EN	WT	6 CP
M-ETIT-104567	Power Electronics	EN	ST	6 CP
M-ETIT-105611	Superconductivity for Engineers	EN	WT	6 CP
Focus Area (Elective: at least 24 credits)				
M-MACH-100501	Automotive Engineering I	DE/EN	WT	8 CP
M-ETIT-100400	Basics of Converter Control	DE	ST	3 CP
M-ETIT-107005	Batteries, Fuel Cells, and Electrolysis	EN	WT	6 CP
M-ETIT-107551	Battery and Fuel Cell Systems	EN	ST	3 CP
M-ETIT-100539	Communication Systems and Protocols	EN	ST	5 CP
M-ETIT-106689	Components of Power Systems	EN	ST	3 CP
M-ETIT-105915	Control of Power-Electronic Systems	DE	ST	6 CP
M-ETIT-100515	Design of Electrical Machines	DE	WT	5 CP
M-ETIT-106690	Digital Real Time Simulations for Energy Technologies	EN	ST	3 CP
M-ETIT-106971	Electric Drives for E-Mobility	EN	ST	4 CP
M-ETIT-105394	Electric Power Transmission & Grid Control	EN	ST	6 CP
M-ETIT-105883	Electrocatalysis	EN	ST	5 CP
M-ETIT-107914	Electrolysis and Electrosynthesis neu	EN	WT	6 CP
M-ETIT-107648	Electrolyzers and their Electrodes	EN	WT+ST	3 CP
M-ETIT-100410	Electronics and EMC	DE	ST	3 CP
M-ETIT-101969	Energy Storage and Network Integration	EN	WT	4 CP
M-ETIT-107444	Hardware/Software Co-Design	EN	WT	6 CP
M-ETIT-105060	High-Voltage Technology	DE	WT	6 CP
M-ETIT-100417	High-Voltage Test Technique	DE	WT	4 CP
M-ETIT-100399	Industrial Circuitry	DE	WT	3 CP
M-WIWI-100498	Introduction into Energy Economics	DE	ST	5 CP
M-WIWI-105403	Liberalised Power Markets	EN	WT	6 CP
M-ETIT-105604	Nano- and Quantum Electronics	EN	ST	6 CP
M-MATH-106972	Numerical Methods with Programming Practice	EN	ST	6 CP
M-ETIT-107497	Optimal Control	EN	WT	6 CP
M-ETIT-100513	Photovoltaics	DE	ST	6 CP
M-ETIT-107533	Power Electronic Systems	EN	WT	6 CP
M-ETIT-104567	Power Electronics	EN	ST	6 CP
M-ETIT-106506	Power System Protection and Automation	DE	WT	3 CP
M-ETIT-100413	Power Systems and Economy	DE	WT	3 CP
M-ETIT-100394	Practical Aspects of Electrical Drives	DE	WT	4 CP
M-ETIT-105595	Pulsed Power Technology and Applications (Lecture)	EN	WT	3 CP
M-ETIT-105606	Quantum Detectors and Sensors	EN	WT	6 CP
M-ETIT-106955	Radio-Frequency Electronics	EN	WT	6 CP
M-ETIT-105916	Real Time Control of Electrical Drives	DE	WT	6 CP
M-WIWI-100500	Renewable Energy-Resources, Technologies and Economics	EN	WT	3 CP
M-ETIT-105629	Seminar Electrocatalysis	DE	WT+ST	3 CP
M-ETIT-107588	Seminar New Power Electronic Systems and Technologies	DE/EN	WT+ST	4 CP
M-ETIT-105615	Seminar on Applied Superconductivity	EN	ST	3 CP
M-ETIT-100524	Solar Energy	EN	WT	6 CP
M-MACH-101924	Solar Thermal Energy Systems	EN	WT	4 CP

ID	Name	Language	Term offered	Credits
M-ETIT-106684	Superconducting Magnet Technology	EN	ST	4 CP
M-ETIT-105521	Superconducting Materials	EN	WT	6 CP
M-ETIT-106683	Superconducting Power Systems	EN	WT	4 CP
M-ETIT-105611	Superconductivity for Engineers	EN	WT	6 CP
M-ETIT-100537	Systems and Software Engineering	EN	WT	6 CP
M-ETIT-107147	Workshop Finite Element Method in Electromagnetics	EN	ST	3 CP
Lab Course (Elective: 1 item)				
M-ETIT-100381	Batteries and Fuel Cells Laboratory	DE/EN	WT	6 CP
M-ETIT-107138	Electric Drives and Power Electronics Lab	DE/EN	ST	6 CP
M-ETIT-107159	Information Systems in Power Engineering Lab	DE/EN	ST	6 CP
M-ETIT-100419	Lab Course Electrical Power Engineering	DE	WT	6 CP
M-ETIT-100468	Lab Course on Nanoelectronics	DE/EN	WT+ST	6 CP
M-ETIT-105614	Lab course on superconducting materials	DE/EN	WT+ST	6 CP
M-ETIT-105605	Lab Course on Superconducting Quantum Electronics	DE/EN	WT+ST	6 CP
M-ETIT-102350	Laboratory Solar Energy	DE/EN	WT+ST	6 CP
M-INFO-105955	Practical Course: Smart Energy System	EN	WT+ST	6 CP
M-ETIT-107135	Robotic Winding Technology for Superconducting Wires Lab	DE/EN	WT	6 CP

7.2.2 Information and Communication Technology

Part of: Field of Specialization

Credits
60

Columns: module code, title, language, semester, credits

Election notes
All modules are listed with English titles, regardless of the course language.

To see English modules with English titles and German modules with German titles (and descriptions) visit the German website.

ID	Name	Language	Term offered	Credits
Fundamentals (Elective: at least 24 credits)				
M-ETIT-106815	Advanced Communications Engineering	EN	WT	6 CP
M-ETIT-107444	Hardware/Software Co-Design	EN	WT	6 CP
M-ETIT-100427	Modern Radio Systems Engineering	EN	WT+ST	6 CP
M-MATH-106972	Numerical Methods with Programming Practice	EN	ST	6 CP
M-ETIT-103270	Optical Networks and Systems	EN	WT	6 CP
M-ETIT-106899	Signal Processing Methods	EN	WT	6 CP
Focus Area (Elective: at least 24 credits)				
M-ETIT-106815	Advanced Communications Engineering	EN	WT	6 CP
M-ETIT-106956	Antennas and Beamforming	EN	WT	4 CP
M-ETIT-107333	Applied Information Theory	EN	WT	6 CP
M-ETIT-105616	Channel Coding: Algebraic Methods for Communications and Storage	EN	ST	3 CP
M-ETIT-105617	Channel Coding: Graph-Based Codes	EN	WT	6 CP
M-ETIT-107440	Coding of Audiovisual Signals	EN	ST	3 CP
M-ETIT-100539	Communication Systems and Protocols	EN	ST	5 CP
M-ETIT-106953	Cyber-Physical Modeling	EN	ST	6 CP
M-ETIT-103450	Digital Signal Processing in Optical Communications – with Practical Exercises	EN	ST	6 CP
M-ETIT-106040	Digital Twin Engineering	EN	WT	4 CP
M-ETIT-100361	Distributed Discrete Event Systems	DE	ST	4 CP
M-ETIT-100566	Field Propagation and Coherence	EN	WT	4 CP
M-ETIT-100449	Hardware Modeling and Simulation	EN	WT	4 CP
M-ETIT-106963	Hardware Synthesis and Optimization	EN	ST	6 CP
M-ETIT-100453	Hardware/Software Co-Design	DE	WT	4 CP
M-ETIT-107444	Hardware/Software Co-Design	EN	WT	6 CP
M-ETIT-103264	Information Fusion	DE	WT	4 CP
M-ETIT-100457	Integrated Intelligent Sensors	DE	ST	3 CP
M-ETIT-106789	IT/OT-Security Seminar	EN	WT	4 CP
M-ETIT-104988	Machine Learning and Optimization in Communications	EN	ST	4 CP
M-ETIT-100454	Microsystem Technology	DE	WT	3 CP
M-ETIT-107007	Microwaves Measurement Techniques	EN	ST	4 CP
M-ETIT-105971	Mobile Communications	EN	WT	4 CP
M-ETIT-106244	Mobile Communications II	EN	ST	3 CP
M-ETIT-100427	Modern Radio Systems Engineering	EN	WT+ST	6 CP
M-ETIT-100430	Nonlinear Optics	EN	ST	6 CP
M-MATH-106972	Numerical Methods with Programming Practice	EN	ST	6 CP
M-ETIT-106974	Optical Engineering and Machine Vision	EN	WT	6 CP
M-ETIT-103270	Optical Networks and Systems	EN	WT	6 CP
M-ETIT-100436	Optical Transmitters and Receivers	EN	WT	6 CP
M-ETIT-100509	Optoelectronic Components	EN	ST	4 CP
M-ETIT-105914	Photonic Integrated Circuit Design and Applications	EN	ST	6 CP
M-ETIT-106673	Practical Machine Learning	DE	ST	6 CP
M-ETIT-105889	Quantum Machine Learning	DE	WT	3 CP
M-ETIT-100420	Radar Systems Engineering	EN	WT	6 CP
M-ETIT-105123	Radio Frequency Integrated Circuits and Systems	EN	ST	6 CP
M-ETIT-103241	Radio Receivers	DE	WT	3 CP
M-ETIT-106955	Radio-Frequency Electronics	EN	WT	6 CP
M-ETIT-105272	Satellite Communications	EN	ST	3 CP

ID	Name	Language	Term offered	Credits
M-ETIT-107886	Semiconductor Quantum Dots for Quantum Photonics neu	EN	WT	6 CP
M-ETIT-107764	Seminar Embedded Systems neu	EN	WT+ST	4 CP
M-ETIT-106970	Seminar Industrial Process and Plant Engineering	EN	ST	4 CP
M-ETIT-100441	Seminar Selected Topics in Communication	DE	WT+ST	4 CP
M-ETIT-107871	Signal Processing for Communication Receivers neu *First usage possible from Apr 01, 2027.*	EN	ST	6 CP
M-ETIT-100443	Signal Processing in Communications *First usage possible until Mar 31, 2027.*	DE	ST	6 CP
M-ETIT-106899	Signal Processing Methods	EN	WT	6 CP
M-ETIT-106675	Signal Processing with Nonlinear Fourier Transforms and Koopman Operators	EN	ST	6 CP
M-ETIT-103042	Spaceborne Radar Remote Sensing	EN	ST	6 CP
Lab Course (Elective: 1 item)				
M-ETIT-107136	Communications Engineering Lab	DE/EN	WT+ST	6 CP
M-ETIT-102266	Digital Hardware Design Laboratory	EN	ST	6 CP
M-ETIT-107294	Mechatronic Measurement Systems Lab	DE/EN	WT	6 CP
M-ETIT-106973	Microwave Engineering Lab	EN	WT+ST	6 CP
M-ETIT-105464	MMIC Design Laboratory	EN	WT+ST	6 CP
M-ETIT-104485	Photonics and Communications Lab	EN	ST	6 CP
M-ETIT-106673	Practical Machine Learning	DE	ST	6 CP
M-ETIT-106633	Signal Processing Lab	EN	ST	6 CP
M-ETIT-107763	System-on-Chip Lab neu	EN	WT	6 CP

7.2.3 Automation, Robotics, and Systems Engineering

Part of: Field of Specialization

Credits
60

Columns: module code, title, language, semester, credits

Election notes
All modules are listed with English titles, regardless of the course language.

To see English modules with English titles and German modules with German titles (and descriptions) visit the German website.

ID	Name	Language	Term offered	Credits
Fundamentals (Elective: at least 24 credits)				
M-ETIT-106953	Cyber-Physical Modeling	EN	ST	6 CP
M-ETIT-106974	Optical Engineering and Machine Vision	EN	WT	6 CP
M-ETIT-107497	Optimal Control	EN	WT	6 CP
M-MACH-107621	Robotic Intelligence: Navigation and Mobile Manipulation neu	EN	ST	6 CP
M-ETIT-106899	Signal Processing Methods	EN	WT	6 CP
M-ETIT-100537	Systems and Software Engineering	EN	WT	6 CP
Focus Area (Elective: at least 24 credits)				
M-ETIT-100539	Communication Systems and Protocols	EN	ST	5 CP
M-ETIT-106953	Cyber-Physical Modeling	EN	ST	6 CP
M-ETIT-106040	Digital Twin Engineering	EN	WT	4 CP
M-ETIT-100361	Distributed Discrete Event Systems	DE	ST	4 CP
M-ETIT-100449	Hardware Modeling and Simulation	EN	WT	4 CP
M-ETIT-106963	Hardware Synthesis and Optimization	EN	ST	6 CP
M-ETIT-107444	Hardware/Software Co-Design	EN	WT	6 CP
M-ETIT-103264	Information Fusion	DE	WT	4 CP
M-ETIT-100367	Information Technology in Industrial Automation Systems	DE	ST	3 CP
M-ETIT-106789	IT/OT-Security Seminar	EN	WT	4 CP
M-ETIT-100434	Laser Metrology	EN	ST	3 CP
M-ETIT-107515	Multivariable Control Systems	EN	WT	6 CP
M-ETIT-105881	Navigation and Localization Techniques	EN	ST	3 CP
M-ETIT-107644	Nonlinear Control Systems	EN	ST	6 CP
M-ETIT-106974	Optical Engineering and Machine Vision	EN	WT	6 CP
M-ETIT-107497	Optimal Control	EN	WT	6 CP
M-ETIT-106673	Practical Machine Learning	DE	ST	6 CP
M-ETIT-106780	Practical Tools for Control Engineers	EN	WT	4 CP
M-ETIT-105594	Process Analysis: Modeling, Data Mining, Machine Learning	DE	ST	3 CP
M-ETIT-104475	Project Management in the Development of Products for Safety-Critical Applications	DE	WT	4 CP
M-ETIT-105889	Quantum Machine Learning	DE	WT	3 CP
M-ETIT-100420	Radar Systems Engineering	EN	WT	6 CP
M-ETIT-107596	Research Seminar Cooperative and Haptic Human-Machine Interactions neu	EN	ST	6 CP
M-MACH-107621	Robotic Intelligence: Navigation and Mobile Manipulation neu	EN	ST	6 CP
M-MACH-107558	Robotic Intelligence: Perception and Scene Understanding	EN	WT	6 CP
M-ETIT-107764	Seminar Embedded Systems neu	EN	WT+ST	4 CP
M-ETIT-106970	Seminar Industrial Process and Plant Engineering	EN	ST	4 CP
M-ETIT-100378	Sensors	DE	ST	3 CP
M-ETIT-106899	Signal Processing Methods	EN	WT	6 CP
M-ETIT-106675	Signal Processing with Nonlinear Fourier Transforms and Koopman Operators	EN	ST	6 CP
M-ETIT-100450	Software Engineering	DE	ST	3 CP
M-ETIT-103042	Spaceborne Radar Remote Sensing	EN	ST	6 CP
M-ETIT-100537	Systems and Software Engineering	EN	WT	6 CP
M-ETIT-100462	Systems Engineering for Automotive Electronics	EN	ST	4 CP
M-ETIT-100546	Test of Embedded Systems in Industrial Contexts	DE	WT	4 CP
Lab Course (Elective: 1 item)				
M-ETIT-102264	Digital Hardware Design Laboratory	DE	ST	6 CP
M-ETIT-100460	Laboratory in Software Engineering	DE	ST	6 CP
M-ETIT-107294	Mechatronic Measurement Systems Lab	DE/EN	WT	6 CP

ID	Name	Language	Term offered	Credits
M-ETIT-106673	Practical Machine Learning	DE	ST	6 CP
M-ETIT-107524	Robotics, Automation, and Systems Control Lab	EN	ST	6 CP
M-MACH-105725	Seamless Engineering	EN	WT	9 CP
M-ETIT-106633	Signal Processing Lab	EN	ST	6 CP
M-ETIT-105073	Student Innovation Lab	EN	WT+ST	15 CP
M-ETIT-107763	System-on-Chip Lab neu	EN	WT	6 CP

7.2.4 Microelectronics, Photonics, and Quantum Technologies

Part of: Field of Specialization

Credits 60

Columns: module code, title, language, semester, credits

Election notes
All modules are listed with English titles, regardless of the course language.

To see English modules with English titles and German modules with German titles (and descriptions) visit the German website.

ID	Name	Language	Term offered	Credits
Fundamentals (Elective: at least 24 credits)				
M-ETIT-106963	Hardware Synthesis and Optimization	**EN**	**ST**	**6 CP**
M-MATH-106972	Numerical Methods with Programming Practice	**EN**	**ST**	**6 CP**
M-ETIT-106954	Quantum Engineering	**EN**	**ST**	**6 CP**
M-ETIT-106955	Radio-Frequency Electronics	**EN**	**WT**	**6 CP**
M-ETIT-107344	Integrated Photonics	**EN**	**WT**	**6 CP**
Focus Area (Elective: at least 24 credits)				
M-ETIT-107364	Analog Circuit Design	**EN**	**WT**	**4 CP**
M-ETIT-106956	Antennas and Beamforming	**EN**	**WT**	**4 CP**
M-ETIT-107365	Digital Circuit Design	**EN**	**ST**	**4 CP**
M-ETIT-100449	Hardware Modeling and Simulation	**EN**	**WT**	**4 CP**
M-ETIT-106963	Hardware Synthesis and Optimization	**EN**	**ST**	**6 CP**
M-ETIT-107363	High-Power Microwave Technology	**EN**	**WT**	**3 CP**
M-ETIT-107007	Microwaves Measurement Techniques	**EN**	**ST**	**4 CP**
M-ETIT-105893	Mixed-Signal IC Design	**EN**	**ST**	**3 CP**
M-ETIT-100427	Modern Radio Systems Engineering	**EN**	**WT+ST**	**6 CP**
M-ETIT-106921	Modern VLSI Technologies	**EN**	**ST**	**6 CP**
M-ETIT-105604	Nano- and Quantum Electronics	**EN**	**ST**	**6 CP**
M-ETIT-100430	Nonlinear Optics	**EN**	**ST**	**6 CP**
M-MATH-106972	Numerical Methods with Programming Practice	**EN**	**ST**	**6 CP**
M-ETIT-100436	Optical Transmitters and Receivers	**EN**	**WT**	**6 CP**
M-ETIT-100484	Optoelectronic Measurement Engineering	**DE**	**ST**	**3 CP**
M-ETIT-105608	Physics, Technology and Applications of Thin Films	**EN**	**WT**	**4 CP**
M-ETIT-107156	Practical Amplifier Design	**EN**	**ST**	**3 CP**
M-ETIT-105606	Quantum Detectors and Sensors	**EN**	**WT**	**6 CP**
M-ETIT-106954	Quantum Engineering	**EN**	**ST**	**6 CP**
M-ETIT-105889	Quantum Machine Learning	**DE**	**WT**	**3 CP**
M-ETIT-105123	Radio Frequency Integrated Circuits and Systems	**EN**	**ST**	**6 CP**
M-ETIT-106955	Radio-Frequency Electronics	**EN**	**WT**	**6 CP**
M-ETIT-107886	Semiconductor Quantum Dots for Quantum Photonics neu	**EN**	**WT**	**6 CP**
M-ETIT-105611	Superconductivity for Engineers	**EN**	**WT**	**6 CP**
Lab Course (Elective: 1 item)				
M-ETIT-100468	Lab Course on Nanoelectronics	**DE/EN**	**WT+ST**	**6 CP**
M-ETIT-105605	Lab Course on Superconducting Quantum Electronics	**DE/EN**	**WT+ST**	**6 CP**
M-ETIT-100470	Laboratory FPGA Based Circuit Design	**DE**	**WT+ST**	**6 CP**
M-ETIT-100478	Laboratory Nanotechnology	**DE/EN**	**WT+ST**	**6 CP**
M-ETIT-106973	Microwave Engineering Lab	**EN**	**WT+ST**	**6 CP**
M-ETIT-105464	MMIC Design Laboratory	**EN**	**WT+ST**	**6 CP**
M-ETIT-104485	Photonics and Communications Lab	**EN**	**ST**	**6 CP**
M-ETIT-107763	System-on-Chip Lab neu	**EN**	**WT**	**6 CP**

7.3 Electives

Credits
24

Columns: module code, title, language, semester, credits

Election notes
All modules are listed with English titles, regardless of the course language.

To see English modules with English titles and German modules with German titles (and descriptions) visit the German website.

ID	Name	Language	Term offered	Credits
Elective Modules (Elective: at most 24 credits)				
M-ETIT-103802	Adaptive Optics	EN	WT	3 CP
M-ETIT-106815	Advanced Communications Engineering	EN	WT	6 CP
M-ETIT-100507	Current Topics of Solar Energy	DE	WT	3 CP
M-ETIT-107364	Analog Circuit Design	EN	WT	4 CP
M-ETIT-106956	Antennas and Beamforming	EN	WT	4 CP
M-CIWVT-108002	Practical Applications and Modelling of Rarefied Gas Flows neu	DE	WT	6 CP
M-ETIT-107333	Applied Information Theory	EN	WT	6 CP
M-INFO-105338	Authentication and Encryption	DE	ST	4 CP
M-MACH-107148	Automotive Vision	EN	ST	6 CP
M-ETIT-107005	Batteries, Fuel Cells, and Electrolysis	EN	WT	6 CP
M-ETIT-107551	Battery and Fuel Cell Systems	EN	ST	3 CP
M-MACH-106903	Biologically Inspired Robots	EN	ST	3 CP
M-MACH-100489	BioMEMS - Microsystems Technologies for Life Sciences and Medicine I	DE	WT	4 CP
M-MACH-100490	BioMEMS - Microsystems Technologies for Life Sciences and Medicine II	DE	ST	4 CP
M-MACH-100491	BioMEMS - Microsystems Technologies for Life Sciences and Medicine III	DE	ST	4 CP
M-ETIT-101834	Business Innovation in Optics and Photonics	EN	WT	4 CP
M-ETIT-105616	Channel Coding: Algebraic Methods for Communications and Storage	EN	ST	3 CP
M-ETIT-105617	Channel Coding: Graph-Based Codes	EN	WT	6 CP
M-MACH-107636	Circular Factory	EN	ST	8 CP
M-ETIT-107798	Clinical Translation and Evaluation of Biomedical Technologies neu	EN	WT	3 CP
M-ETIT-107440	Coding of Audiovisual Signals	EN	ST	3 CP
M-ETIT-100539	Communication Systems and Protocols	EN	ST	5 CP
M-ETIT-106689	Components of Power Systems	EN	ST	3 CP
M-ETIT-107375	Compressed Sensing and Approximate Message Passing: Theory and Applications	EN	WT	3 CP
M-INFO-106190	Computational Imaging	EN	WT	5 CP
M-MACH-105296	Computational Intelligence	DE	WT	4 CP
M-CIWVT-104356	Cryogenic Engineering	EN	WT	6 CP
M-INFO-107435	Cryptographic Protocols II: Universal Composability	EN	WT	3 CP
M-ETIT-106953	Cyber-Physical Modeling	EN	ST	6 CP
M-INFO-107197	Deep Learning and Neural Networks	EN	ST	6 CP
M-INFO-107843	Deep Learning for Computer Vision: Advanced Topics neu	EN	WT	3 CP
M-INFO-107695	Deep Learning for Computer Vision: Advanced Topics	DE/EN	WT	3 CP
M-INFO-107230	Design and Architectures of Embedded Systems (ESII)	EN	WT	3 CP
M-ETIT-100541	Detectores for Applications in Space and Astronomy	DE	WT	3 CP
M-ETIT-105618	The Energy Transition of the Electrical Transmission Grid	DE	ST	3 CP
M-ETIT-107365	Digital Circuit Design	EN	ST	4 CP
M-ETIT-106690	Digital Real Time Simulations for Energy Technologies	EN	ST	3 CP
M-ETIT-103450	Digital Signal Processing in Optical Communications – with Practical Exercises	EN	ST	6 CP
M-ETIT-106040	Digital Twin Engineering	EN	WT	4 CP
M-ETIT-105916	Real Time Control of Electrical Drives	DE	WT	6 CP
M-INFO-100736	Introduction to Video Analysis	DE	ST	3 CP
M-WIWI-100498	Introduction into Energy Economics	DE	ST	5 CP
M-ETIT-106597	Single-Crystal Growth – Crystal Growth Methods and Applications of Crystals for Electronic and Optical Components	DE	WT	3 CP

ID	Name	Language	Term offered	Credits
M-ETIT-106971	Electric Drives for E-Mobility neu	EN	ST	4 CP
M-ETIT-101917	Electric Power Generation and Power Grid	EN	ST	3 CP
M-ETIT-105394	Electric Power Transmission & Grid Control	EN	ST	6 CP
M-ETIT-105883	Electrocatalysis	EN	ST	5 CP
M-ETIT-107914	Electrolysis and Electrosynthesis neu	EN	WT	6 CP
M-ETIT-107648	Electrolyzers and their Electrodes	EN	WT+ST	3 CP
M-ETIT-100386	Electromagnetics and Numerical Calculation of Fields	EN	WT	4 CP
M-ETIT-100410	Electronics and EMC	DE	ST	3 CP
M-CIWVT-104288	Biomass Based Energy Carriers	DE	WT	6 CP
M-ETIT-100413	Power Systems and Economy	DE	WT	3 CP
M-ETIT-101969	Energy Storage and Network Integration	EN	WT	4 CP
M-ETIT-100515	Design of Electrical Machines	DE	WT	5 CP
M-ETIT-105701	Microwave Module Design	DE	WT	3 CP
M-ETIT-101919	Fabrication and Characterisation of Optoelectronic Devices	EN	ST	3 CP
M-MACH-105288	Handling Characteristics of Motor Vehicles I	DE	WT	4 CP
M-ETIT-100566	Field Propagation and Coherence	EN	WT	4 CP
M-ETIT-103241	Radio Receivers	DE	WT	3 CP
M-INFO-100725	Human Brain and Central Nervous System: Anatomy, Information Transfer, Signal Processing, Neurophysiology and Therapy	DE	WT+ST	3 CP
M-BGU-106347	Geodetic Space Techniques for Engineers	DE	WT	4 CP
M-MACH-100501	Automotive Engineering I	DE/EN	WT	8 CP
M-MACH-100502	Automotive Engineering II	DE	ST	4 CP
M-ETIT-100449	Hardware Modeling and Simulation	EN	WT	4 CP
M-ETIT-106963	Hardware Synthesis and Optimization	EN	ST	6 CP
M-ETIT-107444	Hardware/Software Co-Design	EN	WT	6 CP
M-ETIT-107363	High-Power Microwave Technology	EN	WT	3 CP
M-ETIT-100417	High-Voltage Test Technique	DE	WT	4 CP
M-ETIT-105060	High-Voltage Technology	DE	WT	6 CP
M-ETIT-103264	Information Fusion	DE	WT	4 CP
M-ETIT-100367	Information Technology in Industrial Automation Systems	DE	ST	3 CP
M-ETIT-107344	Integrated Photonics	EN	WT	6 CP
M-ETIT-100457	Integrated Intelligent Sensors	DE	ST	3 CP
M-ETIT-105461	Introduction to Automotive and Industrial Lidar Technology	EN	WT	3 CP
M-MACH-107657	Introduction to Microsystem Technology I	EN	WT	4 CP
M-ETIT-106789	IT/OT-Security Seminar	EN	WT	4 CP
M-CIWVT-104354	Refrigeration B - Foundations of Industrial Gas Processing	DE	ST	6 CP
M-INFO-105631	Cryptographic Protocols	DE	ST	5 CP
M-ETIT-100434	Laser Metrology	EN	ST	3 CP
M-ETIT-100435	Laser Physics	EN	WT	4 CP
M-WIWI-105403	Liberalised Power Markets	EN	WT	6 CP
M-ETIT-100485	Lighting Engineering	DE	WT	4 CP
M-INFO-107663	Localization of Mobile Agents	EN	ST	6 CP
M-INFO-100807	Low Power Design	EN	ST	3 CP
M-ETIT-104988	Machine Learning and Optimization in Communications	EN	ST	4 CP
M-WIWI-106604	Machine Learning and Optimization in Energy Systems	EN	WT	4 CP
M-MACH-101923	Machine Vision	EN	WT	8 CP
M-WIWI-105003	Machine Learning 1	DE	WT	5 CP
M-WIWI-105006	Machine Learning 2	DE	ST	5 CP

ID	Name	Language	Term offered	Credits
M-ETIT-105982	Measurement Technology *First usage possible until Mar 31, 2027.*	EN	WT	5 CP
M-ETIT-106672	Medical Image Processing for Guidance and Navigation	EN	WT	9 CP
M-MACH-102714	Microenergy Technologies	EN	ST	4 CP
M-ETIT-107007	Microwaves Measurement Techniques	EN	ST	4 CP
M-MACH-100487	Microactuators	DE	ST	4 CP
M-ETIT-100454	Microsystem Technology	DE	WT	3 CP
M-ETIT-105893	Mixed-Signal IC Design	EN	ST	3 CP
M-ETIT-105971	Mobile Communications	EN	WT	4 CP
M-ETIT-106244	Mobile Communications II	EN	ST	3 CP
M-ETIT-106782	Modeling Physiological Systems	EN	ST	6 CP
M-ETIT-100508	Modelling and Simulation of Electrochemical Systems	DE	ST	3 CP
M-ETIT-100427	Modern Radio Systems Engineering	EN	WT+ST	6 CP
M-ETIT-106921	Modern VLSI Technologies	EN	ST	6 CP
M-INFO-107676	Multimodal Artificial Intelligence	EN	ST	6 CP
M-ETIT-107515	Multivariable Control Systems	EN	WT	6 CP
M-INFO-100825	Pattern Recognition *First usage possible until Sep 30, 2027.*	DE	ST	6 CP
M-ETIT-105604	Nano- and Quantum Electronics	EN	ST	6 CP
M-ETIT-105881	Navigation and Localization Techniques	EN	ST	3 CP
M-CIWVT-105890	NMR Methods for Product and Process Analysis	DE/EN	WT	4 CP
M-ETIT-107644	Nonlinear Control Systems	EN	ST	6 CP
M-ETIT-100430	Nonlinear Optics	EN	ST	6 CP
M-MATH-106972	Numerical Methods with Programming Practice	EN	ST	6 CP
M-ETIT-102311	Numerical Methods for Partial Differential Equations	DE	ST	4 CP
M-ETIT-106974	Optical Engineering and Machine Vision	EN	WT	6 CP
M-ETIT-103270	Optical Networks and Systems	EN	WT	6 CP
M-ETIT-103252	Optical Systems in Medicine and Life Science	EN	ST	3 CP
M-ETIT-100436	Optical Transmitters and Receivers	EN	WT	6 CP
M-ETIT-107497	Optimal Control	EN	WT	6 CP
M-INFO-107229	Optimization and Synthesis of Embedded Systems (ESI)	EN	WT	3 CP
M-ETIT-100486	Optical Technologies for Automotive Applications	DE	ST	3 CP
M-ETIT-100509	Optoelectronic Components	EN	ST	4 CP
M-ETIT-100484	Optoelectronic Measurement Engineering	DE	ST	3 CP
M-ETIT-107455	Organic and Flexible Electronics	EN	WT	3 CP
M-ETIT-100519	Photometry and Radiometry	DE	WT	3 CP
M-ETIT-105914	Photonic Integrated Circuit Design and Applications	EN	ST	6 CP
M-ETIT-100513	Photovoltaics	DE	ST	6 CP
M-CIWVT-103068	Physical Foundations of Cryogenics	EN	ST	6 CP
M-ETIT-105608	Physics, Technology and Applications of Thin Films	EN	WT	4 CP
M-ETIT-107533	Power Electronic Systems	EN	WT	6 CP
M-ETIT-104567	Power Electronics	EN	ST	6 CP
M-ETIT-107888	Power Electronics for the Energy Transition neu	EN	WT	3 CP
M-ETIT-107156	Practical Amplifier Design	EN	ST	3 CP
M-ETIT-106780	Practical Tools for Control Engineers	EN	WT	4 CP
M-ETIT-106673	Practical Machine Learning	DE	ST	6 CP
M-ETIT-100394	Practical Aspects of Electrical Drives	DE	WT	4 CP
M-ETIT-107343	Printed and Thin-Film Electronics	EN	WT	3 CP
M-ETIT-104475	Project Management in the Development of Products for Safety-Critical Applications	DE	WT	4 CP

ID	Name	Language	Term offered	Credits
M-ETIT-105594	Process Analysis: Modeling, Data Mining, Machine Learning	DE	ST	3 CP
M-ETIT-105596	Pulsed Power Technology and Applications (Tutorial)	EN	ST	5 CP
M-ETIT-105595	Pulsed Power Technology and Applications (Lecture)	EN	WT	3 CP
M-ETIT-105606	Quantum Detectors and Sensors	EN	WT	6 CP
M-ETIT-106954	Quantum Engineering	EN	ST	6 CP
M-ETIT-105889	Quantum Machine Learning	DE	WT	3 CP
M-ETIT-100420	Radar Systems Engineering	EN	WT	6 CP
M-ETIT-105123	Radio Frequency Integrated Circuits and Systems	EN	ST	6 CP
M-ETIT-106955	Radio-Frequency Electronics	EN	WT	6 CP
M-ETIT-105915	Control of Power-Electronic Systems	DE	ST	6 CP
M-WIWI-100500	Renewable Energy-Resources, Technologies and Economics	EN	WT	3 CP
M-ETIT-107596	Research Seminar Cooperative and Haptic Human-Machine Interactions	EN	ST	6 CP
M-MACH-107621	Robotic Intelligence: Navigation and Mobile Manipulation neu	EN	ST	6 CP
M-MACH-107558	Robotic Intelligence: Perception and Scene Understanding	EN	WT	6 CP
M-INFO-107123	Robotics II - Humanoid Robotics	EN	ST	3 CP
M-INFO-107130	Robotics III - Sensors and Perception in Robotics	EN	ST	3 CP
M-ETIT-105272	Satellite Communications	EN	ST	3 CP
M-ETIT-107661	Scanning Probe Lithography for Biomedical Applications	EN	ST	3 CP
M-ETIT-100399	Industrial Circuitry	DE	WT	3 CP
M-ETIT-106506	Power System Protection and Automation	DE	WT	3 CP
M-ETIT-107886	Semiconductor Quantum Dots for Quantum Photonics neu	EN	WT	6 CP
M-ETIT-106674	Seminar Advanced Concepts for Flexible and Soft Optoelectronic Devices and Sensors	EN	ST	3 CP
M-ETIT-100441	Seminar Selected Topics in Communication	DE	WT+ST	4 CP
M-ETIT-105321	Seminar Battery II	DE/EN	WT+ST	3 CP
M-ETIT-105322	Seminar Fuel Cell II	DE/EN	WT+ST	3 CP
M-ETIT-105629	Seminar Electrocatalysis	DE	WT+ST	3 CP
M-ETIT-107764	Seminar Embedded Systems neu	EN	WT+ST	4 CP
M-ETIT-106970	Seminar Industrial Process and Plant Engineering neu	EN	ST	4 CP
M-ETIT-107588	Seminar New Power Electronic Systems and Technologies	DE/EN	WT+ST	4 CP
M-ETIT-103447	Seminar Novel Concepts for Solar Energy Harvesting	EN	ST	3 CP
M-ETIT-105615	Seminar on Applied Superconductivity	EN	ST	3 CP
M-ETIT-107839	Seminar Radiation Protection in Medicine neu	EN	WT	3 CP
M-INFO-105884	Seminar: Digital Accessibility and Assistive Technologies	DE/EN	WT	3 CP
M-ETIT-100378	Sensors	DE	ST	3 CP
M-INFO-104877	Safe Human-Robot-Collaboration	DE	WT	3 CP
M-ETIT-107871	Signal Processing for Communication Receivers neu *First usage possible from Apr 01, 2027.*	EN	ST	6 CP
M-ETIT-106899	Signal Processing Methods	EN	WT	6 CP
M-ETIT-106675	Signal Processing with Nonlinear Fourier Transforms and Koopman Operators	EN	ST	6 CP
M-ETIT-100443	Signal Processing in Communications *First usage possible until Mar 31, 2027.*	DE	ST	6 CP
M-ETIT-101971	Single-Photon Detectors	EN	WT	4 CP
M-ETIT-100450	Software Engineering	DE	ST	3 CP
M-ETIT-100524	Solar Energy	EN	WT	6 CP
M-ETIT-100545	Space-Born Microwave Radiometry - Advanced Methods and Applications	EN	ST	3 CP
M-ETIT-103042	Spaceborne Radar Remote Sensing	EN	ST	6 CP
M-MACH-105348	Control Technology	DE	ST	4 CP

ID	Name	Language	Term offered	Credits
M-INFO-107847	Stochastic Information Processing neu	EN	WT	6 CP
M-ETIT-100400	Basics of Converter Control	DE	ST	3 CP
M-ETIT-106684	Superconducting Magnet Technology	EN	ST	4 CP
M-ETIT-105521	Superconducting Materials	EN	WT	6 CP
M-ETIT-105609	Superconducting Nanowire Detectors	EN	ST	4 CP
M-ETIT-106683	Superconducting Power Systems	EN	WT	4 CP
M-ETIT-105611	Superconductivity for Engineers	EN	WT	6 CP
M-ETIT-106026	System Integration and Communication Structures in Industry 4.0 and IoT	DE	WT	3 CP
M-ETIT-100537	Systems and Software Engineering	EN	WT	6 CP
M-ETIT-100462	Systems Engineering for Automotive Electronics	EN	ST	4 CP
M-ETIT-105465	Team Project: Sensors and Electronics	DE	WT+ST	3 CP
M-ETIT-101835	Technical Acoustic	DE	WT	3 CP
M-ETIT-100538	Technical Optics	DE	WT	5 CP
M-INFO-107243	Telematics *First usage possible until Mar 31, 2027.*	EN	WT	6 CP
M-ETIT-100546	Test of Embedded Systems in Industrial Contexts	DE	WT	4 CP
M-INFO-105584	Theoretical Foundations of Cryptography	DE	WT	6 CP
M-MACH-102388	Thermal Solar Energy	EN	WT	4 CP
M-ETIT-100560	Ultrasound Imaging	DE	ST	3 CP
M-ETIT-104493	Verified Numerical Methods	DE	WT	4 CP
M-ETIT-100361	Distributed Discrete Event Systems	DE	ST	4 CP
M-ETIT-100497	Visual Perception in the Automobile	DE	ST	3 CP
M-CIWVT-106680	Water – Energy – Environment Nexus in a Circular Economy: Research Proposal Preparation	EN	ST	5 CP
M-INFO-107113	Wearable Robotic Technologies	EN	ST	4 CP
M-MACH-105107	Machine Tools and Industrial Handling neu	DE	WT	8 CP
M-MACH-105369	Scientific Computing for Engineers	DE	WT	4 CP
M-ETIT-107147	Workshop Finite Element Method in Electromagnetics	EN	ST	3 CP
Lab Course (Elective: between 0 and 1 items)				
M-MACH-107806	Autonomous Driving in Practice neu	EN	WT	6 CP
M-ETIT-100381	Batteries and Fuel Cells Laboratory	DE/EN	WT	6 CP
M-ETIT-107761	Communication Networks and Systems Lab neu	EN	WT+ST	6 CP
M-ETIT-107136	Communications Engineering Lab	DE/EN	WT+ST	6 CP
M-ETIT-102266	Digital Hardware Design Laboratory	EN	ST	6 CP
M-ETIT-102264	Digital Hardware Design Laboratory	DE	ST	6 CP
M-ETIT-107138	Electric Drives and Power Electronics Lab	DE/EN	ST	6 CP
M-ETIT-107159	Information Systems in Power Engineering Lab	DE/EN	ST	6 CP
M-ETIT-100419	Lab Course Electrical Power Engineering	DE	WT	6 CP
M-ETIT-100468	Lab Course on Nanoelectronics	DE/EN	WT+ST	6 CP
M-ETIT-105614	Lab course on superconducting materials	DE/EN	WT+ST	6 CP
M-ETIT-105605	Lab Course on Superconducting Quantum Electronics	DE/EN	WT+ST	6 CP
M-ETIT-100470	Laboratory FPGA Based Circuit Design	DE	WT+ST	6 CP
M-ETIT-100460	Laboratory in Software Engineering	DE	ST	6 CP
M-ETIT-102356	Laboratory Lighting Technology	DE/EN	WT+ST	6 CP
M-MACH-102699	Laboratory Mechatronics	DE	ST	4 CP
M-ETIT-100478	Laboratory Nanotechnology	DE/EN	WT+ST	6 CP
M-ETIT-102350	Laboratory Solar Energy	DE/EN	WT+ST	6 CP
M-ETIT-107294	Mechatronic Measurement Systems Lab	DE/EN	WT	6 CP
M-ETIT-106973	Microwave Engineering Lab	EN	WT+ST	6 CP

ID	Name	Language	Term offered	Credits
M-ETIT-105464	MMIC Design Laboratory	EN	WT+ST	6 CP
M-ETIT-100464	Optical Design Lab	EN	ST	6 CP
M-ETIT-104485	Photonics and Communications Lab	EN	ST	6 CP
M-INFO-105955	Practical Course: Smart Energy System	EN	WT+ST	6 CP
M-ETIT-106673	Practical Machine Learning	DE	ST	6 CP
M-ETIT-106464	Printed Flexible Electronics Lab	EN	WT+ST	6 CP
M-ETIT-107135	Robotic Winding Technology for Superconducting Wires Lab	DE/EN	WT	6 CP
M-ETIT-107524	Robotics, Automation, and Systems Control Lab	EN	ST	6 CP
M-MACH-105725	Seamless Engineering	EN	WT	9 CP
M-ETIT-106633	Signal Processing Lab	EN	ST	6 CP
M-ETIT-105073	Student Innovation Lab	EN	WT+ST	15 CP
M-ETIT-107763	System-on-Chip Lab neu	EN	WT	6 CP

7.4 Interdisciplinary Qualifications

Credits
6

ID	Name	Language	Term offered	Credits
Mandatory				
M-ETIT-105803	Interdisciplinary Qualifications	DE/EN	WT+ST	6 CP

7.5 Additional Examinations

Election notes
§15 Additional Examinations, SPO
(1) Students may also earn Additional Examinations (up to a maximum of 30 LP) from the full range of courses offered by KIT. §3 and §4 of the examination regulations remain unaffected by this provision. These Additional Examinations are not included in the calculation of the overall and module grades. Credit points not considered in the determination of the module grade are listed as Additional Examinations on the Transcript of Records and designated as such. Upon the student's request, the Additional Examinations are included in the Master's diploma and designated as such. Additional Examinations are listed with the grades specified in §7.
(2) Students must declare an examination as an Additional Examination at the time of registration for an examination in a module. Upon the student's request, the assignment of the module may be changed at a later date.

ID	Name	Language	Term offered	Credits
Additional Examinations (Elective: at most 30 credits)				
M-FORUM-106753	Supplementary Studies on Science, Technology and Society	DE	WT+ST	16 CP
M-ETIT-107928	Master's Project neu	DE/EN	WT+ST	6 CP

8 Master‘s thesis registration

8.1 Procedure for admission/registration of the Master’s thesis

Consultation with an examiner of the KIT-Department of Electrical Engineering and Information Technology regarding the topic.

For the preparation of an external Master's thesis, the examiner must sign a confirmation of supervision, which the students receive from the Master's examination board after submission of the approved individual study plan and fulfillment of the requirements.

After the examiner has installed the thesis in the CAMPUS system, the student will receive an automatic e-mail requesting him/her to register for the thesis in CAMPUS. After that the student registers for the thesis in CAMPUS.

The examiner enters the date of assignment and makes the "Examination Master Thesis" visible to the student ("Publication").

The examination board checks whether the prerequisites (see above) have been met. The approved individual study plan must be available at this step at the latest. A later change of the individual study plan is not possible. If all prerequisites are met, the student will receive an e-mail that the thesis has been approved. Work on the thesis before this approval is not permitted.

The student prepares the Master’s thesis and gives the presentation within the processing time. The time of submission will be recorded.

The examiner evaluates the work and enters and releases the grade in the CAMPUS system. The examination board makes the grade visible to the student ("publication"). The student receives an e-mail that the grade of the thesis has been entered in the system.

9 Recognition of external study and examination achievements

The recognition of external achievements taken at other universities is generally possible. Especially a stay abroad is supported by the KIT-department. There are some rules to follow, that are described below.

9.1 Basic regulations

The basic rules for the recognition of external achievements (credits and grades) can be found in the study and examination regulations:

- Bachelor ETIT SPO 2018 of 28.09.2018, §19
- Bachelor ETIT SPO 2023 of 27.04.2023, §19
- Bachelor MIT SPO of 24.07.2023, §19
- Bachelor Medizintechnik SPO of 12.07.2022, §19
- Bachelor Medizintechnik Änderungssatzung of 28.04.2023
- Master ETIT SPO 2018 of 28.09.2018, §18
- Master ETIT SPO 2025 of 22.01.2025, §18
- Master MIT SPO 2025 of 17.01.2025, §18
- Master Biomedical Engineering of 21.05.2025, §18

According to these regulations, the achievements required in the curriculum can also be achieved through recognition of external credits and grades.

External achievements may be acquired as follows:
1. within the higher education system (worldwide)
2. outside the higher education system (at institutions with standardized quality assurance systems; recognition may be denied if more than 50 percent of the curriculum are to be substituted)

Recognition is granted upon application by the student. With regard to the acquired competencies, it must be ensured that there is no significant difference to the achievements or degrees that are to be replaced. The application must be submitted within the first semester after enrollment at KIT.

The examination board is responsible for recognition and crediting and involves the responsible program consultant in the decision. Recognized credits and grades that were not achieved at KIT are shown as "recognized" in the transcript of records.

9.2 Grading

If the external grading system is comparable, the grade of the achievements to be recognized is adopted. If the grading system is not comparable, the grade is converted.

9.3 Procedure

1. **Present the application form and the required documents* to a subject examiner****.
   Important: Recognitions must be applied for at the examination board within the first semester after enrolment
2. In case of equivalence of the acquired competence goals, this is **confirmed with a stamp and signature by the subject examiner**.
3. **Hand in the completed and signed application together with the corresponding transcript of records to the office of the Study Program Service ETIT, MEDT/BME and MIT.**

Note on examinations abroad
If examination achievements from a stay abroad are to be recognized, a learning agreement must be drawn up in advance. It is advisable to discuss planned external examinations with a program consultant with regard to later recognition. You can find further information at https://www.etit.kit.edu/english/internationales.php.

*For the recognition it is required to present documents showing the examination achievements (Certificates, Transcript of Records, excerpts from the module handbook, lecture scripts or similar). In the case of documents that are not available in German or English, an officially certified translation may be requested.

**If you would like to have an achievement recognized instead of a KIT module, please contact the module coordinator (responsible for the module at KIT) for the subject examination. For recognition of examinations in the Elective Modules/ Interdisciplinary Skills/Area of Specialization, please contact one of the program consultants of the KIT-Department of Electrical Engineering and Information Technology.

If you have any further questions, please do not hesitate to contact the Study Program Service Bachelor and Master for ETIT, MEDT/BME and MIT:

bachelor-info@etit.kit.edu, https://www.etit.kit.edu/english/studiengangservice_bachelor_etit_und_mit.php

master-info@etit.kit.edu,
https://www.etit.kit.edu/english/studiengangservice_master_etit_und_mit.php

10 Contact persons and advice

Subject-specific advice:
Program consultants of the department

General advice:
Consultants of the Program Service Bachelor and Master ETIT, MIT, MEDT
(Advice e.g. on study scheduling, examination regulations, individual case problems, applications etc., as well as on organizational procedures at the KIT-Department of Electrical Engineering and Information Technology)

https://www.etit.kit.edu/english/studiengangservice_master_etit_und_mit.php

Tel.: 0721/608-42469, -47516 oder -42746
- „Altes Maschinenbaugebäude“ (Ehrenhof), Building 10.91, 3rd floor, room 223.1

Master programs:
master-info@etit.kit.edu

Bachelor programs:
bachelor-info@etit.kit.edu

Questions to interships (Bachelor):
Internship Office ETIT, Building 11.10 (ETI), room 204,
Mail: praktikantenamt@etit.kit.edu
Please check out the FAQs at first!

Studierendenservice

For organizational questions about studying (application, enrollment, re-registration, graduation documents, certificates, etc.):

https://www.sle.kit.edu/english/wirueberuns/studierendenservice.php

Contact persons concerning your study program:

https://www.sle.kit.edu/english/wirueberuns/studierendenservice_team4.php

Studying abroad

Consultants of the Program Service Bachelor and Master ETIT, MIT, MEDT https://www.etit.kit.edu/english/internationales.php

11 Editors

KIT Department of Electrical Engineering and Information Technology
Karlsruhe Institute of Technology (KIT)
www.etit.kit.edu
Dean of Studies:
Prof. Dr.-Ing. Mike Barth
Module coordination (modulkoordination@etit.kit.edu):
Dr. rer. nat. Andreas Barth

12 Modules

M 12.1 Module: Adaptive Optics [M-ETIT-103802]

Coordinators: Ph.D. Szymon Gladysz
Prof. Dr. Ulrich Lemmer

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-107644	Adaptive Optics	**WT**	**3 CP**	**Gladysz, Lemmer**

Assessment
Type of Examination: Oral examination

Duration of Examination: approx. 30 Minutes

Modality of Exam: The oral exam will be scheduled during the semester break.

Prerequisites
None.

Competence Goal
The students will:

- get familiar with Fourier description of imaging through aberrated optical systems and random media,
- understand the description of aberrations through Zernike modes,
- learn how to analytically compute the effects of turbulence on various optical observables such as image/beam motion, temporal power spectra, Zernike modes, scintillation, etc.,
- understand the effect of noise on various quantities and metrics pertinent to the design of adaptive optical systems,
- understand the advantages and disadvantages of various schemes for wavefront sensing and correction,
- learn how to simulate and design simple adaptive optics systems.

Content
Adaptive optics is a technology of correcting the effect of atmospheric turbulence on images of space objects and on laser beams propagating through random and highly aberrated media such as turbulence, tissue, and the inside of the human eye, to name just a few applications. The course will familiarize the students with theoretical basics of light propagation through random media, principles of wavefront sensing and reconstruction, as well as wavefront correction with deformable mirrors. The students will also receive solid introduction to statistical optics, the Kolmogorov theory of turbulence, practical aspects of turbulence simulation and modelling of adaptive optics.

1. Theory of turbulence (covariances, structure functions, power spectra, inertial range, dimensional argument of Kolmogorov)
2. Fourier optics (point-spread function, modulation transfer function)
3. Statistical optics (characteristic function, probability density function)
4. Sources and description of aberrations (Zernike polynomials, orthogonality, Marechal criterion)
5. Adaptive optics systems (open- and closed-loop systems, error budgets, tip-tilt correction)
6. Wavefront sensing (Shack-Hartmann wavefront sensor, wavefront reconstruction, wavefront-sensorless AO)
7. Wavefront correction (tip-tilt mirrors, deformable mirrors, piezoelectric effect, microelectromechanical systems, electrostatic actuation)
8. Simulation of adaptive optical systems (analytic vs. end-to-end modelling)
9. Propagation of laser beams through atmospheric turbulence (Gaussian beams, Rytov theory, scintillation index, beam wander)
10. Modelling of free-space optical communication systems (aperture averaging, mean signal-to-noise ratio, false-alarm rate and fade probability, bit error-rate)

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
total 90 h, hereof 30 h contact hours and 60 h homework and self-studies

Recommendations
Basic knowledge of statistics.
It is recommended that the students first complete the module Optical Engineering and Machine Vision [M-ETIT-106974] or Optical Engineering [M-ETIT-100456] or Fundamentals of Optics and Photonics [M-PHYS-101927] before they enroll for this course.

Literature
Robert K. Tyson, Principles of Adaptive Optics, CRC Press

Michael C. Roggemann, Byron M. Welsh, Imaging through Turbulence, CRC Press

M 12.2 Module: Advanced Communications Engineering [M-ETIT-106815]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Fundamentals)
Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113676	Advanced Communications Engineering	**WT**	**6 CP**	**Jäkel**

Assessment
The assessment takes place in the form of a written examination lasting 120 min,

Prerequisites
none

Competence Goal
The students are able to analyze and assess properties of communication systems and consider aspects of implementation. They can use mathematical methods in the context of communication systems for understanding involved derivations in the research literature; deriving and autonomously elaborating theoretical results, and checking their viability by simulations.

Content
The module is introducing and deriving results covering, but not being limited to, properties of linear modulation, channel description and diversity schemes, and processing of receiver signals, all based on detailed theoretical concepts. Topics already covered in previous modules are deduced thoroughly and mathematical derivations and reasoning are provided.

Module Grade Calculation
The module grade is the grade of the written exam.

Additional Information
Starting winter term 25/26

Workload

1. Attendance to the lecture: 20 * 1,5 h = 30 h
2. Preparation and review: 20 * 3 h = 60 h
3. Attendance to the tutorial: 6 * 1,5 h = 9 h
4. Preparation and review: 6 * 3,5 h = 21 h
5. Preparation for the exam: 60 h

In total: 180 h = 6 LP

Recommendations
Basics knowledge of communication systems, as, e.g., provided in KIT's Bachelor courses "Grundlagen der Datenübertragung" and "Nachrichtensysteme", is supposed. Furthermore, working knowledge in the areas of system theory and probability theory is assumed.

Teaching and Learning Methods
Lecture: 3 SWS, Exercise: 1 SWS

M 12.3 Module: Analog Circuit Design [M-ETIT-107364]

Coordinators: Prof. Dr. Ivan Peric
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114457	Analog Circuit Design	**WT**	**4 CP**	**Peric**

Assessment
Success control takes place in form of an oral examination (approx. 20 minutes) and a written report with the results of the exercises. Details will be given during the lecture.

Prerequisites
none

Competence Goal
The course aims to provide students with a comprehensive understanding of the principles and techniques of analog circuit design, with a focus on CMOS technology. The learning objectives include:

1. **Understanding Microchip Fabrication**: Students will learn the fundamentals of microchip fabrication
2. **Analog Design Process**: Students will be able to create schematics, perform simulations (DC, AC, transient, noise), and develop layout designs while considering Design Rule Check (DRC) and Layout vs. Schematic (LVS).
3. **Knowledge of MOSFET Functionality**: Students will understand the structure, fabrication, and operation of CMOS transistors (PMOS, NMOS), including potential distributions, charge carrier densities, threshold voltage, operating regions (strong/weak inversion, saturation) and transistor models
4. **Analysis and Design of Simple Circuits**: Students will learn to design and optimize basic circuits such as current sources, current mirrors, voltage amplifiers, and differential amplifiers, including the application of feedback and small-signal models.
5. **Understanding Noise and Optimization**: Students will be able to analyze thermal noise, 1/f noise, and their impact on circuits, as well as optimize amplifiers for signal-to-noise ratio (SNR) and speed.
6. **Clocked Circuits and ADCs**: Students will understand the operation of track-and-hold circuits, clocked comparators, and analog-to-digital converters (flash, time-based, algorithmic), as well as concepts like sampling errors, charge injection, and time-interleaving.
7. **Practical Application through Exercises**: Through practical exercises (e.g., transistor optimization, simulations, layout design), students will develop hands-on skills in using design tools and analyzing circuits.
   By the end of the course, students will be able to design basic analog circuits, analyze their performance, and address practical challenges in chip fabrication and optimization.

Content
This course introduces the fundamentals of **chip design**, starting with microchip fabrication techniques like photolithography and CMOS technologies, and covers the **analog design flow** from schematics to layout. It then dives into **MOSFET physics**, including structure, operating regions, and key features like threshold voltage, saturation, and channel length modulation, both large- and small-signal models. The course explores **basic analog building blocks** such as current mirrors, differential amplifiers, and cascode amplifiers, emphasizing feedback analysis, gain calculation, and frequency response. Students also study **charge-sensitive amplifiers, clocked circuits**, and a few **ADC architectures** like flash, time-based, and algorithmic converters. Special attention is given to **noise analysis**, including thermal and flicker noise, and how to optimize designs for signal-to-noise ratio and speed. Practical skills are reinforced through **hands-on exercises** that cover design and optimization of a charge-sensitive amplifier, ADC simulation, and comparator design from schematics to DRC/LVS verification.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload

1. Attendance time in lectures 30 h
2. Attendance time in exercises 15 h
3. Preparation/follow-up of the same 25 h
4. Exam preparation 50 h

M 12.4 Module: Antennas and Beamforming [M-ETIT-106956]

Coordinators: Prof. Dr.-Ing. Marwan Younis
Prof. Dr.-Ing. Thomas Zwick

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113920	Antennas and Beamforming	**WT**	**4 CP**	**Younis, Zwick**

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal
After successfully participating in this course, students have in-depth knowledge of antennas, antenna systems and beamforming methods. This includes functionality, calculation methods as well as aspects of practical implementation. They are able to understand how typical electromagnetic radiators work and to develop and dimension them with specified properties. Students understand the principle and function of beamforming and the differences between digital, analog and hybrid beamforming. They know the theory, procedures and algorithms for beamforming. They can understand how beamforming is used for radio communication and radar.

Content
The lecture begins with a brief review of the basic knowledge of antennas and antenna arrays from the Bachelor's course. This is followed by a detailed discussion of all major antenna types (functionality, specifics). Furthermore, antenna measurement methods are presented. In the second part, the basic knowledge of noise, radio transmission and radar ambiguities is briefly refreshed, followed by a detailed presentation of the various beamforming algorithms, each with reference to radio communication and radar systems. Aspects such as digital and hybrid beamforming, as well as MIMO and equivalent virtual antenna configuration are explained.

The lecture will be accompanied by exercises. These are discussed in a room exercise and the corresponding solutions are presented in detail.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
The workload includes:

- Attendance study time lecture: 30 h
- Attendance study time exercise: 15 h
- Self-study time including exam preparation: 75 h

A total of 120 h

Recommendations
Knowledge of the basics of radio frequency technology and some basic knowledge on communication and radar systems is recommended.

M 12.5 Module: Applied Information Theory [M-ETIT-107333]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114412	Applied Information Theory	**WT**	**6 CP**	**Jäkel**

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Prerequisites
none

Competence Goal
Students master the methods and concepts of information theory and are able to apply them to analyze communication technology issues.

Students acquire the ability to examine the information content of sources and the flow of information in systems and to evaluate their significance for the realization of telecommunications systems.

Content
The information theory founded by Shannon represents a central starting point for almost all questions of coding and encryption. In order to provide a solid foundation for later considerations, the concepts of information theory are developed at the beginning of the lecture. These are then applied to various areas of communications engineering and signal processing and used to analyze them.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
1. attendance time lecture: 15 * 3 h = 45 h

2. preparation/follow-up lecture: 15 * 6 h = 90 h

3. attendance time exercise: 15 * 1 h = 15 h

4. preparation/post-exercise: 15 * 2 h = 30 h

5. exam preparation and attendance in the exam: included in preparation/follow-up work

Total: 180 h = 6 LP

Recommendations
Previous attendance of the lecture "Wahrscheinlichkeitstheorie" is recommended.

M 12.6 Module: Authentication and Encryption [M-INFO-105338]

Coordinators: Prof. Dr. Jörn Müller-Quade
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-110824	Authentication and Encryption	**ST**	**4 CP**	**Müller-Quade**

Assessment
See Partial Achievements (Teilleistung).

Prerequisites
See Partial Achievements (Teilleistung).

Competence Goal
The student
- can explain the terms confidentiality and authenticity and point out their differences,
- understand the basic security objectives of digital signatures and their relationship to each other and can apply them,
- knows and understands important signature procedures from theory and practice and can explain them,
- understand definitions of actively secure encryption and can explain and apply them,
- can explain procedures for constructing actively secure encryption,
- can understand and apply elementary proof techniques such as reductions and hybrid arguments

Content
The guarantee of authenticity and secrecy is a security requirement that arises in many applications.

Building on the lecture "Theoretical Foundations of Cryptography", this lecture deepens the consideration of cryptographic authentication methods (especially signatures and message authentication codes) and actively secure encryption methods. The focus of the lecture is on various techniques for the construction of digital signature methods as well as the proof of the security properties achieved. For example, the following topics are covered:

- One-time signatures, tree-based signatures and chameleon hash functions
- RSA-based signatures
- Signatures in bilinear groups

In the second part of the lecture, encryption methods that offer security against active attacks are presented. For example, the following constructions are presented:

- Authenticated encryption in the symmetric case
- Methods for constructing actively secure asymmetric encryption

Workload
Attendance time in the lecture: 24 h Preparation/follow-up of the lecture: 31 h Exam preparation and attendance in the lecture: 65 h

Recommendations
Students should be familiar with the contents of the module "Theoretical Foundations of Cryptography"

M 12.7 Module: Automotive Engineering I [M-MACH-100501]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Organisation: KIT Department of Mechanical Engineering

Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
8 CP	graded	Each winter term	1 term	German/English	4	2

ID	Name	Term offered	Credits	Lecturers
Automotive Engineering I (Elective: 1 item)				
T-MACH-102203	Automotive Engineering I	**WT**	**8 CP**	**Gießler**
T-MACH-100092	Automotive Engineering I (in German)	**WT**	**8 CP**	**Gießler**

Assessment
written exam; duration approximately 2 hours

Prerequisites
Only one out of the two moduls "M-MACH-100501 - Grundlagen der Fahrzeugtechnik I" and "M-MACH-102686 - Automotive Engineering I" is allowed.

Competence Goal
The students know the movements and the forces at the vehicle and are familiar with active and passive security. They have proper knowledge about operation of engines and alternative drives, the necessary transmission between engine and drive wheels and the power distribution, so that they can apply their knowledge effectively in actual practise. They have an overview of the components necessary for the drive and have the basic knowledge, to analyze, to judge and to develop the complex system "vehicle".

Content
The module provides an overview of:

1. History and future of the automobile

2. Driving mechanics: driving resistances and driving performances, mechanics of the longitudinal and transverse forces, passive safety

3. Engines: combustion engine, alternative drives (e.g. electric motor, fuel cell)

4. Transmission: clutches (e.g. friction clutch, visco clutch), transmission (e.g. mechanical transmission, hydraulic fluid transmission)

5. Power transmission and distribution: drive shafts, cardon joints, differentials

Workload
1. regular attendance lecture: 15 * 2 * 2 h = 60 h

2. pre and post processing lecture: 15 * 2 * 3 h = 90 h

3. examination preparation and presence in examination: 90 h

In total: 240 h = 8 LP

Literature
1. Mitschke, M./ Wallentowitz, H.: Dynamik der Kraftfahrzeuge, Springer-Verlag, Berlin, 2004

2. Braes, H.-H.; Seiffert, U.: Handbuch Kraftfahrzeugtechnik, Vieweg&Sohn Verlag, 2005

3. Gnadler, R.: Script to the lecture 'Automotive Engineering I'

M 12.8 Module: Automotive Engineering II [M-MACH-100502]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Dr.-Ing. Martin Gießler

Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Automotive Engineering II (Elective: 1 item)				
T-MACH-102117	Automotive Engineering II (in German)	**ST**	**4 CP**	**Gießler**
T-MACH-114435	Automotive Engineering II	**ST**	**4 CP**	**Gießler**

Assessment
Written exam; duration approximately 1,5 h

Prerequisites
Only one out of the two modul components "T-MACH-102117 - Grundlagen der Fahrzeugtechnik II" and "T-MACH-114435 - Automotive Engineering II" is allowed.

Competence Goal
The students have an overview of the modules, which are necessary for the road holding of a motor vehicle and the power transmission between vehicle bodywork and roadway. They have knowledge of different wheel suspensions, the tyres, the steering elements and the brakes. They know different execution forms, the function and the influence on the driving or brake behavior. They can apply their knowledge effectively in actual practise.They are able to develop the appropriate components correctly. They are ready to analyze, to judge and to optimize the complex relationship of the different components under consideration of boundary conditions.

Content
The module provides an overview of:

1. Chassis: Wheel suspensions (rear axles, front axles, kinematics of axles), tyres, springs, damping devices
2. Steering elements: Manual steering, servo steering, steer by wire
3. Brakes: Disc brake, drum brake, comparison of the designs

Module Grade Calculation
The module grade corresponds to the grade of the examination.

Workload
1. regular attendance lecture: 15 * 2 h = 30 h

2. pre and postprocessing lecture: 15 * 3 h = 45 h

3. examination preparation and presence in examnation: 45 h

In total: 120 h = 4 LP

Literature
1. Heißing, B./Ersoy, M.: Fahrwerkhandbuch: Grundlagen, Fahrdynamik, Komponenten, Systeme, Mechatronik, Perspektiven, Vieweg-Verlag, Wiesbaden, 2011

2. Breuer, B./Bill, K.-H.: Bremsenhandbuch: Grundlagen - Komponenten - Systeme - Fahrdynamik, Vieweg-Verlag, Wiesbaden, 2012

3. Gnadler, R.: Script to the lecture 'Automotive Engineering II'

M 12.9 Module: Automotive Vision [M-MACH-107148]

Coordinators: Dr. Martin Lauer
Prof. Dr.-Ing. Christoph Stiller
Organisation: KIT Department of Mechanical Engineering
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-114149	Automotive Vision	**ST**	**6 CP**	**Lauer, Stiller**

Assessment
Type of Examination: written exam
Duration of Examination: 60 minutes

Prerequisites
none

Competence Goal
After having participated in th lecture the participants have gained knowledge on modern techniques of signal processing and artificial intelligence which can be used to evaluate video sequences, to relate the image content to a spatial context and to interpret the content semantically. This comprises, binocular reconstruction, recognition of movements in video sequences, state space modeling and Bayesian filters, and the recognition of road surfaces and object behavior. The participants have learned to analyze the algorithms mathematically, to implement them in software, and to apply them to tasks in autonomous driving and mobile robots. The participants are able to analyze problems in the areas mentioned before and to develop appropriate solutions.

Content
Sensory detection and interpretation of the environment form the basis for generating intelligent behavior. The ability to see opens up completely new perspectives for vehicles and represents a rapidly growing field of research and innovation in automotive technology. The first so-called driver assistance systems have already achieved respectable improvements in terms of comfort, safety and efficiency. However, it will probably take several decades of intensive research before automobiles have a performance comparable to the human visual system.
The lecture is aimed at students of mechanical engineering and related courses,
who wish to acquire an interdisciplinary qualification. It provides a holistic overview of the field of vehicle vision, from the basics of image acquisition and kinematic vehicle models to innovative metrological methods of image processing for seeing vehicles. The derivation of metrological methods of image processing is deepened and illustrated using current, practice-relevant application examples.

Module Grade Calculation
see individual course

Workload
180 hours
composed out of
hours of lecture: 15*3 h = 45 h
preparation time prior to and after lecture: 15*5 h = 75 h
exam preparation and exam: 60 h

Teaching and Learning Methods
Lecture

Literature
TBA

M 12.10 Module: Autonomous Driving in Practice [M-MACH-107806]

Coordinators: Dr.-Ing. Martin Gießler
Kevin Simon

Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-115202	Autonomous Driving in Practice	**WT**	**6 CP**	**Simon**

Assessment
The final grade is based on: 2 written exams (duration about 30 minutes) and the final presentation of the practical task (duration 30 minutes max.), including the final report (2-3 pages). Submission of the homework is mandatory.

The overall impression will be evaluated.

Prerequisites
none

Competence Goal

- Students can configure sensors on the test vehicle, record data and check the quality of the recorded sensor data.
- Students can annotate and segment sensor data and recognize and classify objects using suitable algorithms.
- Students can plan paths and trajectories for driving maneuvers, generate behavioral strategies and evaluate their feasibility.
- Students can implement driving maneuvers using suitable control strategies and control the actuators of the test vehicle.
- Students can implement the entire pipeline - from environmental perception to planning and execution - and test it in a real test vehicle.

Content
The main objective of the event is the practical implementation of the pipeline required for automated/autonomous driving functions with a real test vehicle. This includes recording the environment using various sensors, processing the recorded sensor data (perception), planning driving maneuver and the final execution of the maneuver by the actuators.

- Sensor data acquisition: Setup and data recording of the sensors on the test vehicle
- Perception: data annotation, segmentation of sensor data, object detection
- Maneuver planning: path and trajectory planning, behaviour generation, etc.
- Maneuver execution: vehicle control, implementation of the driving maneuver in the real test vehicle using actuators

Module Grade Calculation
The module grade results of the assessment of the written exams, the presentation, and the final report. Details will be given during the lecture.

Additional Information
Due to capacity limitations, the lecture is limited to 16 students. If necessary, a selection process will be conducted. Places will be allocated based on the students' academic progress (semester level and subject-specific programming skills).

Workload
The workload includes:

1. Attendance time in lectures: 60 h
2. Work on the project: 120 h

Total: 180 h

M 12.11 Module: Basics of Converter Control [M-ETIT-100400]

Coordinators: Dr.-Ing. Andreas Liske
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100717	Basics of Converter Control	ST	3 CP	Liske

Assessment
The success control takes place as part of an online examination in text form with face-to-face supervision (§5, statutes for conducting online examinations at KIT) lasting 60 minutes.

The exam takes place on KIT computers in a pool room.

Prerequisites
none

Competence Goal
A real power converter consists of significantly more parts than the actual power electronics topology. Students gain an overview of all the components of a real power converter. They learn about the typical measurement methods used in power converter technology and are able to dimension measurement circuits themselves. They learn about the power semiconductors typically used and how they are controlled, as well as the circuits commonly used to protect them. In addition, they gain an understanding of how the sequence control and modulation of the respective topologies are implemented in practice.

Content
This module aims to bridge the gap between theoretical considerations regarding power converter topologies and their actual implementation in a specific power converter device. Therefore, the content does not explicitly cover topologies, but rather deals with all components necessary for the actual functioning of a power converter, such as measuring devices, implementation of sequence control and modulation, and teaching the specific control of power semiconductors and their protective circuits. In addition to the theoretical consideration of the respective functional blocks, the specific dimensioning and implementation are also covered.

Module Grade Calculation
The module grade is the grade of the online examination.

Workload
14x lecture (attendance) of 2 h each = 28 h

Exam preparation = 60 h

Total approx. 88 h (corresponds to 3 CP)

M 12.12 Module: Batteries and Fuel Cells Laboratory [M-ETIT-100381]

Coordinators: Dr.-Ing. Andre Weber
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100708	Batteries and Fuel Cells Laboratory	**WT**	**6 CP**	**Weber**

Assessment
Success is assessed in the form of other types of examinations.

The grade is made up of the grades for the colloquium before each experiment (20%), attendance and participation (30%) and the experiment reports (50%).

Prerequisites
none

Competence Goal
Students are able to design test procedures for batteries and fuel cells, carry out appropriate tests and document the results.

Content
The practical course consists of 8 experiments. The experiments cover the design and operation of various fuel cell types and systems. In the course of the practical course, knowledge of operational management, measurement procedures and measurement data evaluation is imparted. The experimental investigations are carried out on (pre-)series products from well-known manufacturers (Ballard Nexa Power Module, Idatech FCS 1200) as well as on test benches specially developed for research purposes. Other experiments deal with the electrical characterization and modelling of batteries.

The duration of the experiments is between ½ and 1 day. Following the experiment, approximately the same amount of time is required to evaluate the data obtained. In addition, approx. 5 h of preparation and 6 - 8 h for the creation of the test protocol should be planned. In order to be able to concentrate on the execution of the tests during the practical course, the participants receive test documents in advance. These consist of a short basic chapter, preparatory questions and the actual description of the experiment. Furthermore, information on the systems and measuring devices used is distributed in the form of data sheets and manuals.
The participants must familiarize themselves with the theory, the measuring methods and devices used and the operation of the fuel cell systems before carrying out the test. In addition to the introduction to the test setup, there will be a short knowledge test on the day of the test. A protocol must be prepared for each experiment.

Module Grade Calculation
The grade is made up of the grades for the colloquium before each experiment (20%), attendance and participation (30%) and the experiment reports (50%).

Workload
1. preparation time trials: 8 * 5 h = 40 h
2. Performance of experiments: 8 experiments, in total 44 h
3. Evaluation of experimental data: 8 * 5 h = 40 h
4. Preparation of experimental protocols: 8 * 7 h = 56 h

Total: 180 h = 6 CP

Recommendations
The contents of the lectures "Batteries and Fuel Cells" and "Battery and Fuel Cell Systems" are assumed to be known. Students who have not (yet) attended these lectures must familiarize themselves with the contents in advance.

M 12.13 Module: Batteries, Fuel Cells, and Electrolysis [M-ETIT-107005]

Coordinators: Prof. Dr.-Ing. Ulrike Krewer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Fundamentals)
Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113986	Batteries, Fuel Cells, and Electrolysis	**WT**	**5 CP**	**Krewer**
T-ETIT-114957	Batteries, Fuel Cells, and Electrolysis - Group Project *This item will not influence the grade calculation of this parent.*	**WT**	**1 CP**	**Krewer**

Assessment
Success control takes place in the form of:

1. an ungraded written technical report (approx. 7-10 pages).
2. a graded written examination lasting 120 minutes.

Prerequisites
none

Competence Goal
Students gain an understanding of batteries, fuel cells and electrolysis including their application, design, and behavior. They acquire in-depth knowledge of the transport and charge transfer processes in them, their impact on performance and design, and the characteristics of the most frequent types of batteries, fuel and electrolysis cells. They understand how to analyze and characterize them using measurement methods and modeling. A practical insight into current areas of application and research topics of electrochemical energy storage and conversion allows them to relate the course work to demands of the society and for R&D. They are able to communicate with specialists from related disciplines in the field of (application of) batteries, fuel cells and electrolysis and can actively contribute to the opinion-forming process in society with regard to energy technology issues.

Content
The course introduces batteries, fuel cells and electrolysis and their use for sustainable mobile and stationary energy supply and storage. The course is divided into five sections. The first part covers the role of batteries, fuel cells and electrolysis for renewable energy storage and electrification of the energy system and the present applications. This is followed by a fundamentals part, where the processes in electrochemical cells at open circuit and during operation and their relation to cell performance and behavior are discussed. It contains thermodynamics, kinetics, transport and performance measures. The third part deals with the working principle, design and operation of fuel cells and electrolysis and the particularities of the different cell types. This is followed by a similar part for batteries. Finally, dynamic and stationary methods for characterizing the cells are covered.

Group project

As part of the coursework, student groups work on the design of a battery, fuel cell or electrolyser for a given application during the semester. This includes literature research on cell type, materials and material data as well as the dimensioning and energetic evaluation of the cell. The results are documented in a short technical report.

Module Grade Calculation
The module grade is the grade of the written examination.

Workload

1. Lecture attendance time: 15 * 2 h = 30 h
2. Preparation and follow-up time for lecture: 15 * 5 h = 75 h
3. Exercise attendance time: 7 * 2 h = 14 h
4. Preparation and follow-up time for exercise: 7 * 4 h = 28 h
5. Group work including writing of a report: 33 h
6. Exam preparation and attendance: included in preparation and follow-up time.

Total: 180 h = 6 CP

M 12.14 Module: Battery and Fuel Cell Systems [M-ETIT-107551]

Coordinators: Dr.-Ing. Andre Weber
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114802	Battery and Fuel Cell Systems	**ST**	**3 CP**	**Weber**

Assessment
Success controll takes place in the form of an oral examination lasting approx. 20 minutes.

Prerequisites
none

Competence Goal

- Students can explain the basic principles required for developing battery and fuel cell systems and apply them using practical examples.
- Students can describe the components of fuel cell systems and plan their integration into fuel cell systems.
- Students can compare system concepts for fuel cell systems and evaluate their advantages and limitations.
- Students can analyze battery systems for hybrid and electric vehicles, particularly lithium-ion systems.
- Students can design appropriate charging and balancing strategies.
- Students can explain safety concepts at cell and battery level as well as battery management systems (BMS) and assess their functionality.
- Students can describe alternative electrochemical energy storage systems such as redox-flow batteries and electrolyzers and analyze their potential applications.

Content
The lecture Battery and Fuel Cell Systems covers current developments in the field of fuel cells, electrolyzers and batteries considering system-relevant aspects of the technologies. In the first part of the lecture, fuel cell systems and their components are discussed. The integration of the various types of low- and high-temperature fuel cells into systems and the related requirements for fuel preparation are presented and different system concepts implemented to date are compared. In the second part of the lecture, battery systems for hybrid and electric vehicles are presented and the batteries and cells used in these are discussed. The focus is on lithium-ion battery systems, with charging strategies and circuits for charge equalization, safety concepts at cell and battery level and BMS systems being discussed. In the last part of the lecture, alternative electrochemical energy storage systems such as redox-flow batteries and electrolyzers are presented.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
1. Lecture attendance time: 15 * 2 h = 30 h
2. Preparation and follow-up time for lecture: 15 * 2 h = 30 h
3. Exam preparation and attendance in the same: 30 h

Total: 90 h

Recommendations
The contents of the lecture "M-ETIT-107005 – Batteries, Fuel Cells, and Electrolysis" are assumed to be known. Students can furthermore download a script with useful information on the Ilias site of the lecture “Battery and Fuel Cell Systems”.

M 12.15 Module: Biologically Inspired Robots [M-MACH-106903]

Coordinators: Prof. Dr.-Ing. Arne Rönnau
Organisation: KIT Department of Mechanical Engineering
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-113856	Biologically Inspired Robots	**ST**	**3 CP**	**Rönnau**

Assessment
See partial achievement

Prerequisites
None

Competence Goal
Students are familiar with various design principles of the "bionics" method in robotics and can analyze and evaluate models for kinematics, mechanics, control, perception and cognition.

Students understand the lightweight construction concepts and material properties of natural models. They are familiar with the concepts and methods of lightweight robotics and can describe the resulting effects on the energy efficiency of mobile robot systems.

Students can distinguish between different natural muscle types and how they function. They also know the corresponding artificial muscle systems and can derive the underlying muscle model.

Students know the most important human senses and the associated stimulus processing and information coding. They can derive technological sensors that perform the same function in robotics.

Students can differentiate the function of a central pattern generator (CPG) from a reflex. They can theoretically derive neuro-oscillators and explain how they control the movement of a robot. Furthermore, they can generate walking patterns for six-legged robots based on the "Cruse rules".

Students can distinguish between different types of locomotion and suitable stability criteria for walking movements. They know the most important walking patterns for multi-legged walking robots and can represent these in a gait diagram.

Students know the most important algorithms for machine learning methods and can explain their advantages and disadvantages in robotics.

Students know the subsumption system architecture and can evaluate the advantages of a reactive system architecture. They can combine "behaviors" for biologically inspired robots into behavior networks.

Students can apply Mendel's laws and explain the differences between meitosis and mitosis. They can also explain the basic genetic algorithm.

Students can identify the greatest challenges in the development of innovative, humanoid robot systems and are familiar with possible solutions and successful implementations.

Content

The lecture biologically inspired robots deals intensively with robots whose mechanical design, sensor concepts or control architecture were inspired by nature. In detail, we will look at solutions from nature (e.g. lightweight construction concepts using honeycomb structures, human muscles) and then at robot technologies that utilize these principles to solve similar tasks (lightweight 3D printed parts or artificial muscles in robotics).

After discussing these biologically inspired technologies, concrete robotic systems and applications from current research that successfully utilize these technologies will be presented. In particular, multi-legged walking robots, snake-like and humanoid robots are presented and their sensor and drive concepts are discussed.

The lecture focuses on the concepts of control and system architectures (e.g. behavior-based systems) of these robotic systems, with locomotion being the main focus. The lecture ends with an outlook on future developments and the development of commercial applications for these robots.

Workload

90 working hours, of which approx:

- 30h for attendance time in lectures
- 30h for preparation and follow-up time
- 30h for exam preparation and participation in the oral exam

M 12.16 Module: Biomass Based Energy Carriers [M-CIWVT-104288]

Coordinators: Dr.-Ing. Siegfried Bajohr
Organisation: KIT Department of Chemical and Process Engineering
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-108828	Energy from Biomass	**WT**	**6 CP**	**Bajohr**

Assessment
Learning control is an oral exam lasting approx. 20 minutes.

Prerequisites
None

Competence Goal
The course mediates fundamentals and process engineering aspects of biomass conversion and conditioning processes. The students learn to understand and to evaluate processes for biomass utilization by balancing mass and energy streams. Taking into account regional and global feedstock potentials the students are enabled to choose the most efficient conversion technologies.

Content
Fundamentals on biomass and its production pathways to energy carriers like substitute natural gas (SNG), bio diesel or other fuels.

Production, properties, and characterization of biomass.

Potential and sustainability; energy demand and supply, potentials today and in the future, CO2 emissions and reduction potential.

Utilization and conversion of biogenic oils and fats.

Biochemical conversion to liquid products like alcohols; fermentation to biogas and its upgrading.

Thermochemical conversion of biomass via pyrolysis and gasification; examples for synthesis processes (FT-, CH4-, CH3OH-, DME-synthesis).

Module Grade Calculation
The grade of the oral examination is the module grade.

Workload

- Attendance time (Lecture): 45 hrs
- Homework: 75 hrs
- Exam Preparation: 60 hrs

Literature

- Kaltschmitt, M.; Hartmann (Ed.): Energie aus Biomasse, 2. Aufl., Springer Verlag 2009.
- Graf, F.; Bajohr, S. (Hrsg.): Biogas: Erzeugung – Aufbereitung – Einspeisung, 2. Aufl., Oldenbourg Industrieverlag 2013.

M 12.17 Module: BioMEMS - Microsystems Technologies for Life Sciences and Medicine I [M-MACH-100489]

Coordinators: Prof. Dr. Andreas Guber
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-100966	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine I	**WT**	**4 CP**	**Guber**

Assessment
Written exam (75 min)

Prerequisites
none

Competence Goal
Professional qualification goals:
Students will be able to describe basic microtechnical manufacturing processes (e.g. LIGA, silicon microtechnology, laser micromachining) and analyze their suitability for biomedical applications.
They are able to compare different microfluidic components (e.g. microchannels, micropumps, microfilters) and explain their function in the context of µTAS and lab-on-chip systems.
Students will be able to explain the properties and possible applications of biomaterials and sterilization processes for microsystems in medical technology.
Furthermore, they can evaluate the interactions between microtechnical production and biomedical application and transfer them to selected case studies from the life sciences.
Interdisciplinary qualification goals:
Students will be able to recognize interdisciplinary connections between technology, biology and medicine and argue in a structured manner in discussions. Furthermore, they will be able to critically reflect on current developments and literature in the field of microsystems technology for life science applications and assess their relevance for research and industry.

Content
The lecture will first address relevant microtechnical manufacturing methods. Then,
selected biomedical applications will be presented, as the increasing use of
microstructures and microsystems in Life-Sciences und in medicine leads to improved medico-technical
products, instruments, and operation and analysis systems.

Introduction into various microtechnical manufacturing methods: LIGA, Micro milling, Silicon Micromachining, Laser Microstructuring, µEDM, Metal-Etching
Biomaterials, Sterilisation.
Examples of use in the life science sector: basic micro fluidic strucutures: micro channels, micro filters, micromixers, micropumps, microvalves, Micro and nanotiter plates, Microanalysis systems (µTAS),
Lab-on-chip applications.

Module Grade Calculation
Written exam

Workload
Literature: 20 h

Lessions: 21 h

Preparation and Review: 50 h

Exam preparation: 30 h

Teaching and Learning Methods
Lecture

Literature
Menz, W., Mohr, J., O. Paul: Mikrosystemtechnik für Ingenieure, VCH-Verlag, Weinheim, 2005

M. Madou
Fundamentals of Microfabrication
Taylor & Francis Ltd.; Auflage: 3. Auflage. 2011

M 12.18 Module: BioMEMS - Microsystems Technologies for Life Sciences and Medicine II [M-MACH-100490]

Coordinators: Prof. Dr. Andreas Guber
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-100967	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine II	**ST**	**4 CP**	**Guber**

Assessment
Written exam (75 min)

Prerequisites
None

Competence Goal
Professional qualification goals:
Students will be able to describe modern microfluidic systems (e.g. Lab-CD, microarrays, biochips) and analyze their functional principles in the context of biomedical applications. They will be able to explain and evaluate biohybrid cell-chip systems and their use in tissue engineering and drug testing.
Students will be able to explain the structure and function of microprocess engineering components (e.g. microreactors, microfluidic measuring cells) and assess their use in spectroscopic investigations.
They can examine specific microsystems technology solutions for medical applications in anesthesia, intensive care medicine and infusion therapy and compare their principles of action.
They will be able to describe micro- and nanoscale technologies (e.g. in nanosurgery or neuroprosthetics) in a differentiated manner and assess their significance for future therapy concepts.

Interdisciplinary qualification goals:
Students will be able to recognize and reflect on interdisciplinary connections between technology, biology and medicine and argue for them in scientific discussions.
Furthermore, they are able to analyze current scientific publications on applications of microsystems technology in the life sciences and critically discuss their relevance.
They will be able to reflect on technological developments in the field of microsystems technology with regard to ethical, social and regulatory aspects.

Content
The lecture will first shortly address some relevant microtechnical manufacturing methods. Then,
selected biomedical applications will be presented, as the increasing use of
microstructures and microsystems in Life-Sciences und in medicine leads to improved medico-technical
products, instruments, and operation and analysis systems.

Examples of use in Life-Sciences and biomedicine: Microfluidic Systems:
LabCD, Protein Cristallisation
Microarrys
Tissue Engineering
Cell Chip Systems
Drug Delivery Systems
Micro reaction technology
Microfluidic Cells for FTIR-Spectroscopy
Microsystem Technology for Anesthesia, Intensive Care and Infusion
Analysis Systems of Person´s Breath
Neurobionics and Neuroprosthesis
Nano Surgery

Module Grade Calculation
Written exam

Workload
Literature: 20 h

Lessions: 21 h

Preparation and Review: 50 h

Exam preparation: 30 h

Teaching and Learning Methods
Lecture

Literature
Menz, W., Mohr, J., O. Paul: Mikrosystemtechnik für Ingenieure, VCH-Verlag, Weinheim, 2005

Buess, G.: Operationslehre in der endoskopischen Chirurgie, Band I und II;
Springer-Verlag, 1994

M. Madou
Fundamentals of Microfabrication

M 12.19 Module: BioMEMS - Microsystems Technologies for Life Sciences and Medicine III [M-MACH-100491]

Coordinators: Prof. Dr. Andreas Guber
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-100968	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine III	**ST**	**4 CP**	**Guber**

Assessment
Written exam (75 min)

Prerequisites
none

Competence Goal

- Students are able to describe relevant microtechnical production methods, explain their functional principle and evaluate their use for biomedical and biotechnological applications as well as for BioMEMS.
- Students are able to analyze the use of microstructures and microsystems in life sciences and (bio)medicine using specific examples and assess their influence on medical technology and biotechnology products and systems.
- Students can explain examples of applications in the life sciences and biotechnology, such as microfluidic systems (e.g. lab-on-a-chip, digital microfluidics), biochips, tissue engineering, organ-on-a-chip, drug delivery systems, respiratory gas analysis and neuroprosthetics (e.g. artificial retina, exoskeletons) and compare their technical requirements.
- Students can explain examples of applications in minimally invasive surgery, neurosurgery, interventional cardiology and vascular therapy as well as NOTES and surgical robotics and compare their technical requirements.
- Students will be able to describe and apply the key requirements of the Medical Devices Act and the principles of quality management for medical technology products.

Content
Examples of use in minimally invasive therapy
Minimally invasive surgery (MIS)
Endoscopic neurosurgery
Interventional cardiology
NOTES
OP-robots and Endosystems
License of Medical Products and Quality Management

Workload
Literature: 20 h

Lessions: 21 h

Preparation and Review: 50 h

Exam preparation: 30 h

Literature
Menz, W., Mohr, J., O. Paul: Mikrosystemtechnik für Ingenieure, VCH-Verlag, Weinheim, 2005

Buess, G.: Operationslehre in der endoskopischen Chirurgie, Band I und II;
Springer-Verlag, 1994

M. Madou
Fundamentals of Microfabrication

M 12.20 Module: Business Innovation in Optics and Photonics [M-ETIT-101834]

Coordinators: Prof. Dr. Werner Nahm
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-104572	Business Innovation in Optics and Photonics	**WT**	**4 CP**	**Nahm**

Assessment
Type of Examination: examination of another type

Duration of Examination: 4 group presentations à 20 minutes (approx.)

Modality of Exam: The exam consists of four group presentations. 2nd day: Technology Presentation. 3rd day: Development plan presentation. 4th day: Business Canvas presentation. Final presentation: Business pitch. The overall impression is rated.

Prerequisites
none

Competence Goal
The student has an understanding how innovative concepts for optical and photonics products are transferred into a successful business development. The student knows about and makes first The student develops practical experience of how to analyse, develop, and pitch innovative business ideas based on optics & photonics technologies, specifically in the field of spectroscopy and hyperspectral imaging. Through a series of short lectures followed by extensive group work over the first week of the block course the student acquires hands-on experience of business development as lived in a technology start-up environment. The students are given a basic introduction to spectroscopy and are challenged to propose innovative businesses enabled by evolving technology and market needs. To assist in this process the students are introduced to the innovation methodology “Deep Tech Exploration” and primed on themes from intellectual property management to effective pitching.

The students organize themselves into teams, and work cooperatively within the team to develop and deliver the complete business pitch.

In summary the students:

- learn innovation methodologies to assess and optimize the desirability, viability, and feasibility of business ideas.
- are introduced to methods to assess and generate relevant intellectual property.
- get to apply the knowledge gained with extensive group work developing their own business idea and pitch.

Content
This course is instructed by ZEISS employees, active within the ZEISS corporate innovation ecosystem.

Monday: Introduction into spectroscopy and the deep tech exploration method
• Technology introduction, state-of-the-art
• Scenario Introduction: Found your own start-up
• Overview Deep Tech Exploration methodology / Select your use case

Tuesday: Deep dive – deep tech exploration
• Methods and ideation techniques
• Fundamental principles in Innovation / Value Proposition
• Define your tech product

Wednesday: Business case development
• How to protect your business idea with IP?
• Develop your business idea with the help of the Business Model Canvas
• Writing a Business plan

Thursday: Storytelling and pitching
• Entrepreneurial fincance: tech product problem sizing / ballparking
• Storytelling
• Preparing an elevator pitch

Friday-Thursday of second week: Final pitch preparation
• Self-organized by groups

Friday of second week: Final pitch

Module Grade Calculation
The module grade results of the assessment of the presentations. Details will be given during the lecture.

Workload
Total 120 h, thereof 34 h contact hours and 86 h preparation, homework, and self-studies

Recommendations
Good knowledge in optics & photonics. Personal motivation and interest for getting deeper into business development aspects, methods and tools. Commitment to active, regular and coninuous participation in the group work.

M 12.21 Module: Channel Coding: Algebraic Methods for Communications and Storage [M-ETIT-105616]

Coordinators: Prof. Dr.-Ing. Laurent Schmalen
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111244	Channel Coding: Algebraic Methods for Communications and Storage	**ST**	**3 CP**	**Schmalen**

Assessment
The exam is held as an oral exam of approx. 20 min.

The module grade of the module corresponds to the grade of the oral exam.

Competence Goal
The students are able to analyse and assess problems of algebraic channel coding. They can apply methods of algebraic coding theory in the context of communication systems for data transmission and data storage and are able to assess their implementation.Additionally, they will get knowledge to current research topics and research results.

Content
This course focuses on the formal and mathematical basics for the design of coding schemes in digital communication systems. These include schemes for data transmission, data storage and networking. The course starts by introducing he necessary fundamentals of algebra which are then used to derive codes for different applications. Besides codes that are important for data transmission appliations, e.g., BCH and Reed-Solomon-Codes, we also investigate codes for the efficient storage and reconstruction of data in distributed systems (locally repairable codes) and codes that increase the throughput in computer networks (network codes). Real applications are always given to discuss practical aspects and implementations of these coding schemes. Many of these applications are illustrated by example code in software (python/MATLAB).

Module Grade Calculation
The module grade of the module corresponds to the grade of the oral exam.

Workload

1. Attendance to the lecture: 15 * 2 h = 30 h
2. Preparation and review: 15 * 4 h = 60 h
3. Preparation for the exam: included in preparation and review
4. In total: 90 h = 3 LP

Recommendations
Knowledge of basic engineering as well as basic knowledge of communications engineering.

Previous attendance of the lectures "Communication Engineering I" and "Probability Theory" is recommended.

M 12.22 Module: Channel Coding: Graph-Based Codes [M-ETIT-105617]

Coordinators: Prof. Dr.-Ing. Laurent Schmalen
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111245	Channel Coding: Graph-Based Codes	**WT**	**6 CP**	**Schmalen**

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 30 minutes in which preparatory tasks are solved.

Prerequisites
none

Competence Goal
Students will be able to understand and apply advanced and modern methods of channel coding. They get to know various tools of modern coding theory for the analysis and optimization of coding schemes, conceptual design approaches of error correction building blocks as well as applications in digital communications (for example, 5G). Additionally, they will get knowledge to current research topics and research results.

Content
The course expands on the topics dealt with in the lecture “Verfahren der Kanalcodierung”. The focus is on modern methods that have been brought into practice in the past few years and that achieve the capacity limits postulated by Shannon. For this purpose, known techniques have to be extended and new methods have to be learnt additionally. The lecture introduces the theoretical limits very quickly and follows with a discussion on the basic concepts of channel coding, including block codes. Based on this, modern error correction methods like LDPC codes, spatially coupled codes, and Polar codes are treated in depth. The lecture ends with a view on the application of channel coding in classical and distributed storage scenarios and in computer networks. Many of the applications are illustrated with example implementations in software (python/MATLAB).

Module Grade Calculation
The modul grade is the grade of the oral exam.

Workload
- Lecture attendance time: 15 * 3 h = 45 h
- Presence time Exercise: 15 * 1 h = 15 h
- Lecture preparation / revision: 15 * 3 h = 45 h
- Exercise: 15 * 1 h = 15 h
- Exam preparation and attendance: 60 h

Total workload: approx. 180 h = 6 LP

Recommendations
Previous attendance of the lectures "Communication Engineering I" and "Probability Theory" is recommended. Knowledge from the lecture "Applied Information Theory" can be helpful. Previous attendance of the lecture “Verfahren der Kanalcodierung” can be helpful, but is not necessary.

M 12.23 Module: Circular Factory [M-MACH-107636]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
8 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-114974	Circular Factory	ST	8 CP	Lanza

Assessment
written exam, duration 120 minutes

Prerequisites
none

Competence Goal
The students ...

- can name dimensions of circularity and circular economy methods (e.g., repair, refurbish, recycle) and describe them in detail.
- can describe challenges in planning and control circular factories, including remanufacturing networks and disassembly systems.
- are able to apply guidelines for designing circular products.
- distinguish data acquisition techniques for metrologically assessing returned products and apply uncertainty-driven product modeling in circular production systems.
- have methodical knowledge on learning from human observation and disassembly automatization and apply this knowledge to new problem cases.
- can describe reprocessing methods, including reconditioning and material characterization.
- understand the challenges in intralogistics for circular products.

After completing this course, students are able to understand the challenges of establishing a circular economy. They are also able to evaluate possible solutions and assess them in relation to these challenges. In particular, students are ultimately able to understand circular production in a circular factory holistically and to relate it to existing concepts in industrial practice.

Content

The module provides a comprehensive overview of the principles, methods, and technologies of circular production and the circular economy within industrial engineering. It conveys a well-founded and practice-oriented understanding of how circularity can be systematically implemented across products, processes, and factories. At the beginning of the module, fundamental concepts of sustainability, circular economy, and circularity dimensions are introduced, including strategies such as repair, refurbishment, remanufacturing, and recycling.

Building on these fundamentals, the module addresses the planning and control of circular factories, with particular emphasis on remanufacturing networks, disassembly systems, and reprocessing operations. Key challenges related to uncertainty, product variety, and quality variability of returned products are discussed. Methods for data acquisition and metrological assessment of end-of-life and returned products are presented, alongside uncertainty-driven product modeling approaches for circular production systems.

Further content focuses on the design of circular products, including design guidelines that enable reuse, disassembly, and material recovery. In addition, the module covers learning from human observation and approaches to disassembly automation, enabling students to transfer methodical knowledge to new and unfamiliar problem cases. Reprocessing methods such as reconditioning and material characterization are systematically explained, and challenges in intralogistics for circular products are analyzed.

Based on examples from current research and industrial practice, the module highlights the holistic interaction between product design, production systems, logistics, and information flows in circular factories. By integrating these perspectives, students develop a comprehensive understanding of circular production and are able to evaluate solution approaches in relation to technical, economic, and organizational challenges encountered in industrial applications.

Module Grade Calculation

The module grade corresponds to the grade of the written examination.

Workload

regular attendance: 62 hours
self-study: 178 hours
In total: 240 h

Teaching and Learning Methods

Lecture and excursions

M 12.24 Module: Clinical Translation and Evaluation of Biomedical Technologies [M-ETIT-107798]

Coordinators: Prof. Dr. Werner Nahm
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-115194	Clinical Translation and Evaluation of Biomedical Technologies		**3 CP**	**Nahm**

Assessment
The examination takes place in form of a written examination lasting 90 minutes.

Prerequisites
none

Competence Goal
The students:

- Understand the principles of clinical translation of biomedical technologies from preclinical development to clinical evaluation
- Are able to relate technical properties of biomedical technologies to requirements for clinical evaluation and study design
- Differentiate between study designs used in clinical research, including observational studies and randomized controlled trials
- Understand the formulation of clinical research questions, hypotheses, and endpoints
- Understand sources of bias and confounding in clinical studies and methods for their minimization
- Understand basic statistical concepts relevant to the design and interpretation of clinical studies
- Understand the principles of Good Clinical Practice (GCP), trial conduct, and study governance
- Recognize ethical principles in clinical research, including informed consent, patient protection, and risk-benefit assessment
- Are able to interpret, critically appraise, and discuss clinical studies in the context of biomedical engineering applications

Content
The module addresses the increasing importance of clinical evidence in the development and evaluation of biomedical technologies. It is designed to provide students with an introduction to the clinical translation and evaluation of biomedical technologies, with a focus on the role of engineers in the generation, analysis, and interpretation of clinical evidence. The module emphasizes the interface between engineering development and clinical evaluation, enabling students to assess how technical design decisions influence clinical study design and outcomes.

The module gives an overview of the systematic transfer of biomedical technologies from preclinical development to clinical investigation, with a focus on the generation and evaluation of clinical evidence.

A central aspect of the module is the methodological framework of clinical studies. This includes the formulation of research questions and hypotheses, the definition of endpoints, the selection of appropriate study designs, as well as the identification and mitigation of bias and confounding. Basic statistical concepts required for study design and data interpretation are introduced.

Furthermore, the module introduces principles of clinical research methodology, Good Clinical Practice, and biomedical ethics relevant to biomedical engineering applications, enabling students to interpret and critically appraise clinical studies, with a focus on assessing the validity and applicability of clinical evidence in biomedical engineering contexts.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
The workload includes:

- attendance in lectures: 15 * 2 h = 30 h
- preparation / follow-up: 15 * 2 h = 30 h
- preparation of and attendance in examination: 30 h

A total of 90 h = 3 CP

Teaching and Learning Methods
Lecture (2 SWS)

Literature
Lecture notes and selected scientific publications will be provided during the course.

M 12.25 Module: Coding of Audiovisual Signals [M-ETIT-107440]

Coordinators: Prof. Dr.-Ing. Laurent Schmalen
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114615	Coding of Audiovisual Signals	**ST**	**3 CP**	**Schmalen**

Assessment
Success control takes place in the form of an oral examination of approx. 20 min.

Prerequisites
Knowledge of basic engineering mathematics including integral transformations (e.g. M-MATH-101731 - Höhere Mathematik I, M-MATH-101732 - Höhere Mathematik II, M-MATH-101738 - Höhere Mathematik III) and probability theory (e.g. M-ETIT-102104 - Wahrscheinlichkeitstheorie) as well as basic knowledge of communications engineering (e.g. M-ETIT-106372 - Signale und Systeme, M-ETIT-106338 - Grundlagen der Datenübertragung).

Competence Goal
Students will be able to understand and apply the methods and tools of source coding. Students will learn a variety of tools for quantizing signals, transforming them into efficient storage and lossless compression methods. They continue to learn the theoretical limitations of source coding and can classify various practical methods based on the theoretical limits of their performance. With the help of numerical methods you can solve problems of source coding yourself.

Content
The course extends the topics covered in the lecture in Communication Technology I. The focus here is on methods that emerge when considering source coding. For this purpose, partially known techniques have to be extended, in some cases new methods have to be learned. Source coding is an indispensable tool in communications engineering, on the one hand to compactly represent and prepare multimedia signals for transmission and, on the other hand, to use storage capacity efficiently and economically. Source coding provides the direct link between the system user and the actual data transmission. The first part of the lecture deals with lossless source coding techniques, such as those used to reduce file size in popular zip format, but also more general lossless methods for the transmission of high quality signals. The second part is devoted to source coding of multimedia signals and, in particular, regards the source coding of audio and video signals. Different methods of quantization of multimedia signals are discussed and then shown how the quantized signals can be encoded in order to obtain as compact a representation as possible. In addition to predictive methods, transformation coding is also described. All of the methods are described in terms of their use in modern methods of source coding such as MP3, JPEG, H264. Many of the applications are illustrated with example implementations in software (python / MATLAB).

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
Attendance lecture: 15 * 2 h = 30 h

Preparation / Postprocessing Lecture: 15 * 2 h = 30 h

Exam preparation and presence: 30 h

Total: 90 h = 3 LP

Recommendations
Previous visit to the lecture "Telecommunications I", "Probability Theory" and "Signals and Systems" is recommended. Knowledge from the lectures "Applied Information Theory" is helpful, but not necessary.

M 12.26 Module: Communication Networks and Systems Lab [M-ETIT-107761]

Coordinators: Prof. Dr.-Ing. Peter Rost
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-115146	Communication Networks and Systems Lab	WT+ST	6 CP	Rost
T-ETIT-115147	Communication Networks and Systems Lab - Reports	WT+ST	0 CP	Rost

Assessment
The success control takes place in the form of

- a coursework (through reports on each experiment) and
- a final oral examination .

The reports are graded as "pass" or "fail." A "pass" grade is required for admission to the oral examination. One report is approx. 15 pages long.

The oral exam lasts approx. 25 min.

Prerequisites
none

Competence Goal
Students are enabled to understand telecommunications protocols and systems. They are enabled to comprehend the structure and function of communication networks, i.e., both wired and mobile communications, using simple experimental setups. They will be able to understand the requirements and design principles of communication systems. Specifically, experiments are conducted on the following topics:

1. Structure and functionality of Ethernet networks
2. Structure and functionality of IP-based networks, including necessary transport protocols and auxiliary functions
3. Structure and functionality of mobile networks
4. Acquisition and analysis of typical mobile communication parameters

Content
The workshop consists of 10 experiments:

- Introduction with simple examples from the area of networks and systems (mostly simulative)
- Setting up an Ethernet-based network with multiple wired and wireless bridges connected to form a network
- Setting up an IP-based network and implementing typical auxiliary protocols such as ARP and ICMP
- Implementing typical transport protocols (TCP, UDP), configuring them, and evaluating their performance characteristics
- Setting up a cellular connection using a cellular modem and control commands via a connected computer. Observing the behavior of the modem in different setups.
- Measurement and recording of typical features of a mobile phone connection, e.g. received signal power and quality.
- Building and analyzing a map showing the different quality indicators in a limited area.
- Comparison and synthesis of different maps to understand measurements on different frequency bands and using different setups.
- Building an ML algorithm based on measured values to predict quality indicators of the mobile network. This part is divided into a part of building the algorithm and conducting experiments to evaluate its performance.

Module Grade Calculation
The module grade is the grade of the oral examination.

Additional Information
Attendance is compulsory during all laboratory sessions, including the introductory session. Compulsory attendance is necessary both for carrying out the work in the team on site and for the practical teaching of techniques and skills that cannot be learned in pure self-study.

Workload

1. Attendance of workshop incl. experiments: 10x6 = 60 h
2. Prepration of experiments: 10x5 = 50 h
3. Review of experiments / report: 10x6 = 60 h
4. Preparation und execution of oral exam: 10 h
   Overall: 180 h = 6 LP

Recommendations
Previous attendance of the lecture "Nachrichtensysteme" and “Mobile Communications” is recommended.

M 12.27 Module: Communication Systems and Protocols [M-ETIT-100539]

Coordinators: Dr.-Ing. Jens Becker
Prof. Dr.-Ing. Jürgen Becker

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
5 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-101938	Communication Systems and Protocols	**ST**	**5 CP**	**Becker, Becker**

Assessment
The examination consists of a written examination of 120 min.

Prerequisites
none

Competence Goal
The students are able to:

- know basic communication systems and to name them
- categorize different communication systems in regards to possible constraints
- name basic mechanisms of communication systems
- carry out these mechanisms
- choose valid mechanisms suitable under given constraints
- design a communication system adhering to constraints, specifications and be able to choose suitable methods, components, and subsystems
- know current communication systems and know about their properties, mechanisms and application.

Content
The lecture will present the physical and technical basics for the design and construction of communication systems. Procedures and technical implementations for communication between electronic devices are presented. This includes, among other things, modulation methods, line model, arbitration, synchronization mechanisms, error correction mechanisms, multiplexing, communication systems, bus systems and on-chip communication. On the basis of selected practical examples, the application of the lecture contents in real systems is demonstrated.

- Information: Definition, Representation, Communication
- Physics: Media, Signals, Mathmatical Descriptions, Line Coupling & Termination, AD Conversion & Sampling, Line Codes, Modulation
- Data Transmission: Definition & Requirements, Transmission Channels, MultiUse of Channels, Multiplexing, Multiple Senders (Arbitration), Multiple Receivers (Addressing), Classification, Interfaces
- Bus Systems: Definitions, Protocols, Transmission of Dataframes, Classification
- Error Protection: Fundamentals, Errors, Error Detection/Correction: Error Handling
- Topologies: physical, logical, examples
- Networks: networks vs. busses, structure, Network specific topologies, routing, OSI Model, TCP/IP, Ethernet
- Classification of Com.Systems
- Real World Systems: Automotive Busses, PC Busses, Field Busses, Networks

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
The workload includes:

1. Attendance in 15 lectures an 7 exercises: 33 h
2. Preparation / follow-up: 66 h (2 h per unit)
3. Preparation of and attendance in examination: 24 h + 2 h

A total of 125 h = 5 LP

Recommendations
Knowledge of the basics from the lecture "Digitaltechnik"is helpful.

M 12.28 Module: Communications Engineering Lab [M-ETIT-107136]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114159	Communications Engineering Lab	**WT+ST**	**6 CP**	**Jäkel**

Assessment
The examination consists of the participation in the experiments and an oral examination of approx. 30 min. The overall impression is rated.

Prerequisites
none

Competence Goal
Students are able to apply methods of signal processing and communications engineering in the implementation of communication systems.

They are able to carry out communications engineering calculations and use the tools required for simulations methodically and appropriately. This enables students to classify the components involved in a communication system in terms of their performance and to understand their interaction in an overall system.

Content
The practical course consists of 11 experiments and covers the following topics:

Introduction to Python, DFT, the sampling theorem, filter design and multirate filters, stochastic signals, digital modulation methods, source coding, channel coding, GNU Radio and Software Defined Radio, OFDM, synchronization algorithms and optimization.

Module Grade Calculation
The module grade results of the participation in the experiments and an oral examination. Details will be given during the lecture.

Additional Information
Attendance is compulsory during all laboratory sessions, including the introductory session. Compulsory attendance is necessary both for carrying out the work in the team on site and for the practical teaching of techniques and skills that cannot be learned in pure self-study.

Workload
- Attendance time practical course: 11 * 4 h = 44 h
- Lecture preparation and follow-up: 11 * 8 h = 88 h
- Exam preparation and attendance of exame: 48 h

Total: 180 h

Recommendations
Previous attendance of the lectures "Signals and Systems" and "Communications Engineering I".

M 12.29 Module: Components of Power Systems [M-ETIT-106689]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113445	Components of Power Systems	**ST**	**3 CP**	**Leibfried**

Assessment
The examination takes place in form of an oral examination lasting approx. 20 minutes.

Prerequisites
none

Competence Goal
Students know the main components of electrical networks and how they interact. Students understand the structure of cable systems in AC and DC voltage technology. The procedure for insulation coordination and overvoltage protection in the high-voltage grid is introduced. They will be able to reproduce key relationships. Students are familiar with the future challenges and trends for selected components and transmission technologies of electrical grids.

Content

- Substations
  - o Types of Substations
  - o Basic Requirements ans Standardization
  - o Air Insulated Switchgears
  - o Gas Insulated Switchgears
- Principle of Inductive Equipments
  - o Magnetic Field in an Iron Circuit
  - o Basic Design of Transformers
- Transformers
  - o Overview
  - o Design and Components of Power Transformers and Reactors
- Overhead Transmission Lines
  - o Development of overhead lines system voltages
  - o Grid Development with OVH Transmission Lines
  - o Parts of an Overhead Line
  - o Comparison DC and AC OVH Transmission Lines
  - o Effects of OHL on Environment
- Cables
  - o Development of Cable Lines System Voltages
  - o Grid Development with cable systems
  - o Parts of Cables Systems
  - o Comparison DC and AC Cables Systems
  - o Offshore Cables Systems
  - o Effects of Cables on Environment
- Insulation Arresters
  - o Insulation Coordination
  - o Surge Arresters
- Circuit Breaker and Disconnectors
  - o Circuit Breakers
  - o Disconnectors
- Power Cable Accessories and Power Line Monitoring
  - o Accessories
  - o Power Line Monitoring
- Application of Power Electronics in Power System
  - o Development of Power Electronics
  - o Fundamental Principles of PE
  - o Application of PE in Power System
- Energy Innovation and Trends

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
The workload includes:

1. attendance in lectures: 30 h
2. preparation / follow-up and preparation of and attendance in examination: 60 h

A total of 90 h = 3 CR

M 12.30 Module: Compressed Sensing and Approximate Message Passing: Theory and Applications [M-ETIT-107375]

Coordinators: Prof. Dr.-Ing. Laurent Schmalen
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114487	Compressed Sensing and Approximate Message Passing: Theory and Applications		**3 CP**	**Fengler, Schmalen**

Assessment
Success control takes place in the form of an oral examination of approx. 30 min.

Prerequisites
Knowledge of probability theory is required (e.g. "M-ETIT-102104 – Wahrscheinlichkeitstheorie").

Competence Goal
Students will learn how to design and analyze efficient algorithms for sparse signal reconstruction, understand performance limits via phase transitions, and explore applications in imaging, communications, and machine learning. By the end of the course, students will be equipped to apply these tools to real-world problems and enter independent research in related areas.

Content
This course covers compressed sensing (CS) and approximate message passing (AMP), two powerful frameworks at the intersection of signal processing, statistical inference, and optimization. Starting from the fundamentals of sparse signal recovery, the course introduces students to key concepts in high-dimensional statistics, signal processing and Bayesian inference that underlie CS and AMP. Special attention will be given to the AMP algorithm and its variants, which provide an iterative solution to a wide set of linear inverse problems.

Topics include: Sparsity and underdetermined linear systems, Convex and greedy recovery algorithms (e.g., Basis Pursuit, Orthogonal Matching Pursuit),Introduction to high-dimensional probability and random matrices, Approximate message passing and state evolution, Bayesian CS and inference in graphical models, Applications in imaging, communications, and machine learning.

Module Grade Calculation
The module grade is the grade of the oral examination.

A bonus can be earned by successfully completing programming exercises. The exact criteria for awarding a bonus will be announced at the beginning of the lecture period. If the grade in the exam is between 4.0 and 1.3, the bonus improves the grade by one grade (0.3 or 0.4). Bonus points do not expire and are retained for any examinations taken at a later date. The final assessment of the bonus performance is carried out by the examiner and demonstrably documented.

Workload

1. Attendance to the lecture: 20 * 1,5 h = 30 h
2. Preparation and review: 20 * 3 h = 60 h
3. Preparation for the exam: included in preparation and review

In total: 90 h = 3 LP

Recommendations
Some familiarity with statistics or machine learning is helpful but not required. Knowledge from the modules "M-ETIT-100444 – Angewandte Informationstheorie" and "M-ETIT-102103 – Nachrichtentechnik I" is helpful.

M 12.31 Module: Computational Imaging [M-INFO-106190]

Coordinators: Prof. Dr.-Ing. Jürgen Beyerer
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
5 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-112573	Computational Imaging	**WT**	**5 CP**	**Meyer**

Assessment
See partial achievements (Teilleistung)

Prerequisites
See partial achievements (Teilleistung)

Competence Goal
Qualification goal: Students are able to model questions of machine vision optically and algorithmically and to process them using holistic optimization.

Learning objectives: Students know
- the essential components of machine vision, their optical modelling and suitable coding methods in the sense of computational imaging,
- methods for emitting, capturing and processing light fields for applications in photography and industrial image processing,
- the concept of light transport analysis, corresponding modelling, capturing and processing methods and
- approaches to holistic modelling and optimization of optical image capturing and processing systems.

Content
Digital image acquisition and processing have revolutionized various fields of applications, e.g., medical imaging or automated visual inspection. Yet, the design of most such systems is still based on the separate and individual optimization of the employed illumination, image acquisition and image processing components. By following a holistic approach for system design, modelling and optimization, computational imaging methods yield superior performance with respect to the state of the art. After introducing the students into relevant basics of optics and signal theory, the lecture will thoroughly cover various topics of computational imaging. Accompanying practical exercises will complement the theoretical part of the lecture. The course will enable students to adequately model artificial vision problems in the sense of computational imaging in order to obtain holistically optimal solutions.

Workload
Lecture with 2 SWS + 1 SWS exercise
5 ECTS corresponds to approx. 150 hours
approx. 30 hours lecture attendance,
approx. 15 hours exercise attendance,
approx. 90 hours post-processing and working on the exercises
approx. 30 hours Exam preparation

Literature
- Ayush Bhandari, Achuta Kadambi, Ramesh Raskar, Computational Imaging, MIT Press, 2022.
- Jürgen Beyerer, Fernando Puente León, Christian Frese, Machine Vision, Springer, 2015.
- Joseph. W. Goodman, Introduction to Fourier Optics. 4. Auflage W. H. Freeman, 2017.

M 12.32 Module: Computational Intelligence [M-MACH-105296]

Coordinators: apl. Prof. Dr. Ralf Mikut
apl. Prof. Dr. Markus Reischl
Organisation: KIT Department of Mechanical Engineering
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-114110	Computational Intelligence in Mechanical Engineering	**WT**	**4 CP**	**Meisenbacher, Mikut, Reischl**

Assessment
see individual course

Prerequisites
None

Competence Goal
The students are able to apply the fundamental methods of computational intelligence (fuzzy logic, artificial neural networks, evolutionary algorithms) efficiently. They know the basic mathematical foundations and are able to transfer these methods to practical applications.

Content

- Terms and definitions Computational Intelligence, application fields and examples
- Fuzzy logic: fuzzy sets; fuzzification and membership functions; inference: T-norms and -conorms, operators, aggregation, activation, accumulation; defuzzification methods, structures for fuzzy control
- Artificial Neural Nets: biology of neurons, Multi-Layer-Perceptrons, Radial-Basis-Function nets, Kohonen maps, training strategies (Backpropagation, Levenberg-Marquardt)
- Evolutionary Algorithms: Basic algorithm, Genetic Algorithms and Evolution Strategies, Evolutionary Algorithm GLEAM, integration of local search strategies, memetic algorithms, application examples
- Deep Learning: History, Architectures, Training strategies, Interpretability and Explainable AI, Use Cases

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
The workload includes:

1. attendance in lectures and exercises: 15*2 h = 30 h
2. preparation / follow-up: 60 h
3. preparation of and attendance in examination: 30 h

A total of 120 h

Teaching and Learning Methods
Lecture

M 12.33 Module: Control of Power-Electronic Systems [M-ETIT-105915]

Coordinators: Dr.-Ing. Andreas Liske
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111897	Control of Power-Electronic Systems	**ST**	**6 CP**	**Liske**

Assessment
The assessment takes place as part of an online examination in text form with face-to-face supervision (§5, statutes for conducting online examinations at KIT) lasting 60 minutes.

The exam takes place on KIT computers in a pool room.

Prerequisites
none

Competence Goal
Students are able to design PI controllers for cascade control in the typical applications of power electronic systems (current, voltage, torque and speed control loops). They are familiar with standard design methods (absolute value optimum, symmetrical optimum) and the frequency characteristic method. They know about the necessity of control loop extensions such as feedforward control, feedforward control and anti-windup measures. They know the common methods for modeling power electronic systems, as well as DC machines and three-phase machines. Students are familiar with the space vector representation and its application in the control of converters in three-phase networks, as well as synchronous and asynchronous machines. They are familiar with the control methods of rotor flux-oriented control and know the usual methods for operating three-phase machines (field weakening, MTPA).

Content
Power electronic circuits are increasingly penetrating all areas of application in electrical energy technology. This ranges from the necessary conversion of electrical energy from decentralized, regenerative energy sources to energy supply networks and energy consumers such as electrical drive technology.

Increasing the quality, reliability and energy efficiency of these power electronic systems is largely achieved by controlling the electrical energy quickly and precisely and adapting it to the load.

The lecture presents the control methods that enable the regulation of current, voltage, speed or torque. The application of the methods and their effect on the system behavior are discussed using technically relevant solutions from practice. Single-phase DC/DC controllers, three-phase "Active Front End" and drive solutions with DC machines, synchronous machines and asynchronous machines serve as practical teaching examples.

Module Grade Calculation
The module grade is the grade of the online examination.

Workload
56h = 21x V + 7x Ü à 2h

21h = 21x post-processing of V à 1 h

12h = 6x preparation of exercises à 2 h

80h = preparation for the exam

Total = 169h = (corresponds to 6 LP)

M 12.34 Module: Control Technology [M-MACH-105348]

Coordinators: Hon.-Prof. Dr. Christoph Gönnheimer
Dr.-Ing. Philipp Gönnheimer

Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-105185	Control Technology	**ST**	**4 CP**	**Gönnheimer**

Assessment
Written Exam (60 min)

Prerequisites
None

Competence Goal
The students ...

- are able to name the electrical controls which occur in the industrial environment and explain their function.
- can explain fundamental methods of signal processing. This involves in particular several coding methods, error protection methods and analog to digital conversion.
- are able to choose and to dimension control components, including sensors and actors, for an industrial application, particularly in the field of plant engineering and machine tools. Thereby, they can consider both, technical and economical issues.
- can describe the approach for projecting and writing software programs for a programmable logic control named Simatic S7 from Siemens. Thereby they can name several programming languages of the IEC 1131.

Content
The module control technology gives an integral overview of available control components within the field of industrial production systems.
The first part of the module deals with the fundamentals of signal processing and with control peripherals in the form of sensors and actors which are used in production systems for the detection and manipulation of process states.
The second part handles with the function of electric control systems in the production environment. The main focus in this chapter is laid on programmable logic controls, computerized numerical controls and robot controls. Finally the module ends with the topic of cross-linking and decentralization with the help of bus systems.
The module is very practice-oriented and illustrated with numerous examples from different branches.

The following topics will be covered:

- Signal processing
- Control peripherals
- Programmable logic controls
- Numerical controls
- Controls for industrial robots
- Distributed control systems
- Field bus
- Trends in the area of control technology

Workload
regular attendance: 21 hours
self-study: 99 hours

Teaching and Learning Methods
Lecture

M 12.35 Module: Cryogenic Engineering [M-CIWVT-104356]

Coordinators: Prof. Dr.-Ing. Steffen Grohmann
Organisation: KIT Department of Chemical and Process Engineering
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-108915	Cryogenic Engineering	**WT**	**6 CP**	**Grohmann**

Assessment
The examination is an oral examination with a duration of about 30 minutes (section 4 subsection 2 number 2 SPO).

Prerequisites
None

Competence Goal
Understanding the principle and modelling of regenerative cryocoolers; Understanding and applying of essential engineering methods and components for the conception and design of low-temperature plants and cryostat systems; Understanding of laboratory measurement principles, assessing and applying of sensors and instruments for cryogenic measurement tasks and analysing of measurement uncertainties

Content
Cryogenic applications; Regenerative cooling with cryocoolers; Fundamentals of low-temperature plant and cryostat design, including fluid mechanics and heat transfer, thermal contacts and thermal insulation, cryogenic pumping of gasses, regulations, design components and safety; General principles of measurement and uncertainties as well as cryogenic temperature, pressure and flow measurement

Module Grade Calculation
The grade of the oral examination is the module grade.

Workload

- Attendance time (Lecture): 45 h
- Homework: 45 h
- Exam Preparation: 90 h

M 12.36 Module: Cryptographic Protocols [M-INFO-105631]

Coordinators: Prof. Dr. Jörn Müller-Quade
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
5 CP	graded	Each summer term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-111261	Cryptographic Protocols	**ST**	**5 CP**	**Müller-Quade**

Assessment
See Partial Achievements (Teilleistung).

Prerequisites
See Partial Achievements (Teilleistung).

Competence Goal
The student

- knows and understands the primitives and protocols presented in the lecture, their (game-based) security definitions and proofs
- knows the real/ideal security notion and can independently analyze the security of protocols in it
- knows and understands basic schemes for secure multi-party computation and their advantages and disadvantages can apply the schemes for secure multi-party computation

Content
While classical cryptography is concerned with ensuring authenticity and secrecy against external adversaries, there are now also a large number of interactive protocols between mutually distrusting parties.
Building on the lecture "Theoretical Foundations of Cryptography", this lecture introduces such basic primitives, protocols and corresponding security notions.
In the first part of the lecture, basic (interactive) building blocks such as commitment schemes, secret sharing, zero-knowledge proof systems and oblivious transfer are introduced. Game-based notions and the real/ideal security notion are used to prove security.
Building on this, the second part introduces more complex protocols for the secure joint evaluation of arbitrary functions on secret inputs. Both protocols based on secret sharing and so-called "garbled circuits" are covered.
First, security against so-called passive adversaries, who honestly follow the protocol and merely attempt to learn additional information, is considered. Building on this, security against active adversaries, who may deviate from the protocol at will, is considered.

Additional Information
The event takes place as an "inverted classroom". The material is presented in the form of videos, which are then interactively deepened in the subsequent face-to-face event.

Workload
Attendance time for inverted classroom videos: 24 h
Preparation/follow-up: 36 h
Attendance time in the exercise: 24 h
Preparation/follow-up of the same: 36 h
Exam preparation and attendance in the same: 30 h

= 150 h

Recommendations
Students should be familiar with the contents of the module "Theoretical Foundations of Cryptography".

M 12.37 Module: Cryptographic Protocols II: Universal Composability [M-INFO-107435]

Coordinators: Prof. Dr. Jörn Müller-Quade
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-114602	Cryptographic Protocols II: Universal Composability	**WT**	**3 CP**	**Müller-Quade**

Assessment
See partial achievements (Teilleistung)

Prerequisites
See partial achievements (Teilleistung)

Competence Goal
The student
- understands the problems that occur in protocol composition and can explain them using examples.
- knows and understands the frameworks, security concepts and their properties presented in the lecture and can explain, relate and prove
these.
- knows and understands the protocols and proofs presented in the lecture and can reproduce them.
- can independently apply the techniques presented in the lecture, for example to prove or demonstrate the security or insecurity of simple protocols.

Content
In the lecture "Kryptographische Protokolle", methods and building blocks for secure multi-party computation were presented. The security was shown for a single execution and under sequential composition.

In reality, however, protocols are executed concurrently - both multiple instances of the same protocol (concurrent composition) and multiple, different protocols (general composition) that were designed independently of each other.

This protocol composition is not adequately covered by classical security concepts: it may be that the execution of a single protocol instance can be proven to be secure. However, if the same protocol is executed several times simultaneously, all security may be lost.

In this lecture, we re-visit the topics of the “Kryptographische Protokolle” lecture with a focus on composability. The first part of the lecture therefore introduces the concept of universally composable (UC) security, which depicts the setting of general composition in which arbitrary protocols are executed concurrently. An important insight here is that UC security can only be achieved with the help of trust assumptions, so-called "setups". In the second part of the lecture, important protocols that fulfill this strong security are presented, for example for commitments or general secure multi-party computation. At the end of the lecture, further composing security notions and related protocols are considered, which are weaker than UC security, but without setups.

Workload
90 h
Lectures: 24 h
Self-study during semester: 31 h
Preparation for the exam: 35 h

Recommendations
Students should be familiar with the contents of the modules "Theoretische Grundlagen der Kryptographie" (Theoretical Foundations of Cryptography) and „Kryptographische Protokolle“ (Cryptographic Protocols I).

M 12.38 Module: Current Topics of Solar Energy [M-ETIT-100507]

Coordinators: Prof. Dr.-Ing. Michael Powalla
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100780	Current Topics of Solar Energy	**WT**	**3 CP**	**Powalla**

Assessment
Success contraol takes placein form of a 30-minute presentation and the preparation of a short written paper. The overall impression is rated.

Prerequisites
none

Competence Goal
- The students independently develop a knowledge base on a specific topic area of solar energy with prioritization and compression of knowledge.

- The students create professional presentation slides in common Office software (e.g. MS PowerPoint). Essential components of a high-quality scientific slide composition (title slide, table of contents, ... summary, font size, image quality, ...) are discussed.

- Students pay attention to the correct integration of citations and the use of presentation technology and live streaming technology.

- Students use online presentation techniques (MS Teams, ZOOM).

- Students pay particular attention to time management and presentation structures during presentations.

- Students are able to give appealing and comprehensible presentations and use body language and rhetoric appropriately.

- Students are able to manage discussions in a frontal position and master appropriate responses to "difficult questions".

- Students are able to write a short summary of the presentation, taking into account important principles for writing a scientific or technical report (text structure, citation of sources, etc.).

- Students can contribute constructively as an active listener and participant in discussions.

- Students can analyze, structure and formally describe knowledge about topics in the field of solar energy.

- Students are able to critically evaluate different presentations.

Content
The module is an in-depth seminar on solar energy/photovoltaics. In the seminar, students give presentations on various solar energy topics. This involves expanding their specialist knowledge and applying presentation techniques. Students are expected to attend regularly, give a 30-minute presentation and prepare a short written paper on the topic.

The course is preferably held as a face-to-face event, but can also be held online and as a hybrid event.

The course is designed for students of engineering and natural sciences who have already successfully attended a basic lecture on photovoltaics/solar energy. However, prior knowledge is not a prerequisite. The number of participants is limited. The course will be held in German, but English lectures will also be accepted in exceptional cases.

Own suggestions for topics are expressly welcome. Current lecture topics will be provided.

The module covers the following topics:

Subject area I: Solar power and integration into the energy industry

From climate protection to economic efficiency

Subject area II: Technology and application

From integration in buildings to the energy yield of large-scale systems

Subject area III: Photovoltaics - fundamentals and materials science

From new solar cell materials, components and manufacturing processes

Module Grade Calculation
The module grade is made up of the assessment of an approx. 30-minute specialist presentation and the preparation of a short written paper. Details will be given during the lecture.

Additional Information
Attendance at the event is compulsory. Compulsory attendance is necessary both for carrying out the work in the team on site and for the practical teaching of techniques and skills that cannot be learned purely through self-study.

Workload

1. Attendance time in lectures, exercises: 15*2 h = 30 h
2. Preparation/follow-up of the same: 15*0.5 h = 7.5 h
3. Preparation of own presentation incl. research and preparation of slides 45 h
4. Short written elaboration 7.5 h

Total: 90 h

Recommendations

- Knowledge of the basics of solar energy is helpful.
- The content of the lectures can be physical or engineering, but also economic.

M 12.39 Module: Cyber-Physical Modeling [M-ETIT-106953]

Coordinators: Prof. Dr.-Ing. Mike Barth
Prof. Dr.-Ing. Sören Hohmann

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Fundamentals)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113908	Cyber-Physical Modeling		**6 CP**	**Barth, Hohmann**

Assessment
The examination takes place in the form of a written examination lasting 90 min.

Prerequisites
none

Competence Goal

- The students are familiar with the concepts of Cyber-Physical System.
- Students understand the need for advanced methods and services in the field of automation.
- Students can validate different information models and ontologies for their applicability in CPS.
- Students will be able to model data, information and knowledge or extract them from existing systems.
- The students know suitable modeling tools and their application.
- The students understand the general model concept as well as the characteristics of physical and data-based modeling and can describe their differences.
- They can structure complex systems and systematically analyze dependencies of subsystems.
- They can explain the general procedure of physical and data-based modeling, apply it to technical systems, and analyze the results.
- They can apply causal and non-causal modeling approaches and distinguish between them.
- Students have gained an understanding of generalized, cross-domain, physical relationships and can develop models for electrical, mechanical, pneumatic and hydraulic systems.
- They can describe the relationship between generalized, cross-domain, physical models and basic procedures of physical-based control and explain their advantages / limitations based on basic knowledge of control engineering.
- The students can estimate and judge the effects of disturbances and real conditions on the identification results.

Content
This course aims at engineering students that focus on a system-based engineering curriculum, including architectures, modeling & simulation for Cyber Physical Systems. The module is designed to teach students the theoretical and practical aspects of Digital Twins and their interconnection with their physical counterpart. It encompasses fundamental topics along the complete process of modeling technical systems. For this purpose, it includes the conception and construction of digital twins including their model components. In terms of modeling and simulation of physical systems, two major areas will be covered: On the one hand, physical-based modeling techniques which derive formal model equations based on analyzing the physical first principles of technical systems. This includes generalized equivalent circuits and variational analysis (Euler-Lagrange of the first kind). On the other hand, data-based identification techniques will be covered which are used to identify concrete model parameters for a given technical system from experimental data sets. When combining the identification with an initial, non-physical, structural set up of model equations, the complete process is often referred to as data-based modeling or black-box modeling. Both modeling areas base on available information about the physical system which is structured in Meta- and Information-Models. Examples that are covered in this lecture are Metamodels, e.g. AutomationML or the asset administration shell principles. Also, semantic web principles and ontologies will be part of the lecture content.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload

1. attendance in lectures an exercise: 3+1 SWS (60 h)
2. pre-/postprocessing of the lecture (90 h)
3. preparation of and attendance in the exam: (30 h)

A total of 180 h = 6 CR

Recommendations
Interest in Modeling and Simulation of modern Cyber-Physical Systems in combination with concepts of digital twins, system architectures and Co-Simulation.
Sound understanding of engineering mechanics, electrical, mechatronic systems / physics / Software-Engineering should be fulfilled to successfully attend the lecture, exercise tasks / case studies, and exam.

Teaching and Learning Methods
lecture (3 SWS), excercise (1 SWS)

M 12.40 Module: Deep Learning and Neural Networks [M-INFO-107197]

Coordinators: Prof. Dr. Jan Niehues
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-114219	Deep Learning and Neural Networks	**ST**	**6 CP**	**Niehues**

Assessment
See partial achievements (Teilleistung)

Prerequisites
See partial achievements (Teilleistung)

Competence Goal
Students will learn about the structure and function of different types of neural networks.

Students should learn the methods for training the various networks and their application to problems.

Students should learn the areas of application of the different types of networks.

Given a concrete scenario, students should be able to select the appropriate type of neural network.

Content
This module introduces the use of neural networks for the solution of solving various problems in the field of machine learning, such as classification, prediction, control or inference. or inference. Different types of neural networks are covered and their areas of application are illustrated using examples.

Workload
Lecture with 4 SWS, 6 CP.
6 LP corresponds to approx. 180 hours, of which
approx. 60 hours lecture attendance
approx. 90 hours post-processing (self-study)
approx. 30 hours exam preparation (self-study)

Recommendations
Prior successful completion of the core module "Cognitive Systems" is recommended.

M 12.41 Module: Deep Learning for Computer Vision: Advanced Topics [M-INFO-107695]

Coordinators: Prof. Dr.-Ing. Rainer Stiefelhagen
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-115078	Deep Learning for Computer Vision: Advanced Topics	**WT**	**3 CP**	**Stiefelhagen**

Assessment
See partial achievements.

Prerequisites
See partial achievements.

Competence Goal
Students gain an understanding of advanced deep learning methods and tasks, especially with regard to their applications in image processing (computer vision). Students are able to apply deep learning methods for special tasks, such as learning with little training data, changes in the data domain or uncertainty determination.

Content
Deep convolutional neural networks (CNNs) achieve excellent results in many areas of computer vision, but face challenges in real-world applications, such as dependence on costly annotated training data, high computing power or difficult traceability of decision paths. While for a long time the development of recognition algorithms was primarily driven by high recognition rates on large and cleanly annotated data sets, today application-relevant goals such as learning with little training data, explainability, uncertainty estimation or domain adaptation are becoming increasingly important.

The lecture deals with advanced network architectures, learning methods and research areas in the field of deep learning for computer vision. Among others, the following topics will be covered:

- Overview of deep learning, convolutional neural networks (CNN), problems of modern architectures
- Interpretability and explainability of CNNs
- Uncertainty in deep learning
- Learning with little training data
- Efficient architectures
- Advanced architectures (transformers, graph neural networks)
- Synergies of computer vision and language models
- Generative Adversarial Networks (GANs)
- Continuous learning

Module Grade Calculation
See partial achievements.

Workload
Attendance of the lectures: approx. 20 hours

Preparation and follow-up of the lecture: approx. 30 hours

Exam preparation: approx. 40 hours

Total: approx. 90 hours

Recommendations
Knowledge of deep learning basics is strongly recommended.

M 12.42 Module: Deep Learning for Computer Vision: Advanced Topics [M-INFO-107843]

Coordinators: Prof. Dr.-Ing. Rainer Stiefelhagen
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-115257	Deep Learning for Computer Vision: Advanced Topics	**WT**	**3 CP**	**Stiefelhagen**

Assessment
See partial achievements.

Prerequisites
See partial achievements.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-INFO-107695 - Deep Learning for Computer Vision: Advanced Topics must not have been started.

Competence Goal
Students gain an understanding of advanced deep learning methods and tasks, especially with regard to their applications in image processing (computer vision). Students are able to apply deep learning methods for special tasks, such as learning with little training data, changes in the data domain or uncertainty determination.

Content
Deep convolutional neural networks (CNNs) achieve excellent results in many areas of computer vision, but face challenges in real-world applications, such as dependence on costly annotated training data, high computing power or difficult traceability of decision paths. While for a long time the development of recognition algorithms was primarily driven by high recognition rates on large and cleanly annotated data sets, today application-relevant goals such as learning with little training data, explainability, uncertainty estimation or domain adaptation are becoming increasingly important.

The lecture deals with advanced network architectures, learning methods and research areas in the field of deep learning for computer vision. Among others, the following topics will be covered:

- Overview of deep learning, convolutional neural networks (CNN), problems of modern architectures
- Interpretability and explainability of CNNs
- Uncertainty in deep learning
- Learning with little training data
- Efficient architectures
- Advanced architectures (transformers, graph neural networks)
- Synergies of computer vision and language models
- Generative Adversarial Networks (GANs)
- Continuous learning

Module Grade Calculation
See partial achievements.

Workload
Attendance of the lectures: approx. 20 hours

Preparation and follow-up of the lecture: approx. 30 hours

Exam preparation: approx. 40 hours

Total: approx. 90 hours

Recommendations
Knowledge of deep learning basics is strongly recommended.

M 12.43 Module: Design and Architectures of Embedded Systems (ESII) [M-INFO-107230]

Coordinators: Prof. Dr.-Ing. Jörg Henkel
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-114254	Design and Architectures of Embedded Systems (ESII)	**WT**	**3 CP**	**Henkel**

Assessment
See partial achievements (Teilleistung)

Prerequisites
See partial achievements (Teilleistung)

Competence Goal
The student learns methods for mastering complexity and applies these methods to the design of embedded systems. He/she evaluates and selects specific architectures for embedded systems. Furthermore, the student receives an introduction to current research topics.

Content
Nowadays, it is possible to integrate several billion transistors on a single chip and thus realize complete SoCs (systems-on-chip). The trend towards being able to use more and more transistors continues unabated, meaning that the complexity of such systems will also continue to increase. Computers will increasingly be ubiquitous, i.e. they will be integrated into the environment and will no longer be perceived as computers by humans. Examples include sensor networks, electronic textiles and many more. However, the physically possible complexity will not be readily achievable in practice, as there is currently a lack of powerful design processes capable of handling this high level of complexity. Powerful ESL tools ("Electronic System Level Design Tools") and novel architectures will be required. The focus of this lecture is therefore on high-level design methods and architectures for embedded systems. Since the power consumption of (mostly mobile) embedded systems is of crucial importance, one focus of the design methods will be on the design with regard to low power consumption.

Workload
90 hours:
30 hours of lectures (15 weeks*2 hours)
30 hours of self study (15 weeks*2 hours)
15 hours of deepening (coding low level C and embedded C) to grasp the theory with practical
15 hours exam preparation

M 12.44 Module: Design of Electrical Machines [M-ETIT-100515]

Coordinators: Prof. Dr. Martin Doppelbauer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
5 CP	graded	Each winter term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100785	Design of Electrical Machines	**WT**	**5 CP**	**Doppelbauer**

Assessment
Success control takes place in the form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal
Students are able to derive all the specifications required for the design of an electrical machine from the framework data of the target application. On this basis, they can develop the electromagnetic design of a suitable electric machine using analytical and numerical methods.

Content
The lecture teaches the basics of calculating and designing electrical machines.

Particular emphasis is placed on the generation of rotating fields and forces, the various windings and the magnetic circuit, from which the various inductances are then calculated.

Separate chapters deal with numerical field calculation, the system equations of rotating field machines and the calculation of losses and efficiencies.

The lecture concludes with two chapters on the calculation of harmonic effects using harmonic field theories, including magnetic noise.

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
14x lecture + 7x exercise 1.5 h each = 31.5 h

14x follow-up of the lecture à 1 h = 14 h

7x preparation of the exercise à 3 h = 21 h

Preparation for the exam = 80 h

Total = 146.5 h (corresponds to 5 CP)

Recommendations
Module: Elektrische Maschinen und Stromrichter

M 12.45 Module: Detectores for Applications in Space and Astronomy [M-ETIT-100541]

Coordinators: Prof. Theo Scherer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100761	Detectors for Applications in Space and Astronomy	**WT**	**3 CP**	**Scherer**

Assessment
Sucess control takes place in form of an overall oral examination lasting approx. 20 minutes.

Prerequisites
none

Competence Goal
After completing the module, students are able to assign radiation sources and their function to an electromagnetic spectrum of astrophysical objects and can explain the structure and operating modes of detectors for the detection of visible light, radio waves, microwaves, IR, THz radiation, X-rays and g-radiation. At the same time, they are able to explain the technology of the construction (functionality), manufacture and operation of such detectors. The transfer of this knowledge enables students to tackle their own detector developments. In addition, they will learn about readout electronics, the cryogenics required to cool the elements and system integration in radio antennas and satellites (earthbound and in space) and will be able to transfer this knowledge to new detector systems to be developed in their later professional life. Classical and new detector principles are taught in the same way.

Content
The lecture imparts knowledge about the function, production and system integration of modern integrated detector circuits for the frequencies used and to be detected in astronomy and space travel in the range from 1 GHz to 5 THz. Fast semiconducting components (HEMTs, Schottky diodes, etc...) as well as superconducting integrated measuring systems based on SIS-Josephson mixers or so-called hot electron bolometers (HEBs) are used for this purpose. The structure widths of these components are in the micrometer or nanometer range, depending on the application. The lecture also covers system integration in satellites or earthbound telescopes in detail using existing instruments worldwide. The function and design of X-ray detectors for future space missions based on TES/SQUID will be explained as well as modern kinetic inductance detectors (KIDs), WIMP and neutrino detectors for the field of astroparticle physics and cosmology.

- Astrophysical radiation sources in space, frequency ranges.
- Semiconductor detectors.
- SIS mixers for radio telescopes.
- Hot electron bolometer (HEB).
- System integration and high-frequency electronics (readout circuits, amplifiers, filters, etc...).
- Filter MEMS.
- Existing instruments worldwide.
- Future large-scale projects (SOFIA, HERSCHEL, ALMA).
- Detectors for X-rays (TES/SQUID) and astroparticle physics.
- Kinetic inductance detectors (KID).
- Neutrino and WIMP detectors.

Module Grade Calculation
The module grade results from the oral examination.

Workload
The workload in hours is broken down below:

1. attendance time in the lecture 18 h
2. preparation and follow-up work 24 h
3. exam preparation and attendance 70 h

Recommendations
Basic knowledge of physics and electrical engineering

M 12.46 Module: Digital Circuit Design [M-ETIT-107365]

Coordinators: Prof. Dr. Ivan Peric
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114458	Digital Circuit Design	**ST**	**4 CP**	**Peric**

Assessment
Success control takes place in form of an oral examination (approx. 20 minutes) and a written report with the results of the exercises. Details will be given during the lecture.

Prerequisites
none

Competence Goal
The course "Design of Digital Circuits" is intended for students of electrical engineering, computer science, and related fields who wish to gain in-depth knowledge of digital circuits and their design processes.

Upon completion of the course, students will be able to design digital circuits at both the gate- and system level, utilize CMOS technologies, perform synthesis and place and route using Cadence tools.

The course provides an in-depth understanding of digital circuit technology and its practical applications. Participants will learn the implementation of digital logic circuits and modern design and synthesis processes. Special emphasis is placed on practical implementation using CMOS technology. The course covers both combinational and sequential circuits, memory elements, and digital design flows.

Content
This course covers the fundamentals of **digital electronics and chip design**, beginning with Boolean algebra, logic gates, and circuit realization. It introduces CMOS technology, explaining the structure and behavior of NMOS and PMOS transistors, **CMOS inverters**, and the design of logic gates like NAND, NOR, and multiplexers. Students learn about **combinational and sequential circuits**, including flip-flops, latches, memory cells, and their timing constraints such as setup and hold times. Various **memory architectures** are discussed, such as SRAM, DRAM, Flash, and emerging technologies like FeRAM/MRAM. The course also explores **adders, counters, and shift registers. Digital communication** topics include serial/parallel transmission, PLLs, and clock data recovery. The **digital design flow** is presented in detail, including synthesis, floorplanning, placement, and routing with emphasis on reliability and performance. Throughout the course, students apply concepts using **Cadence tools and Verilog** to design, simulate, and layout digital circuits in practical exercises.

In the exercises, digital design will be performed using Cadence software tools. The circuits will be designed and simulated using the Verilog language. A netlist will be generated using the synthesis tool, and the layout of a chip will be created using the Place & Route tool.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload

1. Attendance time in lectures 30 h
2. Attendance time in exercises 15 h
3. Preparation/follow-up of the same 25 h
4. Exam preparation 50 h

M 12.47 Module: Digital Hardware Design Laboratory [M-ETIT-102266]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-104571	Digital Hardware Design Laboratory	**ST**	**6 CP**	**Becker**

Assessment
Success control takes place in the form of another type of examination and consists of an oral examination of approx. 15 min. as well as exercises in form of laboratory reports and/or oral interrogations during the laboratory. The overall impression is rated.

Prerequisites
The module ETIT-102264 Praktikum Entwurf digitaler Systeme must not have been started or completed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-ETIT-102264 - Digital Hardware Design Laboratory must not have been started.

Competence Goal
The students

- know the practical usage of FPGAs
- are able to efficiently use modern hardware development tools
- know how to describe hardware in VHDL
- can self dependently draft and implement VHDL-Components based on given specifications
- are able to practically apply common concepts and principles in hardware development (e.g. pipelining)

Content
Grouped in teams of two, the students are introduced to the design of complex hardware/software systems. The laboratory takes place in weekly 4 hour laboratory sessions. During the first few sessions, the students are introduced to the implementation of VHDL-components, the usage of modern synthesis and simulation tools as well as basic knowledge on FPGAs.

Based on those fundamentals, students develop the different components of an image processing system in the second part of the laboratory. This includes implementation and testing steps for the individual components as well as the integration to an overall system. Finally, the hardware system can be realized on FPGA-Hardware and tested with live camera images.

Module Grade Calculation
The module grade is composed of the result of the oral examination and the effected performance during the laboratory sessions (e.g. reports, oral interrogations, etc.). etails will be given during the lecture.

Workload
The amount of work is distributed as follows:

- time of presence during the laboratory sessions: 11 sessions with 4h = 44h
- Preparation and wrap-up: 6h per laboratory session = 66h
- Preparation for the examination: 40h

In total 150h (25h per credit point).

Recommendations
Previous knowledge in design and design automation for electronic systems (e.g. from the lectures HSO, No. 2311619 or HMS, No. 2311608) is recommended.

M 12.48 Module: Digital Hardware Design Laboratory [M-ETIT-102264]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-104570	Digital Hardware Design Laboratory	**ST**	**6 CP**	**Becker**

Assessment

- Success is assessed in a final oral examination (approx. 30 min) and during the laboratory sessions on the basis of test protocols.
- The oral exams are taken in groups of two students. Each session takes approx. 30 minutes with approx. 15 minutes per student.
- During the laboratory the students have to write seven reports with around 1 page per report.

Prerequisites
The module M-ETIT-102266 Digital Hardware Design Laboratory must not have been started or completed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-ETIT-102266 - Digital Hardware Design Laboratory must not have been started.

Competence Goal
Within this course, students will:

- - get familiar with the practical application of FPGAs,
- - use modern development tools,
- - learn to describe digital hardware in VHDL,
- - independently design and implement VHDL components based on specified requirements,
- - apply common concepts and principles of hardware development, such as pipelining, in practice.

Content
During the internship, students work in teams of two to learn how to design complex hardware/software systems. The framework consists of weekly 4-hour lab sessions. In the first few lab sessions, students learn how to implement VHDL components, use modern synthesis and simulation tools, and perform basic operations with FPGAs in introductory exercises.

Based on these fundamentals, students in the second project-oriented part of the internship gradually build the various components of an image processing system as a VHDL description. This includes the implementation and testing steps for the individual components as well as their successive integration into an overall system. Finally, the overall system can be implemented on FPGA hardware and tested using live camera data.

Module Grade Calculation
The module grade is made up of the result of the oral examination and the work performed in the practical course (e.g. test protocols, oral examinations, etc.). Details will be given during the lecture.

Workload
Distribution of the workload:

- Attendance time in the course: 11 laboratory sessions of 4h each = 44h
- Preparation and follow-up: 6h per lab session = 66h
- Exam preparation: 40h

  150h in total. This corresponds to 6LP of 25h each.

Recommendations
Previous knowledge of the design and design automation of electronic systems (e.g. courses HSO, No. 2311619 or HMS, No. 2311608) is recommended.

M 12.49 Module: Digital Real Time Simulations for Energy Technologies [M-ETIT-106690]

Coordinators: Prof. Dr.-Ing. Giovanni De Carne
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113449	Digital Real Time Simulations for Energy Technologies		**3 CP**	**De Carne**

Assessment
The examination takes place in form of other types of examination. It consists of an assessment from an exercise on HiL and an oral overall examination (approx. 15 minutes) explaining the exercise results. The overall impression is evaluated.

Prerequisites
none

Competence Goal
To give bachelor's and master's degree students an overview of the need, concept, implementation and execution of Hardware in the Loop (HiL) testing. At the end of the course:

- The students will be able to understand the setup of HiL systems (Device Under Tests, Real Time simulator, I/O, power amps for PHiL, interfacing principles and techniques)
- The students will be able to devise HiL test cases, creating models with an understanding of the trade-off between simulation fidelity and computational resources.
- The students can interface Device Under Tests, running Real Time simulations and executing tests.
- The students are able to perform an independent HiL project and deliver HiL experimental results.

Content
Lesson 1: Introduction

- Overview of control system development process (V-cycle and variants).
- Real-time simulation concept – What does it mean simulating in real time?
- Basic concepts of Real Control Prototyping and Control-Hardware in the Loop.
- Basic concept of Power-HiL

Lesson 2: Introduction to real time simulation (1/2)

- Initial definitions for real time simulations: Wall-clock time, simulation time, hard real-time and soft real-time.
- Off-line simulation software solvers
- Numerical integration, DAE solvers, numerical stability. Complexities induced by switches.
- Distinctions between state-space based and nodal approach-based solvers for CPU
- Difference between CPU and FPGA modeling
- Numerical examples & practical implementation of solvers

Lesson 3: Modelling in real time: grid modelling

- Modeling for HiL, level of model detail vs computational performance
- problems with parallelization due to latencies in transferring data, I/O latency considerations.
- Introducing transmission line decoupling, stublines, ITM and other techniques
- Introducing the State-Space Nodal approach

Lesson 4: Modelling in real time: power electronics

- Modelling components for fast transients (e.g, switches), average models, full switching models
- Interface FPGA-based with CPU-based modelling
- Practical examples and applications

Lesson 5: Modelling in real time: multi-time-scale networks

- Modeling for HiL: different time-scales phasor/RMS, EMT, fast EMT on FPGA.
- Stability issues associated with multi-time-scale hybrid simulation (e.g. RMS/EMT or EMT on CPU and FPGA.
- Multi-time-scale simulation with phasor/EMT/FPGA

Lesson 6: Rapid Control Prototyping (RCP)

- Main Concept and benefits from RCP, Applications
- Generating a code from Simulink and implementation in the real time simulator
- FPGA-based RCP, Datalogging
- Demonstration in classroom

Lesson 7: Controller Hardware In the Loop

- Definition of HIL and testing opportunities
- Analog and digital I/O in real time simulators
- Designing an HiL test – identifying controller functionality to be tested, creating appropriate model and test sequences, identifying potential failures and testing in failure/off-design conditions, use of test automation

Lesson 8: Power Hardware In the Loop

- Introduction to PHIL
- PHIL equipment: power amplifiers. 2Q/4Q applications, specifying power amplifiers
- Interface algorithms between Real Time Simulation and the Device Under Test
- Stability vs. interface accuracy, current state of the art
- Impedance-based stability analysis

Module Grade Calculation
The module grade results of the assessment of an exercise and the oral exam. Details will be given during the lecture.

Workload
The workload includes (2 SWS):

attendance in lectures and exercises: 15*2 h = 30 h
preparation / follow-up: 15*4 h = 60 h
final project, preparation of and attendance in examination: 30 h
A total of 120 h = 4 CR

Recommendations

- Good knowledge of power electronics, linear control theory, and power systems is required.
- Good knowledge of Matlab/Simulink simulation environment is required.

M 12.50 Module: Digital Signal Processing in Optical Communications – with Practical Exercises [M-ETIT-103450]

Coordinators: Prof. Dr.-Ing. Sebastian Randel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-106852	Digital Signal Processing in Optical Communications – with Practical Exercises	**ST**	**6 CP**	**Randel**

Assessment
The exercise sheets and the oral questionnaire are used to rate other types of examinations. The overall impression is assessed. Duration about 20 minutes.

Prerequisites
Basic knowledge of optical communication systems, as given, for example, in the modules "Optical Networks and Systems", "Optoelectronic Components", or "Optical Transmitters and Receivers".

The module "M-ETIT-103815 - Digitale Signalverarbeitung für optische Kommunikationssysteme" must not have been started.

Competence Goal

- The students understand the functioning of modern optical communication systems, which combine electro-optical technologies with digital signal processing.
- You are able to independently implement and test algorithms from digital signal processing as well as suitable simulation and test environments in a suitable scripting language (e.g. Matlab or. Python).
- Furthermore, they can estimate the influence of interfering effects occurring in the glass fiber such as chromatic dispersion and polarization mode dispersion.
- You are also able to estimate the complexity and power consumption of the resulting logic circuits.

Content

- The module deals with algorithms from digital signal processing that are used in broadband optical communication systems. Practical exercises in which the students implement algorithms independently form an essential part of the module.
- In lectures there will be an introduction to the development of digital coherent transmitters and receivers. Building on this, essential function blocks such as the dispersion compensation, the adaptive equalization of polarization mode dispersion as well as carrier and clock recovery are discussed.
- In the exercises, these function blocks are to be implemented in software (Matlab, Octave).
- In addition, individual examples show how digital signal processing algorithms are described in hardware (Hardware Description Language - HDL) and how their complexity scales.

Module Grade Calculation
The exercise sheets and the oral questioning are used to rate other types of examinations. The overall impression is assessed.

Workload
Approximately 180h workload of the student. The workload includes:

30h - attendance in lectures

30h - exercises

80h - preparation / follow-up

40h - written exercises and exam

Recommendations
Knowledge digital signal processing is helpful.

M 12.51 Module: Digital Twin Engineering [M-ETIT-106040]

Coordinators: Prof. Dr.-Ing. Mike Barth
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-112224	Digital Twin Engineering	**WT**	**4 CP**	**Barth**

Assessment
The examination takes place in form of other types of examination. It consists of a model library developed in the course of a semester-long project in the modeling language Modelica and a presentation of the library lasting 25 minutes. The quality of the model library is evaluated within the framework of the criteria: documentation, formal correctness, functionality, usability, HMI and modeling level of detail. The presentation is evaluated as an additional aspects. The overall impression is evaluated.

Prerequisites
none

Competence Goal

- The students will be able to analyze, structure and formally describe problems in the area of object-oriented physical system modeling.
- The students will be able to understand, apply and further develop the Modelica modeling language.
- The students are able to transfer bidirectionally acting systems into a model.
- The students are able to transfer physical equations into the modeling environment.
- The students are able to critically evaluate the different numerical integration methods for their applicability and to use them sensibly.
- The students are able to create system models and co-simulations using functional mockup units.
- The students will be able to implement a real system at the appropriate modeling depth for the task.
- The students will be able to abstract real system properties and, if necessary, decide whether they need to be modeled.
- The students know suitable simulation tools and their application.

Content

- This module is designed to provide students with the theoretical and practical aspects of object-theoretic equation-based modeling.
- This module also provides a definition of the digital twin and its aspects of the management shell.
  - In this context, a classification of simulation models in the I4.0 VWS takes place.
- Both system simulation in the Open Modelica Editor (OME) and co-simulation with Functional Mockup Units (FMU) will be covered.
- Students create a new model library of a mechatronic system in a semester-long project (teams of 3-4 students).
- The module provides an overview of modern system simulation methods based on bidirectional flow and potential modeling.
- Beyond theoretical and practical modeling, the module imparts the knowledge about practice-relevant modeling levels or depths.
- Furthermore, quality standards for simulation models with focus on the engineering of plants/systems are discussed.

Module Grade Calculation
The assessment of the developed model library and the presentation of the library will be included in the module grade. More details will be given at the beginning of the course.

Workload
The workload includes:

1. attendance in lectures an exercises: 10*1,5 h = 15 h
2. preparation / follow-up: 15*2 h = 30 h
3. Implementation of the model library: 60 h
4. preparation of and attendance in the final presentation: 15 h

A total of 120 h = 4 CR

M 12.52 Module: Distributed Discrete Event Systems [M-ETIT-100361]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100960	Distributed Discrete Event Systems	**ST**	**4 CP**	**Heizmann**

Assessment
Success control takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal

- Students have sound knowledge in the field of discrete-event systems. They can distinguish discrete-event systems from discrete-time and continuous-time systems and describe their properties.
- Students have mastered the basic concepts of Markov theory, the theory of queueing systems and Max-Plus algebra for the analysis and construction of distributed discrete-event systems.
- Students are able to systematically analyse and formally describe tasks relating to distributed discrete-event systems, synthesize possible solutions and assess the suitability of the various theoretical approaches for solving them.

Content
The module deals with the basics of describing and analyzing discrete-event systems. The content of the lecture consists of the following topics:

- System modeling
  - State model
  - Model classes
  - Modeling of discrete event systems, automata
- Graph Theory
  - Basic concepts
  - Description of graphs by matrices
- Markov theory
  - Stochastic state models of discrete event systems
  - Discrete-time Markov chains
  - Continuous-time Markov chains
  - Event processes
  - Semi-Markov processes
- Queueing systems
  - Event processes and number of customers
  - Queuing systems with Poisson distributed customer flows
  - Queuing systems with general distributions
  - Queueing networks
- Max-Plus Algebra
  - Max-plus algebraic structure
  - Max-plus description of digraphs
  - Eigenvalue of the adjacency matrix

Module Grade Calculation
The module grade is the grade of the written examination.

Workload

1. Attendance time in lectures and exercises: 45h
2. Preparation and follow-up of lectures and exercises: 40h
3. Exam preparation and attendance: 35h

Total 120 h

Recommendations
Knowledge of probabilities and measurement technology (e.g. from the Bachelor's module "Mess- und Regelungstechnik") and system theory (e.g. from the Bachelor's module "Signale und Systeme") is helpful.

Teaching and Learning Methods
A lecture and a practice are offered in the module.

M 12.53 Module: Electric Drives and Power Electronics Lab [M-ETIT-107138]

Coordinators: Prof. Dr. Martin Doppelbauer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114162	Electric Drives and Power Electronics Lab	**ST**	**6 CP**	**Doppelbauer**

Assessment
Success control takes place in the form of other types of examination. It consists of one oral examination per experiment (approx. about 20 min. per person). The overall impression is assessed.

Prerequisites
none

Competence Goal
Students are able to connect power converters and electrical machines to the electrical grid and operate them correctly. They implement current control in a rotating coordinate system. They analyze and document the operating characteristics of direct current, induction and synchronous machines through measurements. You will know and operate measuring devices with which characteristic values, characteristic curves and time curves of electrical and mechanical variables are recorded and saved.

Content
The aim of the practical course is to use selected examples to guide students in applying and deepening the theoretical knowledge acquired in lectures in practice. In almost all experiments, the students deal with the combination of analog and digital electrical signal processing, control engineering methods, a power electronic actuator and an electrical machine to be driven. Specifically, the following 8 experiments are carried out:

- Experiment SoC:
  “Space vector transformation and current control with digital signal processing system (system on chip)”
- Experiment LH:
  “Power semiconductors - measurement of static and dynamic properties of an IGBT and a SiC MOSFET”
- Experiment PSM:
  “Permanently excited synchronous machine - speed control with subordinate current control in the constant flux and field weakening range”
- Experiment FAM:
  “Field-oriented control of the three-phase induction machine”
- Experiment DAB:
  “Getting to know topology, modulation methods and modeling”
- PV experiment:
  “Operation of solar modules at the point of maximum energy yield and integration of a lithium-ion storage system”
- MMC experiment:
  “Implementation of a cascaded MMC control system consisting of energy and current controllers”
- VASM experiment:
  “Measurement of the induction machine on the test bench to determine the machine parameters”

Module Grade Calculation
The assessments of the oral examinations are included in the module grade. Further details will be provided at the beginning of the course.

Workload
180h

- Attendance time in the internship with interview: 40 h
- Preparation time: 125 h
- Follow-up time: 15h

Recommendations
The courses

- Regelung elektrischer Antriebe und

- Leistungselektronik

should have been completed or at least heard in parallel to the practical course.

M 12.54 Module: Electric Drives for E-Mobility [M-ETIT-106971]

Coordinators: Prof. Dr. Martin Doppelbauer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113936	Electric Drives for E-Mobility	**ST**	**4 CP**	**Doppelbauer**

Assessment
The success control takes place in the form of a written examination of 120 minutes.

Prerequisites
none

Competence Goal
The students will be able to assess the structure, performance and behavior of full-electric and hybrid drive systems for all types of traction applications from pedelecs to cars, utility vehicles, railroads and even large propulsion systems in ships. They can critically evaluate the different drive systems and components.

The students will be able to understand the latest developments in electric drive technology and future possibilities.

Content
This module is designed to provide students with the theoretical and practical aspects of electric drives for electric traction applications.

Table of content:

- Overview: Electric Drives in Hybrid and Electric Vehicles
- Fundamentals of Rotary Field Machines
- Fundamentals of Power Electronics
- Design of Synchronous Machines
- Design of Induction Machines
- Noise, Vibration and Harshness (NVH)
- Thermodynamics of Electric Machines

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
The workload includes (3 SWS):

1. attendance in lectures 15*2 h = 30 h
2. Attendance in exercises: 15*1 h = 15 h
3. preparation / follow-up: 15*3 h = 45 h
4. preparation of and attendance in examination: 30 h

A total of 120 h

Recommendations
Basic knowledge in the fields of electric machines (types and operation of three-phase machines), electric power systems (complex alternating current theory, three-phase systems, equivalent circuit diagrams) and electronic drives (three-phase bridge circuit, modulation) is strongly recommended (e.g. as given in the BSc-modules "M-ETIT-107133 - Lineare Elektrische Netze", "Elektromagnetische Felder", "M-ETIT-106419 - Elektrische Energietechnik" and "M-ETIT-102124 - Elektrische Maschinen und Stromrichter").

Basic knowledge in the field of hybrid and electric vehicles is helpful (e.g. as given in the BSc-module components “T-ETIT-113612 - Hybrid and Electric Vehicles” or "T-ETIT-100784 – Hybride und elektrische Fahrzeuge").

M 12.55 Module: Electric Power Generation and Power Grid [M-ETIT-101917]

Coordinators: Dr.-Ing. Bernd Hoferer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-103608	Electric Power Generation and Power Grid	**ST**	**3 CP**	**Hoferer**

Assessment
Type of Examination: oral exam
Duration of Examination: approx. 20 minutes

Prerequisites
none

Competence Goal
The students

- are familiar with characteristics of different types of power generation
- are able to evaluate the performance of different types of power generation
- comprehend the challenges in power transmission systems due to volatile power generation.
- can derive solutions for a future power generation pool and power grid
- are able to calculate the efficiency factor of power generation systems
- know how to apply mathematical concepts like load flow calculation and short-circuit calculations

Content
I. Energy resources and energy consumption
II. Conversion of primary energy in power plants; thermo-dynamical fundamental terms, processes in steam power plants; steam power plants components; flue gas cleaning
III. Synchronous machines
IV. Thermal power plants (fossil-fueled steam generation, nuclear-fueled steam generation)
V. Renewable energy generation (hydro-electric, wind, solar)
VI. Transmission systems (AC power transmission, DC power transmission)
VII. Load flow calculations

Module Grade Calculation
The module grade is the grade of the oral exam.

Additional Information
The lecture will take place again in summer term 2026.

Workload
total 90 h, hereof 30 h contact hours and 60 h homework and self-studies

Literature
Schwab; Electric energy systems;
Fink, Beaty; Standard handbook for electrical engineers

M 12.56 Module: Electric Power Transmission & Grid Control [M-ETIT-105394]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Fundamentals)
Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-110883	Electric Power Transmission & Grid Control	**ST**	**6 CP**	**Leibfried**

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-ETIT-105394 - Electric Power Transmission & Grid Control must not have been started.

Competence Goal
Students are familiar with the functionality and physical basics as well as the components of AC and DC of electric power transmission systems. They will be able to calculate transmission characteristics and carry out a basic design. They are also familiar with the functioning of grid control.

Content
The lecture initially deals with the characteristics and stability of electrical energy transmission. A central chapter deals with HVDC technology as a method for transmitting high power. FACTS elements, which are used to make energy transmission more flexible, are then dealt with. Finally, the dynamics of power plants and grids are discussed.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
The workload includes:

1. attendance in lectures and exercises: 30 + 30 h = 60 h
2. preparation / follow-up: 120 h

A total of 180 h = 6 CR

Recommendations

- Basic Knowledge in electrical network analysis
- Basic Knowledge about the functionality of electric grid components
- Basic Knowledge about the calculations of three-phase systems
- Basic Knowledge about symmetrical components, Park-transform and Clark-transform

M 12.57 Module: Electrocatalysis [M-ETIT-105883]

Coordinators: Prof. Dr. Ulrike Krewer
Dr. Philipp Röse

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
5 CP	graded	Each summer term	1 term	English	4	4

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111831	Electrocatalysis	**ST**	**5 CP**	**Röse**

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal
Students have a well-grounded knowledge of electrocatalytic energy technologies for the conversion and storage of electrical energy in chemicals (Power-to-X). They know the functional principle of state-of-the-art electrocatalysts in fuel cells and electrolysis and understand the underlying electrochemical and physical processes. Participation in the course enables the students to assess and understand the relationship between electrode structure and their selectivity, performance and stability. Furthermore, the students learn the theoretical basics of experimental methods that are relevant for the investigation of model electrodes and technical cells.

Content
Lecture:

- **Basics, concepts and definitions within the Power-to-X context:** Catalysis and electrocatalysis; activity and selectivity; fundamentals of electrochemical processes, elementary steps involving adsorbed intermediates.

- The role of intermediates: Electron transfer without intermediates, multi-electron transfer with intermediates; differences in adsorption energies of intermediates and active surfaces

- **Theoretical treatment of electron transfer reactions:** Tunneling processes at electrodes; electron transfer reactions (Marcus theory); role of electrode material on rate of electrode reaction.

- Measurement methods for the investigation of electrocatalytic reactions: Determination of the effective surface; Determination of the activity of electrochemically active species; Determination of the selectivity; Operando measurement methods

- Technically important electrocatalytic reactions and processes: The oxygen reduction reaction (ORR) and evolution reaction (OER); the chlorine evolution reaction.

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
attendance in lectures: 30 * 45 min. = 22,5 h

attendance in exercises: 15 * 45 min. = 11,25 h

preparation and follow up of the lectures and practice: 76.25 hours (approx. 1.75 hours per lecture or exercise)

preparation of examination and attendance in examination: 40 h

A total of 150 h = 5 CR

Recommendations
The participation of the module "Electrochemical Energy Technologies" is helpful.

M 12.58 Module: Electrolysis and Electrosynthesis [M-ETIT-107914]

Coordinators: TT-Prof. Dr. Johanna Schröder
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-115334	Electrolysis and Electrosynthesis		**6 CP**	**Schröder**

Assessment
The examination takes place in form of a written examination lasting 90 minutes.

Prerequisites
none

Competence Goal
The students are able to identify characteristics of electrochemical biomass conversion and electrosynthesis electrolyzers in contrast to water electrolysis designs. In addition, the students are able to describe the formation of shunt currents in electrolyzer stacks and the consequential problems in designing efficient industrial systems. The students translate this knowledge to shunt current challenges in modern electrochemical biomass conversion and electrosynthesis systems.

Content

- This module provides an overview of electrochemical biomass conversion and electrosynthesis as competing reaction compared to water oxidation in electrolysis.
- Furthermore, this module gives knowledge about shunt currents formed when building electrolyzer stacks, as well as the consequences and design strategies in the above-mentioned electrolysis systems.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload

1. attendance in lecture and exercises: 15*2 h + 15*2h = 60 h
2. Preparation/follow-up: 15*6 h = 90 h
3. preparation of and attendance in examination: 30 h

A total of 180 h = 6 CR

Recommendations
The content of the modules “Electrocatalysis” and “Electrochemical Energy Technologies” are helpful but not required.

M 12.59 Module: Electrolyzers and their Electrodes [M-ETIT-107648]

Coordinators: TT-Prof. Dr. Johanna Schröder
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114993	Electrolyzers and their Electrodes	**WT+ST**	**3 CP**	**Schröder**

Assessment
The success control takes place in form of an other type of examination and consists of a written report on the scientific topic (max. 20 pages) and a presentation followed by a discussion, 15 minutes each. The overall impression is rated.

Prerequisites
none

Competence Goal
After completing the seminar, the students will be able to independently familiarize themselves with an engineering topic in the field of electrolysers and their electrodes, identify relevant aspects in the associated literature, and summarize them in the form of a written report and a presentation.

Content
In this seminar, students independently work on a scientific topic from current research fields on electrolyzers and their electrodes and prepare a written report and present it in order to improve their skills in literature research, scientific writing, and presentation in front of a scientific audience. First, there will be an introduction to literature research, scientific writing, and presentation techniques to guide the students towards independence. Then students will select a topic related to electrolyzers and their electrodes for their literature research and prepare a scientific report and presentation on this topic.

Module Grade Calculation
The assessment of the written report and the presentation will be included in the module grade. Further details will be provided at the beginning of the course.

Workload

1. Attendance in the seminar: 15*2 h = 30 h
2. Preparation of seminar report: 30 h
3. Preparation of seminar presentation: 30 h

Total: 90 h = 3 LP

Recommendations
Knowledge of the fundamentals of batteries, fuel cells, electrolysis, or electrocatalysis is helpful but not necessary.

Teaching and Learning Methods
seminar 2 SWS

M 12.60 Module: Electromagnetics and Numerical Calculation of Fields [M-ETIT-100386]

Coordinators: Dr. Yongbo Deng
Prof. Dr. Ulrich Lemmer

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100640	Electromagnetics and Numerical Calculation of Fields	**WT**	**4 CP**	**Deng, Lemmer**

Assessment
Success control is carried out in the form of a written test of 120 minutes.

Prerequisites
none

Competence Goal
Students with very different background in electromagnetic field theory will be brought to a high level of comprehension. They will understand the concept of electric & magnetic fields and of electric potential & vector potential and they will be able to solve simple problems of electric & magnetic fields using mathematics. They will understand the equations and solutions of wave creation and wave propagation. Finally the student will have learnt the basics of numerical field calculation and be able to use software packages of numerical field calculation in a comprehensive and critical way.

The student will

- be able to deal with all quantities of electromagnetic field theory (E, D, B, H, J, M, P, ...), in particular: how to calculate and how to measure them,
- derive various equations from the Maxwell equations to solve simple field problems (electrostatics, magnetostatics, steady currents, electromagnetics),
- be able to deal with the concept of field energy density and solve practical problems using it (coefficients of capacitance and coefficients of inductance),
- be able to derive and use the wave equation, in particular: to solve problems how to create a wave and calculate solutions of wave propagation through various media,
- be able to outline the concepts, the main application areas and the limitations of methods of numerical field calculation (FDM, FDTD, FIM, FEM, BEM, MoM, TLM)
- be able to use one exemplary software package of numerical field calculation and solve simple practical problems with it.

Content
This course first gives a comprehensive recap of Maxwell equations and important equations of electromagnetic field theory. In the second part the most important methods of numerical field calculation are introduced.

Maxwell's equations, materials equations, boundary conditions, fields in ferroelectric and ferromagnetic materials

electric potentials, electric dipole, Coulomb integral, Laplace and Poisson's equation, separation of variables in cartesian, cylindrical and spherical coordinates

Dirichlet Problem, Neumann Problem, Greens function, Field energy density and Poynting vector,

electrostatic field energy, coefficients of capacitance, vector potential, Coulomb gauge, Biot-Savart-law, magnetic field energy, coefficients of inductance magnetic flux and coefficients of mutual inductance, field problems in steady electric currents,

law of induction, displacement current

general wave equation for E and H, Helmholtz equation

skin effect, penetration depth, eddy currents

retarded potentials, Coulomb integral with retarded potentials

wave equation for potential and Vector potential and A, Lorentz gauge, plane waves

Hertzian dipole, near field solution, far field solution

transmission lines, fields in coaxial transmission lines

waveguides, TM-waves, TE-waves

finite difference method FDM

finite difference - time domain FDTD, Yee ´s algorithm

finite difference - frequency domain

finite integration method FIM

finite element method FEM

boundary element method BEM, Method of Moments (MOM), Transmission LIne Matrix Methal (TLM),

solving large systems of linear equations

basic rules for good numerical field calculation

The lecturer reserves the right to alter the contents of the course without prior notification.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
Each credit point corresponds to approximately 25-30 hours of work (of the student). This is based on the average student who achieves an average performance. The workload includes:

Attendance time in lectures (3 h 15 appointments each) = 45 h

Self-study (4 h 15 appointments each) = 60 h

Preparation / post-processing = 20 h

Total effort approx. 125 hours = 4 LP

Recommendations
Fundamentals of electromagnetic field theory.

Literature
Matthew Sadiku (2001), Numerical Techniques in Electromagnetics.
CRC Press, Boca Raton, 0-8493-1395-3
Allen Taflove and Susan Hagness (2000), Computational electrodynamics: the finite-difference time-domain method.
Artech House, Boston, 1-58053-076-1
Nathan Ida and Joao Bastos (1997), Electromagnetics and calculation of fields.
Springer Verlag, New York, 0-387-94877-5
Z. Haznadar and Z. Stih (2000), Electromagnetic Fields, Waves and Numerical Methods.
IOS Press, Ohmsha, 1 58603 064 7
M.V.K. Chari and S.J. Salon (2000), Numerical Methods in Electromagnetism, Academic Press, 0 12 615760 X

M 12.61 Module: Electronics and EMC [M-ETIT-100410]

Coordinators: Dr. Martin Sack
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	1	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100723	Electronics and EMC	**ST**	**3 CP**	**Sack**

Assessment
Success is assessed in an overall oral examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
Students are familiar with coupling mechanisms and possible coupling paths for interference signals in electronic circuits and systems, as well as measures for interference suppression and the functionally reliable design of such systems.

Content
Based on the coupling mechanisms for interference signals, the lecture shows various coupling paths for interference, the effects of interference coupling on the circuit function as well as measures for suppression and the functionally reliable design of systems.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
Attendance time: 30 h

Self-study time: 60 h

Total 90 h = 3 LP

M 12.62 Module: Energy Storage and Network Integration [M-ETIT-101969]

Coordinators: Prof. Dr.-Ing. Giovanni De Carne
apl. Prof. Dr. Francesco Grilli
Prof. Dr.-Ing. Mathias Noe

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-104644	Energy Storage and Network Integration	**WT**	**4 CP**	**Noe**

Assessment
Type of Examination: Oral exam

Duration of Examination: approx. 30 minutes

Prerequisites
see course (Teilleistung)

Competence Goal
Students understand the different types of energy storage and apply their knowledge for the selection and principal dimensioning of relevant energy storage tasks.

Furthermore, students can reflect the state-of-the-art of most important energy storage types, their fundamental characteristics and viability at given boundary conditions; they are enabled to elaborate and apply basic integration issues dependent on the grid structure for the different network types.

Practical work: The students are able to analyse real applications of energy storage and calculate basic design examples for the various storage options.

The students are able to discuss topic-related aspects in English using the technical terminology of the field of study.

Content
The lecture provides an overview of the different storage types and their fundamental integration into the power supply grid.

Thereby, within the scope of this lecture, the necessity and the motivation for converting and storing energy will be given. Starting from the definition of fundamental terms different physical and chemical storage types along with their theoretical and practical basis are described. In particular, the decoupling of energy production and energy consumption, and the provision of different energy scales (time, power, density) will be discussed. Furthermore, the challenge of energy transport and re-integration into the different grid types is considered.

1. Motivation for the need of energy storage in energy systems

a. National and international situation

b. Storage motivation

2. Terms and definitions

a. Different energy types

b. Definitions energy content

c. Definitions energy- and power density

3. Thermal energy storage

a. Classification

b. Sensitive heat storage

c. Latent heat storage

d. Reaction heat storage

4. Mechanical energy storage

a. Flywheels

b. Compressed air

c. Pumpes storage systems

5. Electrodynamic energy storage

a. Main principles

b. Capacitive and inductive storage

6. Electrochemical energy storage

a. Working principles

b. Batteries

c. Fuel Cells

7. Electric Power Systems

a. Storage tasks

b. Storage íntegration

c. Planning reserves

The obligatory **practical work** (23689) is related to real applications of energy storage and to basic design examples for the various storage options.

The lecturer reserves the right to alter the contents of the course without prior notification.

Course material will be available on ILIAS. The link to ILIAS and Up-to-date information will be available via the ITEP-homepage prior to the beginning of the semester (https://www.itep.kit.edu/148.php).

Module Grade Calculation
The module grade is the grade of the oral exam.

Additional Information
Exam and Lecture will be held in English.

Workload
Approximately 120h workload of the student. The workload includes:

45h - attendance in lectures an exercises

45h - preparation / follow-up

30h - preparation of and attandance in examination

Recommendations
Basic knowledge in the fields of Electrical Engineering and Thermodynamics is helpful.

M 12.63 Module: Fabrication and Characterisation of Optoelectronic Devices [M-ETIT-101919]

Coordinators: Prof. Dr. Ulrich Wilhelm Paetzold
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-103613	Fabrication and Characterisation of Optoelectronic Devices	**ST**	**3 CP**	**Paetzold**

Assessment
Type of Examination: written exam

Duration of Examination: 120 Minutes

Modality of Exam: One written exam offered at the end of each semester.

Prerequisites
None

Competence Goal
The students build knowledge on process technology for the fabrication of a range of optoelectronic devices, including LEDs, solar cells, laser diodes, photodiodes, etc. They learn to compare the advantages of different technological approaches, including their economic boundary conditions. This is a technologiccal-based course where students will use their prior fundamental knowledge to gain a firm grasp on the fabrication sequences and characterisation (optical, electrical, electronic, materials) steps that are required to realise the above devices.

While fulfilling the learning targets, the students

- possess the basic knowledge about the working principles of optoelectronic devices;
- comprehend the boundary conditions for the design of optoelectronic devices and have a good understanding of the challenges in microfabrication
- are familiar with different lithographic techniques, including e-beam lithography, optical lithography, multiple-photon lithography, X-ray lithography, etc.
- comprehend the different techniques that are available for thin-film deposition of dielectrics, metals and semiconductors
- understand what role micro-optics can play in such devices
- be able to determine the most promising characterisation techniques for evaluating material quality, electronic properties, as well as optical and electrical performance.
- Exposure to different dry- and wet-etching processes to help realise device structures
- have an understanding of the economic implications of the chosen technologies and their compatibility with highthroughput production

Content
I. Overview: Opto-electronic Devices

II. Thin-film growth and deposition

- epitaxial growth of III-V semiconductors, as well as Si and Ge
- chemical vapour deposition (CVD) based processes, including atomic layer deposition (ALD)
- physical vapour deposition (PVD) based processes, including evaporation (thermal and e-beam) and sputtering (DC and RF)

III. Lithographic techniques

- e-beam lithography, optical lithography, laser interference lithography, two-photon lithography, X-ray lithography

IV. Etching processes

- wet- and dry-etching processes for semiconductors, dielectrics and metals

V. Micro-optics

- micro-optic design in opto-electronic devices

VI. Characterissation:

- materials properties (electron microscopy, crystallinity, bonding energies, elemeental concentrations, layer thicknesses ...)
- electronic properties (dopant profiling, mobility, minority carrier lifetimes, resistivity, bandgap measurements, ...)
- optical (spectrophotometry, photoluminescence, ...)
- electrical (current-voltage measurements, quantum efficiency / spectral response, ...)

VII. Excursion (TBA)

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
Total 90h, hereof 30h contact hours (24h lecture,
6h problem class), and 60h homework and selfstudies

Recommendations
Semiconductor fundamentals

Literature
TBD

M 12.64 Module: Field Propagation and Coherence [M-ETIT-100566]

Coordinators: Prof.Dr.Dr.h.c. Wolfgang Freude
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100976	Field Propagation and Coherence	**WT**	**4 CP**	**Freude**

Assessment
Type of Examination: oral exam

Duration of Examination: approx. 20 minutes

Modality of Exam: Oral examination, usually one examination day per month during the summer and winter terms. An extra questions-and-answers session will be held for preparation if students wish so.

Prerequisites
none

Competence Goal
Presenting in a unified approach the common background of various problems and questions arising in general optics and optical comunications

The students

- knwo the common properties of counting of modes, density of states and the sampling theorem
- comprehend the relationship between propagation in multimode waveguides, mode coupling, MMI and speckles
- can analyze propagation in homogeneous media with respect to system theory, antennas, and the resolution limit of optical instruments
- understand that coherence as a general concept comprises coherence in time, in space and in polarisation
- comprehend the implication of complete spatial incoherence, and what is the radiation efficiency of a source with a diameter smaller than a wavelength (the mathematical Hertzian dipole, for instance)
- can assess when can two incandescent bulbs form an interference pattern in time
- know under which conditions a heterodyne radio receiver, which is based on a non-stationary interference, actually works

Content
The following selection of topics will be presented:

- Light waves, modes and rays: Longitudinal and transverse modes, sampling theorem, counting and density of modes ("states")
- Propagation in multimode waveguides. Near-field and far-field. Impulse response and transfer function. Perurations and mode coupling. Multimode interference (MMI) coupler. Modal noise (speckle)
- Propagation in homogeneous media: Resolution limit. Non-paraxial and paraxial optics. Gaussian beam. ABCD matrix
- Coherence of optical fields: Coherence function and power spectrum. Polarisation, eigenstates and principal states. Measurement of coherence with interferometers (Mach-Zehnder, Michelson). Self-heterodyne and self-homodyne setups

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
total 120 h, hereof 45 h contact hours (30 h lecture, 15 h problem class), and 75 h homework and self-studies

Recommendations
Minimal background required: Calculus, differential equations and Fourier transform theory. Electrodynamics and field calculations or a similar course on electrodynamics or optics is recommended.
Solution of the problems on the exercise sheet, which can be downloaded as homework each week, is highly recommended. Also, active participation in the problem classes and studying in learning groups are strongly advised.

Literature
Detailed lecture notes as well as the presentation slides can be downloaded from the IPQ lecture pages. Additional reading:

Born, M.; Wolf, E.: Principles of optics, 6. Aufl. Oxford: Pergamon Press 1980

Ghatak, A.: Optics, 3. Ed. New Delhi: Tata McGraw Hill 2005

Hecht, E.: Optics, 2. Ed. Reading: Addison-Wesley 1974

Hecht, J.: Understanding fiber optics, 4. Ed. Upper Saddle River: Prentice Hall 2002

Iizuka, K.: Elements of photonics, Vol. I and II. New York: John Wiley & Sons 2002

Further textbooks in German (also in electronic form) can be named on request

M 12.65 Module: Geodetic Space Techniques for Engineers [M-BGU-106347]

Coordinators: Prof. Dr.-Ing. Hansjörg Kutterer
Dr.-Ing. Michael Mayer

Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-BGU-111169	Geodetic Space Techniques, Prerequisite *This item will not influence the grade calculation of this parent.*	**WT**	**1 CP**	**Kutterer**
T-BGU-112871	Geodetic Space Techniques for Engineers, Exam	**WT**	**3 CP**	**Kutterer**

Assessment
Assessment consists of an oral examination (SPO §4 (2) 2; T-BGU-112871 - Geodetic Space Techniques for Engineers, Examination) lasting 20 minutes and an accomplishment (SPO §4 (3); T-BGU-111169 - Geodetic Space Techniques for Engineers, Prerequisite). Successful accomplishment of the prerequisite is mandatory for taking the exam. Details regarding the assessments can be found in the descriptions of the individual module components.

Prerequisites

- M-BGU-104571 - Satellite Geodesy for Engineers and
- M-BGU-107250 - Geodetic Reference Systems and Geodetic Space Techniques for Engineers

must not have been started

Competence Goal
Students explain the principles and applications of geodetic space methods (e.g., satellite methods, VLBI). They name current satellite missions and describe their respective primary objectives. They illustrate the importance and potential of geodetic space-based methods for addressing geodetic and geoscientific questions. Students name the Keplerian elements and derive them from initial conditions. They represent the position of a satellite in the Earth-fixed equatorial coordinate system. Students calculate the corresponding ground track and sky plot from Keplerian elements. They identify the objectives of various satellite missions and describe the associated orbital parameters. Students are able to calculate satellite positions from ephemerides.

Content
Lecture: Fundamentals of Celestial Mechanics (Keplerian motion, Keplerian elements, perturbating forces, and orbital perturbations). Overview of observational methods (atmospheric disturbances, GNSS, laser ranging to satellites and the Moon (SLR, LLR), very long-baseline interferometry (VLBI), satellite altimetry, microwave systems, gravity field missions), data analysis methodology. Selected satellite missions. Overview of applications in geodesy, geosciences, oceanography, and meteorology.

Exercise: Applications of the Kepler problem (ground track, sky plot, satellite visibility). Satellite orbits. Satellite position from ephemerides. Orbital perturbations, GNSS online services.

Module Grade Calculation
The module grade corresponds to the grade on the exam T-BGU-112871 - Geodetic Space Techniques for Engineers, Exam

Additional Information
A comparable module is offered to BSc students in Geodesy and Geoinformatics, carrying a credit load of 4 LPs. Due to the increased workload for students in other disciplines (e.g., MSc ETIT, familiarization with reference systems), the credit value has been increased to 5 LPs. The additional LP results in an increased workload for self study (75h → 105h).

The assessment criteria described in the study program guidelines (SPO) for Geodesy and Geoinformatics and Electrical Engineering and Information Technology (SPO §4) are identical.

Workload
Total: 120 hours (MSc ETIT: 150 hours)

Attendance: 45 hours

- Lectures, including module exams taken

Self study: 75 hours (MSc ETIT: 105 hours)

- Deepening of course content through independent review of lecture material at home
- Completion of exercises (required)
- Deepening understanding of course content through appropriate literature and internet research
- Preparation for the in-semester module exam

Recommendations
Prior knowledge of Advanced Mathematics I and II is recommended.

Literature
Bauer, M. (2018): Vermessung und Ortung mit Satelliten. 7te Auflage, Wichmann, Berlin, Offenbach.

Hofmann-Wellenhof, B.; Lichtenegger, H.; Wasle, E. (2008): GNSS – Global Navigation Satellite Systems: GPS, GLONASS, Galileo & more, Springer.

Seeber, G. (2003): Satellite Geodesy. Foundation, Methods and Applications, 2nd ed., De Gruyter, Berlin.

Teunissen, P.; Montenbruck, O. (Eds.) (2017): Springer Handbook of Global Navigation Satellite Systems. Springer Nature, Cham/Schweiz.

Torge, W.; Müller, J. (2012): Geodesy. 4th ed, De Gruyter, Berlin, Boston.

M 12.66 Module: Handling Characteristics of Motor Vehicles I [M-MACH-105288]

Coordinators: Dr.-Ing. Martin Gießler
Dr.-Ing. Hans-Joachim Unrau

Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-105152	Handling Characteristics of Motor Vehicles I	**WT**	**4 CP**	**Weitz**

Assessment
Oral examination, duration: approximately 30 minutes.

Prerequisites
none

Competence Goal
The students know the basic connections between drivers, vehicles and environment. They can build up a vehicle simulation model, with which forces of inertia, aerodynamic forces and tyre forces as well as the appropriate moments are considered. They have proper knowledge in the area of tyre characteristics, since a special meaning comes to the tire behavior during driving dynamics simulation. Consequently they are ready to analyze the most importent influencing factors on the driving behaviour and to contribute to the optimization of the handling characteristics.

Content
1. Problem definition: Control loop driver - vehicle - environment (e.g. coordinate systems, modes of motion of the car body and the wheels)

2. Simulation models: Creation from motion equations (method according to D'Alembert, method according to Lagrange, programme packages for automatically producing of simulation equations), model for handling characteristics (task, motion equations)

3. Tyre behavior: Basics, dry, wet and winter-smooth roadway

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
The workload includes:

1. attendance in lectures and exercises: 15*2 h = 30 h
2. preparation / follow-up: 60 h
3. preparation of and attendance in examination: 30 h

A total of 120 h

Teaching and Learning Methods
Lecture

M 12.67 Module: Hardware Modeling and Simulation [M-ETIT-100449]

Coordinators: Dr.-Ing. Jens Becker
Prof. Dr.-Ing. Jürgen Becker

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100672	Hardware Modeling and Simulation	**WT**	**4 CP**	**Becker, Becker**

Assessment
Achievement is examined in the form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal
After completing this module, students will be familiar with different hardware description languages and their applications in various abstraction levels. They will gain knowledge of the SPICE Hardware Description Language and become proficient in building and deriving the analog matrix for spice simulation. In the realm of digital design, they will develop a comprehensive understanding of the hardware description language VHDL, encompassing the VHDL Standard and its extensions, such as VHDL 2008, the 9-valued logic, and the VHDL-AMS standard. Furthermore, students will achieve a profound comprehension of simulator principles, particularly the delta cycle model. They will also grasp the fundamentals of fault simulations for testing fabricated circuits and learn to derive test vectors. Additionally, students will acquire an understanding of higher-level hardware construction languages like Chisel and SystemC.

Content
In order to address the complexity of modern chips during development, it is essential to utilize modern hardware description languages. This course offers insights into the various levels of abstraction in these languages. It starts by covering the fundamentals of analog description using SPICE and then progresses through VHDL, VHDL-AMS, and Verilog. Additionally, the course introduces more abstract languages like Chisel and SystemC.

Topics covered in the course are:

- Design Process
- Basics of Modeling and Simulation
- Low Level Modeling
- VHDL
  - VHDL-AMS
  - 9-valued logic
  - Delta cycle simulation
  - Fault simulation
- Verilog
- Chisel
- SystemC

Module Grade Calculation
The module grade results from the grade of the written examination.

Workload
The workload is covered by:

1. Participating in lectures and tutorials: 33h
2. Preparing and wrap up of the above named units: 66h
3. Exam preparation and presence: 21h

Sum: 120h = 4 LP

M 12.68 Module: Hardware Synthesis and Optimization [M-ETIT-106963]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Fundamentals)
Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113922	Hardware Synthesis and Optimization	ST	6 CP	Becker

Assessment
The examination takes place within the framework of an oral overall examination (approx. 30 minutes)

Prerequisites
none

Competence Goal
Students know the basic steps required for the automated design of optimized digital circuits. They are able to classify them in the Y-chart and assess their complexity.

They will be able to name and explain the most important approaches for these design steps and evaluate them with regard to optimality and computational effort. This includes the ability to use algorithms for these approaches, e.g. selected graph algorithms, metaheuristics such as simulated annealing. The students are also able to determine their respective runtime complexities.

In addition, they can solve given problems from the field of design automation by selecting a suitable approach based on certain optimization criteria and applying it to the respective problem.

Content
The module focuses on teaching the formal and methodological foundations for the automated design of optimized electronic systems. The relevant scientific and methodological properties of the methods used are discussed and their implementation in industrial practice is also taught.

The following topics are covered:

- Graph Algorithms and Complexity
- High-Level Synthesis
- Algorithms for Scheduling, Allocation and Binding Problems
- Register-Transfer-Level Synthesis
- Retiming Algorithms
- Logic Optimization
- Technology Mapping for Standard Cells and FPGAs
- Physical Design
- Placement of Standard Cells with ILP and Simulated Annealing
- Global and Detailed Routing

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
The workload includes (4 SWS):

1. attendance in lectures and exercises: 60 h
2. preparation / follow-up: 60 h
3. preparation of and attendance in examination: 60 h

A total of 180 h = 6 CR

Recommendations
Basic knowledge in the field of digital circuits, e.g. as taught in the course “Digital Technology” (2311615) is helpful.

M 12.69 Module: Hardware/Software Co-Design [M-ETIT-107444]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Field of Specialization / Information and Communication Technology (Fundamentals)
Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114623	Hardware/Software Co-Design	**WT**	**6 CP**	**Harbaum**

Assessment
The examination takes place within the framework of an oral overall examination (approx. 25 minutes).

Prerequisites
none

Competence Goal
By attending the lecture Hardware/Software Co-Design, students learn the necessary multi-criteria methods and hardware/software target architectures. The theoretical foundations for the interlinked design of hardware and software parts of a system are presented. Attending the lecture contributes to the understanding of these methods of hardware/software co-design and enables students to apply what they have learned to novel problems.

Students learn about the main target architectures and are able to name their advantages and disadvantages with regard to their applicability in hardware/software co-design. Students will become more familiar with various methods for assessing design quality and will be able to apply these in the early phases of system design. Furthermore, students have an overview of partitioning methods for HW/SW systems, can classify them and know the respective advantages and disadvantages of the methods. Students will be able to select and apply a suitable method for typical HW/SW partitioning problems. In addition, their practical application is demonstrated using various current software and hardware components

By attending the course, students will have a cross-component understanding of the topic of co-design. Furthermore, attending the course enables students to apply the methods presented to problems independently. Tools that are introduces in the course of the lecture can be used for this purpose. The accompanying exercises are intended to consolidate the knowledge acquired in the lectures. Selected topics are repeated and students learn how to apply the methods for modern system design using theoretical and practical examples.

Attending the lecture enables students to independently categorize current scientific work, e.g. theses, and to work on them using the latest methods.

Content
Hardware-software co-design is the simultaneous and interlinked design of hardware and software parts of a system. Most modern embedded systems (examples are cell phones, automotive and industrial control systems, game consoles, home cinema systems, network routers) consist of cooperating hardware and software components. Enabled by rapid advances in microelectronics, embedded systems are becoming increasingly complex with diverse application-specific criteria. The use of appropriate computer-aided design tools is not only necessary to handle the increasing complexity, but also to reduce design costs and design time. The lecture Hardware Software Co-Design deals with the necessary multi-criteria methods and hardware/software target architectures:

- Target architectures for hardware/software systems:
  - General Structure
  - Classification
  - General-Purpose Processors (GPP)
    - Architecture: Accumulator Machine, Stack Machine, Register-based Machine
    - Performance Enhancement: Pipelining, Superscalarity / Out-of-Order Execution, Very Large Instruction Word (VLIW), Single Instruction Multiple Data (SIMD), Caches, Multiple Instruction Multiple Data (MIMD)
  - Specialized Processors
    - Microcontroller (µC), Digital Signal Processors (DSP), Graphics Processing Units (GPU)
  - Bus, Multicore & Network-on-Chip (NoC)
  - Application-specific Instruction-set Processors (ASIP)
  - Field Programmable Gate Arrays (FPGA)
  - System-on-Chip (SoC)
  - Chiplets
  - Machine Learning on Embedded Devices
- Estimation of the design quality
  - Levels of Abstraction
  - System Synthesis
  - Graph Models for Control/Data Flow
  - Parameters of Estimation Methods
    - Accuracy, Fidelity
  - Estimation of Hardware Metrics
    - Clock Period, Clock Slack
  - Estimation of Software Metrics
  - MIPS, FLOPS, WCET, Cache Conflict Graph, Profiling, Tracing
  - Estimation of Communication Metrics
- Hardware/software partitioning methods
  - Classification of partitioning algorithms
  - Iterative and constructive heuristics
    - Hierarchical clustering, Tabu Search, Kernighan Lin, Fiduccia Mattheyses, Simulated Annealing, Genetic Algorithms

Module Grade Calculation
The module grade is the grade of the oral exam.

Additional Information
New Module with 6 CR, replacing M-ETIT-100453 (4 CR)

Last examination for 4 CR in the first attempt in winter term 25/26

Workload
The workload includes (4 SWS):

1. attendance in lectures and exercises: 60 h
2. preparation / follow-up: 50 h
3. preparation of and attendance in examination: 70 h

A total of 180 h = 6 CR

Recommendations
Basic knowledge in the fields of digital technology and information technology is helpful.

Teaching and Learning Methods
lecture (3 SWS), exercise (1 SWS)

M 12.70 Module: Hardware/Software Co-Design [M-ETIT-100453]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100671	Hardware/Software Co-Design	**WT**	**4 CP**	**Harbaum**

Assessment
Success control takes place in the form of an oral examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
By attending the Hardware/Software Co-Design lecture, students learn the necessary multi-criteria methods and hardware/software target architectures. Attending the lecture contributes to the understanding of these methods of hardware/software co-design and enables students to apply what they have learned to novel problems.

Students learn about the main target architectures and are able to name their advantages and disadvantages with regard to their applicability in hardware/software co-design. Students will become familiar with various methods for assessing design quality and will be able to apply these in the early phases of system design. Furthermore, students have an overview of partitioning methods for HW/SW systems, can classify them and know the respective advantages and disadvantages of the methods. Students will be able to select and apply a suitable method for typical HW/SW partitioning problems.

By attending the course, students will have a cross-component understanding of the topic of co-design. Furthermore, attending the course enables students to apply the methods presented to problems independently. Tools that are introduced in the course of the lecture can be used for this purpose.

Attending the lecture enables students to independently classify current scientific work, e.g. theses, and to work on them using the latest methods.

Content

- The lecture presents the theoretical principles of the interlinked design of hardware and software parts of a system. In addition, their practical application is demonstrated using various current software and hardware components.
- The accompanying exercises are intended to consolidate the knowledge acquired in the lectures. Selected topics are repeated and students learn how to apply the methods for modern system design using theoretical and practical examples.
- Hardware-software co-design is the simultaneous and interlinked design of hardware and software parts of a system. Most modern embedded systems (examples are cell phones, automotive and industrial control systems, game consoles, home cinema systems, network routers) consist of cooperating hardware and software components. Enabled by rapid advances in microelectronics, embedded systems are becoming increasingly complex with diverse application-specific criteria. The use of appropriate computer-aided design tools is not only necessary to handle the increasing complexity, but also to reduce design costs and design time. The lecture Hardware Software Co-Design deals with the necessary multi-criteria methods and hardware/software target architectures:
    - Target architectures for hardware/software systems
        - Processor design: pipelining, superscalarity, VLIW, SIMD, cache, MIMD
        - General-purpose processors (GPP), microcontrollers (µC), digital signal processors (DSP), graphics processors (GPU), application-specific instruction set processors (ASIP), field programmable gate arrays (FPGA), system-on-chip (SoC), bus systems, multicore and network-on-chip (NoC)
    - Assessment of the design quality
        - Hardware and software performance
    - Hardware/software partitioning methods
        - Iterative and constructive heuristics

Module Grade Calculation
The module grade is the grade of the oral examination.

Additional Information
Will be changed to 6 CR in winter term 25/26.

Last examination for 4 CR in the first attempt in winter term 25/26

Workload
The workload includes:

1. attendance time in 14 lectures, 7 exercises: 31.5 hrs

2. preparation and follow-up of the same: 63 hours (3 hours per unit)

3. exam preparation and attendance: 20 hours preparation and 0.5 hours exam

Recommendations
Knowledge of the basics of digital technology and information technology is helpful.

M 12.71 Module: High-Power Microwave Technology [M-ETIT-107363]

Coordinators: Prof. Dr.-Ing. John Jelonnek
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114453	High-Power Microwave Technology	**WT**	**3 CP**	**Jelonnek, Marek**

Assessment
The success control is carried out as part of an oral overall examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
The course provides a comprehensive insight into high-power microwave technology. By the end of the course, students will understand the principles of high-power microwave and sub-THz frequency generation, transmission, and diagnostics. They will be introduced to the major types of electron tubes and their key components and become familiar with the major applications of microwaves at high and highest power levels. Considered applications include satellite communication, radar technology, THz spectroscopy, and industrial processes using microwaves, such as dielectric heating, microwave plasmas, particle accelerators, and plasma heating of magnetically confined nuclear fusion devices.

Content
"High-power microwave technology" covers the generation, transmission, diagnostics and applications of microwaves at high power levels and at frequencies ranging from a few hundred Megahertz up to several hundred Gigahertz.

The focus of the lecture is on electron tubes, which are the major sources for the generation of high power over a wide range of frequencies and powers. The lecture will discuss the fundamental mechanisms of microwave generation and introduce the major types of microwave tubes. The preferred application areas for each tube type are explained and the different possibilities of high-power transmission and diagnostics are presented. Considered applications include satellite communication, terahertz (THz) spectroscopy, radar technology, particle accelerators, and fusion research.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
The workload includes:

Attendance study time lecture / exercise: 30 h

Self-study time including exam preparation: 60 h

A total of 90 h = 3 LP

Recommendations
Fundamental knowledge of electromagnetic wave generation, transmission and diagnostics is helpful.

M 12.72 Module: High-Voltage Technology [M-ETIT-105060]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-110266	High-Voltage Technology	**WT**	**6 CP**	**Leibfried, Suriyah**

Competence Goal
Students can determine electric fields using numerical methods or graphically, generate high voltages in the laboratory, dimension, design and calculate AC voltages and impulse voltages. They know the properties of insulating materials in the field space and the processes that lead to breakdown in gases, liquids and solids. They know the most important technical insulating materials and can use them in the context of insulation coordination.

Content
Generation of high voltages in the laboratory, electric fields, dielectrics in the field space, gas discharge physics, breakdown in liquids and solids, technical insulating materials, insulation coordination.

M 12.73 Module: High-Voltage Test Technique [M-ETIT-100417]

Coordinators: Dr.-Ing. Rainer Badent
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-101915	High-Voltage Test Technique	**WT**	**4 CP**	**Badent**

Assessment
Success is assessed in an overall oral examination (approx. 20 minutes) on the selected course.

Prerequisites
none

Competence Goal
The student can measure partial discharges, carry out on-site tests, test cables and accessories. They can operate and design computer-based test systems. He can create the necessary requirements for the accreditation of test laboratories.

Content
This course familiarizes students with issues of high-voltage testing technology, calibration and the contents of international test standards for electrical power engineering products.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
Attendance time lecture: 30 h

Exercise attendance time: 15 h

Self-study time: 67.5 h

Exam preparation and attendance in the same: offset in preparation/follow-up work

Total 112.5 h = 4 LP

M 12.74 Module: Human Brain and Central Nervous System: Anatomy, Information Transfer, Signal Processing, Neurophysiology and Therapy [M-INFO-100725]

Coordinators: Prof. Dr.-Ing. Tamim Asfour
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-101262	Human Brain and Central Nervous System: Anatomy, Information Transfer, Signal Processing, Neurophysiology and Therapy	**WT+ST**	**3 CP**	**Asfour, Spetzger**

Assessment
See Partial Achievements (Teilleistung).

Prerequisites
See Partial Achievements (Teilleistung).

Competence Goal
After successfully completing the course, students will have a basic understanding and basic information about the structure and complex functioning of the brain and the central nervous system (CNS). They will learn the basics of neurophysiology and sensory functions and, in particular, the technically complex modern diagnostics of the brain and CNS. In addition, students learn the basics of modern treatment modalities for various neurological and neurosurgical diseases and are able to understand the application and use of different medical equipment and devices. The overview of modern neuro-medicine allows interdisciplinary recognition and analysis of problems and allows the implementation of theoretical models in a concrete practical context.

Content
The course provides an overview of modern neuro-medicine and provides a basic understanding of neuroanatomy, sensory and neurophysiology, which represents an important interface to the innovative research areas of neuroprosthetics (optical, acoustic prostheses). There is also a close connection to motor systems in robotics. There are further links to the fields of imaging and image processing and intraoperative support systems. Practical relevance is established and specific application examples in medical diagnostics and therapy are presented.

Workload
Lecture with 2 SWS, 3 LP.
3 LP corresponds to approx. 90 hours, of which
approx. 12 * 2h = 24 hrs. attendance time Lecture approx. 12 * 3h = 36 hrs. preparation and follow-up time Lecture approx. 30 hrs. exam preparation and attendance in the same

Recommendations
See Partial Achievements (Teilleistung).

M 12.75 Module: Industrial Circuitry [M-ETIT-100399]

Coordinators: Dr.-Ing. Andreas Liske
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100716	Industrial Circuitry	**WT**	**3 CP**	**Liske**

Assessment
Success is assessed in an overall oral examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
Graduates know the structure and manufacturing process of printed circuits in order to be able to design and commission electronic circuits themselves. They are familiar with the relevant circuit properties of electronic components available on the market and are able to design, commission and evaluate a practical circuit concept.

Content
This module is intended to provide students with basic knowledge of circuit technology, as this is an absolute prerequisite for being able to perform a task with electronic aids. In this module, basic principles of circuit design, manufacturing processes, passive and active electronic components, analog and digital circuits as well as programmable logic circuits are presented, also using examples.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload

1. Attendance time lecture (2SWS): 15 appointments * 2 h = 30 h
2. Preparation and follow-up time for lecture: 15 appointments * 2 h self-study = 30 h
3. Exam preparation and attendance in the same: 30 h
4. Total: 90 h = 3 CP

M 12.76 Module: Information Fusion [M-ETIT-103264]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-106499	Information Fusion	**WT**	**4 CP**	**Heizmann**

Assessment
The assessment takes the form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal

- Students have a sound knowledge of different methods for specifying uncertain knowledge and for processing it for the purpose of information fusion.
- Students are familiar with different concepts of information fusion with regard to their prerequisites, model assumptions, methods and results.
- Students are able to analyze and formally describe information fusion tasks, synthesize possible solutions and assess the suitability of different information fusion approaches for solving them.

Content
For many information acquisition tasks, it is not possible to fully and robustly capture the properties of interest of a scene or process with a single sensor or information source. In such cases, one possible solution is to use several sensors that capture different aspects of the scene. The use of heterogeneous sensors with different sensor principles allows the evaluation of several physical properties of the scene. In addition, non-sensory information (e.g., in the form of a priori knowledge or physical models) may also be available that needs to be considered when determining scene properties of interest.

This lecture introduces concepts, architectures and methods of information fusion. Mathematical concepts for linking sensor data and information from different sources are presented.

The contents include in detail

- Prerequisites for fusion
- Specification of uncertain information
- Preprocessing for information fusion, registration
- Fusion architectures
- Probabilistic methods: Bayesian fusion, Kalman filter, tracking
- Formulation of fusion tasks using energy functionals
- Dempster-Shafer theory
- Fuzzy fusion

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
Total: approx. 120h, of which

1. attendance time in lectures: 34h
2. preparation and follow-up of lectures: 34h
3. exam preparation and attendance: 52h

Recommendations
Knowledge of the basics of stochastics is helpful.

Teaching and Learning Methods
Lecture and exercise

M 12.77 Module: Information Systems in Power Engineering Lab [M-ETIT-107159]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114183	Information Systems in Power Engineering Lab	**ST**	**6 CP**	**Leibfried**

Assessment
Success control takes place in the form of other types of examination and is assessed in form of 3 experiments.

Prerequisites
none

Competence Goal
Students have a basic understanding of how to use common calculation programs in the fields of network calculation, field calculation and automation and control. They are able to carry out basic calculations in the respective sub-areas and are familiar with the underlying theory.

Content
The focus of the lecture is to impart sound knowledge in the field of field calculation using the finite element method, load flow and short flow calculation, as well as the implementation of control programs for PLC systems. The theoretical basics of the sub-areas are taught and the practical application is practiced using common programs based on case studies.

Module Grade Calculation
The module grade results of the assessment of the 3 experiments.

- 20 points are awarded for each experiment (max. 10 for preparation and max. 10 for performance).
- This results in a total of 60 points.
- At least 27 points must be achieved in order to pass the module.

Workload
The workload is 180 hours and is made up as follows:

- Attendance time 40 h
- Self-study time 140 h

Recommendations
Basic knowledge on high-voltage technology, calculation of electrical networks and energy transmission and network control, computer skills

M 12.78 Module: Information Technology in Industrial Automation Systems [M-ETIT-100367]

Coordinators: Dr.-Ing. Peter-Axel Bort
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100698	Information Technology in Industrial Automation Systems	**ST**	**3 CP**	**Bort**

Assessment
Success control takes place in fom of an oral examination (20-25 minutes).

Prerequisites
none

Competence Goal
After completing the course, graduates will have a holistic basic understanding of modern automation technology, from simple sensor/actuator systems, programmable logic controllers and control systems to cloud-based technologies. They are familiar with the interfaces to information technology and the interaction of the individual disciplines, as well as their use in automation technology. Graduates have an understanding and a feel for the various aspects of reliability and functional safety in automation technology. They know the possible applications and limitations of central information technology tools and modeling tools, as well as methods of artificial intelligence in automation technology.

Content
The lecture provides an introduction to modern automation systems ranging from simple PLC controllers, control systems and manufacturing execution systems (MES) to enterprise resource planning (ERP) systems. A wide variety of industries, technologies and standards are considered which are used in such complex systems.

Another focus is on the area of plant project planning, system integration and networking, right through to cloud-based solutions. Various modeling approaches and tools for project planning are presented, as well as the special features of system integration in plant automation, such as the high number of different interfaces, the different life cycles of individual components, subsystems and plant parts or the extreme demands on the functional safety and availability of the plants.

Economic aspects play a central role in all considerations. Using numerous practical examples, students should develop their own feeling for the economic impact of engineering decisions from the developer's and operator's point of view. In this context, topics such as asset management and strategies for plant project planning and control are covered.

The lecturer reserves the right to deviate from the content given here in the context of the current lecture without special notice.

Module Grade Calculation
The module grade is the grade of the oral examination.

Additional Information
There will be 7 lecture dates. These will be announced in the lecture.

Workload
The workload is broken down as follows:

- Attendance time lecture: 7 * 4 h = 28 h
- Preparation/follow-up lecture: 7 * 4 = 28 h
- Exercise attendance time: 0 h
- Preparation and follow-up work for the exercise (PLC programming with Codesys): 4 h
- Exam preparation and attendance in exam: 30 h (alternatively: offset in preparation/follow-up)
- Total: 90 h -> 90/30 LP = 3 LP

M 12.79 Module: Integrated Intelligent Sensors [M-ETIT-100457]

Coordinators: Prof. Dr. Wilhelm Stork
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100961	Integrated Intelligent Sensors	**ST**	**3 CP**	**Stork**

Assessment
Achievement will be examined in an oral examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
The lecture is intended to give students an insight into the wide range of possible applications of intelligent sensor systems and their economic significance.

The students

- Know the most important terms and processes for the development and manufacture of integrated intelligent sensors and are able to assess their advantages and disadvantages.
- Are able to describe the common sensor principles.
- Can select suitable methods for recording different physical variables using IIS.
- Are familiar with the basic processes for manufacturing microsystem sensors
- Have a broad understanding of the structure and functionality of microsystem sensors.
- Have the ability to communicate with experts in sensor technology.
- Are able to critically evaluate different processes.

Content
In the lecture, applications of various microtechniques for sensor technologies, such as microoptics or micromechanics, are presented using current examples from industry and research. The main topics of the lecture are microsensors with integrated signal processing ("smart sensors") for applications in the automotive and manufacturing industries as well as in environmental protection and biomedical technology.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
1. attendance time in lectures, exercises: 30 h

2. preparation and follow-up of the same: 30 h

3. exam preparation and attendance in the same: 30h

M 12.80 Module: Integrated Photonics [M-ETIT-107344]

Coordinators: Prof. Dr.-Ing. Christian Koos
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Fundamentals)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114418	Integrated Photonics		**6 CP**	**Koos**

Assessment
The assessment takes place in the form of an oral examination (approx. 25 minutes); appointments individually on demand.

Prerequisites
none

Competence Goal
At the end of the course, students

- have a refreshed and deepened understanding of the basic principles of light-matter-interaction and wave propagation in dielectric media,
- know and understand the Lorentz model for frequency-dependent material properties and can use this knowledge to quantitatively analyze the dispersive properties of optical media using Sellmeier relations and scientific databases,
- understand and can quantitatively describe the formation and propagation of surface plasmon polaritons (SPP),
- know basic structures of integrated optical waveguides and understand the formation of guided modes in these structures,
- can quantitatively describe the propagation of signals in optical waveguides under the influence of dispersion,
- are familiar with state-of-the-art material systems and technology platforms of integrated photonics,
- understand basic analytical methods for modeling of photonic circuits such as eigenmode expansion (EME) methods, scattering-matrix formalisms, or coupled-mode theory and can apply these methods to specific use cases,
- understand the concepts and limitations of widely used computational techniques for integrated optical devices such as numerical mode solvers or finite-difference-time-domain (FDTD) simulators,
- know and can quantitatively analyze widely used building blocks and passive devices of integrated photonics, comprising, e.g., multi-mode interference (MMI) couplers, directional couplers, and waveguide gratings, as well as Mach-Zehnder interferometers (MZI), arrayed-waveguide gratings (AWG), lattice filters, and ring resonators.
- understand lasers and optical amplifiers and are familiar with the associated models, e.g., based on rate equations, and their predictions for dynamic behavior
- know material systems and device concepts used for electro-optic modulators
- understand photodetectors and the origin and quantitative description of noise in optoelectronic receiver systems.

Content
This course is geared towards engineering students that want to get a deeper insight into the vividly growing field of integrated photonics. The lecture and the associated tutorial provide an advanced understanding of the physical concepts and associated mathematical models of photonic integrated circuits, of the underlying material platforms and fabrication technologies, and of related application in optical communications, optical sensing, or microwave photonics. The concepts explained in the lecture are widely applicable also in fields outside integrated optics and can be a perfect complement to a variety of topics and adjacent fields such as communications engineering and high-speed data transmission, radio-frequency (RF) electronics, sensor system engineering and automation, photovoltaics, or quantum technologies.

The course covers the following aspects:

1. Review of fundamentals of wave propagation and light-matter interaction in photonics: Maxwell's equations in optical media, wave equation and plane waves; material dispersion; Lorentz and Drude model of refractive index; Sellmeier equations
2. Plasmonics: Fundamentals of surface-plasmonic polariton (SPP) propagation
3. Integrated optical waveguides: Basic structures; formation of guided modes; mathematical description; orthogonality of modes
4. Propagation of optical signals in waveguides: Group velocity; dispersion; outlook to nonlinear optical effects
5. Optical fibers: Step-index fibers and associated modes; micro-structured fibers
6. Overview of material systems and integration platforms: Silicon photonics, silicon-nitride-based photonic circuits; III-V compound semiconductors; thin-film lithium niobate; polymer-based photonic circuits
7. Eigenmode expansion (EME) method: Concept and applications
8. Numerical methods in integrated photonics: Basics of numerical mode solvers and finite-difference-time-domain (FDTD) simulators
9. Scattering-matrices (S-matrices) for optical devices: Definition and properties of S-matrices
10. Coupled-mode theory and building blocks of integrated photonics: Multi-mode interference (MMI) devices; directional couplers; waveguide gratings
11. Selected passive devices: Mach-Zehnder interferometers (MZI); arrayed-waveguide gratings (AWG); lattice filters; ring resonators
12. Lasers and optical amplifiers: Material systems; quantitative model and rate equations; dynamic behaviour of lasers
13. Electro-optic modulators: Material systems, device concepts, and figures of merit
14. Photodetectors and noise in optoelectronic receiver systems: Photodetector concepts and implementations; direct and coherent detection; shot noise; thermal noise; noise figures

Module Grade Calculation
The module grade is the grade of the oral exam under consideration of any bonuses that may apply

The module grade is the grade of the oral exam under consideration of bonuses based on the problem sets that are solved during the term – details will be given during the lecture. A bonus of 0.3 or 0.4 grades will be granted on the final mark of the oral exam, except for grades worse than 4.0. Bonus points do not expire and are retained for any examinations taken at a later date.

Workload
The workload amounts to approximately 180 h (6 CP), comprising the following items:

Attendance of lectures and tutorials: 15 × (2 h +2 h) = 60 h
Preparation and follow-up of lectures: 15 × 3 h = 45 h
Preparation and follow-up of tutorials: 15 × 3 h = 45 h
Preparation of oral exam: 30 h

Recommendations
Basics understanding of the underlying topics such as electromagnetic fields and waves, semiconductors and solid-state electronics, and advanced calculus. These skills can, e.g., be acquired in the modules "Höhere Mathematik I-III", "Elektromagnetische Felder und Wellen", "Festkörperelektronik und Bauelemente", and „Fundamentals of Photonics" or in comparable lectures.

M 12.81 Module: Interdisciplinary Qualifications [M-ETIT-105803]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Interdisciplinary Qualifications

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	pass/fail	Each term	1 term	German/English	4	6

Election Notes
For self assignment of taken interdisciplinary qualifications of HoC, FORUM, STK, or SPZ the courses ('Teilleistungen') with the title 'Self Assignment-...' have to be selected according to the grading scale, graded or ungraded. Title and credits of the achievement are adopted.
Students can access the module via the menu item "Exam Registration and Unregistration" at the Study Portal.

ID	Name	Term offered	Credits	Lecturers
Interdisciplinary Qualifications (Elective: at least 6 credits)				
T-MACH-105721	Engineer's Field of Work	ST	2 CP	Doppelbauer, Geimer
T-ETIT-111316	Introduction to the Scientific Method (Seminar, German)	WT+ST	1 CP	Nahm
T-GEISTSOZ-115075	Foundations of Technology Ethics	ST	2 CP	Derpmann
T-WIWI-100796	Industrial Business Administration	WT	3 CP	Fichtner
T-ETIT-111317	Introduction to the Scientific Method (Seminar, Englisch)	WT+ST	1 CP	Nahm
T-ETIT-115047	Master's Seminar on Scientific Study Design in Biomedical Engineering	WT+ST	1 CP	Loewe, Nahm, Spadea
T-MACH-105442	Intellectual Property Rights and Strategies in Industrial Companies	WT+ST	4 CP	Zacharias
T-ETIT-114434	Practical Introduction to Research Software Engineering	WT+ST	1 CP	Houillon
T-ETIT-108820	Seminar Project Management for Engineers	ST	3 CP	Noe
T-ETIT-111369	Strategy Derivation for Engineers	WT	3 CP	Arndt
T-ETIT-100754	Seminar Creating a Patent Specification	ST	3 CP	Stork
T-ETIT-100797	Educational Development for Student Teachers - Basic Level	WT+ST	2 CP	
T-ETIT-112616	Self-booking-HOC-SPZ-FORUM-STK-ungraded	WT+ST	6 CP	
T-ETIT-112617	Self-booking-HOC-SPZ-FORUM-STK-ungraded	WT+ST	6 CP	
T-ETIT-112618	Self-booking-HOC-SPZ-FORUM-STK-ungraded	WT+ST	6 CP	
T-ETIT-112619	Self-booking-HOC-SPZ-FORUM-STK-graded	WT+ST	6 CP	
T-ETIT-112620	Self-booking-HOC-SPZ-FORUM-STK-graded	WT+ST	6 CP	
T-ETIT-112621	Self-booking-HOC-SPZ-FORUM-STK-graded	WT+ST	6 CP	

Assessment
Dependend on the selected course

Prerequisites
none

Competence Goal
The “Interdisciplinary Qualifications” module consists of a wide range of elective courses. The qualification goals outlined here represent overarching and general competencies that can typically be acquired through the courses offered. They are supplemented by specific competencies, which are listed here as examples and are, in some cases, further detailed in the module components.

Upon completion of the module, students will be able to:

- use scientific methods (e.g. hypothesis formation, patent research).
- to use digital tools for efficient work (e.g. Unix shell, Python, Git).
- apply project management skills to structure complex tasks (e.g. resource planning, documentation).
- methodically analyze problems, plan, and implement solutions (e.g. hypothesis formation, risk analysis).
- effectively communicate knowledge, work in teams, and engage in interdisciplinary discussions (e.g. presentation skills, foreign language proficiency).
- act in a socially responsible manner within teams, lead projects, and help shape socially relevant issues (e.g. ethics of technology, sustainability).

Content
See "module component" or the course descriptions on the websites of the HoC (https://studium.hoc.kit.edu/hocampus/index.php/en), FORUM (https://www.forum.kit.edu/english), STK (https://www.stk.kit.edu/english) or the Language Center (https://www.spz.kit.edu/english).

Module Grade Calculation
The module is ungraded. The module is passed with successful assessment of the coursework.

Workload
180 h; the distribution between in-person and self-study components depends on the course selected

M 12.82 Module: Introduction into Energy Economics [M-WIWI-100498]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
5 CP	graded	Each summer term	1 term	German	4	5

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-102746	Introduction to Energy Economics	**ST**	**5 CP**	**Fichtner**

Assessment
The assessment consists of a written exam according to Section 4(2), 1 of the examination regulation.

Prerequisites
None

Competence Goal
The student is able to

- characterize and judge the different energy carriers and their peculiarities,
- understand contexts related to energy economics.

Content
1. Introduction: terms, units, conversions
2. The energy carrier gas (reserves, resources, technologies)
3. The energy carrier oil (reserves, resources, technologies)
4. The energy carrier hard coal (reserves, resources, technologies)
5. The energy carrier lignite (reserves, resources, technologies)
6. The energy carrier uranium (reserves, resources, technologies)
7. The final carrier source electricity
8. The final carrier source heat
9. Other final energy carriers (cooling energy, hydrogen, compressed air)

Workload
The total workload for this course is approximately 165.0 hours. For further information see German version.

M 12.83 Module: Introduction to Automotive and Industrial Lidar Technology [M-ETIT-105461]

Coordinators: Prof. Dr. Wilhelm Stork
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111011	Introduction to Automotive and Industrial Lidar Technology	WT	3 CP	Stork

Assessment
The examination consists of an oral exam and a short oral presentation. The overall impression is rated.

Prerequisites
none

Competence Goal

- The students are able to explain the basic principles of a lidar sensor
- The students can explain all relevant components of a lidar sensor and put them in context
- The students can explain different forms of execution and make a meaningful choice depending on the requirements
- The students can describe lidar sensors theoretically using the lidar equations and explain the interactions based on this theory
- The students are able to assess the eye safety of a system
- The students are able to suggest possible sensor concepts for different applications or to evaluate existing concepts

Content
In this course the functionality of a lidar sensor is explained and then put into context with relevant use cases. Typical criteria for the evaluation of the performance are then presented. In the following the concept of the sensor is presented in detail and all relevant components are introduced individually. Afterwards they are qualitatively related to each other and the whole system is quantitatively examined by means of the lidar equation. Finally, the interaction of the components is further considered to present meaningful combinations and design solutions. The eye safety of lidar sensors is always explicitly considered. The course concludes with a colloquium in which the students will give short presentations on what they have learned. This repetition is intended to repeat and deepen what has been learned and to lead to a discussion of open question

Module Grade Calculation
The module grade results of the assessment of the oral exam and the short oral presentation. Details will be given during the lecture.

Workload

1. participation in the lectures 12h - 8 dates á 1,5h
2. preparation and postprocessing 14 h (2h for VL dates 1-7)
3. preparation of the short lecture (16h)
4. preparation and participation in the oral exam : 48h

Recommendations
Basics of optics / optical technologies are helpful (e.g. optical engineering, optoelectronic, technical optics)

M 12.84 Module: Introduction to Microsystem Technology I [M-MACH-107657]

Coordinators: Dr. Vlad Badilita
Prof. Dr. Jan Gerrit Korvink

Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-114100	Introduction to Microsystem Technology I	**WT**	**4 CP**	**Badilita, Korvink**

Assessment
Written exam: 60 min

Prerequisites
None

Competence Goal
The students are able to:

- Explain the fundamental concepts of microsystems engineering and assess their relevance for modern technical applications.
- Describe the physical principles and relevant materials of microsystems engineering and evaluate their suitability for different applications.
- Explain key basic technologies of microsystems engineering and understand their role in the fabrication of microelectronic and micromechanical structures.
- Analyze fundamental processes of silicon microtechnology and relate them to concrete components and systems.
- Understand scientific literature in the field of MEMS, classify key technological advances, and gain an overview of the current state of the art in technology and research.

Content

- Introduction to microsystems engineering and its applications
- Materials and physical fundamentals
- Fabrication processes and basic technologies of microsystems engineering (e.g. lithography, etching, deposition, in particular silicon micromachining)
- Examples of microsystems devices and their applications
- Key scientific advances in MEMS technologies
- Current state of the art and emerging trends in microsystems engineering

Workload
Time of attendance: 15 * 1,5 h = 22,5 h
Preparation and follow up: 15 * 5,5 h = 82,5 h
Exam Preaparation and Exam: 15 h
Total: 120 h = 4 LP

Literature
M. Madou
Fundamentals of Microfabrication
Taylor & Francis Ltd.; Auflage: 3. Auflage. 2011

Mikrosystemtechnik für Ingenieure, W. Menz und J. Mohr, VCH Verlagsgesellschaft, Weinheim 2005

M 12.85 Module: Introduction to Video Analysis [M-INFO-100736]

Coordinators: Prof. Dr.-Ing. Jürgen Beyerer
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-101273	Introduction to Video Analysis	**ST**	**3 CP**	**Beyerer**

Assessment
See Partial Achievements (Teilleistung).

Prerequisites
See Partial Achievements (Teilleistung).

Competence Goal
After attending the lecture and working through the sources mentioned and discussed, students will have an overview of classical and current methods from various areas of image sequence analysis. These range from motion detection and correspondence formation to the estimation of three-dimensional structures from motion, the detection and tracking of objects in image sequences and the interpretation of visually observable actions and behavior.

Students analyze problems posed to them in the field of image sequence evaluation and evaluate known methods and groups of methods for their suitability to solve the problems and thus select suitable methods and procedures.

Content
Image sequence analysis as a sub-area of machine vision refers to the automatic derivation of statements about the scene depicted in an image sequence and its development over time. The derived statements can be made available to the human user or directly transferred to the actions of technical systems. When analyzing image sequences, it is possible to include movements as part of the temporal change of the observed scene in the derivation of statements compared to the observation of individual images.

The subject of the lecture is initially the determination of an existing movement in the scene from the images of an image sequence. Both change-based and correspondence-based methods are covered. The use of motion estimation between individual images of an image sequence is then illustrated using examples such as mosaic formation, the
determination of scene structures from motion and object detection based on motion cues.

The lecture focuses on object detection and, in particular, object tracking methods, which are used to automatically determine motion traces in the image and to estimate the three-dimensional motion of scene objects. The estimated two- and three-dimensional traces form the basis for methods that link the quantitative information about an observed scene with qualitative concepts. This is discussed using the example of action recognition in image sequences. The use of the conceptualization of image sequence evaluation results to inform the human user as well as for automatic conclusions within an image evaluation system is illustrated using examples.

Workload
Total: approx. 90h, of which
1. Attendance time in lectures: 23h
2. Preparation and follow-up of the same: 23h
3. Exam preparation and attendance: 44h

Recommendations
See Partial Achievements (Teilleistung).

M 12.86 Module: IT/OT-Security Seminar [M-ETIT-106789]

Coordinators: Prof. Dr.-Ing. Mike Barth
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113648	IT/OT-Security Seminar	**WT**	**4 CP**	**Barth**

Assessment
The examination takes place in the form of an oral examinationof approx. 25 min.

Prerequisites
none

Competence Goal
The students:
- know the definitions of terms and use-cases in the IT/OT-Security Domain
- know security requirements of both: the industrial information technology perspecitiv as well as the production related operational technology domain
- can apply basic cryptographic mechanisms with focus on industrial IT networks
- know protection goals of IT/OT-security
- know various aspects of system security (buffer overflow, return-oriented programming, ...)
- can differentiate between classic information technology (IT) and operational technology (OT) in an industrial environment
- are familiar with attacks on industrial automation and control systems (Industrial Control Systems - ICS)
- are familiar with various concepts (defense-in-depth, security by design, ...) and specific security mechanisms (Public-Key-Infrastucture, network segmentation, ...) of OT security
- are familiar with current international security standards for ICS, in particular IEC 62443
- know the different roles involved and their challenges in the life cycle of ICS
- know and understand the concept of a risk analysis for security
- can evaluate the quality of security mechanisms and architectures for industrial systems
- know typical industrial communication protocols and can analyze and evaluate their security mechanisms

Content
- Industrial control and automation systems (ICS) are widely used in numerous domains and industries. They play a crucial role in areas such as industrial production, the process industry, critical infrastructures such as energy and water management, building automation and medical devices.

- In recent years, the frequency of vulnerabilities and attacks on these systems has increased, especially since the emergence of Stuxnet in 2014. As a result, the protection of ICS has become increasingly important.

- Compared to conventional IT systems, ICS have different boundary conditions and requirements. In particular, the focus is on availability and maintaining functional safety. Therefore, classic approaches to information security cannot be applied to industrial control systems without adaptation.

- This module first provides basic knowledge of security. Building on this, concepts, mechanisms and standards for the specific domain of ICS are introduced. This includes, for example:

o Defense-in-Depth concepts
o Risk-based approaches
o IEC 62443
o Structure and operation of cyber security management systems
o Security engineering
o Use of security information and event management systems in the industrial environment
o Secure use of Industry 4.0 technologies such as OPC UA

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
The workload includes:

1. attendance in seminar lectures and exercises: 12*2 h = 24 h
2. preparation / follow-up of seminar lectures: 12*3 h = 36 h
3. implementation of challenges and exercises: 12*3 h = 36 h
4. preparation of exam: 24 h.

A total of 120 h = 4 CR

Recommendations
Enjoy working with networked software systems in the production and industrial IT environment. Curiosity in the interplay between attackers and defenders as well as a general affinity to software related topics.

M 12.87 Module: Lab Course Electrical Power Engineering [M-ETIT-100419]

Coordinators: Prof. Dr. Martin Doppelbauer
Prof. Dr.-Ing. Thomas Leibfried

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Electrical Power Systems and Electromobility (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100728	Lab Course Electrical Power Engineering	**WT**	**6 CP**	**Badent, Doppelbauer, Leibfried**

Assessment
Performance is assessed through alternative assessments as part of an ongoing evaluation process. The internship comprises eight experiments, each of which is individually assessed. Each experiment includes a written component with experiment-specific questions (approximately 15 minutes) followed by an oral component (approximately 45 minutes). The overall impression is evaluated.

The final grade is calculated as the arithmetic mean of the eight individual grades.

Prerequisites
none

Competence Goal
Students can calculate and use asynchronous machines, transformers, uncontrolled rectifier circuits, variable-speed drive systems and high-voltage generators. They can carry out partial discharge measurements.

Content
Building on the basic lectures on electrical machines, power electronics and electrical energy systems, students gain an insight into the fundamental systems of electrical power engineering.

Module Grade Calculation
The module grade includes the answers to questions from each experiment. Further details will be provided at the beginning of the course.

The overall grade is calculated as the arithmetic mean of the eight component grades.

Additional Information
Joint event of the IEH and ETI.

Workload
The workload is 180 hours and is made up as follows:

- Attendance time 40 h
- Self-study time 140 h

Recommendations
Electrical machines and power converters, electrical energy systems

M 12.88 Module: Lab Course on Nanoelectronics [M-ETIT-100468]

Coordinators: Prof. Dr. Sebastian Kempf
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Lab Course)
Field of Specialization / Electrical Power Systems and Electromobility (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100757	Lab Course on Nanoelectronics	**WT+ST**	**6 CP**	**Kempf**

Assessment
The control of success takes place in form of the evaluation of a written report (approx. 10-20 pages) which introduces the topic, discusses the execution of the lab course and the scientific results puts the results into the overall context.

Prerequisites
none

Competence Goal
After successful completion of the module, students will be familiar with elementary processes of microsystems and thin-film technology and will be able to optimize the fabrication of thin-film structures independently and without external guidance. In addition, they will be able analyze and critically evaluate their results using adequate measuring tools. By working on the practical course in small groups, students also acquire or improve their teamwork skills.

Content
The students learn the basic procedures and processes for the fabrication of integrated circuits as they are also used in industry. After an introduction, they work on specified tasks in the clean room and technology laboratory of the Institute for Micro- and Nanoelectronic Systems and work independently on a set of tasks agreed upon in advance with the supervisor. In detail, the students learn the following methods or processes:

- Fabrication of thin films and multilayer systems by sputtering and thermal vapor deposition.
- Fotolithography
- Characterization of the manufactured devices at room temperature and low temperatures.
- Independent analyses, measurements and evaluations of characteristic quantities of the fabricated structures such as critical temperature, residual resistance ratio, current-voltage characteristics, etc.

The results are subsequently summarized by the students in a final report, put into context and critically discussed.

Module Grade Calculation
The module grade is the grade of the written report.

Additional Information
Two weeks block course in lecture-free time

Workload
A workload of approx. 180h is required for the successful completion of the module. This is composed as follows:

- Preparation of the lab course: 20h
- Discussion and lab course planning with supervisor: 10h
- Attendance time in the lab course: 70h
- Preparation of the written report: 80h

Recommendations
Successful completion of the module M-ETIT-103451 - Thin Films: technology, physics and application I or M-ETIT-105608 - Physics, Technology and Applications of Thin Films is recommended.

M 12.89 Module: Lab course on superconducting materials [M-ETIT-105614]

Coordinators: Prof. Dr. Bernhard Holzapfel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111242	Lab course on superconducting materials	**WT+ST**	**6 CP**	**Kempf**

Assessment
The control of success takes place in form of the evaluation of a written report (approx. 10-20 pages) which introduces the topic, discusses the execution of the lab course and the scientific results puts the results into the overall context.

Prerequisites
None

Competence Goal
After successful completion of the module, students will know how elementary synthesis and characterization methods of superconducting materials work and will be able to synthesize and characterize superconducting materials in thin film and bulk form independently and without external guidance. They know how to interpret measured parameters and relate them to the structural and superconducting properties of superconducting materials. Furthermore, the students understand elementary aspects of the underlying synthesis and characterization techniques of superconducting materials at variable temperatures and have an insight into the realization of concrete applications of superconducting materials. By working on the practical course in small groups, students also acquire or improve their teamwork skills.

Content
Superconducting energy and electronic applications play an important role in many areas of research, society and industry today. Superconducting magnets for medical MRI applications, or modern high-performance power engineering components such as superconducting motors and cables are just a few examples. In all superconducting applications, the specific properties of the underlying superconducting materials are critical to the performance and potential range of applications of the corresponding superconducting electrical engineering components.

Against this background, students in this module learn the basic synthesis and characterization techniques of superconducting materials and how to use them to produce superconducting materials and characterize their essential application-relevant superconducting properties independently and without outside guidance. In concrete terms, the students synthesize superconducting HTSL thin films or solid materials in consultation with the supervisor and characterize their structural and superconducting properties. For example, phase purity and growth textures can be analyzed by X-ray methods, and superconducting transition temperatures and temperature-dependent critical current densities can be analyzed by magnetic and transport measurements, respectively. The students characterize the superconductors they have synthesized and compare the results with literature values. In this environment, the students also gain insight into the methods of cryogenic engineering, which plays an essential role in the field of superconducting materials.

Module Grade Calculation
The module grade is the grade of the written report.

Workload
A workload of approx. 180h is required for the successful completion of the module. This is composed as follows:

- Preparation of the lab course: 20h
- Discussion and lab course planning with supervisor: 10h
- Attendance time in the lab course: 70h
- Preparation of the written report: 80h

Recommendations
Successful completion of the modules "Superconductivity for engineers" and " Superconducting Materials " is recommended.

M 12.90 Module: Lab Course on Superconducting Quantum Electronics [M-ETIT-105605]

Coordinators: Prof. Dr. Sebastian Kempf
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Lab Course)
Field of Specialization / Electrical Power Systems and Electromobility (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111233	Lab Course on Superconducting Quantum Electronics	**WT+ST**	**6 CP**	**Kempf**

Assessment
The control of success takes place in form of the evaluation of a written report (approx. 10-20 pages) which introduces the topic, discusses the execution of the lab course and the scientific results puts the results into the overall context.

Prerequisites
none

Competence Goal
After successful completion of the module, students will know how elementary components of quantum electronics work and will be able to design, build and characterize quantum electronic circuits independently and without external guidance using these components. They know how to interpret measured parameters and characteristics and how to relate them to circuit properties. Furthermore, the students understand elementary aspects of the characterization of quantum electronic circuits at low temperatures and have an insight into the required interconnection technology as well as the realization of specific applications using quantum electronic circuits. By working on the practical course in small groups, students also acquire or improve their teamwork skills.

Content
Today, superconducting quantum electronics plays an important role in many areas of research, society and industry. For example, quantum computers have been shown to outperform classical computers, and diagnostic systems based on superconducting quantum interference detectors in the field of medical technology have become an indispensable part of everyday clinical practice.

Against this background, students will learn the basic operation of elementary components of superconducting quantum electronics (Josephson junctions, SQUID, superconducting wiring, etc.) and how to design, build and characterize quantum electronic circuits independently and without external guidance using these components. In fact, students characterize Josephson tunnel junctions, superconducting quantum interference detectors or superconducting microwave resonators in consultation with the supervisor and build circuits for a specific application using these elements. For example, a quasi-primary thermometer for low temperatures or an nA current sensor can be realized. Students characterize the circuits they build and compare the results to the design parameters. In this environment, students also gain a brief insight into the methods of low-temperature engineering, which plays an essential role in the field of superconducting quantum electronics.

Module Grade Calculation
The module grade is the grade of the written report.

Workload
A workload of approx. 180h is required for the successful completion of the module. This is composed as follows:

- Preparation of the lab course: 20h
- Discussion and lab course planning with supervisor: 10h
- Attendance time in the lab course: 70h

Preparation of the written report: 80h

Recommendations
Successful completion of the modules "Quantum detectors and sensors" and "Nano- and quantum electronics" is recommended.

M 12.91 Module: Laboratory FPGA Based Circuit Design [M-ETIT-100470]

Coordinators: Prof. Dr. Sebastian Kempf
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100759	Laboratory FPGA Based Circuit Design	**WT+ST**	**6 CP**	**Kempf**

Assessment
The success control takes place in form of another type of examination, consisting of the evaluation of the three projects and a final report (30-40 pages). The overall impression is rated.

Prerequisites
none

Competence Goal
Students acquire the ability to analyze, structure and formally describe the problem. Furthermore, students will be able to transform the formal descriptions into logical functions and implement these in the programmable FPGA using the development environment. In their experimental work, students will be able to critically evaluate the results they have achieved and modify them if necessary.

Content
The students get to know the development environment for FPGA and acquire the knowledge to implement logical functions in programmable circuits. The following sub-projects are worked on in detail:

- Introduction to the integrated development environment *Altera Quartus II* based on the creation of convolutional coders.
- Creation of simulation stimuli and comparison of the simulation results of the created encoders.
- Creation of digital filters by means of advanced graphic design using the integrated tools of the development environment.
- Programming and measurement of the created filters.
- Creation of parameterized digital filters in VHDL considering different implementation variants.

- Comparison and discussion of the need for logic cells and the performance of the filters.

Module Grade Calculation
The assessment of the three projects and the final report are included in the module grade. Further details will be provided at the beginning of the course.

Workload
The workload in hours is broken down below:

1. attendance time in the internship 48 h
2. preparation/follow-up 82 h
3. preparation of the final report 50 h

M 12.92 Module: Laboratory in Software Engineering [M-ETIT-100460]

Coordinators: Prof. Dr.-Ing. Eric Sax
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100681	Laboratory in Software Engineering	ST	6 CP	Sax

Assessment
Success is assessed by means of two oral examinations (5-10 min.) during the laboratory, the submitted software project and a final presentation (5-10 min.) with a question and answer session. The overall impression will be assessed.

Prerequisites
none

Competence Goal
After completing the module, students can independently design and implement an application-oriented software project in the field of embedded systems. This includes analyzing the problem, designing the system and solution, implementing it, and conducting extensive testing within a simulation environment. Their knowledge of UML, object-oriented programming, service-oriented architecture, virtualization, and the C++ and Python programming languages has been deepened.

Students can derive the necessary specifications from a given user story and create a system design based on these specifications. Using current UML standards, they can develop and clearly visualize software architectures. They are also capable of successfully managing projects as part of a team, including task coordination, constructive conflict resolution, and evaluating and presenting their own results.

Content
Under guidance, students first acquire foundational knowledge in virtualization and service orientation, using Docker/ Podman and ROS2 as examples. Building on this, they independently design and implement service-oriented software to develop an automated driving function. This includes processing sensor data to control the vehicle's actuators within a simulation environment.

The work is carried out in teams of three to four students, who independently pursue and work on the lab's objectives. Professional development tools are used throughout the lab, including the CarMaker simulation environment.

Module Grade Calculation
The assessment of the oral questions, the software project and the final presentation with Q&A session are included in the module grade. Further details will be provided at the beginning of the course.

Workload
1. attendance time in laboratory appointments: 48 hours

2. preparation/follow-up: 96 hours

3. preparation of the presentation: 10 hours

4. preparation of the oral examination: 10 hours

Total: 164 hours

M 12.93 Module: Laboratory Lighting Technology [M-ETIT-102356]

Coordinators: Prof. Dr. Cornelius Neumann
Dr.-Ing. Klaus Trampert

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-104726	Laboratory Lighting Technology	**WT+ST**	**6 CP**	**Neumann**

Assessment
Success is assessed on the basis of a total of four experiments. The overall impression is graded.

Prerequisites
none

Competence Goal
Students have practical experience in using photometric measurement equipment and methods for determining the photometric and electrical properties of lamps and luminaires. They also have basic experience in the simulation of luminaires with CAE tools.

They will be able to assess the plausibility of measurement results and estimate the influence of the measurement method on the uncertainty of the result.

They also have the competence to summarise the results in written form and to interpret the knowledge gained from the measurements scientifically and to explain the physical and photometric properties of lamps and luminaires.

Content
This module is designed to teach students the theoretical and practical aspects of laboratory work in the field of lighting technology by means of independently performed practical experiments. In the four experiments, the handling of real measurement technology is trained on the institute's scientific equipment. The module also teaches the skills required to write a scientific report and the rules for the appropriate visualisation of data sets.

The working titles of the experiments are:

1. thermal spectral behaviour of LEDs
2. far-field goniophotometry
3. near-field goniophotometry
4. simulation of optical systems

Module Grade Calculation
Oral examinations and the assessment of written assignments are included in the module grade. Further details will be provided at the beginning of the course.

Workload
Due to the self-administration of the small groups:

1 x 5 h are required for organizational tasks. This includes attending the information event, attending 2 safety briefings (laser and clean room) and making individual appointments between the experiment supervisor and the small group.

The workload for the 4 experiments in the module is calculated as follows:

4 x 5 h familiarization with the topic and literature study on the basics including preparation for the admission test.

4 x 8 h attendance at the institute

4 x 10 h data preparation and visualization

4 x 16 h Writing an individual report on the basis of the measurement data and the research question for the experiment.

4 x 1 h final discussion on the experiment with feedback on the report

4 x 4 h Improvement of the report based on the feedback on the report

Total hours required = 181 h = 6 CR

Recommendations
Knowledge of the theoretical principles of the individual experiments is helpful. It is highly recommended to attend the module after attending the lectures relevant to the subject, as knowledge of the theoretical basics is helpful but not mandatory. If the basics from the corresponding modules are not present, a longer preparation time for the respective experiment is needed.

Helpful modules: Light technology, optoelectronic measurement technology, photometry and radiometry

M 12.94 Module: Laboratory Mechatronics [M-MACH-102699]

Coordinators: Prof. Dr. Veit Hagenmeyer
Prof. Dr.-Ing. Christoph Stiller

Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	pass/fail	Each summer term	1 term	German	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-105370	Laboratory Mechatronics	**ST**	**4 CP**	**Hagenmeyer, Stiller**

Assessment
The laboratory course is offered exclusively as ungraded course work. The assessment consists of a group colloquium at the beginning of the individual specialization phases (Part 1). During the group phase (Part 2), students must also successfully complete various major assignments and present their results in a final presentation.

Prerequisites
None

Competence Goal
Students are able to apply their knowledge of mechatronics and computer vision in practice using a sample system. They can create an automated object recognition system, control a robotic arm to transport the recognised objects, and establish communication between the different systems.

In addition, they must develop recursive algorithms to solve various tasks.

Content
Part I
Control, Programming, and Simulation of Robotic Systems
Computer Vision for Object Recognition and Conversion into a Robot KOS

Part II
In a group project, students must solve various tasks of varying difficulty levels. This involves recognizing different objects and moving them with the robotic arm according to the rules of the game.

Module Grade Calculation
The module is not graded. Passing the module is 100% tied to the performance assessment of the partial performance.

Workload

1. Attendance time Lecture: 5 * 7 h = 35 h
2. self-study: 17 * 5 h = 85 h

Total: 120 h = 4 LP

Teaching and Learning Methods
Seminar

M 12.95 Module: Laboratory Nanotechnology [M-ETIT-100478]

Coordinators: Prof. Dr. Ulrich Lemmer
Dr.-Ing. Klaus Trampert

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100765	Laboratory Nanotechnology	**WT+ST**	**6 CP**	**Lemmer**

Assessment
Success control takes place in the form of a total of four tests. The overall impression is rated.

Prerequisites
none

Competence Goal
The students have practical experience in the use of measuring devices and manufacturing processes in nanotechnology and the methods for determining the physical and optical properties of optoelectronic components with functional nanotechnology components.

They can evaluate measurement results with regard to their plausibility and assess the influence of the measurement method on the uncertainty of the result.

You will also be able to reproduce the results in written form and interpret the knowledge gained from the measurements scientifically and explain the physical properties and the influence of the nanotechnology components.

Content
This module is designed to teach students the theoretical and practical aspects of laboratory work in the field of nanotechnology by means of independently conducted practical experiments. In the four experiments, students practise working with real measurement technology using the scientific equipment of the institute. The module also teaches students how to write a scientific report and the rules for the meaningful visualization of data sets.

The working titles of the experiments are

1. Production and characterization of an OLED
2. Optical mask lithography
3. Fabrication and characterization of an electrochromic device
4. Nanoimprint lithography and scanning electron microscopy

Module Grade Calculation
Oral examinations and the assessment of written assignments are included in the module grade. Further details will be provided at the beginning of the course.

Workload
Due to the self-administration of the small groups:

1 x 5 h are required for organizational tasks. This includes attending the information event, attending 2 safety briefings (laser and clean room) and making individual appointments between the experiment supervisor and the small group.

The workload for the 4 experiments in the module is calculated as follows:

4 x 5 h familiarization with the topic and literature study on the basics including preparation for the admission test.

4 x 8 h attendance at the institute

4 x 10 h data preparation and visualization

4 x 16 h Writing an individual report on the basis of the measurement data and the research question for the experiment.

4 x 1 h final discussion on the experiment with feedback on the report

4 x 4 h Improvement of the report based on the feedback on the report

Total hours required = 181 h = 6 CR

Recommendations
Knowledge of the theoretical principles of the individual experiments is helpful. It is advisable to attend the module after attending the subject-relevant courses, as knowledge of the theoretical basics is helpful but not mandatory. If the basics from the corresponding modules are not available, this means a longer preparation time for the respective experiment.

Helpful modules: Solid state electronics

M 12.96 Module: Laboratory Solar Energy [M-ETIT-102350]

Coordinators: Prof. Dr. Ulrich Wilhelm Paetzold
Prof. Dr. Bryce Sydney Richards

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Electrical Power Systems and Electromobility (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-104686	Laboratory Solar Energy	**WT+ST**	**6 CP**	**Trampert**

Assessment
Success is assessed on the basis of a total of four experiments. The overall impression is graded.

Prerequisites
none

Competence Goal
Students have practical experience in using photometric measurement equipment and methods for determining the photometric and electrical properties of lamps and luminaires. They also have basic experience in the simulation of luminaires with CAE tools.

They will be able to assess the plausibility of measurement results and estimate the influence of the measurement method on the uncertainty of the result.

They also have the competence to summarise the results in written form and to interpret the knowledge gained from the measurements scientifically and to explain the physical and photometric properties of lamps and luminaires.

Content
This module is designed to teach students the theoretical and practical aspects of laboratory work in the field of solar technology, particularly in the field of photovoltaics, by means of independently conducted practical experiments. In the four experiments, the handling of real measurement technology is trained on the institute's scientific equipment. The module also teaches the skills required to write a scientific report and the rules for the appropriate visualisation of data sets.

The working titles of the experiments are:

1. Light Beam Induced Current (LBIC) measurement in solar cells
2. Optical and electrical modelling of thin-film solar cells
3. Quantum efficiency measurements on solar cells
4. Outdoor measurements of PV modules

Module Grade Calculation
Oral examinations and the assessment of written reports are included in the module grade. Further details will be provided at the beginning of the course.

Workload
Due to the self-administration of the small groups:

1 x 5 h are required for organizational tasks. This includes attending the information event, attending 2 safety briefings (laser and clean room) and making individual appointments between the experiment supervisor and the small group.

The workload for the 4 experiments in the module is calculated as follows:

4 x 5 h familiarization with the topic and literature study on the basics including preparation for the admission test.

4 x 8 h attendance at the institute

4 x 10 h data preparation and visualization

4 x 16 h Writing an individual report on the basis of the measurement data and the research question for the experiment.

4 x 1 h final discussion on the experiment with feedback on the report

4 x 4 h Improvement of the report based on the feedback on the report

Total hours required = 181 h = 6 CR

Recommendations
Knowledge of the theoretical background of each experiment is recommended. It is strongly recommended that you attend this module after attending the relevant lectures, as knowledge of the theoretical background is important but not strictly necessary.

M 12.97 Module: Laser Metrology [M-ETIT-100434]

Coordinators: Prof. Dr. Marc Eichhorn
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100643	Laser Metrology	**ST**	**3 CP**	**Eichhorn**

Assessment
The exam will be taken as an oral examination (about 20 minutes). The individual appointments for examination are offered regularly at two previously determined dates.

Prerequisites
none

Competence Goal
The students understand the fundamental properties of laser light and possess the knowledge necessary to understand the metrologically obtainable information, understand the basics of various detectors as well as their limits and have the knowledge necessary to understand a multitude of laser metrological setups, mainly for interferometry, Moiré methods, distance and velocity measurements and absorption as well as scattering techniques.

Content
In the module several aspects of laser diagnostics will be discussed, beginning with the fundamental properties of laser light and the related metrologically useful information. In addition beam diagnostics and interferometric setups in general, as well as Moiré methods in particular, will be discussed. Further topics of the lecture will be commonly used setups, mainly for laser distance and velocity measurements along with widely used absorption and scattered light methods.

1. Laser diagnostics - theoretical considerations (laser beam properties, coherence, spectral emission of lasers, mode structure and selection, coherence length)
2. Metrological accessible information (propagation in homogeneous and isotropic, in inhomogeneous and in anisotropic media)
3. Beam diagnostics (photoelectric detectors, information theory, granulation properties of laser light)
4. Laser-Interferometer (fundamentals, two-beam Interferometer, interferometry applications in plasma physics, two- and multiwavelength-interferometry, laser gyroscopes)
5. Moiré technique (Moiré deflectometry, Fresnel- and Fraunhofer diffraction, applications and evaluation of the Moiré technique)
6. Laser range measurements (fundamentals, atmospheric influence on propagation, optical distance measurement techniques, accuracy, sensitivity, heterodyne detection, selected heterodyne detection schemes, tomoscopy)
7. Laser velocity measurement techniques (Doppler principle, measuring flow velocities using Doppler effect, the two-focus technique or laser anemometry; time-resolved imaging particle-trace anemometry)
8. Absorption and scattering techniques (absorption techniques, LIDARs, scattering processes in laser diagnostics, spontaneous scattering techniques, spectroscopic techniques, stimulated scattering, nonlinear optical laser light scattering techniques)

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
About 90 h in total, consisting of

30 h lectures

60 h recapitulation and self-studies

Recommendations
Solid mathematical background, basic knowledge in physics. Steady participation in the lecture as well as thorough preparation based on the scriptum is highly recommended.

Literature
M. Eichhorn, *Laser metrology* - Scriptum

A. E. Siegman, *Lasers* (university Science Books)

B. E. A. Saleh, M. C. Teich, *Fundamentals of Photonics* (Wiley-Interscience)

M 12.98 Module: Laser Physics [M-ETIT-100435]

Coordinators: Prof. Dr. Marc Eichhorn
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100741	Laser Physics	**WT**	**4 CP**	**Eichhorn**

Assessment
The exam will be taken as an oral examination (about 20 minutes). The individual appointments for examination are offered regularly at two previously determined dates.

Prerequisites
none

Competence Goal
The students understand the fundamental relations and basics of a laser. They obtain the knowledge necessary for understanding and designing lasers (laser media, optical resonators, pumping schemes) and understand the basics and schemes for pulse generation in a laser. They have the knowledge needed for a multitude of lasers: gas, solid-state, fiber and disc lasers from the visible up to the mid-Infrared spectrum.

Content
Within the module the physical basics of lasers, the fundamental processes of light amplification and the formalisms necessary to describe lasers and laser resonators are covered. The generation of laser pulses and various laser architectures as well as realisations are presented in detail.

The exercises specifically discuss the topics of laser description, theoretical background as well as the realization of different laser designs. The tasks of the exercise will be handed out at the end of each lecture as well as uploaded to the lecture website and are to be solved for the following exercise, in which the solution will be discussed.

Contents:

1 Quantum-mechanical fundamentals of lasers

1.1 Einstein relations and Planck´s law

1.2 Transition probabilities and matrix elements

1.3 Mode structure of space and the origin of spontaneous emission

1.4 Cross sections and broadening of spectral lines

2 The laser principles

2.1 Population in version and feedback

2.2 Spectroscopic laser rate equations

2.3 Potential model of the laser

3 Optical Resonators

3.1 Linear resonators and stability criterion

3.2 Mode structure and intensity distribution

3.3 Line width of the laser emission

4 Generation of short and ultra-short pulses

4.1 Basics of Q-switching

4.2 Basics of mode locking and ultra-short pulses

5 Laser examples and their applications

5.1 Gas lasers: The Helium-Neon-Laser

5.2 Solid-state lasers

5.2.1 The Nd^{3+}-Laser

5.2.2 The Tm^{3+}-Laser

5.2.3 The $Ti^{3+}:Al_2O_3$ Laser

5.3 Special realisations of lasers

5.3.1 Thermal lensing and thermal stress

5.3.2 The fiber laser

5.3.3 The thin-disk laser

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
About 120 h in total, consisting of

30 h lectures

15 h tutorial

75 h recapitulation and self-studies

Recommendations
Steady participation in the lecture as well as thorough preparation based on the scriptum is higly recommended.

Literature
M. Eichhorn, Laser physics (Springer)
M. Eichhorn, Laserphysik (Springer)
A. E. Siegman, Lasers (University Science Books)
B. E. A. Saleh, M. C. Teich,Fundamentals of Photonics (Wiley-Interscience)
F. K. Kneubühl, M. W. Sigrist, Laser (Teubner)

M 12.99 Module: Liberalised Power Markets [M-WIWI-105403]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-107043	Liberalised Power Markets	**WT**	**6 CP**	**Fichtner**

Assessment
Success is monitored in the form of a written examination.

Prerequisites
See course description.

Competence Goal
Upon successful completion of the module, students will be able to:

1. Knowledge and Understanding

- Describe the historical evolution of power markets and forecast potential future developments within the electricity value chain.
- Explain the fundamental principles of market design, specifically the concepts of unbundling, central versus decentralized dispatch, and the distinction between short-term and long-term market models.
- Identify and define the various segments of the electricity market, including day-ahead, intraday, forwards, and futures markets, as well as specialized markets for ancillary services and emission rights.

2. Analysis and Application

- Analyze the technical and economic interdependencies between grid operation and congestion management in a liberalized environment.
- Differentiate between various market mechanisms and their roles in integrating renewable energies into the power system.
- Examine the causes and consequences of market flaws and market failures, and evaluate the role of regulation in mitigating these issues.

3. Evaluation and Synthesis

- Critically assess the presence of market power by applying quantitative and qualitative indicators, and discuss effective strategies for its reduction.
- Evaluate the impact of different regulatory frameworks on market efficiency and security of supply.
- Synthesize complex information to develop perspectives on future market structures and the shifting roles of actors within the electricity value chain.

Content

1. Power markets in the past, now and in future

2. Designing liberalised power markets

2.1. Unbundling Dimensions of liberalised power markets

2.2. Central dispatch versus markets without central dispatch

2.3. The short-term market model

2.4. The long-term market model

2.5. Market flaws and market failure

2.6. Regulation in liberalised markets

3. The power (sub)markets

3.1 Day-ahead market

3.2 Intraday market

3.3 (Long-term) Forwards and futures markets

3.4 Emission rights market

3.5 Market for ancillary services

3.6 The “market” for renewable energies

3.7 Future market segments

4. Grid operation and congestion management

4.1. Grid operation

4.2. Congestion management

5. Market power

5.1. Defining market power

5.2. Indicators of market power

5.3. Reducing market power

6. Future market structures in the electricity value chain

1. Power markets in the past, now and in future

2. Designing liberalised power markets

2.2. Unbundling Dimensions of liberalised power markets

2.3. Central dispatch versus markets without central dispatch

2.4. The short-term market model

2.5. The long-term market model

2.6. Market flaws and market failure

2.7. Regulation in liberalised markets

3. The power (sub)markets

3.1 Day-ahead market

3.2 Intraday market

3.3 (Long-term) Forwards and futures markets

3.4 Emission rights market

3.5 Market for ancillary services

3.6 The “market” for renewable energies

3.7 Future market segments

4. Grid operation and congestion management

4.1. Grid operation

4.2. Congestion management

5. Market power

5.1. Defining market power

5.2. Indicators of market power

5.3. Reducing market power

6. Future market structures in the electricity value chain

Workload
The total workload for this module is approximately 180 hours.

- Attendance: Lectures (2 SWS × 15 weeks): 30 h
- Attendance: Exercises (2 SWS × 15 weeks): 30 h
- Self-Study: Preparation and follow-up of course material: 75 h
- Exam Preparation and Participation: 45 h

M 12.100 Module: Lighting Engineering [M-ETIT-100485]

Coordinators: Prof. Dr. Cornelius Neumann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100772	Lighting Engineering	**WT**	**4 CP**	**Neumann**

Assessment
Success is assessed in the form of an overall oral examination (approx. 20 minutes) on the selected courses, which together fulfill the minimum LP requirement.

Prerequisites
none

Competence Goal
Students gain an overview of the basics and application of lighting technology, light generation and light measurement technology. They learn that the focus of applications is on people and their perception.

They will be able to assess the influence of different lighting applications on people, define application-specific light sources and assess optical systems in applications.

The highly topical nature of the course allows students to follow current market and research developments. They are prepared to work on the topics in research and application.

Students are able to assess and evaluate the consequences of specific developments in lighting technology.

Students deepen their knowledge and application skills through the calculation and joint discussion of practice applications.

Content
Lighting technology is a combination of physics, electrical engineering and physiology. Physics describes the objective side of light as radiation, electrical engineering deals with the technical generation of light and physiology describes the subjective perception of light. Another important focus is photometry, i.e. the measurement of light according to human perception.

- Motivation: Focus on human
- Perception of light
- Basic parameters of lighting technology
- The human eye
- Basics of color perception
- What is light and how is it produced?
- Ambassadors of the atoms
- Thermal radiators
- Gas discharge
- LED
- Manipulation of light
- Fundamentals of optical systems
- Exemplary applications
- Measurement of light

The lecturer reserves the right to deviate from the content given here in the context of the current lecture without special notice.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload

1. attendance time lecture: 15 * 2 h = 30 h
2. preparation and follow-up time lecture: 15 * 1 h = 15 h
3. attendance time exercise: 15 * 2 = 30 h
4. preparation and follow-up time for exercise: 15 * 1 h = 15 h
5. exam preparation and attendance: 30 h

Total: 120 h = 4 LP

M 12.101 Module: Localization of Mobile Agents [M-INFO-107663]

Coordinators: Prof. Dr.-Ing. Uwe Hanebeck
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-115014	Localization of Mobile Agents	**ST**	**6 CP**	**Hanebeck**
T-INFO-115015	Localization of Mobile Agents - Pass	**ST**	**0 CP**	**Hanebeck**

Assessment
See Partial Achievements (Teilleistung).

Prerequisites
See Partial Achievements (Teilleistung).

Competence Goal
Participants in this course will gain a broad overview of probabilistic machine learning tools and learn how to creatively combine them to solve a wide variety of problems.

Content
This module provides a systematic introduction to the field of localization methods. To facilitate entry, the module is divided into four central topics. Dead reckoning (1) deals with the step-by-step determination of a vehicle's position from dynamic parameters such as speed or steering angle, as well as intertial measurement unit (IMU) data. Localization with the aid of measurements to external, known landmarks is part of static localization (2). Specifically, signal times of arrival (multilateration and GPS), distance measurements, and angle measurements (bearings) are covered. In addition to closed-form solutions for special measurements, the method of least squares for fusing arbitrary measurements is introduced. This also includes nonlinear optimization methods and certifiable global optimization. Dynamic localization (3) deals with the combination of dead reckoning and static localization using the Kalman filter, which is used in virtually all object tracking scenarios and other state estimation problems. General estimator properties are also discussed. The course concludes with simultaneous localization and mapping (SLAM) (4), which allows localization even when the location of landmarks is partially unknown. Thereby the measurement association problem is also addressed.

Workload
Per week:
2 SWS Presence
2h Follow-up
6h Digital exercise with computation tasks
2h exam preparation
= 12h/week und 180h/semester

Recommendations
Knowledge of a higher-level programming language with mature libraries for scientific and numerical computing (e.g., Julia, Matlab, Python) is an advantage.

M 12.102 Module: Low Power Design [M-INFO-100807]

Coordinators: Prof. Dr.-Ing. Jörg Henkel
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-101344	Low Power Design	**ST**	**3 CP**	**Henkel**

Assessment
See partial achievements (Teilleistung)

Prerequisites
See partial achievements (Teilleistung)

Competence Goal
Students are made aware of various low power design optimizations employed in state-of-the-art embedded devices. This involves optimization techniques that incorporate embedded machine learning algorithms to enhance system performance. At the end of the lecture, the students will be able to recognize the challenges involved in crafting efficient low power designs and how to tackle them.

Content
The lecture provides an overview of design methods, synthesis tools, estimation models, software techniques, operating system strategies, scheduling algorithms, embedded machine learning methods, etc., with the aim of minimizing the power consumption of embedded devices without compromising their performance. Both the research-relevant and industry-prevalent topics at different level of abstractions (from circuit to system) are discussed in this lecture.

Workload
Attendance time: 30 hours (2 SWS × 15 weeks)

Self-study: 45 hours (1.5 × 2 SWS × 15 weeks)

Exam preparation: 15 hours

Total: 90 hours (3 ECTS)

M 12.103 Module: Machine Learning 1 [M-WIWI-105003]

Coordinators: Prof. Dr.-Ing. Johann Marius Zöllner
Organisation: KIT Department of Business and Economics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
5 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-106340	Machine Learning 1 - Basic Methods	**WT**	**5 CP**	**Zöllner**

Assessment
Success is assessed in the form of a written examination (60 minutes) (in accordance with §4(2), 1 SPO) or in the form of an oral examination (approx. 20 minutes) (in accordance with §4(2), 2 SPO).

The examination is offered every semester and can be repeated at any regular examination date.

Prerequisites
None

Competence Goal

- Students gain knowledge of the basic methods in the field of machine learning.
- Students understand advanced concepts of machine learning and their application.
- Students can classify, formally describe and evaluate methods of machine learning.
- Students can use their knowledge to select suitable models and methods for selected problems in the field of machine learning.

Content
The course prepares students for the rapidly evolving field of machine learning by providing a solid foundation, covering core concepts and techniques to get started in the field. Students delve into different methods in supervised, unsupervised, and reinforcement learning, as well as various model types, ranging from basic linear classifiers to more complex methods, such as deep neural networks. Topics include general learning theory, support vector machines, decision trees, neural network fundamentals, convolutional neural networks, recurrent neural networks, unsupervised learning, reinforcement learning, and Bayesian learning.

The course is accompanied by a corresponding exercise, where students gain hands-on experience by implementing and experimenting with different machine learning algorithms, helping them to apply machine learning algorithms on real world problems.

By the end of the course, students will have acquired a solid foundation in machine learning, enabling them to apply state-of-the-art algorithms to solve complex problems, contribute to research efforts, and explore advanced topics in the field.

Workload
The total workload for this module is approximately 150 hours.

- Attendance: Lectures (2 SWS × 15 weeks): 30 h
- Attendance: Exercises (1 SWS × 15 weeks): 15 h
- Self-Study: Preparation and follow-up of course material: 60 h
- Exam Preparation and Participation: 45 h

Literature
Further reading

- Machine Learning - Tom Mitchell
- Deep Learning - Ian Goodfellow, Yoshua Bengio, Aaron Courville
- Pattern Recognition and Machine Learning - Christopher M. Bishop
- Artificial Intelligence: A Modern Approach - Peter Norvig and Stuart J. Russell
- Reinforcement Learning: An Introduction - Richard S. Sutton and Andrew G. Barto

Further (specific) literature on individual topics will be given in the lecture.

M 12.104 Module: Machine Learning 2 [M-WIWI-105006]

Coordinators: Prof. Dr.-Ing. Johann Marius Zöllner
Organisation: KIT Department of Business and Economics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
5 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-106341	Machine Learning 2 - Advanced Methods	**ST**	**5 CP**	**Zöllner**

Assessment
The assessment of this course is a written examination (60 min) according to §4(2), 1 of the examination regulation or an oral exam (approx. 20 min) following §4, Abs. 2, 2 of the examination regulation.

The exam takes place every semester and can be repeated at every regular examination date.

Prerequisites
None

Competence Goal

- Students gain knowledge of the basic methods in the field of machine learning.
- Students understand advanced concepts of machine learning and their application.
- Students can classify, formally describe and evaluate methods of machine learning.
- Students can use their knowledge to select suitable models and methods for selected problems in the field of machine learning.

Content
The subject area of machine intelligence and, in particular, machine learning, taking into account real challenges of complex application domains, is a rapidly expanding field of knowledge and the subject of numerous research and development projects.

The lecture "Machine Learning 2" deals with modern advanced methods of machine learning such as semi-supervised and active learning, deep neural networks (deep learning, CNNs, GANs, diffusion models, transformer, adversarial attacks) and hierarchical approaches, e.g. reinforcement learning. Another focus is the embedding and application of machine learning methods in real systems.

The lecture introduces the latest basic principles as well as extended basic structures and elucidates previously developed algorithms. The structure and the mode of operation of the methods and methods are presented and explained by means of some application scenarios, especially in the field of technical (sub) autonomous systems (vehicles, robotics, neurorobotics, image processing, etc.).

Workload
The total workload for this module is approximately 150 hours.

- Attendance: Lectures (2 SWS × 15 weeks): 30 h
- Attendance: Exercises (1 SWS × 15 weeks): 15 h
- Self-Study: Preparation and follow-up of course material: 60 h
- Exam Preparation and Participation: 45 h

Literature

- Deep Learning - Ian Goodfellow
- Artificial Intelligence: A Modern Approach - Peter Norvig and Stuart J. Russell
- Machine Learning - Tom Mitchell
- Pattern Recognition and Machine Learning - Christopher M. Bishop
- Reinforcement Learning: An Introduction - Richard S. Sutton and Andrew G. Barto
- Deep Learning - Ian Goodfellow, Yoshua Bengio, Aaron Courville

M 12.105 Module: Machine Learning and Optimization in Communications [M-ETIT-104988]

Coordinators: Prof. Dr.-Ing. Laurent Schmalen
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	4

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-110123	Machine Learning and Optimization in Communications	**ST**	**4 CP**	**Schmalen**

Assessment
Written examination of 120 minutes.

Prerequisites
Knowledge of basic engineering mathematics probability theory as well as basic knowledge of communications engineering (e.g. "M-ETIT-102103 – Nachrichtentechnik I" and "M-ETIT-102104 – Wahrscheinlichkeitstheorie").

Competence Goal
The students will be able to apply the methods and tools of machine learning, artificial intelligence and optimization in communications engineering. You will learn various tools and solution methods of machine learning, artificial intelligence and numerical optimization, and you can use these tools to solve telecommunications problems.

Content
The course broadens the questions dealt with in the lecture Communication Engineering I. The focus here is on methods that arise when considering communication networks. For this purpose, partially known techniques have to be extended, in some cases new methods have to be learned. The first part of the lecture deals with modern methods of machine learning, e.g. deep neural networks, and uses examples to show how they are used in communication networks. The second part of the lecture considers numerical optimization methods and their application to telecommunication questions. In the exercise concrete questions from practice are considered and solved together with the students. The focus of the problems lies in the field of communications engineering. Many of the applications are illustrated with example implementations in software (python).

Module Grade Calculation
The module grade is the grade of the written examination.

A bonus can be earned by successfully participating in the tutorial session. The exact criteria for awarding a bonus will be announced at the beginning ft he lecture period. If the grade in the exam is between 4.0 and 1.3, the bonus improves the grade by one grade (0.3 or 0.4). Bonus points do not expire and are retained for any examinations taken at a later date. The final assessment ft he bonus performance is carried out by the examiner and demonstrably documented.

Workload
- Lecture attendance: 15 * 2 h = 30 h
- Presence time exercise: 15 * 1 h = 15 h
- Lecture preparation/-revision phase: 15 * 2 h = 30 h
- Exercise preparation/-revision phase: 15 * 1 h = 15 h
- Exam preparation and attendance: 30 h

Total workload: approx. 120 h

Recommendations
Knowledge from the modules "M-ETIT-100444 – Angewandte Informationstheorie" and "M-ETIT-105982 – Measurement Technology" is helpful.

M 12.106 Module: Machine Learning and Optimization in Energy Systems [M-WIWI-106604]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-113073	Machine Learning and Optimization in Energy Systems	**WT**	**4 CP**	**Fichtner**

Assessment
The assessment of this module is a written examination (60 min) or an oral exam (approx. 30 min) depending on the number of participants.

Prerequisites
None.

Competence Goal
Participants know about the most common optimization and machine learning approaches for the application in energy systems. They understand the basic principles of the methods and are able to apply them for solving important problems of future energy systems with high shares of renewable energy sources.

Content
In the beginning, the essential transition of the energy system into a smart grid and the need for methods from the field of optimization and machine learning are explained. The course can be subdivided into an optimization part and a larger machine learning part. In the optimization part, the basics of optimization approaches that are used in energy systems are shown. Further, heuristic methods and approaches from the field of multiobjective optimization are introduced. In the machine learning part, the most important methods from the field of unsupervised learning, supervised learning and reinforcement learning are introduced and their application in future energy systems are investigated.

Amongst the considered applications are power plant dispatch, intelligent heating with heat pumps, charging strategies for electric vehicles, clustering of energy data for energy system models and electricity demand and renewable generation forecasting.

We also offer a voluntary computer exercise that deepens the understanding of the methods and applications covered in the lecture. The students will have the opportunity to solve problems from the energy domain by using optimization and machine learning approaches implemented in the programming language Python.

The course's general focus is on the application of the methods in the energy field and not on the mathematical details of the different approaches.

Module Grade Calculation
The module grade is the grade of the written or oral exam.

Workload
The total workload for this module is approximately 120 hours:

- Attendance: 30 hours
- Self-study: 45 hours
- Exam preparation: 55 hours

M 12.107 Module: Machine Tools and Industrial Handling [M-MACH-105107]

Coordinators: Prof. Dr.-Ing. Jürgen Fleischer
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
8 CP	graded	Each winter term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-110962	Machine Tools and High-Precision Manufacturing Systems	**WT**	**8 CP**	**Fleischer**

Assessment
Oral exam (40 minutes)

Competence Goal
The students

- are able to assess the use and application of machine tools and high-precision manufacturing systems and to differentiate between them in terms of their characteristics and design.
- can describe and discuss the essential elements of machine tools and high-precision manufacturing systems (frame, main spindle, feed axes, peripheral equipment, control unit).
- are able to select and dimension the essential components of machine tools and high-precision manufacturing systems.
- are capable of selecting and evaluating machine tools and high-precision manufacturing systems according to technical and economic criteria.

Content
The module gives an overview of the construction, use and application of machine tools and high-precision manufacturing systems. In the course of the module a well-founded and practice-oriented knowledge for the selection, design and evaluation of machine tools and high-precision manufacturing systems is conveyed. First, the main components of the systems are systematically explained and their design principles as well as the integral system design are discussed. Subsequently, the use and application of machine tools and high-precision manufacturing systems will be demonstrated using typical machine examples. Based on examples from current research and industrial applications, the latest developments are discussed, especially concerning the implementation of Industry 4.0 and artificial intelligence.
Guest lectures from industry round off the module with insights into practice.

The individual topics are:

- Structural components of dynamic manufacturing Systems
- Feed axes: High-precision positioning
- Spindles of cutting machine Tools
- Peripheral Equipment
- Machine control unit
- Metrological Evaluation
- Maintenance strategies and condition Monitoring
- Process Monitoring
- Development process for machine tools and high-precision manufacturing Systems
- Machine examples

Workload
1. Presence time lecture/exercise: 15 * 6 h = 90 h
2. Pre- and post-processing time lecture/exercise: 15 * 9 h = 135 h
3. Exam preparation and presence in the same: 15 h
In total: 240 h = 8 LP

Teaching and Learning Methods
Lecture, exercise, field trip

M 12.108 Module: Machine Vision [M-MACH-101923]

Coordinators: Dr. Martin Lauer
Prof. Dr.-Ing. Christoph Stiller

Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
8 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-105223	Machine Vision	**WT**	**8 CP**	**Lauer, Stiller**

Assessment
Type of Examination: written exam
Duration of Examination: 60 minutes

Prerequisites
None

Competence Goal
After having participated in th lecture the participants have gained knowledge on modern techniques of machine vision and pattern recognition which can be used to evaluatecamera images. This especially includes techniques in the areas of gray level image analysis, analysis of color images, segementation of images, describing the geometrical relationship between the image and the 3-dimensional world, and pattern recognition with various classification techniques. The participants have learned to analyze the algorithms mathematically, to implement them in software, and to apply them to tasks in video analysis. The participants are able to analyze real-world problems and to develop appropriate solutions.

Content

The lecture on machine vision covers basic techniques of machine vision. It focuses on the following topics:

image preprocessing

edge and corner detection

curve and parameter fitting

color processing

image segmentation

camera optics

pattern recognition

deep learning

Image preprocessing:

The chapter on image processing discusses techniques and algorithms to filter and enhance the image quality. Starting from an analysis of the typical phenomena of digital camera based image capturing the lecture introduces the Fourier transform and the Shannon-Nyquist sampling theorem. Furthermore, it introduces gray level histogram based techniques including high dynamic range imaging. The disussion of image convolution and typical filters for image enhancement concludes the chapter.

Edge and corner detection:

Gray level edges and gray level corners play an important role in machine vision since gray level edges often reveal valueable information about the boundaries and shape of objects. Gray level corners can be used as feature points since they can be identified easily in other images. This chapter introduces filters and algorithms to reveal gray level edges and gray level corners like the Canny edge detector and the Harris corner detector.

Curve and parameter fitting:

In order to describe an image by means of geometric primitives (e.g. lines, circles, ellipses) instead of just pixels robust curve and parameter fitting algorithms are necessary. The lecture introduces and discusses the Hough transform, total least sum of squares parameter fitting as well as robust alternatives (M-estimators, least trimmed sum of squares, RANSAC)

Color processing:

The short chapter on color processing discusses the role of color information in machine vision and introduces various models for color understanding and color representation. It concludes with the topic of color consistency.

Image Segmentation:

Image segmentation belongs to the core techniques of machine vision. The goal of image segmentation is to subdivide the image into several areas. Each area shares common properties, i.e. similar color, similar hatching, or similar semantic interpretation. Various ideas for image segmentation exist which can be used to create more or less complex algorithms. The lecture introduces the most important approaches ranging from the simpler algorithms like region growing, connected components labeling, and morphological operations up to highly flexible and powerful methods like level set approaches and random fields.

Camera optics:

The content of an image is related by the optics of the camera to the 3-dimensional world. In this chapter the lecture introduces optical models that describe the relationship between the world and the image including the pinhole camera model, the thin lens model, telecentric cameras, and catadioptric sensors. Furthermore, the lecture introduces camera calibration methods that can be used to determine the optical mapping of a real camera.

Pattern recognition:

Pattern recognition aims at recognizing semantic information in an image, i.e. not just analyzing gray values or colors of pixels but revealing which kind of object is shown by the pixels. This task goes beyond classical measurement theory and enters the large field of artificial intelligence. Rather than just being developped and optimized by a programmer, the algorithms are adapting themselves to their specific task using training algorithms that are based on large collections of sample images.

The chapter of pattern recognition introduces standard techniques of pattern recognition in the context of image understanding like the support vector machine (SVM), decision trees, ensemble and boosting techniques. It combines those classifiers with powerful feature representation techniques like the histogram of oriented gradients (HOG) features, locally binary patterns (LBP), and Haar features.

Deep learning:

Throughout recent years standard pattern recognition technqiues have more and more been outperformed by deep learning techniques. Deep learning is based on artificial neural networks, a very generic and powerful form of a classifier. The lecture introduces multi layer perceptrons as the most relevant form of artificial neural networks, discusses training algorithms and strategies to achieve powerful classifiers based on deep learning including deep auto encoders, convolutional networks, and multi task learning, among others.

Workload
240 hours, omposed out of

hours of lecture: 15*4 h = 60 h

preparation time prior to and after lecture: 15*6 h = 90 h

exam preparation and exam: 90 h

Teaching and Learning Methods
Lecture

Literature
Main results are summarized in the slides that are made available as pdf-files. Further recommendations will be presented in the lecture.

M 12.109 Module: Master's Project [M-ETIT-107928]

Coordinators: Prof. Dr.-Ing. Mike Barth
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	pass/fail	Each term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-115350	Master's Project	**WT+ST**	**6 CP**	**Barth**

Assessment
Success control takes place in the form of another type of examination. It consists of a short report of the project results, a presentation of the results.

The overall impression is rated.

Prerequisites
none

Competence Goal
Upon successful completion of the module, students will be able to

- address complex, open-ended problems in the divisions of electrical engineering and information technology within a specified timeframe, both independently and as part of a team, using scientific methods. In doing so, they apply the in-depth theoretical and methodological knowledge acquired during their master's program and consistently adhere to the rules of good research practice.
- systematically research the current state of science and technology regarding a given research question, identify relevant scientific publications, critically evaluate them, and derive a precise problem statement and project objectives from them.
- select appropriate scientific methods, procedures, and tools (e.g., modeling, simulation, design, analysis and evaluation procedures, data processing, software and hardware tools), justify their selection, and apply them in a targeted manner to solve the problem, taking into account the framework conditions (e.g., resources, time, risks).
- When developing technical solutions, take into account societal, safety-related, and/or ethical aspects (e.g., security of networked systems, data protection, sustainability, impacts on the workplace and society) and incorporate these aspects into their evaluation and reasoning in a transparent manner.
- Critically interpret the results achieved in the project, reflect on their validity, limitations, and potential sources of error, and derive alternative approaches and further research questions.
- document their work and results in a clearly structured, academically sound short written report (e.g., in the format of a conference paper) using precise technical terminology and appropriate forms of presentation (e.g., mathematical formulations, circuit diagrams and block diagrams, algorithm specifications, and diagrams) in a clear and comprehensible manner.
- Communicate the results of the project in a technically sophisticated presentation to a group (e.g., a specialized audience from academia and, if applicable, industry) in a manner tailored to the audience and easy to understand; answer questions; and respond constructively to technical criticism. In doing so, they can consciously select and use appropriate presentation techniques and media (e.g., slides, demonstrators, simulations, posters).

In addition to the subject-specific qualification goals, students gain in-depth knowledge and experience in the areas of project management (e.g., project planning, milestone definition, risk and resource management, documentation of project progress) as well as self-management and time management in a research and development environment. They are able to independently plan, prioritize, and coordinate tasks within a team, reflect on and adapt work processes, and collaborate constructively with team members. This also includes in-depth knowledge and methods of various presentation and communication techniques, including the written, oral, and visual presentation of technical and scientific content, as well as collaborative work in interdisciplinary teams.

Content
Under the guidance of an instructor, students work on a project topic—agreed upon and defined in consultation with the examiner—using scientific methods and adhering to the rules of good research practice, which addresses a problem within the scope of the master's program or the respective research areas of the supervising department.

Module Grade Calculation
The module is ungraded. The module is passed with successful assessment of the coursework.

Additional Information
The work is conducted on an independent basis at one of the participating institutes. Laboratory introduction is mandatory.

This module can only be credited as Additional Examination (Zusatzleistung).

Workload

1. Attendance time: 15*6 h = 90 h
2. Independent work on the project: 60 h (the distribution between attendance and independent work may vary)
3. Preparation of the report on the project results and the presentation: 60 h

Total: 180 h = 6 LP

M 12.110 Module: Master's Thesis [M-ETIT-107191]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Master's Thesis

Credits	Grading	Recurrence	Duration	Language	Level	Version
30 CP	graded	Each term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114213	Master's Thesis	**WT+ST**	**30 CP**	**Hiller**

Assessment
The Master's Thesis module has 30 credits. It consists of the Master's Thesis and a presentation. The presentation must be carried out within the processing time in accordance with SPO Section §14(4).

Prerequisites
According to SPO Section §14(1) the prerequisite for admission to the Master's Thesis module is that the student has successfully completed module examinations totaling 75 credits.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. You need to have earned at least 75 credits in the following fields:
   - Electives
   - Field of Specialization
   - Interdisciplinary Qualifications

Competence Goal
After successful completion of the module, students are able to work independently on a challenging task in the field of electrical engineering or information technology within a given period of time using scientific methods and in compliance with the rules of good scientific practice under guidance and by applying the theoretical and methodological knowledge acquired in the Master's program. Students are able to do research, analyze and abstract the information and to compile and recognize basic principles and laws from less structured information. Students are able to get an overview of a problem, select complex scientific methods and procedures and use them to find solutions or show further potentials. In principle, this also takes into account social and/or ethical aspects.

The students are able to interpret and evaluate their results. They are also able to document their results in a clearly structured, written elaboration using the appropriate technical terminology. Furthermore, students are able to present and defend their results in front of a scientific audience. They have also deepened their problem-solving skills and their competence in transferring the theoretical and methodological knowledge of electrical engineering and information technology into concrete applications.

In addition to the subject-related qualification goals, the students also gain knowledge and experience in the areas of project management as well as self and time management. This also includes knowledge and methods of various presentation techniques.

Content
The students work independently and with scientific methods on a research topic agreed with the subject examiner, which deals with a problem from the area of the Master's program in Electrical Engineering and Information Technology.

Module Grade Calculation
The Master's Thesis is examined by at least one university professor of the KIT Department of Electrical Engineering and Information Technology or a habilitated member of the KIT Department and another examiner (SPO Section §14(7)). As a rule, one of the examiners is Person who has awarded the work in accordance with SPO Section §14(2). If the assessment of these two people does not match, the examination board determines the grade of the Master's Thesis in the context of the assessment of these two persons; additionally it can order another reviewer The assessment must be completed within eight weeks of the submission of the Master's Thesis respectively. SPO Section §14 regulates further details.

The module grade consists of the Master's Thesis and a presentation (SPO §14(1a)).

Additional Information
The Master's Thesis is written in English. At the student's request, the examiner may authorize the Master's Thesis to be written in German (SPO Section §14(4)).

M 12.111 Module: Measurement Technology [M-ETIT-105982]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules) (Usage until 3/31/2027)

Credits	Grading	Recurrence	Duration	Language	Level	Version
5 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-112147	Measurement Technology	**WT**	**5 CP**	**Heizmann**

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
M-ETIT-102652 - Messtechnik (German version) must not have started.

Competence Goal

- Students have a sound knowledge of the theoretical foundations of measurement technology, including modeling of measurement systems, consideration of nonlinearities, stochastic deviations and stochastic signals, acquisition of analog signals, and frequency and rotational speed measurement.
- Students are proficient in the approaches to measurement system design in terms of model assumptions, methods, and achievable results.
- Students are able to analyze and formally describe measurement technology tasks, synthesize possible solutions for measurement systems and assess the properties of the solution obtained.

Content
The module deals with the formal, methodical and mathematical fundamentals for the analysis and design of measurement systems. Focal points of the course are

- Measurement systems and deviations (including scales, the SI systems, modeling of measurement systems)
- Curve fitting (approximation, interpolation)
- Stationary behavior of measurement systems (characteristic curve, errors of the characteristic curve, nonlinearities, adjustment)
- Stochastic measurement errors (probabilistic analysis, samples, statistical test methods, statistic process control, error propagation)
- Stochastic processes (correlational measurements, spectral description of stochastic signals, system identification, matched filter, Wiener filter)
- Digitization of analog signals (sampling, quantization, analog-digital converters, digital-analog converters)
- Frequency and rotational speed measurement (generalized frequency concept, digital speed measurement, detection of direction)

Module Grade Calculation
The module grade is the grade of the written examination.

Additional Information
The module will be offered for the **last time in WiSe 25/26 as a video recording** with additional speaking lessons for exam preparation. The first examination can be taken for the last time in winter semester 26/27.

MSc MIT (SPO 2015)
The module will be replaced by "Communication Systems and Protocols" (5 LP, summer term) as of summer term 2026. Students who want to register for "Communication Systems and Protocols" before 01.04.2026 have to contact the MIT Study Program Service

MSc ETIT (SPO 2018)
The module can still be used in the Basic or Compulsory Modules of the Fields of Specialization. Alternatively, the module can be skipped.

Workload
The workload includes:

1. attendance in lectures and exercises: 34 h
2. preparation / follow-up of lectures and exercises: 51 h
3. preparation of and attendance in examination: 65 h

total: 150 h = 5 CR

Recommendations
Basic knowledge in the fields of “Probability Theory” as well as “Signals and Systems” is helpful.

M 12.112 Module: Mechatronic Measurement Systems Lab [M-ETIT-107294]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Lab Course)
Field of Specialization / Automation, Robotics, and Systems Engineering (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114341	Mechatronic Measurement Systems Lab	**WT**	**6 CP**	**Heizmann**

Assessment
The success control is carried out in the form of a oral examination of approx. 20 min.

Prerequisites
none

Competence Goal

- Students have in-depth knowledge of different methods for measuring objects, especially surfaces.
- Students master different procedures for the metrological recording of objects and know the relevant requirements, procedures and results.
- Students are able to implement procedures for evaluating sensor data from (surface) measuring devices and to evaluate the quality of the measurement result.

Content
A large number of different measuring methods and systems can be used for the quality inspection of technically manufactured objects and their surfaces. Examples are white light interferometry, confocal microscopy and systems based on focus variation. The measurement methods and systems naturally differ in terms of the physical measurement principle used, but also in terms of the evaluation of the raw sensor data recorded.

In this internship, different systems for the metrological recording of (technical) surfaces are presented and their properties are characterized. In the test dates, the students themselves create procedures and algorithms for processing the sensor data in order to obtain information about the desired geometric and / or optical properties of the examined surface. The algorithms obtained are evaluated on the basis of sensor data from exemplary objects and characterized in terms of the quality of the measurement statements achieved.

Module Grade Calculation
The module grade is the grade of the oral exam.

Additional Information
The submission of the protocols of all experiments is required for admission to the examination. The quality of the protocols is assessed; for the admission to the exam, the quality must be acceptable. Attendance of all laboratory dates including the introductory event is required. Already in the event of a single unexcused absence, admission to the examination will not be granted.

Workload
Total: approx. 170 hours, of which

1. Attendance time in introductory session: 2 hours
2. Preparation of the test dates: 36 h
3. Attendance time in test appointments (8 appointments of 4 hours each): 36 hours
4. Follow-up of the test dates, Creation of the minutes: 36 h
5. Exam preparation and attendance in the same: 60 h

Recommendations
Knowledge in measurement technology as well as basic knowledge of programming (e.g. in Matlab, Python) are helpful.

M 12.113 Module: Medical Image Processing for Guidance and Navigation [M-ETIT-106672]

Coordinators: Prof. Dr.-Ing. Maria Francesca Spadea
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113425	Medical Image Processing for Guidance and Navigation	**WT**	**9 CP**	**Spadea**

Assessment
The examination takes place within the framework of an oral overall examination of approx. 30 minutes about the lecture including a presentation and discussion of the project developed during the course. The overall impression is rated.

Prerequisites
none

Competence Goal

- The students will be able to analyze, structure and formally describe problems in the field of image guided surgery and therapy.
- The students can apply the methods form medical image processing, surgical navigation, augmented reality for surgery and therapy, medical data science.
- The student will be able to communicate in English technical language.
- The students are able to perform calculations and use the necessary tools for this in a methodologically appropriate way.
- The students are able to critically evaluate them

Content

- This module is designed to provide students with the theoretical and practical aspects of image guidance for minimally invasive surgery and therapy
- This module gives an overview about current status of technology in operation rooms (OR) and advanced radiotherapy bunkers
- Furthermore, this module gives knowledge about image process for quantitative information extraction
- Table of contents
    - Introduction to the course: minimally invasive surgery and medical data science
    - Git introduction
    - Image characteristics
    - Basic point, histogram and masked based operations
    - Similarity metrics, projections
    - Planning imaging, Dicom format, pre processing pipeline
    - Case study: planning in radiotherapy
    - Path planning
    - Pixel based image segmentation: manual segmentation, threshold, region growing
    - Convolution based segmentation: edge detection, morphological filters
    - Case study: neurosurgery and tractography
    - Image registration
    - Atlas based segmentation: SABS, MABS, atlas selection
    - Rendering and computer graphics
    - In room imaging technology
    - Reference system, notation and transformation
    - Localizing systems, tracking and calibration
    - Case study: patient monitoring in radiotherapy, adaptive treatments
    - Lab demonstration
    - Point based registration
    - Surface registration
    - Image features and descriptors (example with SIFT SURF)
    - Radiomics Features
    - Deep Learning in image processing
    - The role of deep learning in radiotherapy
    - Augmentet reality

Module Grade Calculation

The module grade is the grade of the oral exam.

A bonus can be earned for submitting homework that will be provided during the lecture time.

The exact criteria for awarding a bonus will be announced at the beginning of the lecture period. If the grade in the oral exam is between 4.0 and 1.3, the bonus improves the grade by 0.3 or 0.4.

Bonus points do not expire and are retained for any examinations taken at a later date.

Additional Information

The course is limited to a number of 30 participants due to capacity reasons. If necessary, a selection procedure will be carried out. Places will be allocated taking into account the students' study program (students of “Biomedical Engineering” specialization will be preferred, students from Computer Science Program and interest in medical applications will be preferred) and academic progress. Details will be announced on the lecture website.

Workload
The workload includes:

1. attendance in lectures and exercises: 15*6 h = 90 h
2. preparation / follow-up: 15*8 h = 120 h
3. preparation of and attendance in examination: 60 h

A total of 270 h = 9 CR

Recommendations

- Basic knowledge in the field of medical imaging;
- Knowledge of basic programming concept;
- Familiarity with Linux environment;
- Basic knowledge of linear algebra (transformations);
- Attitude towards teamwork and code management in Git;
- It is recommended to have access to a personal computer or desktop

Teaching and Learning Methods
Lectures in “Medical Image Processing” (3 SWS), Seminars in “In room imaging modalities” (1 SWS), Tutorials/ Demostrations in Medical image processing and navigation (2 SWS)

M 12.114 Module: Microactuators [M-MACH-100487]

Coordinators: Prof. Dr. Manfred Kohl
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-101910	Microactuators	**ST**	**4 CP**	**Kohl**

Assessment
Written exam: 60 min

Prerequisites
none

Competence Goal
- Knowledge of the actuation principles including pros and cons
- Knowledge of important fabrication technologies
- Explanation of layout and function of the microactuators
- Calculation of important properties (time constants, forces, displacements, etc.)
- Development of a layout based on specifications

Content
- Basic knowledge in the material science of the actuation principles
- Layout and design optimization
- Fabrication technologies
- Selected developments
- Applications

The lecture includes amongst others the following topics:

- Microelectromechnical systems: linear actuators, microrelais, micromotors
- Medical technology and life sciences: Microvalves, micropumps, microfluidic systems
- Microrobotics: Microgrippers, polymer actuators (smart muscle)
- Information technology: Optical switches, mirror systems, read/write heads

Workload
lTime of attendance: 15 * 1,5 h = 22,5 h
Preparation and follow up: 15 * 5,5 h = 82,5 h
Exam Preaparation and Exam: 15 h
Total: 120 h = 4 LP

Literature
- Lecture notes
- D. Jendritza, Technischer Einsatz Neuer Aktoren: Grundlagen, Werkstoffe, Designregeln und Anwendungsbeispiele, Expert-Verlag, 3. Auflage, 2008
- M. Kohl, Shape Memory Microactuators, M. Kohl, Springer-Verlag Berlin, 2004
- N.TR. Nguyen, S.T. Wereley, Fundamentals and applications of Microfluidics, Artech House, Inc. 2002
- H. Zappe, Fundamentals of Micro-Optics, Cambride University Press 2010

M 12.115 Module: Microenergy Technologies [M-MACH-102714]

Coordinators: Prof. Dr. Manfred Kohl
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-105557	Microenergy Technologies	**ST**	**4 CP**	**Xu**

Assessment
Oral examination (approx 30 min.)

Prerequisites
none

Competence Goal
The students can:

- describe the energy conversion principles and exemplify them
- explain the underlying concepts of thermodynamics and materials science
- illustrate the layout, fabrication and function of the treated devices
- calculate important properties (time constants, power output, efficiency, etc.)
- develop a layout based on specifications

Content
- Basic physical principles of energy conversion - Layout and design optimization - Technologies - Selected devices - Applications

The lecture includes amongst others the following topics:

- Micro energy harvesting of vibrations using different conversion principles (piezo, electrostatic, electromagnetic, etc.)
- Thermoelectric energy generation
- Novel thermal energy conversion principles (thermomagnetic, pyroelectric)
- Miniature scale solar devices
- RF energy harvesting
- Miniature scale heat pumping
- Solid-state cooling technologies (magneto-, electro-, mechanocalorics)
- Power management
- Energy storage technologies (microbatteries, supercapacito4rs, fuel cells)

Module Grade Calculation
Module grade calculation: The module grade is the grade of the written exam.

Workload
Time of attendance: 15 * 1,5 h = 22,5 h
Preparation and follow up: 15 * 5,5 h = 82,5 h
Exam Preaparation and Exam: 15 h
Total: 120 h = 4 LP

Literature

- Lecture notes (overhead transparencies) „Micro Energy Technologies“
- Stephen Beeby, Neil White, Energy Harvesting for Autonomous Systems, Artech House, 2010
- Shashank Priya, Daniel J. Inman, Energy Harvesting Technologies, Springer, 2009

M 12.116 Module: Microsystem Technology [M-ETIT-100454]

Coordinators: Prof. Dr. Wilhelm Stork
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100752	Microsystem Technology	**WT**	**3 CP**	**Stork**

Assessment
Achievement will be examined in an oral examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
The students

- Know the most important terms and processes in microsystems technology and can assess their advantages and disadvantages.
- Are able to describe the common methods and tools.
- Can select suitable processes for the production of microsystems.
- Have a broad understanding of the structure and functionality of microsystem sensors.
- Have the ability to communicate with experts in microtechnology.
- Are able to critically evaluate various microsystems technology processes.

Content
The methods of microstructure technology from lithography and etching techniques to ultra-precise machining processes are explained and their applications, especially in micromechanics and microoptics, are presented.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
1. attendance time in lectures, exercises: 30 h
2. preparation and follow-up of the same: 30 h
3. exam preparation and attendance: 30 h

M 12.117 Module: Microwave Engineering Lab [M-ETIT-106973]

Coordinators: Prof. Dr.-Ing. Thomas Zwick
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Lab Course)
Field of Specialization / Information and Communication Technology (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113938	Microwave Engineering Lab	**WT+ST**	**6 CP**	**Zwick**

Assessment
In preparation for the laboratory experiments, each laboratory group must work together on a number of tasks as homework before the experiment and submit a single copy to the supervisor immediately before the start of the experiment. The tasks for the experiment itself are completed and recorded during the experiment. The protocol should be handed in to the supervisor immediately after the experiment. Before each experiment is carried out, there is a written or oral examination (approx. 20 minutes, no aids) on the content of the experiment.

Prerequisites
none

Competence Goal
Students will have in-depth knowledge of high-frequency components and systems as well as the functionality of the most important high-frequency measuring devices (network analyzer, spectrum analyzer, noise measurement, power measurement, oscilloscope, antenna measurement). They are also familiar with the use of high-frequency measuring devices and components. They are able to independently select and operate measuring devices based on specific applications and interpret the measurement results. Furthermore, they are able to work together in a self-organized team.

Content
Under the motto: “Practical relevance through state-of-the-art equipment and current problems”, students are offered a modern and technically sophisticated high-frequency laboratory at Master's level. The aim of the experiments is to deepen the theory taught in the lectures in a practical way and to train the handling of high-frequency measuring devices and HF components. In groups of 2-4 students, various experiments are carried out and recorded over 8 afternoons. The order and topics of the experiments may vary.

Module Grade Calculation
The grade for the experiments is made up of the preparation, the protocol and the written or oral assessment of the learning objectives for each experiment. The final grade for the entire laboratory results from the overall impression of the performance. Students who appear unprepared for the respective experiment may not take part in the experiment. The experiment must be repeated at another time.

Workload
The workload includes:

- attendance study time laboratory: 45 h
- test preparation protocols, test preparation: 135 h

A total of 180 h = 6 LP

Recommendations
Knowledge of microwave measurement technology and RF components and systems is helpful.

M 12.118 Module: Microwave Module Design [M-ETIT-105701]

Coordinators: Prof. Dr.-Ing. Thomas Zwick
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111375	Microwave Module Design	**WT**	**3 CP**	**Geist, Zwick**

Assessment
The success control is carried out as part of an oral overall examination (approx. 20 minutes) of the selected courses, which is in total meet the minimum requirement for LP.

Prerequisites
none

Competence Goal
After successful completion of the module, students will be able to design complex microwave modules. This includes in particular complete transmit and receive modules consisting of amplifiers, mixers, filters, signal generation, etc. They have a deep understanding of the technological and circuit aspects as well as for embedding in the overall system. The students are able to understand and describe the operation of the individual components and the systems. They are able to transfer this knowledge to other areas of high-frequency technology and thus to analyze and solve high-frequency technical problems. They are able to apply what they have learned in a practical manner.

Content
Applied course on microwave module design: lines on substrates, connectors, limiters, PIN switches, SIW components and filter design, amplifiers, frequency response compensation, phase noise, signal generation, planar mixers, and receiver noise figure.

In addition, during the lecture examples of microwave module design will be implemented and discussed in detail in a state-of-the-art software environment.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
The workload includes:

Attendance study time lecture: 30 h

Self-study time including exam preparation: 60 h

A total of 90 h = 3 LP

Recommendations
Knowledge of the basics of radio frequency technology and communications technology is helpful.

M 12.119 Module: Microwaves Measurement Techniques [M-ETIT-107007]

Coordinators: Dr.-Ing. Mario Pauli
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114046	Microwaves Measurement Techniques	**ST**	**4 CP**	**Zwick**

Assessment
Success control takes place in the form of an oral examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
The students have an in-depth knowledge of the structure and functioning of microwave measuring devices (signal generator, power measurement, frequency measurement, spectral analyzer, network analyzer). They understand the special features of measuring power, frequency and scattering parameters in the microwave range. They can apply the knowledge they have learned in practice and interpret the measurement results. They can analyze and assess possible sources of error in the measurement. They are able to design measurement setups and to carry out measurements correctly.

Content
This lecture contains all basic areas of today's high-frequency measurement techniques, such as power measurement, frequency measurement, spectral analysis and network analysis. Particular attention is paid to the description of those measurement systems and methods that are used in modern applications.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
The workload includes:

Attendance study time lecture / exercise: 45 h

Self-study time including exam preparation: 75 h

A total of 120 h = 4 CR

Recommendations
Knowledge of the basics of high frequency technology is helpful.

M 12.120 Module: Mixed-Signal IC Design [M-ETIT-105893]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111845	Mixed-Signal IC Design	**ST**	**3 CP**	**Ulusoy**

Assessment
Success control takes place in form of an oral examination (approx. 30 min.)

Prerequisites
none

Competence Goal
- The students acquire the competencies to mixed-signal advanced microelectronics integrated circuits.

- They have a good understanding of circuit design with linear circuits and "switched-capacitor" circuit techniques.

- They can design a sample-and-hold and track-and-hold circuits and discuss how it can improve an A/D converter's performance.

- They can design an A/D or D/A converter to a given performance specification, choosing an overall architecture, number of stages and internal precision.

- They can design phase lock loop (PLL) circuits, including design details and benefits and disadvantages of each type.

- They are familiar with time-to-digital converters and applications.

- They are familiar with the design of low-power circuits.

- They are able to develop test procedures, test structures and test patterns (ATPG - Automatic Test Pattern Generation) for ASICs.

- They have the basic understanding of the printed circuit board design practices, die-attached and high-density interconnection technology in order to connect the final ASIC to other chips and measurement equipment.

Content
This course covers fundamentals of data converters, Nyquist-rate converters, discrete-time signal processing, central concept of oversampling and noise-shaping, and delta-sigma modulators, phasedlocked loops, assembly and testing procedures of such mixed-signal ICs. Intended for engineers working with digital and analog signals, seeking to learn more about mixed-signal (analog plus digital) circuit design, analysis, and application.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
Each credit point corresponds to an approximately 25-30h of workload in average. Based on this, the amount of work for this lecture is calculated as follows:
1. Attendance to the lectures (15*2=30h)
3. Preparation to the lectures (15*2=30h)
4. Preparation to the oral exam (25h)
Total: 85h

Recommendations
Basic knowledge on analog and digital circuits are recommended.

Literature

1. CMOS Analog Integrated Circuits; Razavi; McGraw-Hill Education
2. Principles of Data Conversion System Design; Razavi; Wiley-IEEE Press
3. Time-to-Digital Converters; Stephan Henzler; Springer Series in Advanced Microelectronics
4. VLSI Technology; Sze; McGraw-Hill

M 12.121 Module: MMIC Design Laboratory [M-ETIT-105464]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Lab Course)
Field of Specialization / Information and Communication Technology (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111006	MMIC Design Laboratory	**WT+ST**	**6 CP**	**Ulusoy**

Assessment
The written report (approx. 30-40 slides) and the oral presentation (approx. 40 min.) are used to mark the course. The overall impression is assessed.

Prerequisites
none

Competence Goal
The students have a comprehensive understanding on the design of monolithic microwave integrated circuits.

The students are able to deduce specifications of individual building blocks in a microwave system and are able to connect these with system level considerations.

They are familiar with various IC fabrication technologies, and are able to identify pros and cons of the various state of the art technologies that are available today.

The students are able to perform the design of a complete microwave sub-system from conception to schematic level design and layout design, and are able to apply high-level design verification methods.

The students can apply their theoretical knowledge on RF engineering using modern design tools.

Content
In this laboratory course, the students will be assigned an RF system and will propose a hardware solution that will meet the requirements of the assigned RF system. The students will then perform schematic level design and system-level simulations of the proposed hardware. The laboratory course will be finalized with a layout implementation and verification of the proposed hardware. The students will learn to use state of the art CAD tools for system level simulations, schematic design, electromagnetic simulations, and layout design and verification in modern IC process technologies. Each RF sub-system will be developed by a group of 3-4 students.

Module Grade Calculation
The written report and the oral presentation are used to mark the course. The overall impression is assessed.

Workload
Each credit point corresponds approximately to 30h of the student's workload. Here, the average student is expected to reach an average performance. This contains:

1. Attendance to the laboratory tutorials (10*(3)=30h)
2. Preparation to the laboratory tutorials (10*(2)=20h)
3. Implementation of assigned design tasks after each tutorial (10*(8)=80h)
4. Preparation of report and oral presentation (20h)

Total: 150h

Recommendations
Radio-Frequency Integrated Circuits and Systems, Modern Radio Systems Engineering, Microwave Engineering, Electromagnetics and Numerical Calculation of Fields

M 12.122 Module: Mobile Communications [M-ETIT-105971]

Coordinators: Prof. Dr.-Ing. Peter Rost
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-112127	Mobile Communications	**WT**	**4 CP**	**Rost**

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Prerequisites
none

Competence Goal
Students are enabled to analyze and assess functionalities of mobile communication systems. They learn how to apply and implement fundamental methods of the lecture “Communications Engineering I” in mobile radio networks. Furthermore, students will be enabled to understand requirements and limitations of mobile applications.

Content
At the beginning, this course describes exemplary applications of mobile communications and elaborates on resulting requirements. Based on a solid understanding of those requirements, selected approaches and techniques will be presented that are solving the respective challenges in mobile communication systems. To this end, algorithms as well as system architectures are discussed in order to acquire solid knowledge on the radio network, the core network and the integration with applications and services.

Module Grade Calculation
Grade of the module corresponds to the grade of the oral exam.

Workload

1. Attendance time in lectures: 15 * 2 h = 30 h
2. Preparation and follow-up of lectures: 15 * 2 h = 30 h
3. Attendance time in excercises: 15 * 1 h = 15 h
4. Preparation and follow-up of excercises: 15 * 1 h = 15 h
5. Preparation for the oral exam: 30 h

In total: 120 h = 4 LP

Recommendations
Knowledge of basic engineering as well as basic knowledge of communications engineering and Previous attendance of the lecture "Communication Engineering I" is recommended. Sound English language skills are required.

M 12.123 Module: Mobile Communications II [M-ETIT-106244]

Coordinators: Prof. Dr.-Ing. Peter Rost
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-112679	Mobile Communications II	**ST**	**3 CP**	**Rost**

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Prerequisites
none

Competence Goal
Students are able to analyze and assess functionalities of mobile communication systems. They know how to apply and implement fundamental methods of the lecture “Communications Engineering” in mobile radio networks. Furthermore, students are able to understand requirements and limitations of mobile applications.

This lecture complements the contents of the lecture "Mobile Communications", which mainly deals with aspects of communications access networks. Building on this, the focus of this lecture is on mobile communication architectures, core networks, and specific application scenarios and relevant technologies.

Content
The subject of the lecture is to first introduce a basic mobile communication system architecture including core network and the integration into applications. Based on this, specific core network functions are explained in detail, e.g. user administration, security, quality of service. Finally, specific applications are introduced and it is explained how mobile communication services are integrated in, e.g. industrial networks, connected cars, wide-area IoT applications.

Module Grade Calculation
Grade of the module corresponds to the grade of the oral exam.

Workload

1. Attendance to the lecture: 15 * 2 h = 30 h
2. Preparation and review: 15 * 2 h = 30 h
3. Preparation for the exam: included in preparation and review = 30 h

In total: 90 h = 3 LP

Recommendations
Basic knowledge of communications engineering. For this purpose previous attendance of the modules "M-ETIT-102103 - Communication Engineering I" and “M-ETIT-105971 - Mobile Communications” is strongly recommended.

M 12.124 Module: Modeling Physiological Systems [M-ETIT-106782]

Coordinators: Dr.-Ing. Axel Loewe
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113630	Modeling Physiological Systems	**ST**	**6 CP**	**Loewe**
T-ETIT-114690	Modeling Physiological Systems - Workshop	**ST**	**0 CP**	**Loewe**

Assessment

- The examination takes place in form of a written examination lasting 90 min.
- The submission of the ungraded workshop (2 exercise sheets) tasks before the exam is mandatory.

Prerequisites
none

Competence Goal
The students will be able to

- Describe physiological functional principles of selected organs
- Formalize physiological relationships using engineering methods (e.g. mathematical equations, standardized diagram forms, etc.)
- Implement these models with adequate numerical schemes
- Apply formalized models to develop a deeper understanding of physiological relationships; e.g. by means of simulation studies
- Describe pathomechanisms of selected diseases
- Characterize selected pathologies qualitatively and quantitatively by using physiological models

Content
The module provides knowledge and methods for modeling physiological processes and pathomechanisms. Physiological functional principles are described using the example of 2-3 organ systems and then implemented in mathematical-technical models. The model types of ordinary differential equations, electrical equivalent circuits and control loops are taken up and deepened in practical tasks. The course is deepened both fundamentally by working on theoretical tasks with pen and paper as well as through programming and simulation studies.

At least one clinical disease entity is introduced for each example organ system and examined using modeling and simulation.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
Attendance in lectures and exercises: 22*2h = 44h

Preparation / follow-up: 22*2h = 44h

Preparation of and attendance in examination: 42h

Programming exercises25h+25h = 50h

A total of 180 h = 6 CR

Recommendations
Basic knowledge of

- ordinary differential equations and linear algebra (e.g. Höhere Marthematik I, Höhere Mathematik III),
- system dynamics and control engineering (e.g. Signale und Systeme)
- programming in a scripting language like Python or Matlab (e.g. Informationstechnik I)
- version control using git and GitLab (e.g. Practical Introduction to Research Software Engineering)
- human anatomy & physiology (e.g. Physiologie und Anatomie für die Medizintechnik)

M 12.125 Module: Modelling and Simulation of Electrochemical Systems [M-ETIT-100508]

Coordinators: Dr.-Ing. Andre Weber
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	1	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100781	Modelling and Simulation of Electrochemical Systems	**ST**	**3 CP**	**Weber**

Assessment
Success control takes place in the form of an oral examination lasting approx. 20 minutes.

Prerequisites
none

Competence Goal
The students know models on different scales (elementary kinetics to system model) for the description of electro-chemical systems and are able to use them in the development of batteries and fuel cells.

Content
The modeling of electrochemical systems is a multiscale problem. While the charge transfer at the electrode/electrolyte interface takes place on an atomic scale, system modeling requires highly simplified submodels for the system components that allow real-time simulation of system operation. In the lecture, current electrochemical models for batteries and fuel cells are presented at the various levels, the experimental determination of the model parameters is discussed and examples of model validation are shown.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
1. lecture attendance time: 15 * 2 h = 30 h
2. Preparation and follow-up time for lecture: 15 * 2 h = 30 h
3. Exam preparation and attendance in the same: 30 h

Total: 90 h = 3 CP

Recommendations
The contents of the lecture "M-ETIT-107005 - Batteries, Fuel Cells, and Electrolysis" are assumed to be known. Students who have not (yet) heard this lecture are recommended to work through the lecture notes for this lecture in advance.

M 12.126 Module: Modern Radio Systems Engineering [M-ETIT-100427]

Coordinators: Prof. Dr.-Ing. Thomas Zwick
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Information and Communication Technology (Fundamentals)
Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100735	Modern Radio Systems Engineering	WT+ST	6 CP	Zwick

Assessment
The success control takes place in the form of an oral examination of approx. 20 minutes.

Prerequisites
none

Competence Goal
After attending this course, students will be able to design an analog front end for a radio transmission system at the block diagram level. In particular, the non-idealities of typical components of high-frequency technology and their effects on the overall system performance are part of the knowledge imparted. The students also have an in-depth understanding of wave propagation.

Content
The course gives a general overview of radio transmission systems and their components including the radio channel and wave propagation. A brief repetition of microwave basics is also included. The focus is on the system components realized in analog technology and their non-idealities. Based on the physical functioning of the various system components, parameters are derived that allow an examination of their influence on the overall system performance.

The exercise is closely linked to the lecture and mainly consists of computer-based exercises that allow a visualization of the influences of various non-idealities on the overall system performance and demonstrate the practical system design of modern radio transmission systems.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
The workload includes:

- Attendance study time lecture: 45 h
- Attendance study time computer exercise SystemVue ESL Design Software / MATLAB: 15 h
- Self-study time including exam preparation: 120 h

A total of 180 h

Recommendations
Knowledge of the basics of radio frequency technology and communications technology is recommended.

M 12.127 Module: Modern VLSI Technologies [M-ETIT-106921]

Coordinators: Prof. Dr. Jasmin Aghassi-Hagmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113864	Modern VLSI Technologies	**ST**	**6 CP**	**Aghassi-Hagmann**
T-ETIT-114743	Modern VLSI Technologies Lab	**ST**	**0 CP**	**Aghassi-Hagmann**

Assessment

- Success control takes place in form of an oral examination with a duration of approx. 20 minutes.
- 50% of approx. 10 ungraded lab exercises have to be sucessfully completed before the oral examination is taken.

Prerequisites
none

Competence Goal

- The students will gain distinct knowledge in the field of modern CMOS technologies (FinFETs, High-k Gate-Stacks, Below 20nm nodes, Nanosheets).
- They will gain a good understanding of device physics and how to apply the most important design rules to design physical layouts of components and simple circuits.
- The students will be able to compare the functionality (current, performance, noise) of electrically characterized components with simulated devices, while assessing advantages and disadvantages.
- Students will learn to compare different technologies and to perform technology assessments based on benchmark circuit analyses. Understanding of critical paths in circuits for power-performance assessments.

Content
The lecture introduces CMOS technology with the latest technological innovations (high-k materials, gate-last processes, stress engineering, FinFETs, Gate all around FETs, nanosheets, etc.). A detailed understanding of interactions between novel materials, device architectures, and the functionality of basic components will be studied. In addition to physical and circuit properties (variations, self-heating, noise, performance), so-called layout effects, which play a crucial role in advanced CMOS are introduced. Special emphasis is put on the respresentation of technologies in design systems (electronic design automation) as well as SPICE simulations according to the BSIM (Berkeley Simulation Transistor Models) and PSP (Advanced Surface-Potential-Based MOSFET Model) standards. In addition, the use of industrial software (PDKs) for electrical simulation and circuit design will be introduced and practiced in the Excercises. Finally, highly integrated low power systems and their special requirements, wiring concepts and variation modeling are explained.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
The workload includes:

1. Attendance time in laboratory (15h*2=30h)
2. Attendance time in lectures (15h*2=30h)
3. Preparation/follow-up, lecture and exercises (15h*(2+2)=60h)
4. Preparation, written exercises and oral exam (60h)

Total: 180h

Recommendations
Previous knowledge from the lectures "Festkörperelektronik und Bauelemente" and "Elektronische Schaltungen" is recommended.

M 12.128 Module: Multimodal Artificial Intelligence [M-INFO-107676]

Coordinators: Prof. Dr. Jan Niehues
Prof. Dr.-Ing. Rainer Stiefelhagen
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-115041	Multimodal Artificial Intelligence	**ST**	**6 CP**	**Niehues, Stiefelhagen**

Assessment
See Partial Achievements (Teilleistung).

Prerequisites
See Partial Achievements (Teilleistung).

Competence Goal

- The students know the relevant elements of a multimodal AI .
- The students understand the algorithms and methods of multimodal AI.
- The students are able to understand the different sub-components to develop and analyze a system .
- The students can transfer this knowledge to new applications, as well as analyze and compare different methods.

Content
Due to the successes in research, multimodal AI systems are increasingly integrated into our everyday lives. These are, for example, systems that can understand and generate language and speech or analyze images and videos. Furthermore, advanced video and image generation AI shows leads to new applications

Based on the knowledge of the lecture “Introduction to AI”, the students learn to understand, develop and evaluate multimodal AI Systems.
In order to bring this knowledge closer to the students, the lecture is divided into 5 parts. First, the lecture investigates method of speech and language perception followed by a part about visual perception. The second part deals speech and language generation followed by a part about video processing. Finally, multimodal AI methods are presented.

Workload
Lecture with 3 SWS + 1 SWS exercise , 6 CP.
6 LP corresponds to approx. 180 hours, of which
approx. 45 hours lecture attendance
approx. 15 hours exercise visit
approx. 90 hours post-processing and processing of the exercise sheets (self-study)
approx. 30 hours exam preparation (self-study)

Recommendations
See Partial Achievements (Teilleistung).

M 12.129 Module: Multivariable Control Systems [M-ETIT-107515]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114727	Multivariable Control Systems		**6 CP**	**Hohmann**

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal

- The students are able to describe linear multivariable Systems in time and frequency domain with as well time continous as time discrete models.
- In particular the students are familiar with the state space representation and its transformations in canonical forms.
- The students are able to analyze multivariable system with respect to stability, trajectory plots in state space, controllability, observability and zeros and poles.
- They master the control of linear multivariable systems in frequency domain by series decoupling with frequency control.
- The students are familiar with the basic control structures in time domain (pole placement with preliminary filter).
- The students are able to use different methods for direct pole placement in state space via state feedback like the Ackermann formula, modal synthesis and the complete modal synthesis.
- They are able to synthesize controllers for the decoupling control of multivariable systems in state space.
- The students are able to design time discrete controllers, in particular dead-beat control.
- They are familiar with the problem of determining non measurable state variables using complete and reduced state observers.
- As an extension, they are able to solve a generalized estimation problem in state space including noise by the design of Kalman filters.
- They can apply state feedback methods for the elimination of permanent deterministic disturbances like disturbance feedforward control, use of disturbance models or PI-State space controllers.
- The students can apply complete modal synthesis to design output feedback controllers in state space.
- They are able to extend the state feedback control with dynamical elements in order to create dynamical controllers.

Content
The module is designed to provide students with basic knowledge and skills to analyze linear multivariable dynamic systems (described both in continuous and discrete time) and to design respective controllers. Furthermore the students are enabled to handle poor sensor information by state estimation methods or to eliminate disturbances and uncertainties, for example by dynamic state feedback.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
The workload includes (4 SWS):

1. attendance in lectures and exercises: 15*4 h = 60 h
2. preparation / follow-up: 15*4 h = 60 h
3. preparation of and attendance in examination: 60 h

A total of 180 h = 6 CR

Recommendations
Basic knowledge in the field of control theory is required (for example the contents of the bachelor module M-ETIT-106339 “Mess- und Regelungstechnik”).

Teaching and Learning Methods
lecture (3 SWS), exercise (1 SWS)

M 12.130 Module: Nano- and Quantum Electronics [M-ETIT-105604]

Coordinators: Prof. Dr. Sebastian Kempf
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111232	Nano- and Quantum Electronics	**ST**	**6 CP**	**Kempf**

Assessment
The assessment of success takes place in the form of a written examination lasting 120min. The grade corresponds to the result of the written examination.

Prerequisites
none

Competence Goal
Students will understand the physical limits of CMOS scaling and will be able to analyze the function of conventional nanoelectronic devices. Students will also understand the operation of novel nanoelectronic and quantum electronic devices and will be able to design this kind of devices that are based on quantum mechanical effects. They develop the ability to design nanoelectronic sensors and devices and can understand and analyze the fabrication methods for nano- and quantum electronic devices.

Content
Nanoelectronics deals with integrated circuits whose typical length scale is well below 100nm. In this regime, physical effects, in particular of quantum mechanical origin, occur and strongly influence the scaling of classical microelectronic devices. This ultimately leads to a new form of electronic components as well as novel operation principles. A special form of nanoelectronics is quantum electronics in which quantum mechanical effects are exploited on purpose to build an entirely new class of devices whose performance reaches far beyond any other microelectronics devices. Well-known examples are superconducting digital electronics which enables to build, for example, microprocessors with clock rates exceeding several 100GHz, or the quantum computer, which will lead to a change of paradigms in the field of information processing.

Within this context, the module "Nano- and quantum electronics" intends to give students an overview of the theoretical and practical aspects of nano- and quantum electronics. In particular, it discusses the following topics:

- Limitations of conventional CMOS technology
- Quantum mechanical effects in the field of nano- and quantum electronics (quantized conductance, Coulomb blockade, tunnel effect, etc.)
- Hot-electron effect
- Nano- and quantum-technological manufacturing and analysis methods
- Nanostructure field-effect transistors
- Quantum dots
- Carbon nanotube field-effect transistor
- Resonant tunnel diodes
- Unipolar resonant tunnel transistor
- Single Electron Transistor (SET)
- Josephson junction based analog and digital electronics
- Quantum bits, quantum computers and quantum computing

The tutorial is closely linked to the lecture and deals with special aspects concerning the development of nano- and quantum electronics. In particular, the development and system integration of such devices for various applications is discussed by means of exercises.

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
A workload of approx. 175h is required for the successful completion of the module. This is composed as follows:

- Attendance time in lectures and exercises: 18*1.5h + 6*1.5h = 36h
- Preparation and follow-up of lectures: 21*3h= 54h
- Preparation and follow-up of tutorials: 7*5h= 35h
- Preparation for the exam: 50h

Recommendations
Successful completion of the modules "Superconductivity for Engineers" and „Einführung in die Quantentheorie für Elektrotechniker“ is recommended.

M 12.131 Module: Navigation and Localization Techniques [M-ETIT-105881]

Coordinators: Prof. Dr.-Ing. Thomas Zwick
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111829	Navigation and Localization Techniques	ST	3 CP	Zwick

Assessment
The success control is carried out as part of an oral overall examination (approx. 20 minutes) of the selected courses, which in total meet the minimum requirement for LP.

Prerequisites
none

Competence Goal
The students are able to understand navigation-related problems and develop appropriate solutions. They have a solid understanding of navigation and positioning principles, corresponding location-dependent measurements, parameter and position estimation as well as position tracking algorithms. With suitable performance measures, they can compare different navigation and localization solutions. For example, the students have sound knowledge on GNSS receivers, IMUs, sensor fusion, and radio navigation for aerospace and land-based applications. In case they face advanced problems, they have ideas how to approach them.

Content
Today navigation systems are an integral part of our daily live. Most of us use a global navigation satellite system (GNSS) receiver integrated in our smartphones to find our way in cities and outdoors. Often, these systems are augmented by localization information from radio systems such as WLAN access points, correction information for assisted GNSS (A-GNSS) or inertial measurement units (IMUs) and magnetometers. In the near future navigation technologies become a key enabler for more efficient, safe, and environmentally friendly automated transport, be it autonomous driving cars and trains or automated landing of airplanes and space vehicles. Therefore, this lecture addresses the following topics:

The first chapter will introduce navigation systems, define common terms, and provide an outline of the lecture as well as organizational details.

The second chapter will discuss navigation and positioning principles based on dead reckoning, landmarks, and electromagnetic waves.

Subsequently, Chapter 3 will present first the estimation problem and lower bounds for estimating location dependent parameters and then suitable estimation methods for location dependent parameters, e.g. received power, time of flight, and direction of arrival of radio signals.

Building on Chapter 3, Chapter 4 will explain snapshot position estimation algorithms including triangulation, trilateration, multilateration, and signature matching. To enable comparisons between different methods, performance bounds and measures are discussed.

Chapter 5 will address the continuous position tracking first with a performance bound and second with different tracking filters such as the Kalman filter and particle filter.

Being familiar with navigation and positioning principles, estimation of location dependent parameters and positions as well as position tracking, Chapter 6 elaborates the architecture and functionality of GNSSs, GNSS receivers, and augmentation systems.

In Chapter 7, inertial navigation and the sensor fusion between IMU and GNSS receiver data is discussed including loosely and tightly coupled approaches.

Particular navigation aspects and systems in the aerospace domain are explained in Chapter 8 whereas Chapter 9 explains navigation and localization for railways and road transport as well as indoors.

Finally, Chapter 10 introduces some advanced research topics in navigation and how to approach them. For instance, multipath and non-line-of-sight propagation, mitigation, and exploitation or cooperative and swarm navigation may be discussed. The topics may be adapted based on current research work and discussions throughout the course.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
Each credit point corresponds to approximately 25-30 hours of work (of the student). This is based on the average student who achieves an average performance. The workload includes:

Attendance study time lecture: 30 h

Self-study time including exam preparation: 60 h

A total of 90 h = 3 LP

Recommendations
Basic knowledge of linear algebra, stochastic, radio frequency technology, and communications technology is helpful.

M 12.132 Module: NMR Methods for Product and Process Analysis [M-CIWVT-105890]

Coordinators: apl. Prof. Dr. Gisela Guthausen
Organisation: KIT Department of Chemical and Process Engineering
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-111843	NMR Methods for Product and Process Analysis	**WT**	**4 CP**	**Guthausen**

Prerequisites
None

Competence Goal
Students will be able to

- explain the physical principles of nuclear magnetic resonance (NMR) and describe the underlying phenomena.
- identify various areas of application for NMR and assess its usefulness in the fields of bioengineering/chemical engineering & process engineering.
- understand and critically evaluate the functioning and usefulness of the NMR method using concrete examples.

Content
An overview of applications of nuclear magnetic resonance (NMR) will be given together with the basic description of this analytical tool. In the focus of the lectures are typical applications of NMR in chemical and bio engineering. The understanding of this versatile analytical method will be developed on the basis of dedicated examples.

Module Grade Calculation
The module grade ist the grade of the oral examination.

Additional Information
If required, the module can be offered in English.

Workload

- Attendance time (Lecture): 30 h
- Revision course: 30 h
- Exam Preparation: 60 h

Literature
Textbooks: Kimmich and Callaghan; further reading will be announced during the lectures.

M 12.133 Module: Nonlinear Control Systems [M-ETIT-107644]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114989	Nonlinear Control Systems	**ST**	**6 CP**	**Kluwe**

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal

- The students are able to describe nonlinear systems in time domain, in particular they are familiar with the state space representation.
- The students are able to analyze the behavior of nonlinear systems on the basis of the respective trajectory plots in state space.
- They master the stability analysis of equilibrium points with the fundamental Lyapunov theory and its extensions.
- They are familiar with the passivity concept in nonlinear systems.
- They are as well familiar with dissipativity as the generalization of passivity.
- They are able to design controllers with control Lyapunov functions.
- They master the controller design with the back-stepping principle.
- They are able to design sliding-mode control in nonlinear systems.
- They are able to synthesize passivity-based controllers.
- They are able to use design compensation control methods with feedback linearization on the basis of relative degree.
- They can design controllers based on the concept of flatness.

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
The workload includes (4 SWS):

1. attendance in lectures and exercises: 15*4 h = 60 h
2. preparation / follow-up: 15*4 h = 60 h
3. preparation of and attendance in examination: 60 h

A total of 180 h

Recommendations
Basic and advanced knowledge knowledge in the field of control theory is recommended (for example the contents of the bachelor module M-ETIT-106339 “Mess- und Regelungstechnik” and the master module M-ETIT-107515 “Multivariable Control Systems”)

Teaching and Learning Methods
Lecture (3 SWS) + exercises (1 SWS)

M 12.134 Module: Nonlinear Optics [M-ETIT-100430]

Coordinators: Prof. Dr.-Ing. Christian Koos
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-101906	Nonlinear Optics	**ST**	**6 CP**	**Koos**

Assessment
Type of Examination: oral exam

Duration of Examination: approx. 30 Minutes

Modality of Exam: The oral exam is offered continuously upon individual appointment.

Prerequisites
none

Competence Goal
The students

- understand and can mathematically describe the effect of basic nonlinear-optical phenomena using optical susceptibility tensors,
- understand and can mathematically describe wave propagation in nonlinear anisotropic materials,
- have an overview and can quantitatively describe common second-order nonlinear effects comprising the electro-optic effect, second-harmonic generation, sum- and difference frequency generation, parametric amplification and optical rectification,
- have an overview and can quantitatively describe the Kerr effect and other common third-order nonlinear effects, comprising self- and cross-phase modulation, four-wave mixing, self-focussing, and third-harmonic generation,
- have an overview and can describe nonlinear-optical interaction in active devices such as semiconductor optical amplifiers
- conceive the basic principles of various phase-matching techniques and can apply them to practical design problems,
- conceive the basic principles electro-optic modulators, can apply them to practical design problems, and have an overview on state-of-the art devices,
- conceive the basic principles third-order nonlinear signal processing and can apply them to practical design problems.

Content

1. The nonlinear optical susceptibility: Maxwell´s equations and constitutive relations, relation between electric field and polarization, formal definition and properties of the nonlinear optical susceptibility tensor,
2. Wave propagation in nonlinear anisotropic materials
3. Second-order nonlinear effects and devices: Linear electro-optic effect / Pockels effect, second-harmonic generation, sum- and difference-frequency generation, phase matching, parametric amplification, optical rectification
4. Third-order nonlinear effects and devices: Nonlinear refractive index and Kerr effect, self- and cross-phase modulation, four-wave mixing, self-focussing, third-harmonic generation
5. Nonlinear effects in active optical devices

Module Grade Calculation
The module grade is the grade of the oral exam.

There is a bonus system based on the problem sets that are solved during the tutorials: During the term, 3 problem sets will be collected in the tutorial and graded without prior announcement. If for each of these sets more than 70% of the problems have been solved correctly, a bonus of 0.3 grades will be granted on the final mark of the oral exam.

Workload
Approx. 180 h – 30 h lectures, 30 h exercises, 120 h homework and self-studies

Recommendations
Solid mathematical and physical background, basic knowledge in optics and photonics.

Literature
R. Boyd. Nonlinear Optics. Academic Press, New York, 1992.
E.H. Li S. Chiang Y. Guo, C.K. Kao. Nonlinear Photonics. Springer Verlag, 2002
G. Agrawal, Nonlinear Fiber Optics, Academic Press, San Diego, 1995.

M 12.135 Module: Numerical Methods for Partial Differential Equations [M-ETIT-102311]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-104595	Numerical Methods for Partial Differential Equations	**ST**	**4 CP**	**Hohmann**

Assessment
Success control takes place ni form of an overall oral examination (approx. 20 minutes).

Prerequisites
none

Competence Goal

- Students are familiar with the concepts and structures of partial differential equations as well as the basic methods and algorithms for their numerical treatment.
- They are familiar with all aspects from modeling and the development of numerical methods to algorithmic implementation and concrete programming, e.g. in MATLAB.
- Students are proficient in the application of computer-aided calculation methods to practical tasks.
- They are able to derive and practically implement a discretization of a partial differential equation and to assess and numerically check the convergence behaviour.

Content

- Examples of partial differential equations from the natural sciences
- Dirichlet boundary value problem for the Poisson equation
- Wave equation
- Heat conduction equation
- Basic concepts of functional analysis
- Separation of variables for some elementary partial differential equations
- Numerical solution methods -- Finite elements
  - Variational methods
  - Finite element method
  - Error estimation
  - Realization of finite element methods
- - Numerical methods in electrodynamics
  - Maxwell equations, modeling
  - Consideration in the frequency domain, eigenvalue problems
  - Finite elements for Maxwell's equations
  - Error estimation

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
The workload includes

1. Attendance time in lecture/exercise (2+1 SWS: 45h1.75 LP)
2. Preparation/follow-up of lecture/exercise (60h2 CP)
3. Preparation/attendance time for oral examination (7.5h0.25 CP)

Recommendations
Knowledge of the following modules is recommended:

Higher Mathematics I-III in the Bachelor

"M-MATH-106972 - Numerical Methods with Programming Practice" or "M-MATH-105831 Numerical Methods"

M 12.136 Module: Numerical Methods with Programming Practice [M-MATH-106972]

Coordinators: Prof. Dr. Wolfgang Reichel
Organisation: KIT Department of Mathematics
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Fundamentals)
Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Electrical Power Systems and Electromobility (Fundamentals)
Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Field of Specialization / Information and Communication Technology (Fundamentals)
Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MATH-111700	Numerical Methods - Exam	**ST**	**5 CP**	**Kunstmann, Liao, Reichel**
T-MATH-113937	Numerical Methods - Workshop *This item will not influence the grade calculation of this parent.*	**ST**	**1 CP**	**Kunstmann, Liao, Reichel**

Assessment
Success control takes the form of a written examination (120 minutes)and mandatory participation in the programming workshop. Successful participation in the workshop is confirmed by signing the attendance sheet provided at each practice session.

Prerequisites
none

Competence Goal
Students who pass the module are familiar with basic concepts and ways of thinking on the topic of numerical mathematics. They know different procedures for solving linear and nonlinear problems in numerical mathematics. They are furthermore able to use numerical methods for solving problems from applications in an independent, critical, and needs-based way. The students are capable to implement the numerical procedures they have learned in programming workshop.

Content
In the lecture basic ideas and numerical methods for the following topics will be presented:

- systems of linear equations, Gauss-algorithm, LR-decomposition, Cholesky decomposition
- eigenvalue problems, von-Mises iteration
- linear optimization (also called linear programming)
- error analysis
- Newton's method
- quadrature, Newton-Cotes formulas
- numerical solution of initial value problems, Runge-Kutta methods
- finite difference method for solving boundary value problems
- finite elements

Module Grade Calculation
The module grade is the grade of the written exam.

Additional Information
The workshop is held twice during the semester, offering students the opportunity to earn an additional credit point (+1) upon successful participation. Students are expected to work on the programming exercises on their own laptops prior to the workshop. During the workshop, solutions to the programming exercises are discussed with the students.

Workload
Approximately 180h workload. The workload includes:

45h - attendance in lectures, exercises and examination

4h - attendance in workshop

131h – self studies:

- follow-up and deepening of the course content
- solving problem sheets
- literature study and internet research on the course content
- preparation for the module examination
- preparation of workshop

M 12.137 Module: Optical Design Lab [M-ETIT-100464]

Coordinators: Prof. Dr. Wilhelm Stork
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100756	Optical Design Lab	**ST**	**6 CP**	**Stork**

Assessment
The examination consists of an oral exam (approx. 20 min).

Prerequisites
none

Competence Goal
The students can apply previous theoretical knowledge in optics to design optical systems based on ray tracing, using a typical optics design software.

The students can apply typical analysis methods to evaluate the imaging performance of optical systems.

The students can recognize aberrations in optical systems and apply methods to compensate them.

Content
The students participating in this lab are given the opportunity to gain practical experience in the use of software tools commonly used in industry for the design of optical elements and systems. Thus improving their knowledge in optical engineering.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
Approximately 162 h workload of the student.

The workload includes:

1. attendance in lectures an exercises: 36 h
   - 9 excercises of 4 h
2. preparation / follow-up: 51 h
   - preparation 9x3 h
   - writing lab reports: 8x3 h
3. preparation of and attendance in examination: 75h

Recommendations
Basic knowledge in optics. The participation in the course Optical Engineering is strongly adviced.

M 12.138 Module: Optical Engineering and Machine Vision [M-ETIT-106974]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Fundamentals)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113941	Optical Engineering and Machine Vision	**WT**	**6 CP**	**Heizmann**

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal

- Students have a sound knowledge of the fundamentals (physical basics of optics, optical imaging, image sensors) and procedures of optical engineering and machine vision.
- Students are proficient in diverse methods for optical imaging, image acquisition, pre-processing and image evaluation and can characterize them based on their prerequisites, model assumptions and results.
- Students are able to analyze and structure optical engineering and machine vision tasks, synthesize possible solutions from optics principles and image processing methods and assess their suitability.

Content
Optical engineering and machine vision are collective terms for using optical signals to solve tasks of information retrieval for technical and other application. They comprise the propagation of light in optical systems, the acquisition of image signals using optical imaging and cameras, the processing of the recorded image signals using (digital) image processing and the evaluation of the image data to obtain useful information from the recorded images.

The module teaches the basics, procedures and exemplary applications of optical engineering and image processing.

The module include in detail:

- Optical Imaging
  - Imaging with a pinhole camera, central projection
  - Imaging using a (single) lens
- Color
  - Photometry
  - Color perception and color spaces
  - Filters
- Sensors for Image Acquisition
  - Point, line and area sensors
  - CCD, CMOS sensors
  - Line-scan cameras
  - Color sensors and color cameras
  - Infrared cameras
  - Quality criteria for image sensors
- Methods of Image Acquisition
  - Measuring optical properties
  - 3D shape capturing
- Image Signals
  - Mathematical model of image signals
  - Systems theory
  - Two-dimensional Fourier transform
  - Noise of digital imaging sensors (EMVA 1288)
  - Principal component analysis (Karhunen-Loève transform)
- Preprocessing and Image Enhancement
  - Simple image enhancement methods
  - Reduction of systematic errors
  - Attenuation of random disturbances

The following topics do not apply for KSOP students:

- Light
  - Light as an electromagnetic wave
  - Light as a quantum phenomenon
  - Interaction of light and matter
  - Light sources
- Segmentation
  - Region-based segmentation
  - Edge-oriented methods
- Morphological Image Processing
  - Binary morphology
  - Gray-scale morphology
- Texture analysis
  - Types of textures
  - Model-based texture analysis
  - Feature-based texture analysis
- Detection
  - Detection of known objects by linear filters
  - Detection of unknown objects (defects)
  - Detection of straight lines (Radon and Hough transform)

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
The workload includes:

1. attendance in lectures and exercises: 15*4 h = 60 h
2. preparation / follow-up: 15*4 h = 60 h
3. preparation of and attendance in examination: 60 h

A total of 180 h = 6 CR

Recommendations
Basic knowledge of systems theory and signal processing (e.g. from the module “Signals and Systems”) as well as optics is helpful.

Teaching and Learning Methods
lecture (3 SWS) and exercise (1 SWS)

M 12.139 Module: Optical Networks and Systems [M-ETIT-103270]

Coordinators: Prof. Dr.-Ing. Sebastian Randel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Fundamentals)
Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-106506	Optical Networks and Systems	**WT**	**6 CP**	**Randel**

Assessment
Type of Examination: oral exam

Duration of Examination: 20 min (approx.)

Modality of Exam: Oral exams (approx. 20 minutes) are offered throughout the year upon individual appointment.

Prerequisites
none

Competence Goal
The module provides knowledge about optical networks and systems with applications ranging from photonic interconnects, to fiber-to-the-home (FTTH), optical metro and long-haul networks, and automotive and industrial automation. The role of various network layers will be discussed in conjunction with relevant standards and protocols. Physical-layer specifications of relevant photonic components and system design trade-offs will be introduced.

The students

- can describe and compare different optical network architectures and protocols, and evaluate their suitability for specific application scenarios.
- can design optical communication systems tailored to various application requirements, considering technical constraints and performance goals.
- can analyze how constraints such as performance, cost, and energy efficiency influence the selection and innovation of optical technologies.
- can assess the advantages and limitations of optical communication in comparison to electrical and wireless alternatives, and justify technology choices.
- can identify relevant standardization bodies and interpret key aspects of standard documents related to optical communication systems.

Content
Photonic interconnects: rack-to-rack, board-to-board, chip-to-chip, datacenter interconnects, intensity modulation, direct detection, single-mode fiber vs. multi-mode fiber, serial vs. parallel optics, space-division multiplexing vs. wavelength-division multiplexing, Ethernet (10G, 40G, 100G), Fibre Channel, scaling and energy efficiency.

Access neetworks: fiber-to-the-X, passive optical networks (GPON, EPON, NG-PON2, WDM PON), statistical multiplexing vs. point-to-point

Metro- and long-haul networks:

- System-design aspects: dense WDM (ITU grid), optical amplifiers, chromatic dispersion, coherent detection, optical vs. electronic impairment mitigation, capacity limits.
- Wavelength switching: wavelength selective switch (WSS), reconfigurable optical add-drop multiplexer (ROADM).
- Standards and protocols: synchronous optical networking and synchronous digital hierarchy (SONET/SDH), optical transport network (OTN), generalized multi-protocol label switching (GMPLS), software-defined networking (SDN).

Optical networks in automotive and industrial automotion: polymer-optical fiber (POF), MOST Bus, Profibus and Profinet, optical vs. electrical communication links, overcoming bandwidth limitations using digital signal processing.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
total 180 h, here of 45 h lecture, 15 h problems class and 120 h recapitulation and self-studies.

Recommendations
Interest in communications engineering, networking, and photonics.

Teaching and Learning Methods
lecture (3 SWS), exercise (1 SWS)

Literature
Ivan Kaminow, Tingye Li, Alan E. Willner (Editors), Optical Fiber Telecommunications (Sixth Edition), Elsevier
Rajiv Ramaswami, Kumar N. Sivarajan and Galen H. Sasaki, Optical Networks (Third Edition), Elsevier

M 12.140 Module: Optical Systems in Medicine and Life Science [M-ETIT-103252]

Coordinators: Prof. Dr. Werner Nahm
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	6

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-106462	Optical Systems in Medicine and Life Science	**ST**	**3 CP**	**Nahm**

Assessment
Written exam (60 minutes)

Prerequisites
Only one out of the two modules "M-ETIT-100552 - Optische Systeme für Medizintechnik und Life Sciences" and "METIT-103252 - Optical Systems in Medicine and Life Science" is allowed.

Competence Goal
Overall Course Objectives:

This course will allow the students to understand how the basic optical and optoelectronic principles are applied in the design of modern medical devices and routine diagnostic equipment. Besides extending and deepening their expert knowledge in engineering sciences and physics this course will provide profound insight into the applicative, the regulatory and safety and the cost requirements. This will help to be able to understand how the systems are designed to fulfill the requirements.

Furthermore, in this course the students will be introduced into case-based learning. The in-class journal club helps to make the students become more familiar with the advanced literature in the field of study. This interactive format helps to improve the students' skills of understanding and debating current topics of active interest.

Teaching Targets:

The successful participation in this course enables the students to

- derive and formulate system requirements
- layout the system architecture of optical devices
- explain the underlying physical and physiological principles and mechanisms
- elaborate technical and methodological constraints and limitations

present, challenge and debate recent research results

Content
Optical Systems:

- Surgical microscope
- Scanning laser ophthalmoscope (SLO) / Confocal endomicroscope (CEM)
- Optical coherence tomography (OCT) / Optical biometer
- Refractive surgical laser
- Flow-Cytometry

Applied Optical Technologies:

- Magnification and illumination
- Fluorescence and diffuse reflectance imaging
- Confocal laser microscopy
- Low coherence interferometry
- fs-Laser
- Laser scattering (Mie-Therory)

Systems Design and Engineering:

- System architecture

V-Model of Product Development Process

Module Grade Calculation
The module grade is the grade of the written exam.

Additional Information
Language English

Workload
Each credit point corresponds approximately to 30h of the student's workload. Here, the average student is expected to reach an average performance. This contains:

1. Presence during lectures (15 x 1.5 = 22.5h)
2. Preparation and wrap-up of subject matter (57.5h)

Preparation and presentation of one contribution to the in-class journal club (1 x 10h)

Recommendations
Good understanding of optics and optoelectronics.

Literature
M. Kaschke, Optical Devices in Opthalmology and Optometry, Willey-VCH

M 12.141 Module: Optical Technologies for Automotive Applications [M-ETIT-100486]

Coordinators: Prof. Dr. Cornelius Neumann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100773	Optical Technologies for Automotive Applications	**ST**	**3 CP**	**Neumann**

Assessment
Success control takes place in form ofan overall oral examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
In this course, students learn the basics and applications of automotive lighting technology. They know the essential legal requirements, the design principles for signal, headlamp and interior lighting functions and are up to date with the latest knowledge on the subject.

They are able to assess lighting technology designs for vehicle lighting and are prepared to make active contributions to research and development in this field.

Thanks to their knowledge of the current state of development, students are able to assess the influence of vehicle lighting on social aspects such as safety when driving at night.

Content

- Recap: Light & color
- Recap: Light sources
- Signal & rear lights
- Reflectors
- Headlights
- Interior lights
- Manufacturing technology
- History of automotive lighting technology

The lecturer reserves the right to deviate from the content of the current lecture without special notice.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
The workload includes:

1. attendance time in lectures, exercises 30 h

2. preparation/follow-up of the same 30 h

3. preparation for and attendance at exams 30 h.

Recommendations
Knowledge of the module "M-ETIT-100485 - Lichttechnik"

M 12.142 Module: Optical Transmitters and Receivers [M-ETIT-100436]

Coordinators: Prof.Dr.Dr.h.c. Wolfgang Freude
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100639	Optical Transmitters and Receivers	**WT**	**6 CP**	**Freude**

Assessment
Oral examination (approx. 20 minutes). The individual dates for the oral examination are offered regularly.

Prerequisites
none

Competence Goal
The students

- understand the peculiarities of optical communications, and how optical signals are generated, transmitted and received,
- know about sampling, quantization and coding,
- learn the basics about noise on reception,
- understand the properties of a linear and a nonlinear optical fibre channel,grasp the idea of channel capacity and spectral efficiency,
- know about various forms of modulation,
- acquire knowledge of optical transmitter elements,
- understand the function of optical amplifiers,
- have a basic understanding of optical receivers,
- know the sensitivity limits of optical systems, and
- understand how these limits are measured.

Content
The course concentrates on basic optical communication concepts and connects them with the properties of physical components. The following topics are discussed:

- Advantages and limitations of optical communication systems
- Optical transmitters comprising lasers and modulators
- Optical receivers comprising direct and heterodyne reception
- Characterization of signal quality

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
Approx. 180 hours workload for the student. The amount of work is included:

30 h - Attendance times in lectures
30 h - Exercises
120 h - Preparation / revision phase

Recommendations
Knowledge of the physics of the pn-junction

Literature

Detailed textbook-style lecture notes can be downloaded from the IPQ lecture pages.
Grau, G.; Freude, W.: Optische Nachrichtentechnik, 3. Ed. Berlin: Springer-Verlag 1991. In German. Since 1997 out of print. Electronic version available via w.freude@kit.edu.
Kaminow, I. P.; Li, Tingye; Willner, A. E. (Eds.): Optical Fiber Telecommunications VI A: Components and Subsystems +VI B: Systems and Networks', 6th Ed. Elsevier (Imprint: Academic Press), Amsterdam 2013

M 12.143 Module: Optimal Control [M-ETIT-107497]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Fundamentals)
Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Fundamentals)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114701	Optimal Control	**WT**	**6 CP**	**Hohmann**

Assessment
The assessment consists of a written exam (120 min) taking place in the recess period.

Prerequisites
none

Competence Goal

- The students know as well the mathematical basics as the fundamental methods and algorithms to solve constraint and unconstraint nonlinear static optimization problems.
- They can solve constraint and unconstraint dynamic optimization by using the calculus of variations approach and the Dynamic Programming method.
- The students understand and can apply the Bellman Principle, including its theoretical foundations and practical relevance.
- Also they are able to transfer dynamic optimization problem to static problems.
- They are proficient in formulating Optimal Control Problems and applying Model Predictive Control techniques
- The students know the mathematic relations, the pros and cons and the limits of the particular optimization methods.
- They can transfer problems from other fields of their studies in a convenient optimization problem formulation and they are able to select and implement suitable optimization algorithms for them by using common software tools.

Content
The module teaches the mathematical basics that are required to solve optimization problems. The first part of the lecture treats methods for solving dynamic optimization problems, including Dynamic Programming and methods derived from the calculus of variations, such as the Hamilton method.

This is followed by a section on optimal control problems. After presenting techniques to transform dynamic optimization problems into static ones, the final part of the lecture addresses methods for solving static optimization problems, both constrained and unconstrained.

Module Grade Calculation
The module grade is the grade of the written exam.

Additional Information
Successor of "M-ETIT-100531 - Optimization of Dynamic Systems"

Workload
The amount of work includes

1. presence in lecture/exercises/tutorial(optional) (2+2 SWS: 60h)
2. preparation/postprocessing of lecture/exercises (90h)
3. preparation/presence in the written exam (30h)

Teaching and Learning Methods
lecture (2 SWS), excercise (2 SWS)

M 12.144 Module: Optimization and Synthesis of Embedded Systems (ESI) [M-INFO-107229]

Coordinators: Prof. Dr.-Ing. Jörg Henkel
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-114253	Optimization and Synthesis of Embedded Systems (ESI)	**WT**	**3 CP**	**Henkel**

Assessment
See partial achievements (Teilleistung)

Prerequisites
See partial achievements (Teilleistung)

Competence Goal
The student can develop embedded systems. They can specify, synthesize and optimize their own hardware. They learn the hardware description language and are familiar with the special boundary conditions of the design of embedded systems.

Content
The cost-effective and error-free development of embedded systems represents a challenge that should not be underestimated and which is having an ever greater influence on the added value of the overall system. In Europe in particular, the design of embedded systems is playing an increasingly important economic role in many sectors of the economy, such as the automotive industry, so that a number of well-known companies are already involved in the development of embedded systems.

The lecture deals comprehensively with all aspects of the development of embedded systems at hardware, software and system level. This includes diverse areas such as modelling, optimization and synthesis of systems.

Workload
90 hours:
30 hours of lectures (15 weeks*2 hours)
30 hours of self study (15 weeks*2 hours)
15 hours of deepening (coding RTL) to grasp the theory with practical
15 hours exam preparation

M 12.145 Module: Optoelectronic Components [M-ETIT-100509]

Coordinators: TT-Prof. Dr. Tobias Huber-Loyola
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-101907	Optoelectronic Components	**ST**	**4 CP**	**Huber-Loyola**

Assessment
Type of Examination: oral exam

Duration of Examination: approx. 30 minutes

Modality of Exam: Oral examination, usually one examination day per month during the Summer and Winter terms. An extra questions-and-answers session will be held if students wish so.

Prerequisites
none

Competence Goal
Comprehending the physical layer of optical communication systems. Developing a basic understanding which enables a designer to read a device´s data sheet, to make most of its properties, and to avoid hitting its limitations.

The students

- understand the components of the physical layer of optical communication systems
- acquire the knowledge of operation principles and impairments of optical waveguides
- know the basics of laser diodes, luminescence diodes and semiconductor optical amplifiers
- understand pin-photodiodes
- know the systems´sesitivity limits, which are caused by optical and electrical noise

Content
The course concentrates on the most basic optical communication components. Emphasis is on physical understanding, exploiting results from electromagnetic field theory, (light waveguides), solid-state physics (laser diodes, LED, and photodiodes), and communication theory (receivers, noise). The following components are discussed:

- Light waveguides: Wave propagation, slab waveguides, strip wave-guides, integrated optical waveguides, fibre waveguides
- Light sources and amplifiers: Luminescence and laser radiation, luminescent diodes, laser diodes, stationary and dynamic behavior, semiconductor optical amplifiers
- Receivers: pin photodiodes, electronic amplifiers, noise

Module Grade Calculation
The module grade is the grade of the oral exam.

Additional Information
There are no prerequisites, but solution of the problems on the exercise sheet, which can be downloaded as homework each week, is highly recommended. Also, active participation in the problem classes and studying in learning groups are strongly advised.

Workload
total 120 h, hereof 45 h contact hours (30 h lecture, 15 h problem class), and 75 h homework and self-studies

Recommendations
Minimal background required: Calculus, differential equations, Fourier transforms and p-n junction physics.

Literature

Detailed textbook-style lecture notes as well as the presentation slides can be downloaded from the IPQ lecture pages.

Agrawal, G.P.: Lightwave technology. Hoboken: John Wiley & Sons 2004

Iizuka, K.: Elements of photonics. Vol. I, especially Vol. II. Hoboken: John Wiley & Sons 2002

Further textbooks in German (also in electronic form) can be named on request.

M 12.146 Module: Optoelectronic Measurement Engineering [M-ETIT-100484]

Coordinators: Dr.-Ing. Klaus Trampert
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100771	Optoelectronic Measurement Engineering	**ST**	**3 CP**	**Trampert**

Assessment
Success control takes place in form of an overall oral examination (approx. 25 minutes).

Prerequisites
none

Competence Goal
Students have basic knowledge of the measurement of optical quantities and the procedures and measuring devices required for this. They can analyze the common methods for determining spectrally resolved optical quantities and describe their physical operating principle. You will be able to select the appropriate method for the measurement task from the pool of methods and devices, depending on the desired measured quantity. They are also able to adapt known methods for new tasks, taking into account the strengths and weaknesses of the selected method or device.

Content
The focus of the module is to impart sound knowledge of the methods and devices of optical measurement technology. In particular, spectrally resolved methods. The lecture is structured along the measurement chain starting from the optical quantity via the optical system via the conversion of the optical quantity into the electrical quantity and the processing and interpretation of the electrical measurement signal. The module provides an overview of the existing types of measurement receivers and their physical properties and teaches the ability to select the appropriate type of receiver for the specific application.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
Based on 15 courses per semester, each with 1.5 h presence in the lecture, 2.5 h each Before and after, as well as approx. 2 hours of literature reading and self-exercises, the total workload is 90 hours

Recommendations
Knowledge from Lighting Technology and Technical Optics is an advantage.

M 12.147 Module: Organic and Flexible Electronics [M-ETIT-107455]

Coordinators: Prof. Dr. Gerardo Hernandez Sosa
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114638	Organic and Flexible Electronics	**WT**	**3 CP**	**Hernandez Sosa**

Assessment
The control of success takes place within the framework of an oral overall examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
The students

- understand the electronic and optical characteristics of organic semiconductors.
- know the fundamental differences between organic and conventional inorganic semiconductors.
- have basic knowledge of manufacturing and processing technologies for rigid, flexible and stretchable electronics.
- have knowledge of organic light-emitting diodes, organic solar cells and photodiodes, organic field-effect transistors and physical/chemical sensors based on organic materials.
- have an overview of the possible applications, markets and development lines for these components.
- are able to work in multidisciplinary teams with engineers, chemists and physicists.

Content

1. Introduction
2. Optoelectronic properties of organic semiconductors
3. Organic light-emitting diodes (OLEDs) and applications
4. Organic Solar Cells
5. Organic Photodetectors
6. Organic FETs
7. Physical and chemical sensors
8. Printing technology for flexible and stretchable electronics

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
1. lecture: 21 h
2. recapitulation and self-studies: 42 h
3. preparation of examniation: 27 h

in total: 90 h

Recommendations
Knowledge of semiconductor components

Literature
The corresponding documents are available online in ILIAS (https://studium.kit.edu/)

M 12.148 Module: Pattern Recognition [M-INFO-100825]

Coordinators: Prof. Dr.-Ing. Jürgen Beyerer
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules) (Usage until 9/30/2027)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	German	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-101362	Pattern Recognition	**ST**	**6 CP**	**Beyerer, Zander**

Assessment
See Partial Achievements (Teilleistung)

Prerequisites
See Partial Achievements (Teilleistung)

Competence Goal

- Students have a sound knowledge of the selection, extraction and properties of features used to characterize objects to be classified. Students know how the feature space can be examined, how features can be transformed and how distances in the feature space can be determined. Furthermore, they can normalize features and construct features. In addition, students know how to reduce the dimension of the feature space.
- Students have a sound knowledge of how to select and adapt suitable classifiers for different tasks. They are familiar with Bayesian decision theory, parameter estimation and parameter-free methods, linear discriminant functions, support vector machines and matched filters. Students are also proficient in classification with nominal features.
- Students are able to solve pattern recognition problems, taking into account the efficiency of classifiers and the relationships in the processing chain object - pattern - feature - classifier on a task-specific basis. To this end, students are familiar with the principle for determining the performance of classifiers and the principle of boosting.

Content
Features:

- Feature types
- Examination of the feature space
- Transformation of the features
- Distance measurement in the feature space
- Normalization of features
- Selection and construction of features
- Reduction of the dimension of the feature space

Classifiers:

- Bayesian decision theory
- Parameter estimation
- Parameter-free methods
- Linear discriminant functions
- Support Vector Machine
- Matched Filter, Template Matching
- Classification with nominal features

General principles:

- Vapnik-Chervonenki's theory
- Performance determination of classifiers
- Boosting

Additional Information
Will take place for the last time in the summer term of 2027.

Workload
Total: approx. 180h, of which

Lecture attendance time 31h
Preparation and follow-up 40h
Exercise attendance time 10h
Preparation, solving exercises, follow-up 40h
Exam preparation and attendance 59h

Recommendations
See Partial Achievements (Teilleistung)

M 12.149 Module: Photometry and Radiometry [M-ETIT-100519]

Coordinators: Dr.-Ing. Klaus Trampert
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100789	Photometry and Radiometry	**WT**	**3 CP**	**Trampert**

Assessment
Success control takes place in the form of an overall oral examination (approx. 25 minutes).

Prerequisites
none

Competence Goal
Students have basic knowledge of the measurement of absolute optical quantities and the necessary procedures and calibrations. They can describe the common methods for determining optical quantities. They are able to select the appropriate method for the measurement task from the pool of methods, depending on the desired measurement quantity. They are also able to adapt known methods for new tasks, taking into account the strengths and weaknesses of the selected method. They are able to name the most important influencing variables on the uncertainty of the measurement result and can name methods to quantify this influence in the real measurement task.

Content
The focus of the module is to impart sound knowledge of the methods of light measurement technology including the description of measurement uncertainties. The first essential subject area is the established methods and determination of the photometric quantities luminous flux, luminous intensity, illuminance and luminance and the associated measuring equipment. The second important subject area comprises the recording and description of the measurement uncertainties that occur during the calibration of such systems using the established GUM method.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
Based on 14 courses per semester, the workload is calculated as 1.5 h attendance in the lecture, 3 h preparation and follow-up, as well as a total of approx. 40 h literature research and preparation and 40 h exam preparation = 133 h total workload

Recommendations
Knowledge of the module "M-ETIT-100484 - Optoelektronische Messtechnik" and "M-ETIT-100485 - Lichttechnik" is an advantage.

M 12.150 Module: Photonic Integrated Circuit Design and Applications [M-ETIT-105914]

Coordinators: Prof. Dr.-Ing. Christian Koos
Prof. Dr.-Ing. Sebastian Randel

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111896	Photonic Integrated Circuit Design and Applications	**ST**	**6 CP**	

Assessment

- Part 1 – Solutions of problem sets: We will grade your solutions of the various problem sets and design projects. To this end, please upload your solution via the online teaching platform of your respective institution (see above) before the respective deadline. Please merge all pages into a single pdf file, and please use a scanner. Smartphone made snapshots are often illegible, and in this case your solutions cannot not be evaluated. In case there are any technical difficulties with the platforms, you may also submit your solutions by e-mail to picda@ipq.kit.edu before the respective deadline.
- Part 2 - Presentation of one pre-assigned problem set: At the beginning of the term, design projects will be pre-assigned to groups of participants. Each of these groups will explain their approach and results to lecturers and peer students in a short presentation (approx. 15 min), followed by approx. 10 min of public discussion with peer students and professors, and an individual private interview of each group member (approx. 10 min per person).

The overall impression is rated.

Competence Goal

The students understand the basic principles of photonic component design and can apply them to concrete design tasks of increasing complexity and independence, that they will solve in small groups and present to their peers. Doing so they will learn to translate theoretical knowledge gained during the lecture into actionable knowledge used to solve hands-on design tasks. In addition to design principles, students will learn how to satisfy key requirements for making photonic integrated circuits manufacturable and useable in a system environment, such as corner analysis of manufacturing tolerances, design for testability, design for manufacturability, and packaging. In short, we aim at teaching students the skills for hands-on design of manufacturable and application relevant photonic integrated circuits, preparing them to productively contribute to a design team. In addition, we will convey the most recent trends in the application of photonic integrated circuits and let students design a circuit addressing one of these application spaces, giving them a feeling for both the potential as well as the limitations of the technology, so that they may take informed decisions on what systems to integrate in the future.

Content
Lectures:

- Lecture 1: Introduction to silicon photonics
- Lecture 2: Silicon photonics – technology overview
- Lecture 3: Wave propagation in silicon photonic waveguides
- Lecture 4: Mode expansion and orthogonality
- Lecture 5: Coupled-mode theory
- Lecture 6: Selected passive devices
- Lecture 7: Modulators
- Lecture 8: Photodetectors
- Lecture 9: Optical amplifiers and lasers
- Lecture 10: Test and packaging
- Lecture 11: Optical communications
- Lecture 12: Optical metrology
- Lecture 13: Biophotonics and neurophotonics
- Lecture 14: Integrated quantum optics and optical computing

Design lab:

- Problem Set 1: Mode fields and mode expansion
- Problem Set 2: Coupling efficiency and coupled-mode theory
- Design Project A: Optical filter
- Design Project B: Optical transceiver
- Design Project C: Optical communication link

Module Grade Calculation
The module grade results of the assessment of the solutions of the design projects and problem sets, the presentation of one design project with discussion, and the individual oral interview.

Details will be given during the lecture.

Workload
Each credit point corresponds to approximately 30 hours of work (of the students). This is based on average students who achieve an average performance. The workload includes (e.g. 2 SWS):

1. attendance in lectures an exercises: 15*2 h = 30 h
2. preparation / follow-up: 15*2 h = 30 h
3. preparation of and attendance in examination: 120 h

A total of 180 h = 6 CR

Teaching and Learning Methods
In addition to the teaching of fundamental concepts to the extent necessary to enable students to perform practical designs, the lecture will focus from the start on a specific technology platform (silicon-on-insulator) in which the students will solve design problems of increasing complexity with the design suite Lumerical. As the final hands-on problem, students will design an entire photonic subsystem for an application of their choice, leaving free room for creative thinking and self-driven work. Since each group of students will present one of the solved designed problems to their peers, students will get exposed to solutions found for and practical problems encountered in a variety of design tasks, providing them with a wider experience base to draw on for future design.

Since the class will be taught by lecturers from several Universities, all lectures will be streamed live (with the possibility to interact and to ask questions) and made available online. Design tasks will be performed with the Design Suite Lumerical, for which introductory videos will be made available. An online forum will be provided to allow students to ask questions offline to the lecturers as well as to interact with each other, inside and across Universities.

M 12.151 Module: Photonics and Communications Lab [M-ETIT-104485]

Coordinators: Prof. Dr.-Ing. Christian Koos
Prof. Dr.-Ing. Sebastian Randel

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Lab Course)
Field of Specialization / Information and Communication Technology (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-109173	Photonics and Communications Lab	**ST**	**6 CP**	**Koos, Randel**

Assessment
Success control takes place in the form of other types of examination. Success is assessed by solving the tasks relating to the preparation of the experiment (written and oral) and writing an experiment report.

Prerequisites
none

Competence Goal
The students have gained practical experience with experimental setups and experimental setups. Students are able to handle laboratory equipment/devices and simulation environments for optical data transmission and optical measurement technology. Students are familiar with the organization, preparation and supervision of the necessary practical experiments.

Content
The following topics are covered in the practical course:

- Laser diodes and LEDs
- photodetectors
- optical coherence tomography (OCT)
- Reverse control in fibers
- BPM simulations of integrated optical waveguides
- Ring resonator filters
- Simulations of optical transmitters (-40 GBps)
- Generation, transmission and reception of digitally modulated signals

Module Grade Calculation
The module grade is the grade for the practical course. The grade results from the oral and written tasks.

Recommendations
Successful participation in the module "M-ETIT-100436 – Optical Transmitters and Receivers" and basic knowledge in MatLab

M 12.152 Module: Photovoltaics [M-ETIT-100513]

Coordinators: Prof. Dr.-Ing. Michael Powalla
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-101939	Photovoltaics	**ST**	**6 CP**	**Powalla**

Assessment
Success is assessed in the form of a written examination (2 hours). The module grade is the grade of this written examination.

Prerequisites
Module "M-ETIT-100524 - Solar Energy" must not have been started.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-ETIT-100524 - Solar Energy must not have been started.

Competence Goal
For the lecture Photovoltaics with 3 SWS lecture and 1 SWS exercise, the following competence requirements are specified by the following objectives:

A.Specialist knowledge:

After attending the course, students will be able to:

- understand energy conversion in semiconductors. They will be able to analyze the physical description of light and the interaction of light with solids. Students will gain knowledge about the energy conversion of different forms of energy and the transport of electrical energy in semiconductors and metals. They will be able to describe and mathematically represent the functioning of p/n diodes.
- discuss the associated current technological and production-related issues. In particular, students examine the technical implementation of semiconductor requirements in technical processes. They gain knowledge of the entire value chain (physical principles, material science aspects, production engineering applications and systemic integration)
- understand photovoltaic energy systems in the interaction of all components. The comparison of the systemic integration of off-grid and grid-integrated solar-based energy generation systems helps to explain the components and their design. The system quality, efficiency, costs etc. can be explained with the help of key figures.
- Students quantify the loss mechanisms in the solar cell in the solar converter and the solar systems, in particular to optimize economic and ecological key figures, and learn about operating experience and long-term stability issues.
- understand how different solar cell technologies and solar thermal energy conversion work and how to classify them in an overall energy system

B. Research and problem-solving skills:

The students (after participating in the course)

- are able to think in an interdisciplinary way. Basic skills from physics, electrical engineering and information technology, production engineering and economics are brought together and complement each other to form an overall picture.
- are familiar with methods for analyzing systems composed of discrete components,
- are familiar with state-of-the-art methods of describing energy conversion systems using solar primary energy,

C. Assessment and planning skills:

Students (after participating in the course)

- are able to assess and categorize different solar cell concepts and different solutions for solar power generation,
- recognize the limits and challenges of providing electrical energy from locally and temporally fluctuating sources and can thus initiate new developments,
- scrutinize new concepts in the highly dynamic field of solar electrical energy generation in connection with climate protection and security of supply

D. Personal and social skills:

The students (after participating in the course)

- are familiar with the derivation and origin of the most important physical relationships and recognize the synergy of different scientific disciplines,
- can calculate tasks independently and communicate the results orally and in writing,
- recognize the relevance of technical solutions for climate protection

Content
This module is designed to help students understand energy conversion in semiconductors. Photovoltaic energy systems are dealt with in the interaction of all components and loss mechanisms in the solar cell and in the photovoltaic system are quantified. The functionality of solar thermal energy generation is taught. In addition, the associated current technological and production-related issues are discussed.

Module Grade Calculation
The module grade is the grade of the written examination.

Additional Information
Slides will be provided via Ilias. A summary of the content is also available as a pdf.

Workload
Calculation basis: 15 lecture weeks

1. Lecture attendance time: 23 * 1.5 h = 34.5 h
2. Preparation and follow-up time for lecture: 23 * 2 h = 46 h
3. Exercise 7 x 1.5 h = 10.5 h.
4. Preparation and follow-up time for exercise: 7 x 4 h = 28 h
5. Excursion 10 h
6. Exam preparation and attendance (2 h): 51 h

   Total = 180 h

Teaching and Learning Methods
Lecture and exercise

Literature
List of relevant technical literature.

http://www.erneuerbare-energien.de

http://pveducation.org/pvcdrom.

http://www.sciencedirect.com/science/referenceworks/9780080878737#ancv1

Würfel, Physik der Solarzellen, 2nd edition (Spektrum Akademischer Verlag, Heidelberg, 2000)

Konrad Mertens Photovoltaik: Lehrbuch zu Grundlagen, Technologie und Praxis (Carl Hanser Verlag GmbH & Company KG, 06.08.2018)

M 12.153 Module: Physical Foundations of Cryogenics [M-CIWVT-103068]

Coordinators: Prof. Dr.-Ing. Steffen Grohmann
Organisation: KIT Department of Chemical and Process Engineering
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-106103	Physical Foundations of Cryogenics	**ST**	**6 CP**	**Grohmann**

Assessment
Learning control is an oral examination lasting approx. 30 minutes.

Prerequisites
None

Competence Goal
Understanding of the mechanisms of entropy generation, and the interaction of the first and the second law in thermodynamic cycles; understanding of cryogenic material properties; application, analysis and assessment of real gas models for classical helium I; understanding of quantum fluid properties of helium II based on Bose-Einstein condensation, understanding of cooling principles at lowest temperatures.

Content
Relation between energy and temperature, energy transformation on microscopic and on macroscopic scales, physical definitions of entropy and temperature, thermodynamic equlibria, reversibility of thermodynamic cycles, helium as classical and as quantum fluid, low-temperature material properties, cooling methods at temperatures below 1 K.

Module Grade Calculation
The grade of the oral examination is the module grade.

Workload

- Attendance time (Lecture): 45 h
- Homework: 45 h
- Exam Preparation: 90 h

Literature
Schroeder, D.V.: An introduction to thermal physics. Addison Wesley Longman (2000)

Pobell; F.: Matter and methods at low temperatures. 3rd edition, Springer (2007)

M 12.154 Module: Physics, Technology and Applications of Thin Films [M-ETIT-105608]

Coordinators: Dr. Konstantin Ilin
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111237	Physics, Technology and Applications of Thin Films	**WT**	**4 CP**	**Ilin**

Assessment
Oral examination of approximately 20 minutes

Prerequisites
The modul "M-ETIT-102332 - Thin films: technology, physics and applications" and "Thin Films: Technology, Physics and Applications I" may neither be started nor completed.

Competence Goal
Students should be able to discuss interplay between growth conditions of thin films, physical and geometrical properties of nanostructure made of these films, and performance and suitable areas of application of detectors of radiation based on interaction of these nanostructures with electromagnetic power. The knowledge obtained by students should provide a theoretical basis for the most important steps in development of thin film nanoelectronic devices.

Content
Students will get practically oriented information about technology of thin films including different methods of deposition of thin films like magnetron sputtering, thermal evaporation, pulsed laser ablation, about basics of vacuum technology, and about mechanisms of growth of thin films of different materials at different conditions.

Patterning methods (photo- and e-beam lithography, reactive ion etching, ion milling, and lift-off techniques) suitable for nanometer scale features of electronic devices will be considered in details.

Experimental methods of characterization of material, geometrical, optical, physical, superconducting, electron and phonon properties of thin films, nanostructures made of these films, and devices based on these nanostructures will be discussed.

Consideration of technology and physics of thin film structures will be done on example of development of three types of fast and sensitive detectors of electro-magnetic radiation for applications in optical and THz spectral ranges: superconducting nanowire single-photon detector, hot-electron bolometer, and YBCO ps-fast detector of synchrotron emission. Dependence of detector's performance on their fabrication condition will be analyzed in frame of physical models which describe response mechanisms of the detectors to absorbed radiation.

Practical actualization of the knowledge is possible in frame of Praktikum Nanoelektronik (LVN 23669).

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
A workload of approx. 90 h is required for the successful completion of the module. This is composed as follows

1. attendance time in lecture/exercise 18 h
2. pre-/postprocessing of the lecture 24 h
3. preparation/attendance oral exam 48 h

M 12.155 Module: Power Electronic Systems [M-ETIT-107533]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114751	Power Electronic Systems	**WT**	**6 CP**	**Hiller**

Assessment
Success control takes place in the form of an oral examination lasting approx. 25 minutes.

Prerequisites
none

Competence Goal
Students are familiar with the grid-connected and self-commutated power converter circuits relevant to energy technology applications.
They are able to select power converters for drive technology and grid applications (including high-voltage direct current transmission) and assess their operating characteristics.
They are familiar with the functionality as well as the advantages and disadvantages of the various multi-stage inverter circuits and are able to select the required power semiconductors depending on the electrical requirements and the type of cooling.
Students are also able to design the power semiconductors and passive components of a power converter circuit electrically and thermally.
They are familiar with the normative insulation requirements and can analyze and explain the requirements for the protection of a power converter.

Content
The electrical and thermal design and dimensioning of power converters in drive and energy technology are presented and discussed in detail in this lecture. Based on the terminal behavior of the various power converter topologies, the interactions with other system components are presented and evaluated.
The lecture provides an overview of possible measures to improve the system behavior and deals with the protection of power converter circuits.
The following topics are covered in detail:

- Introduction
- Grid-connected power converters under idealized and real conditions and their most important applications in power engineering
- Self-commutated multilevel converters: Neutral point clamped inverters, floating capacitor inverters, series cell inverters, modular multilevel converters, hybrid circuits, modulation methods
- Semiconductor components for grid-connected and self-commutated power converters, protective devices
- Heat dissipation concepts for power semiconductors and passive components, junction temperature calculations
- Load cycle stability of power semiconductors
- Short-circuit current design for mains and motor side
- Protection concepts
- Insulation coordination, standards
- Transformer, grid connection
- Mains and motor-side filters
- Cable models
- Interaction between inverter and machine (insulation, bearing currents)
- Reliability calculations
- Excursion to converter plant if necessary

The lecturer reserves the right to deviate from the content of the current lecture without special notice.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload

- 22x lecture and 8x exercise à 2 h = 60 h
- 22x follow-up of the lecture à 1 h = 22 h
- 8x preparation of the exercise à 2 h = 16 h
- Preparation for the exam = 81 h
- Examination time = 1 h

  Total = 180 h

Recommendations
Knowledge of the basics of power electronics and electrical machines is helpful, but not essential.

M 12.156 Module: Power Electronics [M-ETIT-104567]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Fundamentals)
Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	6

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-109360	Power Electronics	**ST**	**6 CP**	**Hiller**

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
None

Competence Goal
Students will be familiar with state-of-the-art power semiconductors including their application related features. Furthermore students will be familiar with the circuit topologies for DC/DC and DC/AC power conversion. They know the associated modulation and control methods and characteristics. They are able to analyze the circuit topologies with regard to harmonics and power losses. This also includes the thermal design of power electronic circuits. In addition, they are able to select and combine suitable circuits for given electrical energy conversion requirements.

Content
In the lecture, power electronic circuits for DC/DC and DC/AC power conversion using IGBTs and MOSFETs are presented and analyzed. First, the basic properties of self-commutated circuits under idealized
conditions are elaborated using the DC/DC converter as an example. Then, self-commutated power converters for three-phase applications are presented and analyzed with respect to modulation and their AC
and DC terminal behavior. Based on the real power semiconductor behavior in on- and off-state the device losses are calculated. Furthermore the thermal design of power converters is explained using thermal equivalent circuits of power devices and cooling equipment. The voltage and current stress on the power
semiconductors in switching operation is explained as well as protective snubber circuits allowing a reliable operation within the safe operating area of the devices.

In detail, the following topics are treated:

- Power Semiconductors
- Commutation principles
- DC/DC converters
- Self-commutated 1ph and 3ph DC/AC inverters
- Modulation methods (Fundamental frequency modulation, Pulse width modulation with 3rd harmonic injection, Space vector modulation)
- Multilevel inverters
- Switching behavior in hard and soft switching applications
- Loss calculation
- Thermal equivalent circuits, thermal design
- Snubber circuits.

The lecturer reserves the right to adapt the contents of the lecture to current needs without prior notice.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
14x lecture and 14x exercise à 2 h = 56 h
14x wrap-up of the lecture à 1 h = 14 h
14x preparation of the exercise à 2 h = 28 h
Preparation for the exam = 75 h
Examination time = 2 h
Total = approx. 175 h (corresponds to 6 LP)

M 12.157 Module: Power Electronics for the Energy Transition [M-ETIT-107888]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-115308	Power Electronics for the Energy Transition		**3 CP**	**Hiller**

Assessment
The success control takes place in form of an oral examination (approx. 25 minutes).

Prerequisites
none

Competence Goal

- The students will be able to understand the importance and area of application of power converters for the energy transition technologies of photovoltaics, wind and battery storage
- The students will have the knowledge of relevant circuits used to convert DC to grid-compatible AC power
- The students will be able to describe the importance of converter characteristics, such as grid-forming control, for system stability in a grid with 100 % renewable energy
- The students are able to evaluate approaches for a circular economy for power electronics

Content
This module gives an overview about power electronics needed for the energy transition and system stability aspects in converter-dominated grids

Table of contents:

- Renewable Energy Transformation Scenarios
- Need for Power Electronics
- Single-phase and three-phase PV inverters
- PV systems aspects, PV power plant concepts
- Battery storage integration
- Wind energy converters
- Grid-integration and power system stability with grid-forming converters
- Sustainability aspects of power electronics

Module Grade Calculation
The module grade is the grade of the oral exam.

Additional Information
Place limitation to 16 students due to capacity reasons.

The module takes place in the framework of the EUCOR program.

The course is offered as part of “Eucor – The European Campus” as a joint course with the University of Freiburg. Students are expected to travel.

The course will be organized as follows:

- Start: see Calendar in ILIAS course “Power Electronics for the Energy Transition”
- 4 Online lectures, 1.5 hours each
- 4 In-person lecture sessions, consisting 2 lectures of 1.5 hours each (double session) (possibly hybrid)
  - 2 Lecture sessions at the University of Freiburg, including a lab tour at Fraunhofer ISE
  - 2 Lecture sessions at KIT, Building 11.10, Room ETI, including a lab tour at ETI / Energy Lab

For students of KIT: Travel expenses from Karlsruhe to Freiburg can be reimbursed. Please consult **INTL** regarding the requirements and the reimbursement procedure.

Workload
The workload includes:

1. attendance in lectures: 15*2 h = 30 h
2. preparation / follow-up: 15*2 h = 30 h
3. preparation of and attendance in examination: 30 h

A total of 90 h = 3 CR

Recommendations
Basic knowledge in the field of electrical components (semiconductors, inductors, capacitors) is helpful.

Teaching and Learning Methods
2 SWS lecture

M 12.158 Module: Power System Protection and Automation [M-ETIT-106506]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113164	Power System Protection and Automation	WT	3 CP	Leibfried

Assessment
Success is assessed in an overall oral examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
Students know the main components of protection and control technology in electrical networks and how they interact. Students understand the main protection functions and their areas of application. They are familiar with the structure and function of substations. The role of network control technology in electrical networks is understood. They will be able to reproduce key interrelationships. Students know the future challenges for the electrical grid in general and for protection and control technology in particular and are able to understand them.

Content

- Introduction to protection systems
  - Short-circuit quantities
  - Star point treatment
  - Fault types in the grid
- Planning of grid protection devices
  - Planning of switchgear and substations
  - Protection principles and philosophies
  - Relay planning
- Measurement data acquisition
  - Voltage transformers
  - Current transformers
  - Switching of transformers
  - Developments in transformer technology
- Earth fault detection
  - Planning of compensated networks
  - Earth fault suppression coil
  - Earth fault detection
- Mains protection/ branch circuit protection
  - Overcurrent protection
  - Distance protection
  - Differential protection
- Frequency protection
  - Principle of system protection
  - Underfrequency protection
- Generator protection
  - Design of generator protection systems
  - Short-circuit protection - Stator protection
  - Rotor protection
- Introduction to control technology
  - Structure of grid control systems
  - Communication in grid control technology
  - Basics of telecontrol and station control technology
- Switchgear technology
  - Conventional systems
  - Digital station control technology
- Network control technology
  - Structure of network control technology
  - SCADA
  - HEO functions
- Developments in protection and control technology
  - Digital substations
  - Influence of converters on fault behavior
  - Preventive or curative grid management

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
The workload includes (for one lecture)

1. attendance time in lecture (30 h = 1 CP)

2. self-study time for preparation and follow-up of the lecture including exam preparation (60h = 2 CP)

This results in a total of 90 hours = 3 CP

M 12.159 Module: Power Systems and Economy [M-ETIT-100413]

Coordinators: Dr.-Ing. Bernd Hoferer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	1	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100725	Power Systems and Economy	WT	3 CP	Hoferer

Assessment
Success control takes place in the form of an overall oral examination (20 minutes)

Prerequisites
none

Competence Goal
Students will be familiar with the key characteristics, operational tasks and responsibilities of the interacting areas of the energy industry, generation, trading, sales and grids, including the important legal framework and regulation. The focus is on the transformation of a traditionally centrally organized energy industry into an increasingly decentralized supply structure with special consideration of the expansion of renewable energies, transmission and distribution grids and the necessary storage systems. As a synthesis, students will learn about a possible transformation path on which the German government's climate and environmental policy goals appear achievable by 2045.

Content
This lecture is aimed at undergratuate students and is intended to convey the interrelationships and interactions, particularly in the European energy market. Based on the presentation of fossil energy resources available today, the expected energy demand on earth for the year 2030 is forecast with reference to an Exxon study. Consequences for the type and scope of sensible energy use and the required energy supply are derived from this. The structure, the legal framework and the interaction of the various market participants in the European energy market are described in detail. The treatment of practical examples provides a basic understanding of the complex processes in these markets.

First, the energy demand in Germany and worldwide is presented. Opportunities for targeted energy savings are described in their dimensions. The forecast global energy demand in 2030 is the benchmark for the type and scope of energy generation to be provided. Renewable energy generation plants of the highest efficiency are discussed as a sensible and necessary supplement to fossil energy generation.

The European Union has liberalized the energy market through legislative changes. The transition from a monopoly to a competitive market is described in detail in the lecture. The changes for the market partners, especially for the customers, are presented and newly created structures and processes such as trading on energy exchanges are developed.

The market environment for energy trading and energy sales has changed fundamentally. Today, energy pricing is increasingly subject to national and international influences. Costs for energy generation, energy transportation and, above all, government levies determine the price of energy and cause sales margins to shrink. New products are expected to generate new business and sales.
The deregulation of energy transportation systems is an essential basis for a competitive energy market. Options for the further development of this transportation infrastructure with the aim of guaranteeing all market participants unhindered access at the same prices are discussed in the lecture.

The competitive market requires a very detailed provision of data of all kinds. Energy data management as an indispensable basis for planning, forecasting, production, transportation and billing is described in the lecture in terms of its structure and practical implementation.

Increasing efficiency and improving customer service are goals of current international legislation. They place new demands on future companies in the energy industry and will give rise to new solutions. The previous centrally structured energy industry will be expanded to include decentralized structures in generation and distribution and the products of electricity and gas supply will be increasingly supplemented or replaced by service products.

To implement the energy transition in Germany a model of networked, cellular energy systems with sector coupling is presented, and the necessary systems and components for energy grids and energy storage are discussed in terms of their effective, sustainable and economically successful interaction. This chapter on corporate structures, corporate management and profit and loss acounting rounds off the „Energy Industy“ lecture.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
Attendance time: 30 h

Self-study time: 45 h

Total 75 h = 3 CP

M 12.160 Module: Practical Amplifier Design [M-ETIT-107156]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114179	Practical Amplifier Design	**ST**	**3 CP**	**Ulusoy**

Assessment
Success control takes place in the form of an oral examination of approx. 30 min.

Prerequisites
none

Competence Goal

- The students learn to apply theory of high-speed amplifier design into practical implementation
- They understand critical limitations such as interconnect parasitics, self-heating, power supply and stability, and learn methods how to handle these limitations in practice
- They are familiar with the required fabrication process and the components for the implementation of such amplifiers
- Students acquire hands-on experience on fabrication, assembly and characterization of high-speed amplifiers

Content
In this lecture, the student will be provided with material for theoretical understanding of high-speed amplifier design. The lectures will be organized around the design of a high-speed amplifier, that will be fabricated, assembled and characterized. Following topics will be covered:

- Transistor analysis, and characterization
- Schematic level amplifier design, steady-state analysis and important metrics of amplifier performance
- Layout design and practical limitations with respect to parasitics, interconnections
- Electro/thermal effects and self heating
- Impact of bias circuitry
- Stability analysis
- Assembly and characterization of high-speed amplfiiers

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
The total effort for this lecture is estimated as following:

1. Attendance to the lectures and exercises (15*(2)=30h)
2. Preparation to the lectures and exercises (15*(2)=30h)
3. Amplifier design, assembly, characterization (15h)
4. Preparation to the exam (15h)

Total: 90h

Recommendations
The contents of "M-ETIT-105124 – Radio-Frequency Electronics" is recommanded.

M 12.161 Module: Practical Applications and Modelling of Rarefied Gas Flows [M-CIWVT-108002]

Coordinators: Dr.-Ing. Thomas Giegerich
Organisation: KIT Department of Chemical and Process Engineering
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-115470	Practical Applications and Modelling of Rarefied Gas Flows	**WT**	**6 CP**	**Giegerich**

Assessment
Learning control is an oral examination lasting approx. 20 minutes

Prerequisites
None

Competence Goal
Students will be able to explain basic physical relationships in vacuum science. Building on this, they can design a complex vacuum system correctly and in accordance with specifications.

Content
Basics; vacuum pumps; practical vacuum limits; outgassing and its minimization; cleanliness requirements; vacuum instrumentation; total pressure measurement; residual gas analysis; leak detection; rarefied gas flow; design of vacuum systems; technical specifications; quality in vacuum; examples for large vacuum systems; industrial applications in the process industry.

Module Grade Calculation
The grade of the module ist the grade of the oral exam.

Workload

- Attendance time (Lecture): 60 h
- Homework: 80 h
- Exam Preparation: 40 h

Literature
K. Jousten (Ed.) - Wutz Handbuch Vakuumtechnik, 11. Auflage, Springer, 2013.

M 12.162 Module: Practical Aspects of Electrical Drives [M-ETIT-100394]

Coordinators: Prof. Dr. Martin Doppelbauer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100711	Practical Aspects of Electrical Drives	**WT**	**4 CP**	**Doppelbauer**

Assessment
Success is assessed in a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal
Students understand the function of all components of modern electrical drive systems. They have detailed knowledge of the basic electrical machine types and know the function and physical behavior of loads and other drive components. Students will be able to design electrical drive systems for an application-specific use, taking into account all boundary conditions, and calculate their mechanical and electrical behavior.

Content
The lecture is divided into the following areas

- Drive systems
- electric motors
- Transmission elements
- Drive and load
- Starting, braking, positioning
- Thermal and protection
- Variable speed drives
- Electromagnetic compatibility
- Small drives
- Noise
- Drives with limited movement

Module Grade Calculation
The module grade is the grade of the written examination.

Additional Information
Shift from SoSe to WiSe, does not take place in WiSe24/25 and SoSe25.

Workload
14x lecture + 7x exercises of 1.5 h each = 31.5 h
14x post-processing of lectures à 1 h = 14 h
6x preparation of exercises à 2 h = 12 h
Preparation for the exam = 50 h
Total = 107.5 h (corresponds to 4 CP)

Recommendations
To understand the module, basic knowledge in the field of electrical machines is recommended (acquired, for example, by attending the modules "Electrical Machines and Power Electronics"

M 12.163 Module: Practical Course: Smart Energy System [M-INFO-105955]

Coordinators: Prof. Dr. Veit Hagenmeyer
Organisation: KIT Department of Informatics
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-112030	Practical Course: Smart Energy System	**WT+ST**	**6 CP**	**Jumar**

Assessment
See partial achievements (Teilleistung)

Prerequisites
See partial achievements (Teilleistung)

Competence Goal
After successfully completing the course, students will be able to

- explain and categorize a set of current research issues in modern energy systems
- explain structure and objectives of the KIT Energy Lab and its role in smart grid research
- conduct selected research activities in the field of innovative, application-oriented information, automation and system technology for sustainable energy systems
- isolate a research activity / hypothesis suitable for evaluation in the Smart Energy System Simulation and Control Center
- plan, organize and manage their laboratory / experimental work to fit a given time-frame
- concisely present and defend their research activity and results in front of a peer audience
- structure and writeup their work in research-paper-style
- work in team

Content
As part of the two-week lab course, students work in teams of two on a project topic derived from current research questions of the Smart Energy System Simulation and Control Centre of the Energy Lab (https://www.iai.kit.edu/RPE.php). A list of topics is made available in advance via the Campus Plus Portal. Interested students should indicate their preferences based on which the topics will be assigned.

The group of students typically goes through the following phases: Concept development/experiment preparation, implementation/experiment execution, evaluation/experiment analysis, presentation of results and write-up. The teams will work independently on the assigned project - supervision is provided by research assistants.

The course begins with a joint kick-off event. This includes an organizational introduction, a tour of the Energy Lab and a brief presentation of all project topics. For the following preparation phase students are provided with literature and material to thoroughly familiarize themselves with the topic, research background and literature, plan and prepare experiments etc. In the two lab weeks the experiments are conducted and a scientific writing workshop is held. At the end of the lab course each team has to defend their work in a 10 min presentation followed by 5 minutes of discussion in front of peers and examiners. Additionally, the work needs to be documented in the style of a scientific paper.

Workload
6 credit points corresponds to approx. 180 working hours, of which
- Attendance time / meetings in large and small groups: 10h
- Select and carry out project work: 140h
- Writing a research report and preparing a presentation: 30 hours

Recommendations
- Knowledge of the fundamentals of energy informatics is a prerequisite.
- Knowledge of the fundamentals of electrical engineering and energy technology is required.
- Knowledge of the basics of mechatronics, data analysis and signal processing is helpful.
- Knowledge of power systems or power electronics is helpful.

M 12.164 Module: Practical Machine Learning [M-ETIT-106673]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Prof. Dr.-Ing. Sander Wahls

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Information and Communication Technology (Lab Course)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Lab Course)
Electives (Elective Modules)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113426	Practical Machine Learning	**ST**	**6 CP**	**Heizmann, Wahls**

Assessment
Success is assessed by the submission of the scientific essay (approx. 10 pages) and a 30-minute presentation of the team project.

Prerequisites
none

Competence Goal

- After completing the module, students have in-depth knowledge in the field of machine learning.
- They have in-depth knowledge and an overview of various algorithms and methods in the field of machine learning.
- Students are able to describe different concepts and methods of machine learning and recognize connections between different algorithms.
- They are able to communicate with specialists in related disciplines in the field of machine learning and artificial intelligence and to formulate and evaluate solution approaches for tasks in this area.
- Students will gain practical experience in the field of machine learning through the semester-long team project. In particular, students will benefit from mutual feedback on their theoretical work at the end of the semester.

Content
Remarkable progress has been made in the field of artificial intelligence (AI) in recent years. Machine learning (ML) is a sub-discipline of AI that attempts to develop techniques that enable computers to learn from data. The goal of ML methods is to reliably abstract the underlying model for specific tasks.
This lecture covers the theoretical foundations as well as the basic concepts and techniques of machine learning, with a focus on problem solving and practical application. The course offers the opportunity to explore various ML algorithms and their applications in different areas, including computer vision, natural language processing and data mining.
During the course, you will have the opportunity to work on various application tasks and a group project in which you will apply the concepts you have learned to real-world data sets. You will learn how to use common libraries and tools for ML such as Scikit-Learn, TensorFlow and Keras and apply them to real-world datasets. You will also learn how to evaluate the performance of your models and interpret their results.
The lecture style will be a mix of theory and practical applications, with an emphasis on problem solving and hands-on experimentation. The theoretical part of the lecture will be offered as a block course at the beginning of the semester (early/mid April). Students then have the opportunity to work on a problem from the field of
ML alone or in small groups during the semester and present their results in the form of a scientific essay.
The quality assurance of the essay is carried out through a mutual peer review process in which students benefit from mutual feedback both from a technical point of view and with regard to the presentation of content.
The module covers the fundamentals and concepts of machine learning. Topics covered include the following:

- Introduction to machine learning and its applications.
- Data pre-processing and feature engineering techniques.
- Supervised and unsupervised learning algorithms.
- Deep learning techniques such as Convolutional Neural Networks and Recurrent Neural Networks.
- Transfer learning and Tiny ML.
- Probabilistic ML.
- Evaluation metrics for ML models.
- Hyperparameter tuning and model selection techniques.
- Interpreting the results of ML models.
- ... other interesting topics.

Module Grade Calculation
The module grade results from the team project accompanying the semester and the presentation of the team project with subsequent discussion. The overall impression is assessed. Further details will be provided at the beginning of the course.

Workload

- Attendance of the lectures: approx. 21 hours
- Preparation and follow-up of the lecture: approx. 42 hours
- Team project during the semester: approx. 60 hours
- Peer review of the scientific essays
  and presentation of the team project: approx. 47 hours

Total: approx. 170 hours (6 CP)

Recommendations
Basic knowledge of mathematics and linear algebra (matrices, vectors, etc.)
as well as basic knowledge of Python.

Teaching and Learning Methods
Block lecture (2 SWS) and practical part (by arrangement within the framework of 2 SWS)
The number of semester-accompanying projects is limited to a maximum of 20 per semester.

M 12.165 Module: Practical Tools for Control Engineers [M-ETIT-106780]

Coordinators: Dr.-Ing. Balint Varga
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113628	Practical Tools for Control Engineers	**WT**	**4 CP**	**Varga**

Assessment
The examination takes place in form of other types of examination. It consists of an oral overall examination in the amount of 25 minutes and a homework programming task. The examination includes questions from the lecture slides and the presentation of the homework assignment. The homework must be submitted two weeks before of the oral exam. The overall impression is evaluated.

Prerequisites
none

Competence Goal
1. The students will be able to analyze, structure and formally describe problems in the field of practical control engineering.
2. The students are able to use the necessary tools for software projects with control engineering focus.
3. The students can apply the methods

- Modular software development for control engineering problems
- Model Predictive Controller for practical engineering problems
- Inevitable software engineering tools to able to develop control system

Content

- Practical examples from the control engineering problems and modelling tool
  - Robotics examples
  - Human-machine interaction
  - Automotive
- Control solution concepts for these practical problems
- Software development tool

Module Grade Calculation
The module grade results of the assessment of the oral exam and of the homework programming task. Details will be given during the lecture.

Workload
The workload includes 2 SWS:

1. attendance in lectures and exercises: 15*2 h = 30 h
2. preparation / follow-up: 15*2,5 h = 37,5 h
3. preparation of the homework assignment: 22,5 h
4. preparation of and attendance in examination: 30 h

Sum: 120 h = 4 CR

Recommendations
The contents of the modules “Optimization of Dynamic Systems (ODS)” and “Regelung linearer Mehrgrößensysteme (RLM)” are helpful for the lecture.

M 12.166 Module: Printed and Thin-Film Electronics [M-ETIT-107343]

Coordinators: Prof. Dr. Jasmin Aghassi-Hagmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114417	Printed and Thin-Film Electronics	**WT**	**3 CP**	**Aghassi-Hagmann, Hirtz**

Assessment
The assessment takes place in form of an oral examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
The students

- have a general overview over the topic of Printed Electronics and basic knowledge of the materials, manufacturing methods and applications
- can critically analyze the specific advantages and limits of different methods and approaches of manufacturing printed electronics and can thus evaluate the potential for certain applications in relation to classical electronics
- are able to design basic printed electronic devices and know how to characterize their electronic properties
- are enabled to follow current developments in printed electronics and utilize this knowledge in their own research projects
- are enabled to navigate in applications of multidisciplinary nature (physics, material science, medical engineering)
- have gained relevant expertise for responsible R&D positions in industries such as semiconductor, sensors, automation, instrumentation and measurement technology.

Content
This module gives the student an overview over theoretical and practical aspects of printed and thin-film electronics, including applications. It will introduce the basics of manufacturing methods such as 2D / 3D printing, thin-film methods as CVD, PLD, sputtering, nanolithography, e-beam lithography and lift-off processes. Different classes of materials (e.g. inorganic semiconductors, metals, biomaterials, ceramics) and their properties are discussed. Furthermore, the module will introduce the students to applications of printed electronics in a broad range of fields, as IoT, computing, medical wearables, hybrid sensor devices, bioelectronics, and soft robotics.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload

1. attendance time in lecture: 15*2 h = 30 h
2. preparation/follow-up of the same: 15*2 h = 30 h
3. exam preparation and attendance in the same: 30 h

Total: 90 h = 3 CP

Recommendations
Ideally, this module will be selected by students in combination with “Lab Course Printed and Flexible Electronics”, but this is optional. Basic knowledge in electronic devices, materials of electronics, introductory courses in physics and sensor systems will be beneficial.

M 12.167 Module: Printed Flexible Electronics Lab [M-ETIT-106464]

Coordinators: Prof. Dr. Gerardo Hernandez Sosa
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	pass/fail	Each term	1 term	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113075	Printed Flexible Electronics Lab	**WT+ST**	**6 CP**	**Hernandez Sosa**

Assessment
The examination consists of a written journal article and an oral presentation of the student's work, both given in English. The overall impression is rated.

Competence Goal
The students will gain practical experience in the use of measuring instruments and manufacturing processes for printed electronics technology and the methods for determining the physical and optical properties of the fabricated components. They will be able to fabricate devices such as photodetectors and thin film transistors, and electrically characterize them. Furthermore, they will evaluate measurement results and correlate the fabrication process conditions to the device characteristics. They will be able to critically evaluate their results using adequate measurement tools. In addition, they will have the competence to report the results in written form and to interpret the knowledge gained from the experiments. By working on the practical course in small groups, students also acquire or improve their teamwork skills.

Content
This module is designed to teach students the theoretical and practical aspects of laboratory work in the field of Printed and Flexible Electronics by means of guided and independently conducted practical experiments. In the four experiments, the student will learn to handle real measurement and fabrication technology on the scientific equipment of the institute such as various printers, probe stations, characterization methods and formulation of inks. The module also teaches the competence to write a scientific report, as well as the rules to visualize data sets in a meaningful way.

The working titles of the experiments are:

1) Ink formulation and characterization of Functional Inks
2) Printing optimization of an Inkjet Printer for functional Electronics
3) Fabrication and characterization of a printed sensor device.

Fabrication of a printed thin film transistor device with subsequent electrical characterization

Module Grade Calculation
The module is passed with successful assessment of the written paper and the oral presentation. Details will be given during the lecture. The module is ungraded.

Additional Information
- The lab is limited to a number of 6 participants due to capacity reasons. If necessary, a selection procedure will be carried out. Places will be allocated taking into account the students' academic progress. Details will be announced on the lecture website.
- The Lab course will take place in the clean room Facilities of InnovationLab in Heidelberg. Speyerer str. 4, 69115 Heidelberg where the research laboratories of Prof. Hernandez-Sosa are located.
- The 4 th experiment will take place at KIT Campus North, Institute of Nanotechnology, in the research unit and laboratories of Prof. Jasmin Aghassi-Hagmann.

Attendance of at least 80% is compulsory during the seminar course. Compulsory attendance is necessary for actively contributing to the discussion of the topics presented by all students.

Workload
Due to the self-administration of the groups (max. 3 students):

1 x 5 hours are required for organizational tasks. This includes the attendance of the information event, the attendance of 2 safety briefings (general safety and clean room) as well as the organizational tasks for the individual appointment between the experiment supervisor and the small group.

For the 4 experiments in the module, the workload is calculated as follows:

4 x 5 h familiarization with the topic and literature study on the basics incl. preparation for the entrance examination.

4 x 8 h presence for the execution of experiments at the institute

4 x 1 h discussion of results and learned concepts

4 x 10 h data preparation and visualization

4 x 15 h writing of an individual report on the basis of the measured data and the research question.

4 x 1 h final discussion on the experiment with feedback on the report

4 x 4 h Rework of the report on the basis of the feedback on the report.

Total hours = 181 h = 6 LP

Recommendations
Basic knowledge in the field of conventional and/or organic (opto) electronic or printed devices and sensors is helpful. The course Modern VLSI is recommended but not necessary.

It is recommanded to have started "M-ETIT-100475 Modul: Plastic Electronics / Polymerelektronik"

M 12.168 Module: Process Analysis: Modeling, Data Mining, Machine Learning [M-ETIT-105594]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111214	Process Analysis: Modeling, Data Mining, Machine Learning	**ST**	**3 CP**	**Borchert, Heizmann**

Assessment
Success control takes place in the form of an oral examination lasting approx. 30 minutes.

Prerequisites
none

Competence Goal
Students learn about process engineering issues from the perspective of industrial practice, which are dealt with using physico-chemical modeling and data science methods. Students learn about important relationships in process engineering and can explain these using example processes. They are able to recognize relevant process data and select and apply suitable modelling approaches for their interpretation. Students can carry out practical analyses with process data and apply methods of varying complexity. Students are familiar with the data analysis value chain and are able to select a suitable data analysis method. The focus is on teaching broad methodological knowledge and application using practical examples. Students are referred to specialized in-depth lectures and/or in-depth literature.

Content
Aims of process technology

- Material and energy conversion by means of chemical, mechanical, thermal or biological operations
- Basic operations (selection)
- System examples
- Important variables in process technology (temperature, pressure, composition,...)
- Economic efficiency in the process industry

Data acquisition

- Measured variables and measuring principles (selection)
- Measurement uncertainty

Models of process technology

- Balance equations (selection)
- Constitutive equations (selection)
- Solving balance equations (example in Matlab)
- Parameter uncertainty and estimation
- Data-driven models
- Grey box models / hybrid models

Data analysis

- Requirements for data analysis in the process industry
- Cost-effectiveness and prioritization of process analyses
- Data pre-treatment
- Application of data mining and machine learning
- Online processes

Excursion

- Excursion to BASF Ludwigshafen

Term paper 1: Process model and simulation.

Term paper 2: Identification and analysis.

Homework 3: Predictive maintenance.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
28 hours of teaching,

30 hrs. Homework,

32 hours of preparation and follow-up work, exam preparation and implementation.

Recommendations
Basics in: Mathematics, differential equations, linear algebra, statistics, basic knowledge of Matlab

Literature
Bequette (1998). Process Dynamics: Modeling, Analysis and Simulation. Prentice Hall.

Russel & Novig (2016). Artificial Intelligence - A modern approach. Pearson.

Matlab Documentation (In2019). Mathworks.

M 12.169 Module: Project Management in the Development of Products for Safety-Critical Applications [M-ETIT-104475]

Coordinators: Dr.-Ing. Manfred Nolle
Prof. Dr.-Ing. Eric Sax

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-109148	Project Management in the Development of Products for Safety-Critical Applications	**WT**	**4 CP**	**Nolle**

Assessment
Success is assessed in the form of a written examination lasting 90 minutes.

Prerequisites
none

Competence Goal
Students have a fundamental understanding of all the key terms, methods and processes used in project management, which are applied in the various phases of a project. Students can contribute actively to international projects for the development of technical systems in project management and are also capable of managing smaller projects themselves and leading a project team. They are familiar with the specific requirements wherever product safety is an essential feature. As project managers, students know what is important without being experts in technical matters themselves.

Content
The lecture covers:

1. Terms and fundamental understanding of project management (PM).
2. Segmentation of project implementation into phases with the respective tasks, methods, and processes of PM on the one hand and project realisation on the other.
3. Understanding of different approach models for project realisation, such as plan-based, agile, and hybrid, as well as the implementation of specific requirements that are mandatory for certification in products for safety-critical applications.
4. Understanding and application of typical processes such as
   - Planning / control
   - Organisation / team building / leadership
   - Requirements management
   - Change and configuration management
   - Risk (& opportunity) management
   - Stakeholder management
   - Quality management
   - Contract & claim management
   with references to specific challenges regarding safety

1. understanding of the requirements from the project environment within and outside the organisation initiating the project (norms, standards, processes, certifications, etc.),
2. introduction to social skills such as team building, leading a project team, communication, conflict management, etc.,
3. cultural differences and the resulting challenges in international projects in general.

Examples are demonstrated and explained for the development of products for safety-critical applications.

Exercises in which the acquired knowledge is applied and consolidated:

1. by testing and repeating the knowledge acquired,
2. by carrying out smaller projects,
3. with simulation scenarios and case studies.

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
The workload includes:

1. attendance time in lectures, exercises: 45h

2. preparation and follow-up of the same: 30h

3. exam preparation and participation: 45h

Recommendations
Basic knowledge of hardware and software design is helpful.

Teaching and Learning Methods
Lecture and exercise

M 12.170 Module: Pulsed Power Technology and Applications (Tutorial) [M-ETIT-105596]

Coordinators: Prof. Dr. Georg Müller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
5 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111216	Pulsed Power Technology and Applications (Tutorial)	**ST**	**5 CP**	**Müller**

Assessment
The examination takes place in form of other types of examination. It consists of oral examination and discussioning the amount of 30 min and a written report about the results of the experiments conducted (one report per group) The overall impression is evaluated. Details will be given during the lecture.

Prerequisites
None

Competence Goal
After completing the course students have theoretical and practical experience on the performance of different types of pulse generators and components (e.g. switches), according measurement techniques and data acquisition systems. Additionally, students will know the basics of electroporation process and analytical methods for material characterization (SEM, EDX, XRD).

Content
The course gives an overview of the features and phenomena of pulse power engineering and emerging applications. Modern applications of pulsed power technologies cover a wide range of topics, ranging from applications in the field of renewable energies as a pretreatment method for biomass conversion, material processing for high temperature applications (e.g. concentrated solar power), inertial confinement fusion to medical applications. Beside the electrical engineering aspect, one goal of this course is to provide basic knowledge in bioelectric and in analytical methods for material characterization. Following topics will be addressed:

- Transmission line based generators
- Marx- generators
- Gas filled spark gaps
- Impedance measurements on biological tissue
- Inactivation of biological cells and electro-orientation
- Surface modification by high-power pulsed electron beams
- Material characterization by SEM, EDX and XRD

The tutorials are carried out in groups of two or three students. Since working with high voltages (up to 120 kV) particular emphasis is put on safety. Therefore, part of the tutorial is an obligatory safety briefing conducted at the beginning of the course.

Module Grade Calculation
The module grade results of the assessment of the oral examination and a written report. Details will be given during the lecture.

Workload
After completing the course students have theoretical and practical experience on the performance of different types of pulse generators and components (e.g. switches), according measurement techniques and data acquisition systems. Additionally, students will know the basics of electroporation process and analytical methods for material characterization (SEM, EDX, XRD).

The workload includes:

32 h - attendance in tutorials : 8*4 h=32 h

4 h - safety instruction 1*4 h=4 h

112 h - preparation of each tutorial: 8 x14 h = 112 h

A total of 148 h = 5 CR

Recommendations
Knowledge of the content provided in the lecture `Pulsed Power Technology and Applications` is strongly recommended.

M 12.171 Module: Pulsed Power Technology and Applications (Lecture) [M-ETIT-105595]

Coordinators: Prof. Dr. Georg Müller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111215	Pulsed Power Technology and Applications (Lecture)	**WT**	**3 CP**	**Müller**

Assessment
The success control takes place within the scope of an overall oral examination (approx 20 minutes).

Prerequisites
none

Competence Goal
The students know the common methods of high-power pulse generation including voltage multiplication by stacking, pulsed generators based on transmission lines, different methods of pulse forming and the related measurement technique. Furthermore, students become familiar with actual scientific and industrial applications of pulsed power.

Content

- Introduction: general principles of pulsed power technology
- Basics: static and dynamic breakdown strength of dielectric materials, energy storage (capacitive, inductive, chemical and mechanical), basic circuits and pulse shaping
- Switches: opening and closing switches
- Systems: pulse forming and transmission lines, voltage and power amplification, high-power generators,
- Diagnostics: metrology in pulsed technique (e.g. capacitve/ inductive sensors)
- Applications: surface treatment by charged particle beams, electrodynamic fragmentation, electroporation and bioelectrics, inertial confinement fusion

An excursion to the Institute for Pulsed Power and Microwave Technology (IHM) at KIT CN will give an insight to different pulsed power facilities and it´s specific applications.

Module Grade Calculation
The module grade is the grade of the oral examination.

Additional Information
Following the lecture period the oral exam is offered by appointment.

Workload
Each credit point corresponds to approximately 25-30 hours of work (by the student). Approximately workload corresponds to:

1. Presence time in lectures (2 SWS: 28 h)
2. Preparation / follow-up of the lecture (30 h)
3. Preparation for oral exam and presence (24 h)

Total 82 h equals to 3 credit points.

M 12.172 Module: Quantum Detectors and Sensors [M-ETIT-105606]

Coordinators: Prof. Dr. Sebastian Kempf
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111234	Quantum Detectors and Sensors	**WT**	**6 CP**	**Kempf**

Assessment
The assessment of success takes place in the form of a written examination lasting 120min. The grade corresponds to the result of the written examination.

Prerequisites
None

Competence Goal
Students know the basics and fundamentals of quantum detectors and sensors and understand how quantum technology can be used to design and realize devices those performance reaches far beyond the limits of any classical sensor or detector. They know the basic components of quantum sensors and detectors, in particular in the field of superconducting quantum technology, and are able to analyze the operation of such detectors and sensors on the basis of circuit diagrams. Students are able to develop quantum sensors and detectors for given applications and know how to consider special requirements in a concrete component.

Content
This module provides a comprehensive overview of the basics and physical principles of quantum detectors and sensors and discusses in detail how quantum technology can be used to design and realize detectors and sensors with performance that reaches far beyond the limits of any classical sensor or detector. The discussion includes particularly an introduction to the basic components of quantum sensors and detectors, especially in the field of superconducting quantum technology, and their fabrication. Using simplified circuit diagrams, the functionality and operation of quantum detectors and sensors such as superconducting quantum interference devices, low-temperature detectors, noise thermometers or superconducting radiation detectors is analyzed. Furthermore, methods and simple models are developed allowing to realize quantum sensors and detectors that are matched to given applications. Within this context, typical applications of quantum detectors and sensors are also discussed.

The tutorial is closely related to the lecture and deals with special aspects concerning the development of quantum detectors and sensors. In particular, the development and system integration of quantum detectors and sensors for applications in precision metrology, particle detection or applied sciences is discussed by means of exercises.

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
A workload of approx. 180h is required for the successful completion of the module. This is composed as follows:

- Attendance time in lectures and exercises: 21*1.5h + 7*1.5h = 42h
- Preparation and follow-up of lectures: 21*3h= 63h
- Preparation and follow-up of tutorials: 7*5h= 35h
- Preparation for the exam: 40h

Recommendations
Successful completion of the module "Superconductivity for Engineers" is recommended.

M 12.173 Module: Quantum Engineering [M-ETIT-106954]

Coordinators: Prof. Dr. Sebastian Kempf
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Fundamentals)
Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113909	Quantum Engineering	**ST**	**6 CP**	**Kempf**

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal
The students will be able to analyze, structure and formally describe problems in the field of quantum engineering. They will particularly be able to understand the difference between classical and quantum engineering. They will be able to analyze, design and implement concepts, technology, circuits and algorithms in the fields of quantum computing, quantum communication and quantum sensing. Moreover, the students will be able to critically evaluate existing concepts, methods, technologies, and circuits regarding complexity, suitability and applicability.

Content
This module introduces the emerging field of quantum engineering. For this, the module (1) addresses physics basics and the related mathematical framework, (2) discusses recent realizations of engineered quantum systems for the sub-fields of quantum computing, quantum communication and quantum sensing, (3) introduces hardware designs and practical realizations, (4) summarizes quantum information processing aspects, and (5) discusses real implementations and applications.

The module particularly addresses:

1. Basics of quantum mechanics and the mathematical frameworks (state vector, postulates of quantum mechanics, bra-ket-formalism, superposition, entanglement and Bell inequalities, probability density matrix formalism, pure and mixed states, squeezing)
2. Description of light and photons
3. Decoherence and coherence of quantum systems, quantum errors
4. Basics of quantum computers: quantum gates, gate-based quantum computing architecture
5. Quantum algorithms: quantum random number generator, quantum penny flip, Deutsch-Josza-Algorithm, Grover algorithm, quantum Fourier transform, Shor algorithm, quantum phase estimation, HHL algorithm
6. Quantum error correction: Concepts and architectures
7. Hardware realization of quantum bits: DiVincenzo criteria, Cooper pair box, phase qubit, flux qubit, transmon, quantum dots etc.
8. Examples of recent quantum computing architectures
9. Quantum sensing: Atomic clocks, NV centers, SQUIDs, parametric amplifiers, quantum imaging, interaction free quantum measurements, interferometry at the quantum limit, quantum logic spectroscopy,
10. Quantum communication: Quantum cryptography, quantum key distribution, quantum teleportation, entanglement swapping, quantum repeater and quantum networks

As the lecture intends to include recent advancements of the field of quantum engineering, the actual content might slightly differ from the announced topics.

The exercises are closely related to the lecture and deal with special aspects concerning quantum engineering. Moreover, it deepens the student's knowledge by discussing various examples.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
The workload includes:

1. Attendance in lectures: 15 * 3 h = 45 h
2. Attendance in exercices: 15 * 1 h = 15 h
3. Preparation and follow up of lectures and exercises: 90 h
4. Preparation of and attendance in examination: 30 h

A total of 180 h.

Recommendations
Basic knowledge of quantum mechanics and the related mathematical framework, as taught, for example, within the module “M-ETIT-106264 – Introduction to Quantum Information Processing” is strongly recommended.

Teaching and Learning Methods
“Lecture Quantum Engineering” (3 SWS) and “Exercises to Quantum Engineering” (1 SWS)

M 12.174 Module: Quantum Machine Learning [M-ETIT-105889]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Prof. Dr.-Ing. Sander Wahls

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111838	Quantum Machine Learning	**WT**	**3 CP**	**Heizmann, Wahls**

Assessment
Success control takes place in form of an oral examination lasting approx. 20 minutes.

Prerequisites
none

Competence Goal

- After completing the module, students will have a sound knowledge of quantum machine learning.
- They have in-depth knowledge and an overview of various algorithms and methods in the field of quantum machine learning.
- Students are able to describe different concepts and methods of Quantum Machine Learning and recognize connections between different algorithms.
- They are able to communicate with specialists in related disciplines in the field of machine learning and artificial intelligence and to formulate and evaluate solution approaches for tasks in this area.

Content
Remarkable progress has been made in the field of artificial intelligence (AI) in recent years. Machine learning (ML) is a sub-discipline of AI that attempts to develop techniques that enable computers to learn from data. The aim of ML methods is to reliably abstract the underlying model for specific tasks. Quantum computing describes information processing with devices based on the laws of quantum theory. Based on the success of ML and quantum computing to date, it can be expected that both technologies will play a huge role in digital data processing in the future. It is therefore exciting to find out how these two techniques can be combined to provide better and more reliable solutions for different tasks.
Quantum Machine Learning (QML) is an interdisciplinary field of research that encompasses physics, mathematics, computer science and electrical engineering. It is dedicated to the use of quantum computers to calculate machine learning algorithms. QML methods help to improve classical ML methods by utilizing the advantages of quantum computing. The use of QML not only enables previous tasks to be solved more quickly, but also allows more aspects of the natural world to be integrated into existing AI methods.
The module covers the fundamentals and concepts of quantum machine learning.

Topics covered include the following.

- Basic concepts of quantum mechanics
- From bits to qubits
- Quantum computers and quantum circuits
- Quantum information theory
- Quantum signal processing
- Repetition of classical machine learning
- Quantum algorithms
- Quantum classification and regression
- Quantum deep learning
- ... other interesting topics.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
Lecture attendance: approx. 23 hours
Lecture preparation and follow-up: approx. 23 hours
Exam preparation: approx. 40 hours
Total: approx. 86 hours (3 CP)

Recommendations
Knowledge of machine learning and stochastics
as well as quantum computing is an advantage, but not a prerequisite.

M 12.175 Module: Radar Systems Engineering [M-ETIT-100420]

Coordinators: Prof. Dr.-Ing. Marwan Younis
Prof. Dr.-Ing. Thomas Zwick

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100729	Radar Systems Engineering	**WT**	**6 CP**	**Younis, Zwick**

Assessment
Success control is carried out as part of a written overall examination (120 minutes) of the selected course, which in total meets the minimum requirement for LP.

Prerequisites
none

Competence Goal
Students can name the basic radar principles and explain how they work, their primary uses and their advantages and disadvantages. They are able to characterize the basic characteristics and mechanisms of propagation of electromagnetic waves and to apply the relevant equations. You can evaluate the influence of various system parameters on accuracy, resolution, false alarm rate, etc. and optimize systems. You can describe different radar system configurations (CW, FMCW, pulse, SAR) and apply the relevant radar signal processing methods. They are especially able to use and use the technologies and system configurations for the radars of the future for surveillance, automotive and industrial applications for research and development. In this lecture system technology is specifically taught.

Content
Based on electromagnetic field theory, the lecture teaches the basics of radar principles and their system technology. An insight into the system hardware is given and processing techniques are presented. All relevant, known radar systems (CW, FMCW, pulse and synthetic aperture radar) are described in detail. The system technology for the radars of the future is specifically dealt with. The reflective properties of radar targets are analyzed for their classification. In particular, polarimetry is taught. In this lecture, students learn how system technology contributes to the implementation of a radar system.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
Each credit point corresponds to approximately 25-30 hours of work (of the student). This is based on the average student who achieves an average performance. The workload includes:

Attendance study time lecture: 44 h

Attendance time computer exercise: 16 h

Self-study time including exam preparation: 120 h

A total of 180 h = 6 LP

M 12.176 Module: Radio Frequency Integrated Circuits and Systems [M-ETIT-105123]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-110358	Radio Frequency Integrated Circuits and Systems	**ST**	**6 CP**	**Ulusoy**

Assessment
The success criteria will be determined by an oral examination (approx. 20-30 min.)

Competence Goal

- The students acquire a comprehensive understanding in the design of monolithic integrated circuits for millimeter-wave frequencies, and they can apply the acquired knowledge using modern design tools.
- They have a good understanding of the critical performance parameters of high-frequency circuits such as stability, power gain and efficiency and reflection coefficient.
- They can describe the benefits and disadvantages of modern transistor technologies for millimeter-wave applications.T
- They can identify potential applications of integrated millimeter-wave circuits and understand the specific requirements of each application.
- They are familiar with basic elements of a high-frequency system, which consists of linear and non-linear circuits, low-noise and power amplifiers, as well as oscillators, switches and frequency converting circuits such as frequency multipliers and mixers.

Content
In this lecture the theory and the design methodology of monolithic integrated millimeter-wave circuits will be studied in detail. The focus of the lecture is on the active linear and non-linear circuits in high-frequency frontends up to an application frequency of 300 GHz. In addition to this, fundamental topics such as impedance matching, stability, performance parameters of high-frequency transistors, and properties of active and passive circuit elements will be studied in detail. The operation principal of critical building blocks of a millimeter-wave system will be introduced including low-noise and power amplifiers, mixers, oscillators and switches. In the workshop, the students will have the chance to apply the acquired theoretical knowledge to design a millimeter-wave frontend using state-of-the art integrated circuit technology.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
1. Attendance to the lectures (15*(2)=30h)
2. Attendance to the exercises (15*(2)=30h)
3. Preparation to the lectures and exercises (15*(2+2)=60h)
4. Preparation to the oral exam (40h)

Total: 160h

Recommendations
The lecture materials to „Grundlagen der Hochfrequenztechnik“ and „Halbleiterbauelemente“ are recommended.

M 12.177 Module: Radio Receivers [M-ETIT-103241]

Coordinators: Prof. Dr. Friedrich Jondral
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-106431	Radio Receivers	**WT**	**3 CP**	**Jondral**

Assessment
Success ontrol takes place in the form of an oral examination lasting approx. 20 minutes.

Prerequisites
none

Competence Goal
Students will be able to understand the full functionality of radio receivers, write specifications and design radio receivers from a systems theory perspective.

Content
The course expands on the issues covered in the Communications Engineering I lecture. The focus here is on the detailed treatment of complex receiver technologies, which particularly concern the interaction between analog and digital signal processing.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload

- Attendance time lecture: 15 * 2 h = 30 h
- Lecture preparation/follow-up: 15 * 4 h = 60 h
- Exam preparation and attendance in the same: offset in preparation/follow-up work

Total: 90 h

Recommendations
Knowledge of basic engineering mathematics including integral transformations and probability theory as well as basic knowledge of communications engineering.

M 12.178 Module: Radio-Frequency Electronics [M-ETIT-106955]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Fundamentals)
Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113910	Radio-Frequency Electronics	**WT**	**6 CP**	**Ulusoy**

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal

- The students have a comprehensive understanding of the theory and the basic design methodology of RF and microwave circuits up to 300 GHz.
- They understand the limitations of active and passive circuit elements at high frequencies and their impact on the applications.
- They understand the limitations and how linear network theory is applied at higher frequencies.
- The students can apply the acquired theoretical knowledge to modern RF design problems.

Content
In this lecture, the theory and design methodology of RF electronic circuits will be studied in detail. The focus of the lecture is on the fundamentals of active and passive linear circuits. The important topics are:

- Phasor analysis and resonance,
- Electromagnetic theory, transmission lines and waveguides,
- Impedance matching networks,
- Two-port parameters of RF components and microwave network analysis,
- Feedback and stability analysis,
- High-frequency behavior of basic amplifier circuits, RF amplifiers design techniques,
- Microwave power dividers, couplers and filters

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
The total effort for this lecture is estimated as following:

1. Attendance to the lectures (15*(3)=45h)
2. Attendance to the exercises (15*(1)=15h)
3. Preparation to the lectures and exercises (17*(3+1)=68h)
4. Preparation to the exam (52h)

A total of 180h

Recommendations
Basic knowledge of linear electrical networks and electronic circuits is recommended (e.g. M-ETIT-106417 – Lineare Elektrische Netze; M-ETIT-104465 – Elektronische Schaltungen).

M 12.179 Module: Real Time Control of Electrical Drives [M-ETIT-105916]

Coordinators: Dr.-Ing. Andreas Liske
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111898	Real Time Control of Electrical Drives	**WT**	**6 CP**	**Liske**

Assessment
Success is assessed in an overall oral examination (approx. 30 minutes).

Competence Goal
Students are familiar with common methods for modeling electrical machines (permanent magnet and separately excited synchronous machines, asynchronous machines). In addition to analytical fundamental wave models, they are also familiar with the method of mapping non-linear, real machine behavior using flux maps. They are able to determine the parameters required for the models using methods for offline or online parameter identification. Students will be able to design standard and advanced control methods for electrical drives (current, voltage, speed and torque control). In addition to classic cascade control, this includes in particular flow-map-based control, direct control methods (DSR), model-based predictive control (MPC) and adaptive control methods. The students know how the time sequences in an inverter-controlled drive system are connected, which requirements result from this and what solutions look like in concrete terms. They gain concrete experience with real signal processing hardware and real drive test benches through the concrete implementation of the content covered in the module on test afternoons.

Content
Electric drive technology is increasingly dominated by inverter-fed three-phase machines. Particularly in the field of electromobility, the electric drive train is a central component alongside the battery. The energy efficiency and reliability of electric drives are achieved to a large extent through fast, precise and load-adapted control of the electrical energy.

The lecture presents common and modern control methods that enable highly dynamic and energy-efficient current, voltage, position, speed or torque control. The non-linear modeling required for the control methods of highly utilized, efficient three-phase machines is discussed analytically, based on characteristic diagrams and also using adaptive methods.

The module also focuses on the implementation and real-time signal processing of inverter-controlled drive systems. Hardware requirements, common programming and development workflows as well as specific implementation options are explained using examples and practiced in practical experiments.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
Each credit point corresponds to approx. 25-30 hours of work (by the student). This is based on the average student who achieves an average performance.

56h = 22x lecture à 2h + 3x exercise à 4h

21h = 21x post-processing of lecture à 1 h

12h = 3x preparation of exercises of 4 h each

80h = preparation for the exam

Total = 169 h (corresponds to 6 LP)

M 12.180 Module: Refrigeration B - Foundations of Industrial Gas Processing [M-CIWVT-104354]

Coordinators: Prof. Dr.-Ing. Steffen Grohmann
Organisation: KIT Department of Chemical and Process Engineering
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-108914	Refrigeration B - Foundations of Industrial Gas Processing	**ST**	**6 CP**	**Grohmann**

Assessment
The examination is an oral examination with a duration of about 30 minutes (section 4 subsection 2 number 2 SPO).

Prerequisites
None

Competence Goal
Understanding the principles of different processes for gas liquefaction and gas separation; Analysing processes in order to reveal the sources of energy demand; Applying the principles of thermodynamics of mixtures and analysing the states of fluids in rectification columns; Assessing the potential of technical concepts from a thermodynamic point of view

Content
Gas liquefaction processes, process analyses, refrigerators and mixed-refrigerant cycles, gas separation by low-temperature rectification, air separation and extraction of noble gasses, processing and separation of natural gas, ethylene production, processing of H2-enriched gas mixtures, storage and transport of liquefied gasses

Module Grade Calculation
The grade of the oral examination is the module grade.

Workload

- Attendance time (Lecture): 45 h
- Homework: 45 h
- Exam Preparation: 90 h

M 12.181 Module: Renewable Energy-Resources, Technologies and Economics [M-WIWI-100500]

Coordinators: Prof. Dr. Russell McKenna
Organisation: KIT Department of Business and Economics
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-100806	Renewable Energy-Resources, Technologies and Economics	**WT**	**3 CP**	**Jochem**

Assessment
The assessment consists of a written exam according to Section 4(2), 1 of the examination regulation.

Prerequisites
None

Competence Goal
The student:

- understands the motivation and the global context of renewable energy resources.
- gains detailed knowledge about the different renewable resources and technologies as well as their potentials.
- understands the systemic context and interactions resulting from the increased share of renewable power generation.
- understands the important economic aspects of renewable energies, including electricity generation costs, political promotion and marketing of renewable electricity.
- is able to characterize and where required calculate these technologies.

Content
1. General introduction: Motivation, Global situation
2. Basics of renewable energies: Energy balance of the earth, potential definition
3. Hydro
4. Wind
5. Solar
6. Biomass
7. Geothermal
8. Other renewable energies
9. Promotion of renewable energies
10. Interactions in systemic context
11. Excursion to the “Energieberg” in Mühlburg

Workload
The total workload for this course is approximately 105.0 hours. For further information see German version.

Literature
Elective literature:

- Kaltschmitt, M., 2006, Erneuerbare Energien : Systemtechnik, Wirtschaftlichkeit, Umweltaspekte, aktualisierte, korrigierte und ergänzte Auflage Berlin, Heidelberg : Springer-Verlag Berlin Heidelberg.
- Kaltschmitt, M., Streicher, W., Wiese, A. (eds.), 2007, Renewable Energy: Technology, Economics and Environment, Springer, Heidelberg.
- Quaschning, V., 2010, Erneuerbare Energien und Klimaschutz : Hintergründe - Techniken - Anlagenplanung – Wirtschaftlichkeit München : Hanser, Ill.2., aktualis. Aufl.
- Harvey, D., 2010, Energy and the New Reality 2: Carbon-Free Energy Supply, Eathscan, London/Washington.
- Boyle, G. (ed.), 2004, Renewable Energy: Power for a Sustainable Future, 2ndEdition, Open University Press, Oxford.

M 12.182 Module: Research Seminar Cooperative and Haptic Human-Machine Interactions [M-ETIT-107596]

Coordinators: Dr.-Ing. Balint Varga
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114891	Research Seminar Cooperative and Haptic Human-Machine Interactions	**ST**	**6 CP**	**Varga**

Assessment
Success control takes place in the form of another type of examination. It consists of an the preparation of an research paper in IEEE-conference style (4-6 IEEE pages) and an oral presentation (approx. 15 min). The overall impression is rated.

Prerequisites
none

Competence Goal
Upon completion of the seminar, students will be able to:

- independently develop and test research questions in the field of MPC, Shared Control and haptics methodically.
- conceive, conduct and critically evaluate practical experiments with MPC and robotics systems.
- clearly and methodically document and present complex technical and user-centered research results in research paper form and presentation.

Content

1. Theoretical Foundations and Interactive Learning: The course begins with an initial phase dedicated to building a theoretical foundation in key areas such as Model Predictive Control (MPC), Shared/Cooperative Control, robotics, and haptics. This is achieved through a mix of short lectures, workshops, and discussions of current research papers. A significant emphasis is placed on active learning, where students prepare and present some of topics themselves, moving away from traditional lectures towards an interactive, group-based discourse.
2. Practical, Hands-on Research Project: Following the theory phase, students engage in an intensive, multi-week project in 2-person groups. Each group formulates a specific research question, designs an experimental plan, and then independently implements their project in a laboratory setting. This practical work involves tasks like programming MPC controllers in MATLAB/Python/C++, integrating haptic devices, conducting experiments or simulations, and performing statistical analysis on the collected data, all while receiving regular coaching and peer feedback.
3. Scientific Communication and Conference Simulation: The final phase of the course is focused on the dissemination of research findings in a professional, scientific manner. Students learn the principles of academic writing by preparing a project report in the format of a 4-6 page conference paper (in IEEE format). The seminar culminates in a simulated academic conference where each group presents their work in a 15-minute talk, followed by a Q&A session, thereby mirroring the complete research process from conception to publication.

Module Grade Calculation
The module grade results of the assessment of the written paper and the oral presentation. Details will be given during the lecture.

Additional Information
For capacity reasons, the seminar is limited to 12 students. If necessary, a selection procedure will be carried out. Places will be allocated according to the degree programm, the student's academic progress and waiting time. Details will be announced on the course website.

Workload
The workload includes:

1. Attendance in preparational lectures and seminars (2 SWS): 15*2 h = 30 h
2. Attendance in workshops, on-site experiments, and the in-house conference (2 SWS): 15*2 h = 30 h
3. Formulation and Answering the research question 30 h
4. Practical research implementation: 60 h
5. Writing the research paper and presentation: 30 h

A total of 180 h = 6 CR

Recommendations
The contents of the modules “Optimization of Dynamic Systems (ODS)” or "Optimal Control (OC)", “Regelung linearer Mehrgrößensysteme (RLM)”, and Practical Tool for Control Engineers (PTCE) are strongly recommended, since they are helpful for the seminar.

Teaching and Learning Methods
Preparational lectures 1 SWS, seminars 1 SWS, workshops and on-site experiments 2 SWS

M 12.183 Module: Robotic Intelligence: Navigation and Mobile Manipulation [M-MACH-107621]

Coordinators: Prof. Dr.-Ing. Arne Rönnau
Organisation: KIT Department of Mechanical Engineering
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Fundamentals)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-114952	Robotic Intelligence: Navigation and Mobile Manipulation	**ST**	**6 CP**	**Rönnau**

Assessment
See module component

Prerequisites
See module component

Competence Goal
Students acquire a solid understanding of fundamental principles and current methods in robotics, including navigation, legged locomotion, and manipulation with mobile systems. They can apply suitable techniques from machine learning, simultaneous localisation and mapping (SLAM), decision-making, and collision-free path planning to autonomously traverse complex real-world environments—for example, with a multi-legged robot. A particular emphasis is placed on the direct physical interaction of legged robots with terrain and obstacles. Students learn to design suitable machine learning approaches for locomotion and to train them in dynamic simulation environments, enabling robots to perform movements on simple, flat terrain and to overcome complex obstacles such as stairs.

Furthermore, students gain knowledge of various manipulation techniques, including robotic grasping and object handling. They can analyse and compare both classical manipulation approaches and modern machine-learning-based methods, and evaluate their performance. Combined, these capabilities enable mobile robots to navigate challenging terrain and autonomously manipulate objects at their destination.

Content
The lecture *Robotic Intelligence: Navigation and Mobile Manipulation* provides fundamental and advanced concepts for developing intelligent, autonomous robots for complex environments. The focus is on methods for navigation, legged locomotion, and mobile manipulation, as well as their integration into real robotic systems. Covered topics include sensor fusion, localization and mapping (SLAM), decision-making, and collision-free motion and path planning. Additionally, machine learning techniques for controlling legged robots in challenging terrain and for direct physical interaction with the environment are introduced. The course further addresses concepts of mobile manipulation, enabling robots to safely grasp, handle, or interact with objects and their surroundings. Both classical and modern, learning-based approaches are presented and discussed. By combining these methods, robotic systems can achieve a high level of autonomy. The lecture integrates theoretical foundations with application-oriented methods and enables students to develop and critically assess autonomous robotic systems and their methods for real-world scenarios.

An important component of the course is the practical group work conducted in the *Machine Intelligence and Robotics Lab (MaiRo)*, where students solve tasks using real sensors and robots. They develop both technical competencies and teamwork skills to address complex navigation and manipulation challenges of mobile systems.
The topics include:

- Fundamentals of machine learning and sensor fusion
- Simultaneous Localization and Mapping (SLAM) in unknown environments
- Approaches for motion planning and collision avoidance
- Intelligent decision-making and task planning
- Reinforcement learning for the locomotion of legged robots
- Machine learning approaches for traversing and coping with complex obstacles
- Manipulation of objects in static scenarios
- Mobile manipulation in complex environments

By the end of the course, students can design, implement, and integrate methods for navigation and mobile manipulation. These approaches enable mobile systems to autonomously navigate through complex, dynamic environments, apply learned locomotion strategies, and perform various manipulation tasks.

Module Grade Calculation
Written exam: duration 60 minutes. A grade bonus can be achieved by successfully completing the exercises.

Additional Information
See further details on course organization at: teaching.imi.kit.edu and in ILIAS.

Workload
180 hours total, including approximately:

- **30h** of in-person lecture time
- **110h** of exercises and group work
- **40h** for exam preparation and participation in the written exam

Teaching and Learning Methods
Vorlesung, Übung und Gruppenarbeit

M 12.184 Module: Robotic Intelligence: Perception and Scene Understanding [M-MACH-107558]

Coordinators: Prof. Dr.-Ing. Arne Rönnau
Organisation: KIT Department of Mechanical Engineering
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-114813	Robotic Intelligence: Perception and Scene Understanding	WT+ST	6 CP	Rönnau

Assessment
See module component

Prerequisites
See module component

Competence Goal
The student will understand the fundamental principles and state-of-the-art methods for robotic perception and scene understanding. They will gain the ability to select and apply suitable machine learning and computer vision techniques to interpret complex real-world environments, from raw sensory data to structured scene representations. The student will be able to design perception pipelines that combine multiple sensing modalities, enabling robust object recognition, tracking, segmentation, and 3D scene analysis. Furthermore, they will be able to critically assess and adapt these methods for robotic applications that require situational awareness and informed decision-making.

Content
The lecture *Robotic Intelligence: Perception and Scene Understanding* combines modern AI-based perception methods with the engineering skills needed to apply them in robotics. It covers the full perception chain, from acquiring and calibrating sensor data to interpreting scenes and integrating perception modules into robotic systems, like mobile walking robots.
Students will first build a foundation in machine learning and computer vision, covering image classification, object detection, tracking, and segmentation in both 2D and 3D. The course then explores advanced deep learning architectures, including Vision Transformers, multimodal perception systems, and end-to-end multimodal machine learning pipelines. Special emphasis is placed on techniques for integrating vision, depth, and other sensing modalities, enabling robust perception in challenging real-world environments.

A key part of the course is practical group work in the *Machine Intelligence and Robotics Lab (MaiRo)*, where students solve hands-on tasks using real sensors and robots, building both technical competence and teamwork skills.

Topics include:

- Fundamentals of machine learning and computer vision for robotics
- Vision Transformers and advanced deep learning architectures
- Object detection, tracking, and segmentation (2D and 3D)
- Instance, semantic, and panoptic segmentation
- Multimodal sensing, fusion, and end-to-end multimodal ML
- World models and their role in robotics
- Sensor modeling, calibration, and perception system integration

By the end of the course, students will be able to design, implement, and integrate intelligent robotic perception systems that enable mobile robots to understand and act within complex, dynamic environments.

Module Grade Calculation
Written exam: duration 60 minutes. A grade bonus can be achieved by successfully completing the exercises.

Workload
180 working hours, of which approx:

- 30h for attendance time in lectures
- 110h for exercises and group work
- 40h for exam preparation and participation in the written exam

Teaching and Learning Methods
Lecture, exercise and group work

M 12.185 Module: Robotic Winding Technology for Superconducting Wires Lab [M-ETIT-107135]

Coordinators: Prof. Dr. Tabea Arndt
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	German/English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114158	Robotic Winding Technology for Superconducting Wires Lab	**WT**	**6 CP**	**Arndt**

Assessment
Success control takes place in the form of other types of examination. It takes place in form of 3 experimental works. The overall performance gives the grade.

Prerequisites
The module "Superconducting Magnet Technology" must be passed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-ETIT-106684 - Superconducting Magnet Technology must have been passed.

Competence Goal

- The students have a solid knowledge of architecture and design aspects of windings applied in magnets, coils and energy technology devices.
- The students have a clear understanding of opportunities, benefits and limitations of superconducting windings.
- The students are able to perform the required design estimations.

Content
As the superconducting wires become increasingly mature and powerful, using superconductivity in a variety of applications of electrical engineering is of rising interest and benefit, too. Most components and applications require dedicated coil windings. This module focuses on

- Wire properties to consider in advance
- Specification and design of windings (planar & non-planar geometries) & important sensing (current, voltage, temperature)
- Description of the winding process
- In advance testing of winding process in „digital twin“ environment (RobotStudio to control two 6-axis robots and a 2-axis positioner)
- Preparation of simple coils (by Copper-wire or by High-Temperature Superconductor wire; depending on the realized funding options of the module).
- Vacuum-Pressure-Impregnation of coils to prepare a robust sample device
- Performance test (Voltage vs. Current) immersed in liquid nitrogen
- Detailed report on work performed and achieved results.

Part of the (experimental) work has to be conducted in working groups.

The contents may be adopted without further announcement.
Additional materials for reference and study will be offered partly on ILIAS.

Module Grade Calculation
The assessment of the oral contributions and the written reports are included in the module grade. Further details will be provided at the beginning of the course.

Workload
180h

1. Attendance in teaching/ exercises: 48 h
2. Design/ Preparation/ Documentation/ Clean-up: 120 h
3. Reporting (post-process): 12 h

M 12.186 Module: Robotics II - Humanoid Robotics [M-INFO-107123]

Coordinators: Prof. Dr.-Ing. Tamim Asfour
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-114152	Robotics II - Humanoid Robotics	**ST**	**3 CP**	**Asfour**

Assessment
See partial achievements (Teilleistung)

Prerequisites
See partial achievements (Teilleistung)

Competence Goal
Students can explain the challenges and goals of humanoid robotics research, particularly regarding the implementation of complex sensorimotor and cognitive abilities in humanoid robots. They are familiar with the history of the field and understand how biomechanical models of the human body inform humanoid robot design.

Students have comprehensive knowledge of human and robotic grasping and manipulation. They can analyze human grasping and manipulation strategies, understand taxonomies, and evaluate different computational approaches for grasping and manipulation. They understand the challenges in transferring concepts from human studies to humanoid robots.

Students understand fundamentals of learning from human demonstration and imitation learning. They can explain the learning from demonstration cycle and methods for learning generalized task representation, especially task constraints from demonstrations, and how learned behaviors are reproduced on robots.

Students can describe cognitive and AI-based architectures for humanoid robots, including approaches to address the signal-to-symbol gap and human-inspired memory architectures that enable intelligent robot behavior.

Content
This lecture addresses the implementation of complex sensorimotor and cognitive abilities in humanoid robots, with humans serving as inspiration. The lecture begins by addressing the motivation for humanoid robotics. The history of humanoid robotics and biomechanical models of the human body that inform robot design are covered, along with some mechatronic principles underlying humanoid robot systems.

The first main topic comprehensively addresses grasping and manipulation. After covering fundamentals, the lecture discusses neuroscientific insights and concepts like grasp phases and grasping synergies to reduce control complexity. Taxonomies are presented as structured frameworks for understanding the space of possible constraints in grasping and manipulation. The lecture addresses computational approaches for grasping known, familiar, and unknown objects, covering both classical methods and modern learning-based approaches using deep learning and vision-language-action models.

The second main topic covers learning from demonstration and imitation learning. After introducing the topic, fundamentals are presented including the learning from demonstration cycle. The lecture emphasizes learning task models on both symbolic/semantic and subsymbolic/sensorimotor levels. While methods for capturing human demonstrations are briefly covered, the focus is on semantic segmentation of human demonstrations and learning of task constraints that capture the task and allow generalization. Movement primitives are presented as efficient representations that allow robots to generalize learned behaviors to new situations.

The lecture concludes with cognitive and AI-based architectures for humanoid robots, discussing state-of-the-art approaches, methods to address the signal-to-symbol gap, and human-inspired memory architectures that enable intelligent robot behavior.

Workload
Lecture with 2 SWS, 3 ECTS.
3 ECTS corresponds to approx. 90 hours, thereof:
approx. 15 * 2h = 30h attendance time
approx. 15 * 2h = 30h self-study prior/after the lecture
approx. 30h preparation for the exam and exam itself

Recommendations
Having visited the lecture on Robotics I – Introduction to Robotics is recommended

M 12.187 Module: Robotics III - Sensors and Perception in Robotics [M-INFO-107130]

Coordinators: Prof. Dr.-Ing. Tamim Asfour
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-114155	Robotics III - Sensors and Perception in Robotics	**ST**	**3 CP**	**Asfour**

Assessment
See partial achivements (Teilleistung)

Prerequisites
See partial achivements (Teilleistung)

Competence Goal
Students can explain the main sensor principles used in robotics and distinguish between proprioceptive sensors (encoders, force/torque sensors, IMUs) and exteroceptive sensors (proximity, range, visual, and tactile sensors). They understand and can characterize sensor properties such as resolution, range, accuracy, and bandwidth, and can explain the principles of analog-to-digital conversion. Students are able to propose, analyze, and justify suitable sensor concepts for specific robotic tasks, taking into account trade-offs between different sensor modalities.

Students can apply fundamental perception methods for robotic applications. This includes robot vision techniques such as feature detection and matching, object detection and recognition, and semantic segmentation, including deep learning-based approaches. They can describe and analyze perception methods for manipulation tasks, including object pose estimation, grasp detection, active vision strategies, and haptic exploration. In addition, students can explain point cloud processing methods for registration and surface reconstruction, as well as principles of simultaneous localization and mapping (SLAM).

Content
This lecture complements the lecture “Robotics I” and provides a comprehensive overview of sensors and perception methods in robotics. It is divided into two main parts.

The first part introduces fundamental concepts of perception in robotics, including the distinction between sensation and perception within the perception-cognition-action loop. It covers sensor fundamentals, such as sensor characteristics (resolution, range, accuracy, bandwidth), analog-to-digital conversion, and common sensor classification schemes. Proprioceptive sensors for measuring the robot’s internal state are discussed, including encoders (optical, magnetic, absolute, and incremental), force/torque sensors, and inertial measurement units (IMUs). The lecture then covers exteroceptive sensors, including proximity sensors, range sensors (LiDAR, time-of-flight cameras, ultrasonic sensors), visual sensors (monocular, stereo, RGB-D cameras), and tactile sensing technologies (capacitive, resistive, and optical).

The second part focuses on processing and interpreting sensor data. Topics in robot vision include image representation, feature detection and matching, object detection and recognition, semantic segmentation, and corresponding deep learning approaches. Point cloud processing is covered with emphasis on data structures, registration, surface reconstruction, and semantic segmentation. The lecture also addresses perception for manipulation, covering object pose estimation, grasp detection, visual servoing, active vision, and haptic exploration. It concludes with the simultaneous localization and mapping (SLAM) problem, including EKF SLAM, Graph SLAM, and FastSLAM.

Workload
Lecture with 2 SWS, 3 ECTS.
3 ECTS corresponds to approx. 90 hours, thereof:
approx. 15 * 2h = 30h attendance time
approx. 15 * 2h = 30h self-study prior/after the lecture
approx. 30h preparation for the exam and exam itself

Recommendations
Having visited the lectures on Robotics I – Introduction to Robotics is recommended.

M 12.188 Module: Robotics, Automation, and Systems Control Lab [M-ETIT-107524]

Coordinators: Prof. Dr.-Ing. Mike Barth
Prof. Dr.-Ing. Sören Hohmann

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114740	Robotics, Automation, and Systems Control Lab	**ST**	**6 CP**	**Barth, Hohmann**

Assessment
The examination takes place in the form of other types of examination. It consists of pitches after every thematic block, a final presentation and an oral examination in the amount of 20 minutes. The overall impression is evaluated.

Prerequisites
none

Competence Goal

- Students are enabled to develop approaches to various problems in a group, solve them, and present and defend their results. In addition, they are able to present their approach, thought processes, applied methods, and results in a comprehensible and scientifically precise style
- Students are able to familiarize themselves independently with a complex technical system and its components
- Students can discuss problem-solving strategies with team members in technical language and argue their preferred solution. Further they are able to reflect on the solution approaches used.
- Students can implement and validate automation solutions in various development environments (e.g. MATLAB/ Simulink, ROS, ...). They are further able to apply their solutions across various system boundaries (e.g. automation concepts on Windows, ROS on Linux)
- Students gain insights into various engineering disciplines (e.g. mechanical engineering) and in rapid prototyping processes
- Students are able to use industrial bus systems (e.g. EtherCAT) in an automation context
- Students are able to apply methods that they have previously acquired in lectures on automation, control and robotics

Content
This module is designed to teach students the stated qualification objectives in the fields of automation, robotics and control by means of a complex automation task. The task not only covers the automation concept itself but also underlying aspects like hardware design, communication using an industrial bus system, drive system control and others. To do so, laboratory systems, namely laboratory scale robotic manipulator arms, are provided to the students.

Module Grade Calculation
The module grade results of the assessment of the pitches, presentation and the oral exam. Details will be given during the lecture at the beginning of the semester.

Additional Information
Due to capacity reasons there are only 25 places available. If necessary, a selection procedure will be carried out. Selection criteria are the progress in studies and in the recommended lectures. The application window is from March 1 for March 31 in Campus Plus (https://plus.campus.kit.edu/). For further information and current news, regularly visit the courses website (https://www.irs.kit.edu/Lectures_5404.php).

Workload

1. Introduction to the test stand (15h)
2. Introduction to single-axis motor control (15h)
3. Construction and print of end-effector components (30h)
4. Introduction to ROS (15h)
5. Implementation of hardware-interfaces in ROS (30h)
6. Automation, motion planning and control (30h)
7. Implementation on the real robot (15h)
8. Presentations and preparation (15h)
9. Oral exam and preparation (15h)

A total of 180 h = 6 CR

Recommendations
Knowledge in automation, control and robotics in general. Recommended, but not mandatory, lectures are:

- Practical Tools for Control Engineers – M-ETIT-106780
- Cyber Physical Modeling – M-ETIT-106953
- Seminar Industrial Process and Plant Engineering – M-ETIT-106970
- Optimization of Dynamic Systems – M-ETIT-100531
  (or successor Optimal Control – M-ETIT-107497)
- Regelung linearer Mehrgrößensysteme – M-ETIT-100374
  (or successor Multivariable Control Systems – M-ETIT-107515)
- Nichtlineare Regelungssysteme – M-ETIT-100371
  (or successor Nonlinear Control Systems - in preparation)

Teaching and Learning Methods
LACR – Laboratory Automation Control & Robotics (4 SWS)

M 12.189 Module: Safe Human-Robot-Collaboration [M-INFO-104877]

Coordinators: Prof. Dr.-Ing. Tamim Asfour
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-109911	Safe Human-Robot-Collaboration	**WT**	**3 CP**	**Kurth**

Competence Goal
You will get familiar with the various forms of human-robot collaboration (HRC) and their respective advantages. You will understand the requirements of the Machinery Directive and the relevant standards for the safety of HRC applications. Participants will be able to identify risks and to develop a safety concept for HRC systems.

Course objectives:

- Successful participants are familiar with all relevant aspects of human-robot collaboration, from the planning to the implementation of an HRC application as well as the requirements for safety.
- Successful participants understand the process of risk assessment, the importance of functional safety and predictable misuse.
- Successful participants will understand the different safety functions of robots and their use to reduce the risk to an acceptable residual risk and know to use complementary safety light curtains and laser scanners.
- The participants will know what has to be considered in designing a HRC system and can create a safety concept for this system and check it for completeness.

Content

- Introduction and fundamentals of Human-Robot Collaboration (HRC)
- Different forms of HRC and differentiation to full automation
- Practical examples from series applications
    - Advantages of HRC compared to full automation with robots
- Definition of safety
    - Machinery Directive / Standards
    - Declaration of incorporation / CE conformity
    - Safety Levels
    - Safety requirements in robotics
- Potential hazards in human-robot collaboration
    - bump and squeeze
    - Foreseeable misuse
    - Error in the application
- "Safe (?)" Robots
    - Requirements for collaborative operation according to ISO 10218-1
    - Overview of robots and their security concepts
    - Safely supervised robots
    - Gray technology / yellow technology in the robot controller
    - Safety functions based on position values and on force / torque values
- Safe HRC-Applications
    - Risk assessment
    - HRC cell layout
    - Design of HRC end effectors, peripherals
    - Use of safety features
    - Examples of industrial HRC-solutions
- From planning to the realization of HRC systems
    - HRC compliant engineering
    - Detailing of the HRC concept in the design phase
    - Programming and validation
    - Measurements demonstrating compliance with biomechanical limits
- Biomechanical limits
    - TS 15022
    - transient and quasi-static contact
    - body atlas with limits
- Safe sensor technology for safeguards
    - Basics
    - Laser scanners
    - Light curtains
    - Tactile sensors
    - Safe vision systems
    - Planning and design of safeguards

- Reaction time from the triggering event to the robot reaction

- Necessary safety distances for effectiveness of safeguards

- Safety in mobile robotics

M 12.190 Module: Satellite Communications [M-ETIT-105272]

Coordinators: Prof. Dr.-Ing. Laurent Schmalen
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	2 terms	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-110672	Satellite Communications	**ST**	**3 CP**	**Schmalen**

Assessment
The assessment will be carried out as part of an overall oral exam (about 20 minutes).

Prerequisites
Knowledge of basic engineering mathematics probability theory as well as basic knowledge of communications engineering (e.g. "M-ETIT-102103 – Nachrichtentechnik I" and "M-ETIT-102104 – Wahrscheinlichkeitstheorie")

Competence Goal
Students will be able to understand and apply the basics of satellite communications and navigation. They will be able to design and evaluate a satellite communication link and compute the achievable data rates over the link. They understand the key components of a satellite communication system, and get to distinguish the different types of satellite systems. Additionally, they know about existing satellite systems and novel ideas such as mega constellations

Content
The course covers the following contents:

- Introduction and Historical Notes
- Orbits and Geometry (including also constellations)
- Link BudgetsInformation Transmission Aspects
- Multiple Access and Multiplexing * Multi-Beam Satellite Systems
- Spacecraft Payload Aspects
- Network Aspects
- Overview of Existing Satellite Systems
- Overview of Future Satellite Systems
- Satellite Systems for Navigation

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
Attendance lecture: 15 * 2 h = 30 h

Preparation / Postprocessing Lecture: 15 * 2 h = 30 h

Exam preparation and presence: 30 h Total: 90 h = 3 LP

Total: 90 h = 3 LP

Recommendations
Knowledge from the module "Communications Engineering II" can be helpful, but is not necessary.

M 12.191 Module: Scanning Probe Lithography for Biomedical Applications [M-ETIT-107661]

Coordinators: Dr. Dr. Michael Hirtz
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-115009	Scanning Probe Lithography for Biomedical Applications		**3 CP**	**Hirtz**

Assessment
The examination takes place within the framework of an oral overall examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
The students

- will gain a general overview over scanning probe lithography methods (e.g. dip-pen nanolithography, polymer pen lithography, microcapillary printing and related techniques) for micro/nanofabrication
- will understand the working principles, capabilities of the introduced methods
- will be introduced to the basic biological and biotechnological building blocks for biofunctionalization
- can critically analyze the specific advantages and limits of different methods and approaches of SPL based manufacturing in the biomedical field
- are enabled to follow current developments in SPL methods and utilize this knowledge in their own research projects
- are enabled to navigate in applications of multidisciplinary nature (physics, material science, medical engineering)

Content
This module gives students an overview of the field of scanning probe lithography (SPL) methods with a special focus on their applications in biomedical research and applications. The module will introduce the basic working principles of the main SPL methods (e.g. dip-pen nanolithography (DPN), polymer pen lithography (PPL), microcantilever contact spotting (µCS)) and how these methods are used in particular for biofunctionalization of surfaces and devices. The students will also be introduced to basic concepts in biotechnology in regard to biomaterials, binding and sensing mechanism and the physics and chemistry needed to understand their use in SPL based manufacturing. Biomedical applications in particular in diagnostics and disease monitoring applications will be showcased and critically discussed.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload

1. attendance in lectures and exercises: 15*2 h = 30 h
2. preparation / follow-up: 15*2 h = 30 h
3. preparation of and attendance in examination: 30 h

A total of 90 h

Recommendations
Basic knowledge in biomedical technology is helpful.

Teaching and Learning Methods
2 SWS

M 12.192 Module: Scientific Computing for Engineers [M-MACH-105369]

Coordinators: Dr. Daniel Weygand
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-100532	Scientific Computing for Engineers	**WT**	**4 CP**	**Gumbsch, Weygand**

Assessment
written exam 90 minutes

Prerequisites
none

Competence Goal
The student can

- apply the programming language C++ for scientific computing in the field of materials science
- adapt programs for use on parallel platforms
- choose suitable numerical methods for the solution of differential equations.

Content
1. Introduction: why scientific computing
2. computer architectures
3. Introduction to Unix/Linux
4. Foundations of C++
* progamm organization
* data types, operator, control structures
* dynamic memory allocation
* functions
* class
* OpenMP parallelization
5. numeric /algorithms
* finite differences
* MD simulations: 2nd order differential equations
* algorithms for particle simulations
* solver for linear systems of eqns.

Additional Information
The lecture can not be combined with the lecture "Application of advanced programming languages in mechanical engineering" (2182735).

Workload
egular attendance: 22,5 hours
Lab: 22,5 hours (optional)
self-study: 75 hours

Teaching and Learning Methods
Tutorial

Literature

1. C++: Einführung und professionelle Programmierung; U. Breymann, Hanser Verlag München
2. C++ and object-oriented numeric computing for Scientists and Engineers, Daoqui Yang, Springer Verlag.
3. The C++ Programming Language, Bjarne Stroustrup, Addison-Wesley
4. Die C++ Standardbibliothek, S. Kuhlins und M. Schader, Springer Verlag
5. Numerical recipes in C++ / C / Fortran (90), Cambridge University Press
6. Numerische Mathematik, H.R. Schwarz, Teubner Stuttgart
7. Numerische Simulation in der Moleküldynamik, Griebel, Knapek, Zumbusch, Caglar, Springer Verlag

M 12.193 Module: Seamless Engineering [M-MACH-105725]

Coordinators: Prof. Dr.-Ing. Kai Furmans
Prof. Dr.-Ing. Eric Sax

Organisation: KIT Department of Mechanical Engineering

Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-111401	Seamless Engineering	**WT**	**9 CP**	**Furmans, Sax**

Assessment
Examination of another type. The description of the form of examination can be found in the description of the module component.

Prerequisites
None

Competence Goal
After successful completion of the course, the students are able to model and parameterise the requirements and boundary conditions for typical mechatronic systems. In addition, students learn the ability to select the appropriate procedures, processes, methods and tools for the development of a mechatronic system.

Important core competences in the areas of communication, problem solving and self-organisation are further essential components of the workshop, which enable the students to do reflected work independently and in a team.

Content
This module is designed to teach students how to develop a heterogeneous integrated mechatronic system. In the lecture, students are introduced to a system-oriented, higher-level approach to the description, assessment and development of a mechatronic system.

Parallel to this, the contents taught are applied and deepened in the practical part on hardware that is close to industry. The students learn the systematic development in a simulative environment as well as the transition from simulation to real hardware.

To achieve this, important components of software development in the robotics environment are taught. This includes, among other things, the basics of programming (Python) as well as the handling of the framework "Robot Operating System (ROS)". In addition, students gain insights into the use of sensors and actuators, image processing, autonomous navigation of automated guided vehicles and robotic grasping.

Module Grade Calculation
The evaluation of the colloquia and a final examination are included in the module grade. Further details will be provided at the beginning of the course.

Additional Information
None

Workload

1. attendance time lecture and exercise: 45 h
2. interdisciplinary qualification: 45 h
3. group work project: 130 h
4. colloquia and final event: 30 h
5. exam preparation and presence in the same: 20 h

In total: 270 = 9 LP

Recommendations
None

Teaching and Learning Methods
Lecture, exercise, project.

Literature
None

M 12.194 Module: Semiconductor Quantum Dots for Quantum Photonics [M-ETIT-107886]

Coordinators: TT-Prof. Dr. Tobias Huber-Loyola
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-115306	Semiconductor Quantum Dots for Quantum Photonics		**6 CP**	**Huber-Loyola**

Assessment
The examination takes place within the framework of an oral overall examination (approx. 30 minutes).

Prerequisites
none

Competence Goal
After successful completion of the module, students are able to:

- explain the physical principles of optically active semiconductor quantum dots
- describe and compare relevant epitaxial growth methods
- analyze the electronic and optical structure of semiconductor quantum dots
- apply basic models of light–matter interaction and cavity quantum electrodynamics to quantum-dot systems
- compare and assess nanophotonic structures and device concepts for quantum-dot photonics
- explain relevant nanofabrication and device-processing techniques
- identify and analyze important noise and decoherence mechanisms
- interpret typical experimental data from quantum-dots
- evaluate current applications of optically active semiconductor quantum dots in quantum photonics, including single-photon sources, entangled-photon-pair sources, quantum key distribution, entanglement swapping, spin–photon interfaces, and quantum repeater concepts.

Content
The module provides a comprehensive introduction to optically active semiconductor quantum dots for quantum photonics. The focus is on epitaxially grown quantum dots that emit light and can be integrated into photonic nanostructures and quantum-optical experiments. Electrostatically defined quantum dots in silicon or similar charge-qubit platforms are not covered.

Table of contents:

1. Introduction and motivation
   Semiconductor quantum dots as artificial atoms; comparison with atoms, molecules, color centers, and 2D-material emitters; role in quantum photonics.
2. Materials and epitaxial growth
   III–V and related semiconductor material systems; band alignment; strain; Stranski–Krastanov growth; droplet epitaxy; local droplet etching; site-controlled growth; MBE; MOVPE; quantum-dot morphology and composition.
3. Electronic and optical structure
   Carrier confinement; excitons, biexcitons, charged excitons; fine-structure splitting; spin states; optical selection rules; phonons; Auger and non-radiative effects.
4. Optical spectroscopy of quantum dots
   Micro-photoluminescence; excitation schemes; lifetime measurements; photon-correlation measurements; two-photon interference; entanglement characterization.
5. Cavity quantum electrodynamics and nanophotonic structures
   Density of optical states; Purcell effect; weak and strong coupling; mode volume and quality factor; micropillars, photonic crystal cavities, circular Bragg gratings, nanowires, waveguides, open cavities, hybrid cavities.
6. Nanofabrication and device technology
   Lithography, dry and wet etching, dielectric deposition, metallization, electrical contacts, deterministic positioning, strain tuning, piezoelectric actuators, charge-control structures, photonic integration.
7. Noise and decoherence in quantum dots
   Charge noise, spectral diffusion, spin noise, nuclear-spin fluctuations, phonon coupling, electrical noise, linewidth broadening, blinking, and methods for stabilization.
8. Quantum-optical protocols and applications
   Single-photon sources, entangled-photon-pair sources, indistinguishable photons, quantum key distribution, quantum teleportation, entanglement swapping, spin–photon interfaces, cluster-state generation, quantum repeaters, and integrated quantum photonics.
9. Current research directions
   Telecom quantum dots, deterministic fabrication, high-rate photon sources, hybrid integration, scalable architectures, and comparison with competing platforms.

Module Grade Calculation
The module grade is the grade of the oral exam.

Additional Information
Solution of the problems on the exercise sheet, which can be downloaded as homework each week, is highly recommended. Also, active participation in the problem classes and studying in learning groups are strongly advised.

Workload
The workload includes:

1. attendance in lectures and exercises: 15x4 h = 60 h
2. preparation / follow-up: 15x5 h = 75 h
3. independent study and preparation for the examination: 45 h

A total of 180 h = 6 CR

Recommendations
Basic knowledge of semiconductor physics, optoelectronics, quantum mechanics, and electromagnetic waves is recommended but not mandatory.

Teaching and Learning Methods
Lecture: 3 SWS, 1 SWS

M 12.195 Module: Seminar Advanced Concepts for Flexible and Soft Optoelectronic Devices and Sensors [M-ETIT-106674]

Coordinators: Prof. Dr. Gerardo Hernandez Sosa
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113427	Seminar Advanced Concepts for Flexible and Soft Optoelectronic Devices and Sensors	**ST**	**3 CP**	**Hernandez Sosa**

Assessment
The examination consists of a written journal article and an oral presentation of the student's work, both given in English. The overall impression is rated.

Prerequisites
none

Competence Goal
At the end of the seminar, students will be able to independently familiarize themselves with new research topics, independently search and select relevant scientific literature and summarize the topic in the form of a concise journal-style article as well as an oral presentation. Through the critical evaluation and exposure to current scientific literature, the students will develop a deeper knowledge in the future directions in the research field of flexible and soft electronics. Furthermore, they will develop skills in scientific writing and communication in English language, which are key competences for their future academic and professional career.

Content
The seminar on "Seminar Advanced Concepts for Flexible and Soft Optoelectronic Devices and Sensors" is for students curious in the latest research developments on devices, materials and the physics of novel optoelectronic devices. Discussed topics include but are not limited to: solution processable and printed electronics, flexible and stretchable electronics, wereable sensors, soft robotics, printed optics, biodegradable & sustainable electronics, 3D electronics, etc.

The seminar addresses master students from electrical engineering, physics, mechanical engineering, material science, KSOP and related MSc programs. During the seminar, the students will get the opportunity to familiarize themselves with state-of-the-art research from a selection of topics under the guidance of a mentor and discuss the topic during a presentation in the seminar. The students must attend the seminar regularly, independently present the research topic in a 30-min scientific talk and submit a short scientific review paper (3-5 pages) based on the scientific literature the presentation was based on.

Prof. Dr. Gerardo Hernandez Sosa and Prof. Dr. Jasmin Aghassi-Hagmann will select the topics and guide the discussion.

Module Grade Calculation
The module grade results of the assessment of the written paper and the oral presentation. Details will be given during the lecture.

Additional Information
Attendance of at least 80% is compulsory during the seminar course. Compulsory attendance is necessary for actively contributing to the discussion of the topics presented by all students.

Workload
The workload includes (2 SWS):

- active participation in the preparation sessions and seminar lectures: 22,5 h
- preparation of the seminar presentation: 36 h
- preparation of the written journal article: 31,5 h

Total: 90 h = 3 LP

Recommendations
Basic knowledge in the field of conventional and/or organic (opto) electronic devices and sensors is helpful.

M 12.196 Module: Seminar Battery II [M-ETIT-105321]

Coordinators: Dr.-Ing. Andre Weber
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-110801	Seminar Battery II	**WT+ST**	**3 CP**	**Weber**

Assessment
Success control takes place in form of an examination of another type. It consists of a written paper (20-40 pages) and an oral presentation (approx. 20 min.) of the students work. The overall impression is rated.

Prerequisites
none

Competence Goal
After completion of the seminar the students are able to work independently on an engineering problem in the field of batteries, to analyze the related literature and to present it in the form of a written paper and a presentation.

Content
The seminar "Batteries II" is primarily aimed at master students who are planning to write their master thesis in the field of batteries. In this seminar the participants will work on a scientific topic in the field of batteries. This usually includes a literature study, the compilation of the methods, procedures and results described in the publications as well as a critical evaluation of the same. In individual cases, next to a literature study more practical topics can be in focus.

The results are summarized in a seminar paper and presented in a lecture during the seminar. The grading is based on the written paper as well as the presentation t.

Module Grade Calculation
The assessment of the written paper and the seminar presentation will be included in the module grade. Further details will be given at the beginning of the course.

Workload
1. presence time seminar: 15 * 2 h = 30 h

2. preparation of seminar paper: 30 h

3. preparation of seminar presentation: 30 h

Total: 90 h = 3 LP

M 12.197 Module: Seminar Electrocatalysis [M-ETIT-105629]

Coordinators: Prof. Dr.-Ing. Ulrike Krewer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111256	Seminar Electrocatalysis	WT+ST	3 CP	Krewer

Assessment
The success control takes place in form of a different type of examination. It comprises a written report (maximum 25 pages) on the scientific topic and an oral presentation with subsequent discussion, each lasting aprox. 15 minutes. The overall impresssion will be rated.

Prerequisites
none

Competence Goal
Upon completion of the seminar, students are able to research an scientific-engineering topic in the field of electrocatalysis/electrolysis, identify the essential aspects of the technology, and prepare them in the form of a written report and an oral presentation, which they then present.

Content
In the seminar, students independently work on a scientific topic from the current research areas of electrocatalysis/ electrolysis and present it in order to improve their skills in literature research, scientific writing, and presentation. At the beginning, there will be an introduction to literature research, scientific writing, presentation techniques.

Finally, students select a topic from the field of electrocatalysis/electrolysis for their scientific work and prepare both a written report and a presentation on this topic.

Module Grade Calculation
The module grade is the grade of the written elaboration of the scientific topic and the grade of the presentation. Details will be given during the lecture.

Workload
1. attendance time seminar: 15*2 h = 30 h

2. preparation of seminar paper: 30 h

3. preparation of seminar presentation: 30 h

Total: 90 h = 3 CP

Recommendations
Knowledge of the basics of fuel cells is helpful.

M 12.198 Module: Seminar Embedded Systems [M-ETIT-107764]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Prof. Dr.-Ing. Eric Sax
Prof. Dr. Wilhelm Stork

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-115156	Seminar Embedded Systems	**WT+ST**	**4 CP**	**Becker, Sax, Stork**

Assessment
Success control takes place in the form of other types of examination. The examination consists of:

- Written report (approx. 6 pages)
- 2 peer reviews (approx. 1 page each)
- Presentation (approx. 15 minutes) followed by discussion (approx. 10 minutes)

The overall impression is rated.

Prerequisites
none

Competence Goal
Seminar participants can independently familiarize themselves with a given technical topic, identify all relevant aspects and summarize the results. In this context, students can identify relevant literature in terms of the research question, assess the strengths and weaknesses of existing approaches and methods, and formally evaluate other works according to specified criteria. They can also suggest new aspects in line with the research question. They can present the results of their work concisely in the form of a short text (approx. 6-page paper, usually written in English) and an approx. 15-minute presentation in words and pictures (slides).

Content
In the seminar, students work on a given topic from the field of information processing through literature and internet research under the guidance of research assistants and then present it to the other seminar participants in a short text (approx. 6-page paper, usually written in English) and a 15-minute presentation in words and pictures (slides). The students give each other feedback as part of a peer review and thus experience a part of the scientific publication process.

Module Grade Calculation
The module grade results from the combination of the elaboration, the mutual review, and the presentation. Details will be given during the seminar.

Workload
The workload includes:

1. Independent familiarization with a topic: 50h
2. Writing a scientific article: 40h
3. Preparing a peer review: 10h
4. Preparing and giving a presentation: 20h

Total: 120h = 4 LP

M 12.199 Module: Seminar Fuel Cell II [M-ETIT-105322]

Coordinators: Dr.-Ing. Andre Weber
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-110799	Seminar Fuel Cell II	WT+ST	3 CP	

Assessment
Success control takes place in form of an examination of another type. It consists of a written paper (20-40 pages) and an oral presentation (approx. 20 min.) of the students work. The overall impression is rated.

Prerequisites
none

Competence Goal
After completing the seminar, the students are able to familiarize themselves independently with an engineering question in the field of fuel cells, analyze the associated literature and present it in the form of a written report and a presentation.

Content
The seminar “Fuel Cell Research Projects” is primarily aimed at students who are planning to carry out a scientific thesis in the fuel cell research area.
In this seminar the participants deal with scientific questions in the field of fuel cells. This includes a literature search, the compilation of the methods, processes and results described in the publications as well as a critical evaluation of the same.

The results are summarized in a seminar paper and presented in a lecture during the seminar. The written work and a lecture that has to be given during the event are included in the grading of the thesis.

Module Grade Calculation
The module grade results of the assessment of the written paper and the oral presentation. Details will be given during the lecture.

Workload
1. Presence seminar: 15 * 2 h = 30 h
2. Preparation of seminar paper and lecture: 30 h
3. Preparation of seminar lecture: 30 h

Total: 90 h = 3 LP

M 12.200 Module: Seminar Industrial Process and Plant Engineering [M-ETIT-106970]

Coordinators: Prof. Dr.-Ing. Mike Barth
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113932	Seminar Industrial Process and Plant Engineering	**ST**	**4 CP**	**Barth**

Assessment
The examination will be the seminar presentation at the end of the semester. The criteria are:

- Live presentation of the created CAD and simulation models
- Poster design and usage within the presentation
- Answering the questions from the examiners
- Structure of the talk

Prerequisites
none

Competence Goal
The students:

- are able to create (concept and virtual realization) mechatronic plants and production facilities (e.g. robot cells).
- understand the Engineering-Lifecycle of unique production plants, machines and modules.
- are familiar with modern CAX-Engineering Methods.
- know advanced (cloud-based) CAX-Tool-Chains.
- know about advanced computer-aided design principles, e.g. model- and equation-based 3D-Design.
- are familiar with design definitions, patterns and features for semi-automated engineering tasks, e.g. automatic generation of variants.
- know CAX-information models, e.g. asset administration shell (AAS) or Parasolid.
- can implement new software-based Engineering features and interfaces.
- are able to perform the virtual commissioning of complex systems using advanced CAX-features.

Content

- This module is designed to teach students the theoretical and practical aspects of advanced model-based and computer-aided design of unique systems, e.g.
  - Industrial plants,
  - Production cells (e.g. robot),
  - Production modules and machines.
- This includes every lifecycle phase of an engineering project starting from 3D-CAD Design to advanced system simulations and concluding in virtual commissioning and optimization during operations.
- Introduction to advanced cloud-based 3D-Plant-Engineering (e.g. OnShape).
- Principals of model-based system design (3D-Modeling, Drawings, Feature-based Rule Set) including advanced kinematics and simulation.
- Configuration and use of VR-, AR-Setups for virtual commissioning and operator training setups.
- Concept and implementation of software-based feature scripts (e.g. C++, python) for advanced engineering in the used tool chain.
- Concept and implementation of interfaces to external engineering data-sources (e.g. data tables, Asset Administration Shell via REST-API, ROS, MQTT or OPC UA).

Module Grade Calculation
The module grade is the grade of the final presentation including the aspects named above.

Additional Information
The seminar is limited to a number of 20 participants due to capacity reasons. If necessary, a selection procedure will be carried out. Places will be allocated taking into account the students' academic progress. Details will be announced on the lecture website.

Workload

1. Attendance in Lecture Blocks for Engineering Theory: 10*2 h = 20 h
2. Guided Seminar Work Block: 80 h
3. Preparation of exam: 20 h

A total of 120 h = 4 CR

Recommendations
Enjoyment and interest in industrial engineering and Computer-aided-X Technologies like e.g. 3D CAD, Robotics, Kinematic Simulations, VR, MR and AR-Technologies.

M 12.201 Module: Seminar New Power Electronic Systems and Technologies [M-ETIT-107588]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114862	Seminar New Power Electronic Systems and Technologies	**WT+ST**	**4 CP**	**Hiller**

Assessment
The success control takes place in the form of an examination of another type. It consists of a 15-minute final presentation followed by a discussion and a 2- to 3-page written summary paper. The overall impression is evaluated.

The following aspects are assessed:

Presentation:

- Quality of the presentation (form and content)
- Presentation style (structure, style, content)
- Behavior during the discussion

Paper with a summary of the main content:

- Format, spelling, linguistic style (academic/factual)
- Content, (graphical) presentation of the researched findings
- Relevance and quality of the references used, citation style

Prerequisites
none

Competence Goal
During the seminar, students learn to independently familiarize themselves with previously unknown technologies and applications in the field of power electronics. They will be able to independently research and select relevant scientific literature, which is usually in English, and interpret its content. In addition, students will be able to present the topic in a scientific lecture to an expert audience and summarize the essential content in a compact article in the style of a journal publication.

By critically evaluating and analyzing current scientific literature, students develop a deeper understanding of current development trends in technologies and applications in the field of power electronics. In addition, they develop skills in scientific writing and in presenting new scientific topics. These qualifications represent key competencies for the students' future academic and professional careers.

Content
The seminar “New Power Electronics Systems and Technologies” is aimed at students who are interested in the latest technologies and applications in the field of power electronics.

Participants in the seminar are expected to conduct independent studies on current topics in science and research. In addition to research, the selection of the most relevant findings and their presentation to an expert audience are key components of the seminar.

The specific topics are redefined each semester by the doctoral students and professors at ETI and are based on current developments in research and science. The topics cover a wide range and are presented in detail at the beginning of the seminar. Each participant can usually choose their “favorite topic.”

In past seminars, topics from the following areas were offered, for example:

- Electrical energy storage
- Photovoltaic systems
- Power converter systems in energy transmission and distribution
- Control of power electronic systems
- Drive systems for cars and trucks
- Structure and properties of modern power semiconductors

If there is particular interest, topics suggested by the participating students can also be considered.

The seminar is aimed at master's students in electrical engineering, mechatronics, and biomedical engineering. During the seminar, students have the opportunity to familiarize themselves with current research results in the field of their chosen topic under the guidance of their supervisor and to discuss the topic during a presentation in the seminar.

Students should regularly attend group and individual meetings throughout the seminar. They will independently present the research topic in a 15-minute scientific lecture and prepare a short scientific summary (2-3 pages) based on the scientific literature on which the presentation is based.

The seminar includes participation in the following group and individual sessions:

- Introduction and presentation of possible seminar topics (all seminar participants)
- Discussion and assignment of topics (all seminar participants)
- Tips for literature research (all seminar participants)
- Helpful presentation techniques (all seminar participants)
- Presentation of the structure and content of the presentations (individually with the seminar topic supervisors)
- Discussion and feedback on the final presentation (individually with the supervisors of the seminar topic)
- Rehearsal presentation (individually with the supervisors of the seminar topic)
- Plenary seminar presentations (all seminar participants)

Module Grade Calculation
The module grade is based on the assessment of the final oral presentation and the written summary paper (each according to the criteria mentioned above). Further details will be provided at the beginning of the course.

If the candidate's grade falls between two grades, their participation in the preparatory meetings will be the deciding factor.

Additional Information
For capacity reasons, the number of participants in the seminar is limited to approximately eight students. If necessary, a selection procedure will be carried out. Places will be allocated taking into account the students' academic progress. Details will be announced on the seminar's website.

Presentations and summaries may be given/written in English or German.

Workload
The workload of approx. 110 hours (equivalent to 4 LP) includes:

- Participation in organizational and preparatory meetings: 14 hours
- Literature study on the chosen seminar topic: 40 hours
- Preparation of the seminar presentation, including rehearsal: 32 hours
- Preparation of the final documentation: 16 hours
- Participation in the final presentations of the seminar: 8 hours

M 12.202 Module: Seminar Novel Concepts for Solar Energy Harvesting [M-ETIT-103447]

Coordinators: Prof. Dr. Bryce Sydney Richards
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-108344	Seminar Novel Concepts for Solar Energy Harvesting	**ST**	**3 CP**	**Richards**

Assessment
The examination consists of two components: a written journal article and an oral presentation on a research topic selected by the student under the lecturer's guidance. Topics are agreed upon during the first weeks of the semester. Both the lecture and the presentation are conducted in English. The preparation of the manuscript is completed as homework.

The journal article must be at least 3 pages long and adhere to the template provided by the lecturer. The oral presentation should last approximately 25 minutes, followed by a 20-minute discussion.

The overall impression is rated.

Prerequisites
none

Competence Goal
After completion of the seminar, students are able to independently familiarize themselves with a new research topic, recapitulate the corresponding literature and present the topic in the form of a review journal article as well as an oral overview presentation. Besides the exposure to new scientific research topics, the students will develope their know-how in scientific presentations and scientific writing in English which are key competences for their future (e.g. MSc thesis projects and research).

Content
We are offering an advanced seminar on „Novel Concepts for Solar Energy Harvesting" for students curious in latest research topics on devices, materials and physics of next generation solar energy harvesting. The students will get the opportunity to familiarize themselves with a state-of-the-art research topic of their choice under the guidance of a mentor and present the topic during the seminar. The seminar addresses master students from electrical engineering, physics, mechanical engineering, material science, KSOP and related MSc programs.

Module Grade Calculation
The module grade results of the assessment of the written paper and the oral presentation. Details will be given during the lecture.

Additional Information
The seminar comprises 8–10 sessions of 90 minutes each. Active participation in these sessions is expected. The sessions include three introductory sessions at the beginning of the semester, which will prepare students in relevant methods for delivering presentations and writing academic papers.

Workload

1. participation in the seminar lectures: 20 h
2. preparation of the seminar presentation: 35 h
3. preparation of the journal article: 35 h

Recommendations
Good knowledge of semiconductor components/optoelectronics is desirable.

M 12.203 Module: Seminar on Applied Superconductivity [M-ETIT-105615]

Coordinators: Prof. Dr. Tabea Arndt
Prof. Dr. Bernhard Holzapfel
Prof. Dr. Sebastian Kempf
Prof. Dr.-Ing. Mathias Noe

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111243	Seminar on Applied Superconductivity	**ST**	**3 CP**	**Arndt, Holzapfel, Kempf, Noe**

Assessment
Elaboration of a scientific topic and presentation of a talk on the topic within the seminar of about 30min.

Competence Goal
Students are to familiarize themselves with an unknown scientific topic in the field of applied superconductivity. They independently prepare a presentation on the topic they have chosen and are able to present it to the general audience. In this role, the students will learn to clearly and didactyly communicate scientific topics and to lead a scientific discussion. As audience members, students are also enabled to recognize strengths and weaknesses of a presentation and to give constructive feedback to the person giving the presentation.

Content
In the seminar, students choose a current topic from the fields of

- Superconducting materials
- Superconducting magnet technology
- Superconducting power supply systems
- Superconducting detectors and sensors
- Superconducting Quantum Bits and Quantum Computing

and present this topic in a lecture to the other seminar participants.

Module Grade Calculation
The module grade corresponds to the grade of the oral examination.

Workload
For the successful completion of the module, a workload of approx. 90h is required. This is composed as follows:

1.) Attendance time in the seminar: 12*1.5h = 18h
2.) Preparation and follow-up of the seminar: 12*3h = 36h
3.) Preparation and execution of the presentation with handouts: 36h

Recommendations
None

M 12.204 Module: Seminar Radiation Protection in Medicine [M-ETIT-107839]

Coordinators: PD Dr. Bastian Breustedt
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-115253	Seminar Radiation Protection in Medicine	**WT**	**3 CP**	**Breustedt**

Assessment
The examination consists of a written handout (approx. 2-3 pages) and an oral presentation (approx. 25-35 min.) of the students work. Considered will be the quality of the contents, the style of the presentation, the quality and content of the handout. The overall impression is rated.

Prerequisites
none

Competence Goal

- The students have an overview of the applications of radiation protection in medicine.
- The students are able to understand medical exposures in terms of their magnitude (dose) and risk. They are able to propose appropriate radiation protection measures for the cases considered.
- The students can analyze, structure and formally describe radiation protection issues in medicine.
- The students have an understanding of the radiation protection principles and the system proposed by the International Commission of Radiological Protection. They are able to describe the systems' implementation into legal force via Basic Safety Standards and national regulations. They are able to define the interplay with other fields of regulations (such as the medical devices act).
- The students are able to present cases of use of (ionizing) radiation in medicine and will be able to classify these in the framework of radiological protection and discuss appropriate regulations and practical measures that need to be applied.

Content
The seminar will consist of a block of 3 introductory lectures presenting medical exposures in the system of radiological protection. Applications of radiation in medicine will be presented and their specific issues regarding protection of patient, staff and public will be outlined. The second block will consist of presentations of the students on a dedicated topic, which will be selected by them a the begin of the term. Students shall look into the problem and develop a solution to the issues identified. The presentations shall introduce the topic, state the problems from a radiation protection perspective and introduce solutions and methods applied. The topic will then be discussed within the group. A final lecture will summarize the presentations of the students and offer space for a final discussion on medical radiation protection.

Module Grade Calculation
The module grade is the grade of the seminar contributions. Details will be given during the lecture.

Additional Information
For capacity reasons, the seminar is limited to 20 students. If necessary, a selection procedure will be carried out. Places will be allocated according to the progress of the students (semester and number of attended courses with topics covering radiation medicine). Details will be announced in the first course and on the course homepage.

Workload

1. Attendance in lectures: 15* 2 h= 30 h
2. Preparation / follow up of lectures: 15 * 2 h = 30 h
3. Preparation of Presentation/Handout = 30h
4. A total of 90 h = 3 CR

Recommendations
Knowledge about of the basics of radiation protection is strongly recommanded. Information and material for self-study will be provided in the seminar.

Teaching and Learning Methods
2 SWS

M 12.205 Module: Seminar Selected Topics in Communication [M-ETIT-100441]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100962	Seminar Selected Topics in Communications	**WT+ST**	**4 CP**	**Jäkel**

Assessment
The success controll consists of a written paper (approx. 5 pages in IEEE format) and a presentation (approx. 25 minutes). The overall impression is rated.

Prerequisites
none

Competence Goal
Students can familiarize themselves independently with a subject area and rely on independent time management. They are able to reflect on what they have worked on and to summarize and present it in a comprehensible manner.

Students are proficient in the methods and tools for creating academic texts and presentations.

Content
The participants work their way into a given communication technology issue through independent literature research, summarize the topic in an overview and present it to the other seminar participants in a lecture.

In addition to the technical skills required for familiarization and understanding of the topic, the focus is on communicating scientific facts. A structured and comprehensible presentation of the topic in an article is just as important as a clear layout of the slides and a confident presentation style.

Module Grade Calculation
The module grade results of the assessment of the written paper and the seminar presentation. Further details will be given during the lecture.

Workload
1. independent familiarization with a topic: 60 h

2. writing a scientific article: 40 h

3. preparation and giving a presentation: 20 h

Total: 120 h

M 12.206 Module: Seminar: Digital Accessibility and Assistive Technologies [M-INFO-105884]

Coordinators: Prof. Dr.-Ing. Rainer Stiefelhagen
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-111832	Seminar: Digital Accessibility and Assistive Technologies	**WT**	**3 CP**	**Stiefelhagen**

Competence Goal
Students acquire basic knowledge of

- Basics on the topic of "accessibility"
- Visual impairments, their causes and effects
- existing assistive technologies (AT) for various fields of application - such as AT for everyday life, for mobility support and access to information
- Guidelines for the development of accessible websites and accessible software applications
- Accessible software development
- Accessible document creation
- Current research approaches in the AT division
- In particular on the use of machine vision methods (computer vision) to develop new AT
- Evaluation of assistive technologies
- Writing and presenting conference papers

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Research Practice" and successfully apply these guidelines in the preparation of their scientific work.

Content

Digital accessibility is a topic that affects us all. Accessing information digitally, from childhood to old age. Assistive technologies, such as smartphones, tablets, smartwatches, wearables in general have become a part of our everyday life. Exactly these things should be operable and usable by all people. Regardless of any barriers.

But what are the details behind this? What are the rights and foundations for this? What all has to be done to be "barrier-free"?

This can all be best illustrated by the example of "visual impairment".

According to the World Health Organization, there are approximately 285 million people with visual impairments worldwide, including approximately 39 million people who are blind. The partial or complete loss of vision significantly restricts blind and visually impaired people in their working and social lives. It is difficult for blind and visually impaired people to orient themselves and move around in public spaces without assistance. The reasons for this are problems in perceiving obstacles and landmarks as well as the resulting fear of accidents and orientation difficulties. Other problems in everyday life are: reading texts, recognizing banknotes, food, clothes or finding objects in the household.

For support, blind and visually impaired people can already rely on a number of technical aids. For example, digitized texts can be made accessible through speech output or Braille output devices. There are also various devices made specifically for the blind. The most important aid for improving mobility is by far the cane for the blind. In recent years, some electronic aids for obstacle detection or orientation support have also been developed, but these offer only very limited functionality at a relatively high price and are therefore rather rarely in use.

The seminar is intended to provide insight into topics of IT-based assistive technologies (AT) and, on the other hand, to prepare participants for writing conference articles on the topic. The selection of topics may span a wider range. Such as:

- Legal principles
- Existing assistive technology tools for different application areas
- AT for information access
- New steps of accessible software development
- New basics and techniques for accessible web design (websites and web applications)
- Accessible documents today and tomorrow
- Use of machine vision methods
- Feedback systems and their basics
- Insights into current research topics around the topic of digital accessibility

For the latest information, visit **http://cvhci.anthropomatik.kit.edu/**

M 12.207 Module: Sensors [M-ETIT-100378]

Coordinators: TT-Prof. Dr. Johanna Schröder
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-101911	Sensors	**ST**	**3 CP**	**Schröder**

Assessment
Success is assessed in the form of a written examination lasting 90 minutes.

Prerequisites
none

Competence Goal
Students know the basic properties and functions of the most important industrially and commercially used sensors (temperature, pressure, gas, etc.). They have a basic understanding of the physical and chemical processes of signal formation and can use this knowledge to analyze problems, design and apply sensors and transfer it to other areas of their studies. They are able to communicate with specialists from related disciplines in the field of sensor technology and can actively contribute to the opinion-forming process in society with regard to scientific and technical issues.

Content
The lecture provides the most important basics for understanding commercially available sensors. In addition to the sensor effects, material aspects and the technical realization in components as well as the application of sensors in electrical circuits and systems are discussed. The following are covered: mechanical sensors, temperature sensors, optical sensors, magnetic sensors, ultrasonic sensors, gas sensors, chemical sensors.

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
1. lecture attendance time: 15 * 2 h = 30 h
2. Preparation and follow-up time for lecture: 15 * 4 h = 60 h
3. Exam preparation and attendance in the exam: included in preparation and follow-up time.

Total: 90 h = 3 CP

M 12.208 Module: Signal Processing for Communication Receivers [M-ETIT-107871]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area) (Usage from 4/1/2027)
Electives (Elective Modules) (Usage from 4/1/2027)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-115288	Signal Processing for Communication Receivers	**ST**	**6 CP**	**Jäkel**

Assessment
Exam will be held as 25 mins oral. Beforehand, students are elaborating small problems within a 15 mins. period of preparation.

Prerequisites
none

Competence Goal
The students are able to analyze and assess algorithms of signal processing used in communication systems and to consider aspects of implementation. They can use mathematical methods in the context of signal processing in communication systems for autonomously deriving and elaborating theoretical results, and check their viability by simulations.

Content
The module is introducing and deriving results in the areas of:

- Sampling in communication receivers incl. interpolation, decimation, sub-sampling
- Properties of Fourier and z-transformations
- Basics of discrete-time system theory
- Discrete-time filter in communications
- Estimation basics
- Spectral estimation

Topics already covered in the Bachelor system theory modules are deduced thoroughly and mathematical derivations and reasoning are provided.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload

1. Attendance to the lecture: 20 * 1,5 h = 30 h
2. Preparation and review: 20 * 3 h = 60 h
3. Attendance to the tutorial: 6 * 1,5 h = 9 h
4. Preparation and review: 6 * 3,5 h = 21 h
5. Preparation for the exam: 60 h

In total: 180 h = 6 LP

Recommendations
Basics knowledge of communication systems, as, e.g., provided in KIT's Bachelor course "Grundlagen der Datenübertragung", is supposed, Furthermore, working knowledge in the areas system theory (KIT's Bachelor course "Signale und Systeme") and probability theory (KIT's Bachelor course "Wahrscheinlichkeitstheorie) is assumed.

Teaching and Learning Methods
3 SWS lecture, 1 SWS practical exercise

M 12.209 Module: Signal Processing in Communications [M-ETIT-100443]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area) (Usage until 3/31/2027)
Electives (Elective Modules) (Usage until 3/31/2027)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	German	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100747	Signal Processing in Communications	ST	6 CP	Jäkel

Assessment
Success control takes place in the form of a written examination lasting 120 min.

Prerequisites
none

Competence Goal
The students are able to analyze and assess algorithms of signal processing used in communication systems and to consider aspects of implementation. They can use mathematical methods in the context of signal processing in communication systems for autonomously deriving and elaborating theoretical results, and check their viability by simulations.

Content
Topics already covered in the Bachelor system theory modules are deduced thoroughly and mathematical derivation and reasoning is provided.

Module Grade Calculation
The module grade is the grade of the written exam.

Additional Information
Offered in English during the summer term 2027:
M-ETIT-107871 – Signal Processing for Communication Receivers

Workload

1. Attendance to the lecture: 20 * 1,5 h = 30 h
2. Preparation and review: 20 * 3 h = 60 h
3. Attendance to the tutorial: 6 * 1,5 h = 9 h
4. Preparation and review: 6 * 3,5 h = 21 h
5. Preparation for the exam: 60 h

In total: 180 h = 6 LP

Recommendations
Basics knowledge of communication systems, as, e.g., provided in KIT's Bachelor courses, is supposed, Furthermore, working knowledge in the areas system theory and probability theory is assumed.

M 12.210 Module: Signal Processing Lab [M-ETIT-106633]

Coordinators: Prof. Dr.-Ing. Sander Wahls
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Lab Course)
Field of Specialization / Automation, Robotics, and Systems Engineering (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113369	Signal Processing Lab	**ST**	**6 CP**	**Wahls**

Assessment
Success is assessed in the form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal
After this module, students will have a sound basic knowledge of the main methods of signal processing as well as their areas of application, key parameters and the effects of parameter changes on the behavior of the methods. Students will be able to analyze given signal processing tasks in group work, develop solutions and document their results.

Content
The Digital Signal Processing practical course currently comprises eight experiments designed to familiarize students with the fundamentals of signal processing, in particular some selected measurement methods such as correlation measurement technology and modal analysis as well as Kalman filtering and the fundamentals of image processing. The focus of the experiments to be completed with various programs and devices is to teach students the practical aspects of modern signal processing.

Note: The lecturer reserves the right to include experiments other than those listed here in this practical course without prior notice.

Module Grade Calculation
The module grade is the grade of the written examination.

Additional Information
A prerequisite for admission to the examination is the submission of protocols of all experiments. The quality of the protocols will be assessed; they must be acceptable for admission to the examination.

Compulsory attendance is necessary both for carrying out the work in the team on site and for the practical teaching of skills and hardware experience that cannot be learned in pure self-study.

Workload
The workload results from attending the introductory event (1.5 h), 8 experimental sessions of 4 h each. In addition, the preparation of the experiments is estimated at 8x4 h and the writing of the protocols as well as the follow-up work at 8x4 h. Preparing for the exam and attending it takes about 60 hours. This results in a total workload of approx. 160 hours.

Recommendations
Knowledge of the contents of the modules "Signals and Systems", "Measurement Technology" and "Methods of Signal Processing" is strongly recommended.

M 12.211 Module: Signal Processing Methods [M-ETIT-106899]

Coordinators: Prof. Dr.-Ing. Sander Wahls
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Fundamentals)
Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Fundamentals)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113837	Signal Processing Methods	WT	6 CP	Wahls

Assessment
Written exam, 120 minutes.

Prerequisites
none

Competence Goal
Students can

- choose appropriate estimation methods based on theoretical properties and practical considerations
- determine estimators for specific problems
- can weight the pros and cons of data decomposition methods; apply them to given problems; interpret the results
- understand the advantages and limitations of the considered time-frequency analysis methods
- interpret time-frequency representations
- choose appropriate analysis and synthesis windows/wavelets
- determine time-frequency transforms of given signals

Content
This module introduces students to advanced signal processing methods that are widely employed in engineering. The three main topic areas are

1. Parameter estimation
2. Decomposition of data into components and modes
3. Time-frequency analysis

The following topics are treated:

· Best linear unbiased estimator
· Maximum likelihood estimation
· General Bayesian estimators
· Linear Bayesian estimators
· Principal component analysis
· Independent component analysis
· Dynamic and empirical mode decomposition
· Hilbert spaces and frames
· Short-time Fourier transform
· Wavelets
· Analytic signals
· Wigner-Ville-Distribution
· Huang-Hilbert transform

Illustrating examples from diverse application areas are discussed.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
The workload includes:

1. attendance in lectures and tutorials: 15*4 h = 60 h
2. preparation / follow-up: 15*4 h = 60 h
3. preparation of and attendance in examination: 60 h

A total of 180 h = 6 CR

Recommendations
Familiarity with signals and systems (in particular, Fourier transforms) and probability theory at the Bachelor level is assumed.

M 12.212 Module: Signal Processing with Nonlinear Fourier Transforms and Koopman Operators [M-ETIT-106675]

Coordinators: Prof. Dr.-Ing. Sander Wahls
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113428	Signal Processing with Nonlinear Fourier Transforms and Koopman Operators	ST	6 CP	Wahls

Assessment
The examination in this module consists of programming assessments and a graded written examination of 120 minutes.

The programming assignments are either pass or fail. They must be passed during the lecture period for admission to the written examination.

Prerequisites
none

Competence Goal
Students

- understand the basic theory of linear operator on Hilbert spaces and can analyze simple operators analytically
- know the use cases for selected integrable partial differential equations (PDEs) and can apply them under non-ideal circumstances (small non-integrable terms)
- can determine the PDE corresponding to a given Lax-pair and check if the PDE is actually integrable (i.e. check if the Lax pair is “fake”)
- understand the theory of nonlinear Fourier analysis for selected PDEs and can compute nonlinear (inverse) Fourier transforms numerically and, in simple cases, analytically
- know and implement practical engineering applications of nonlinear Fourier transforms
- understand the theory of the Koopman operator including selected engineering applications
- compute Koopman spectra numerically using data-driven methods and use them in practical engineering applications

Content
This module introduces students to signal processing methods that rely on nonlinear Fourier transforms and Koopman operators. These methods allow us to transform large classes of nonlinear systems such that they essentially behave like linear systems. They can also be used to decompose signals driven by such systems into physically meaningful nonlinear wave components (for example, solitons).

While these methods originated in mathematical physics, there has been a growing interesting of exploiting their unique capabilities in engineering contexts. The goal of this module is to give engineering students a practical introduction to this area. It provides the necessary theoretical background, enables students to apply the methods in practice via computer assignments, and discusses recent research from the engineering literature.

The following topics will be discussed:

- Introduction to linear operators on Hilbert spaces
- Integrable model systems (Korteweg-de Vries equation, Nonlinear Schrödinger equation)
- Lax-integrable systems (representations of Lax pairs, fake Lax pairs, conserved quantities)
- Solution of integrable model systems using nonlinear Fourier transforms (inverse scattering method) and the unified transform method
- Physical interpretation of nonlinear Fourier spectra (in particular, solitons)
- Practical applications of nonlinear Fourier transforms
- Theoretical properties of Koopman operators
- Data-driven computation of Koopman operators (residual dynamic mode decomposition)
- Practical applications of Koopman operators

Module Grade Calculation
The module grade is the grade of the written exam.

Additional Information
Some tutorial sessions will be classically devoted to solving pen and paper problems, but in others students will be working on their practical computer assignments. For the latter, students have to bring their own laptops with Matlab installed. The solutions of the computer assignments must be submitted by the provided deadlines, which are typically one week after the corresponding tutorial has taken place.

Workload
The workload includes:

1. attendance in lectures and tutorials: 15*4 h = 60 h
2. preparation / follow-up: 30*3 h = 60 h
3. finishing programming assignments: 30 h
4. preparation of and attendance in examination: 30 h

A total of 180 h = 6 CR

Recommendations
Familiarity with signals and systems at the Bachelor level (Fourier and Laplace transforms, linear systems, etc.) is assumed.

M 12.213 Module: Single-Crystal Growth – Crystal Growth Methods and Applications of Crystals for Electronic and Optical Components [M-ETIT-106597]

Coordinators: Prof. Dr. Marc Eichhorn
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113293	Single-Crystal Growth – Crystal Growth Methods and Applications of Crystals for Electronic and Optical Components	**WT**	**3 CP**	**Eichhorn**

Assessment
Success control takes place in form of an overall oral examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
Students should be able to interpret phase diagrams and solubility curves of crystals and select suitable crystal growth methods for these crystals. Students will be able to describe the various crystal growth methods and discuss their special features. Students will be able to explain the functionality and physical principles of important commercial crystals and their application.

Content

- This module is designed to teach students the theoretical and practical aspects of single crystal growth and the special physical properties of commercially usable single crystals.
- It covers:
- History of commercial single crystal growth
- Fundamentals of crystallography
- Crystal optics
- Absorption and colors of crystals, crystal field theory
- Phase diagrams
- Growing crystals from aqueous solution
- Melt growth methods (Czochralski, skull melting, flux, Kyropoulos, hot zone)
- Hydrothermal growth processes

Module Grade Calculation
The module grade is the grade of the oral examination.

Additional Information
Block lecture, 5 days x 5 h

Workload

1. Attendance time in lectures, exercises: 5*5 h = 25 h
2. Preparation and follow-up of the same: 5*6 h = 30 h
3. Exam preparation and attendance: 35 h

Total: 90 h = 3 LP

Recommendations
Knowledge of the fundamentals of inorganic chemistry, solid state physics or photonics

M 12.214 Module: Single-Photon Detectors [M-ETIT-101971]

Coordinators: Dr. Konstantin Ilin
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-108390	Single-Photon Detectors	**WT**	**4 CP**	**Ilin**

Assessment
Type of Examination: Oral exam
Duration of Examination: approx. 20 minutes

Prerequisites
none

Competence Goal
After completing the module, students will get basic knowledges on various physical mechanisms underlying optical response of the currently available detectors with the ultimate sensitivity – the single-photon detectors (SPDs) – thereby will be able to explain their functionality in details. The grasp of these knowledges enables students to critically analyze advantages and limitations of different types of SPDs and to make a decision on development of the detection system for particular applications.

Content
The students will get an overview of the modern types of single-photon detectors already widely used in applications and currently developing as well. Basics of the response mechanisms of the detectors and particular areas of their application will be considered as well as the main directions of development and optimization of new types of SPDs and detection systems. In particular the following topics will be addressed:

- Applications of single-photon detectors (SPD)
- Detection system and light-matter interaction
- Basic characteristics of SPDs and experimental methods of their determination
- Photoelectric effect: photomultiplier tubes (PMT); microchannel plate (MCP)
- Semiconducting detectors: photoresistor, PIN photodiode, avalanche photodiode (APD), single-photon avalanche diode (SPAD), visible light photon counter (VLPC), quantum dot field effect transistor (QD-FET)
- Superconducting detectors: transition edge sensor (TES), superconducting tunnel junction (STJ), superconducting nanowire single-photon detector (SNSPD)
- Hybrid detection system

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload

1. Lecture presence time in winter semester– 18 h
2. Exercises presence time – 9 h
3. Pre- /Post-preparation on lectures/exercises- 36 h
4. Preparation to and examination – 57 h

M 12.215 Module: Software Engineering [M-ETIT-100450]

Coordinators: Dr. Clemens Reichmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	4	4

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-108347	Software Engineering	**ST**	**3 CP**	**Reichmann**

Assessment
Success control takes place in the form of an oral examination lasting approx. 25 minutes.

Prerequisites
none

Competence Goal
Students know the terms and processes of systematic software development. They can apply and describe common methods and tools. They are able to compare different solution approaches and assess the respective advantages and disadvantages. They have an extensive understanding of the UML modeling language and can apply it to software engineering problems.

Content
Building on the Systems and Software Engineering (SSE) lecture, software-specific knowledge is deepened. In order to develop students' skills, the aim is to deepen their understanding of the necessity and application of procedures, aids and tools from all areas of software development.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
The workload includes:

1. attendance time in lectures, exercises: 22.5h
2. preparation and follow-up of the same: 22.5h
3. exam preparation and attendance in the same: 30h-45h

Recommendations
Knowledge of Systems and Software Engineering (course 2311605) is helpful.

M 12.216 Module: Solar Energy [M-ETIT-100524]

Coordinators: Prof. Dr. Ulrich Wilhelm Paetzold
Prof. Dr. Bryce Sydney Richards

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100774	Solar Energy	**WT**	**6 CP**	**Paetzold, Richards**

Assessment
Type of Examination: written exam

Duration of Examination: 120 Minutes

Modality of Exam: One written exam at the end of each semester.

Prerequisites

- Knowledge of optoelectronics is a prerequisite, e.g. "M-ETIT-107146 - Bauelemente der Opto- und Nanoelektronik".
- Students are not allowed to take "M-ETIT-100513 - Photovoltaik“ in addition to this one.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-ETIT-100513 - Photovoltaics must not have been started.

Competence Goal
The students:

• understand the basic working principle of pn-junction solar cells,

• learn about the different kinds of solar cells (crystalline and amorphous silicon, CIGS, Cadmium telluride, organic, dye-sensitized solar cells, etc.),

• get an overview over upcoming third-generation photovoltaic concepts,

• receive information on photovoltaic modules and module fabrication,

• develop an understanding of solar cell integration and examples of systems,

• get insight into solar concentration and tandem solar cells for highly efficient energy conversion,

• understand potential synergies with solar thermal technologies

• understand the environmental impact of solar energy technologies.

Die Studentinnen und Studenten können in englischer Fachsprache sehr gut kommunizieren.

Content
I. Introduction: The Sun

II. Semiconductor fundamentals

III. Solar cell working principle

IV. First Generation solar cells: silicon wafer based

V. Second Generation solar cells: thin films of amorphous silicon, copper indium gallium diselenide, cadmium telluride, organic photovoltaics and dye sensitized solar cells

V. Third Generation Photovoltaics: high-efficiency device concepts incl. tandem solar cells

VI. Modules and system integration

VII. Cell and module characterization techniques

VIII. Economics, energy pay-back time, environmental impact

IX. Other solar energy harvesting processes, incl. thermal and solar fuels

X. Excursion

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
Total 180 h, thereof 60h contact hours (45h lecture, 15h problems class), and 120h homework and self-studies

Literature

- A. Smets: Solar Energy: The physics and engineering of photovoltaic conversion, technologies and systems
- P. Würfel: Physics of Solar Cells
- V. Quaschning: Renewable Energy Systems
- C. Honsberg and S. Bowden, PV Education CD-ROM and website, http://www.pveducation.org/pvcdrom

M 12.217 Module: Solar Thermal Energy Systems [M-MACH-101924]

Coordinators: apl. Prof. Dr. Ron Dagan
Organisation: KIT Department of Mechanical Engineering

Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	4

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-106493	Solar Thermal Energy Systems	**WT**	**4 CP**	**Dagan**

Assessment
oral exam of about 30 minutes

Prerequisites
None

Competence Goal
The students get familiar with the global energy demand and the role of solar thermal systems for this demand

learn about improved designs for using efficiently the potential of solar energy

gain basic understanding of the main thermal hydraulic phenomena which support the work on future innovative applications

will be able to evaluate quantitatively various aspects of the thermal solar systems

Content
I. Introduction to solar energy: Energy resources, consumption and costs

II. The sun as an energy resource:

Structure of the sun, Black body radiation, solar constant, solar spectral distribution

Sun-Earth geometrical relationship

III. Passive and active solar thermal applications.

IV. Fundamentals of thermodynamics and heat transfer

V. Solar thermal systems - solar collector-types, concentrating collectors, solar towers. Heat losses and efficiency

VII. Energy storage

The course deals with fundamental aspects of solar energy. Starting from a global energy panorama the course deals with the sun as a thermal energy source. In this context, basic issues such as the sun's structure, blackbody radiation and solar–earth geometrical relationship are discussed. In the next part, the lectures cover passive and active thermal applications and review various solar collector types including concentrating collectors and solar towers and the concept of solar tracking. Further, the collector design parameters determination is elaborated, leading to improved efficiency. This topic is augmented by a review of the main laws of thermodynamics and relevant heat transfer mechanisms.

The course ends with an overview on energy storage concepts which enhance practically the benefits of solar thermal energy systems.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
Total 90 h, hereof 30 h contact hours and 60 h homework and self-studies

Teaching and Learning Methods
Lecture, tutorial

Literature

Foster, Ghassemi, cota,; Solar Energy

Duffie and Beckman; Solar engineering of thermal processes

Holman:, Heat transfer

Heinzel; script to solar thermal energy (in German)

W. Platzer, R. Stieglitz “Solar thermal Energy Systems”

M 12.218 Module: Space-Born Microwave Radiometry - Advanced Methods and Applications [M-ETIT-100545]

Coordinators: Prof. Dr.-Ing. Thomas Zwick
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100810	Space-Born Microwave Radiometry-Advanced Methods and Applications	ST	3 CP	Zwick

Assessment
Success control is carried out as part of an overall oral examination (approx. 20 minutes) of the selected courses, with which the minimum requirement for CP is met.

Prerequisites
none

Competence Goal
The students have a basic knowledge of remote sensing with microwave radiometers on satellites. Applications of microwave radiometry on the ground, on airplanes and satellites. They are familiar with modern methods for the detection of anti-personnel mines, detection of hidden explosives and weapons. They can describe and evaluate the different types of radiometers and are able to apply the theoretical basics.

Content
The term microwave radiometry is the measurement of the natural thermal electromagnetic radiation in our natural environment. It has its origin in the atomic and molecular state transitions in matter at a physical temperature above 0K. It appears as unpolarized, random, broadband radiation (noise) and is dependent on the chemical / physical composition of the body to be imaged, its surface quality, frequency, polarization and the physical temperature.

Microwave radiometry is the logical continuation of photographic imaging in the optical range and radiometry in the infrared wavelength range.

The lecture is interdisciplinary and covers the entire system chain of imaging systems (radiation properties of the measurement object - propagation medium - sensor technology - data analysis) on the ground, on aircraft and satellites.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
Each credit point corresponds to approximately 25-30 hours of work (of the student). This is based on the average student who achieves an average performance. The workload includes:

Attendance study time lecture / exercise: 30 h

Self-study time including exam preparation: 60 h

A total of 90 h = 3 LP

Recommendations
Knowledge of the basics of high frequency technology is helpful.

M 12.219 Module: Spaceborne Radar Remote Sensing [M-ETIT-103042]

Coordinators: Prof. Dr.-Ing. Thomas Zwick
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Information and Communication Technology (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-112857	Spaceborne Radar Remote Sensing - Exam	**ST**	**4 CP**	**Moreira, Prats**
T-ETIT-112858	Spaceborne Radar Remote Sensing - Workshop	**ST**	**2 CP**	**Younis**

Assessment
The assessment takes place in the form of a written examination lasting 120 min. and in the form of reports (other types of examination). Those reports have to be submitted as part of the SAR computer workshop (approx. a total of five workshops). Details will be given during the lecture.

Prerequisites
"M-ETIT-100426 - Spaceborne SAR Remote Sensing" is not allowed to be started or to be completed.

Competence Goal
The students obtain a sound knowledge on the fundamentals, theory and applications of spaceborne radar systems. They understand the principle and function of synthetic aperture radars (SAR). They are able to explain the theory, techniques, algorithms for data processing and system concepts as well as to report on several application examples.

Content
The lecture is interdisciplinary and well suited for students interested in learning different aspects of the entire end-to-end system chain of spaceborne radar systems. Today, Synthetic Aperture Radar (SAR) systems are generating images of the Earth´s surface with a resolution better than 1 meter. Due to their ability to produce high-resolution radar images independent of sunlight illumination and weather conditions, SAR systems have demonstrated their outstanding capabilities for numerous applications, ranging from environmental and climate monitoring, generation of three-dimensional maps, hazard and disaster monitoring as well as reconnaissance and security related applications. We have entered a new era of spaceborne and airborne SAR systems. New satellite systems like TerraSAR-X and TanDEM-X provide radar images with a resolution cell of more than a hundred times better than the one of conventional SAR systems. The lecture will cover all aspects of spaceborne radar systems including an overview of new technologies, applications and future developments.

Supporting the main lecture, exercise assignments are distributed to the students. The exercise solutions are presented and discussed in detail during lecture hall exercises. Further dedicated topics are explained to deepen the understanding of the main lecture contents.

The aim of the computer-workshop is to gain practical experience on radar systems using data and parameter simulations which are based on the evaluation of simplified models.

Module Grade Calculation
The module grade results of the assessment of the exam (4 LP) and the reports (2 LP).

Additional Information
Further information can be found at the internet page of the IHE (https://s.kit.edu/ihe-srrs).

Workload
Each credit point corresponds to approximately 25-30 hours of work (of the student). This is based on the average student who achieves an average performance. Workload (for a lecture)

Attendance time in lectures, exercises: 60 h

Present study time computer exercise: 40 h

Self-study time including exam preparation: 80 h

A total of 180 h = 6 LP

Recommendations
Signal processing and radar fundamentals.

Literature
Lecture viewgraphs, reading material, and literature references can be found on ILIAS at https://s.kit.edu/srrs.

M 12.220 Module: Stochastic Information Processing [M-INFO-107847]

Coordinators: Prof. Dr.-Ing. Uwe Hanebeck
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-115261	Stochastic Information Processing	**WT**	**6 CP**	**Hanebeck**

Assessment
See Partial Achievements (Teilleistung).

Prerequisites
See Partial Achievements (Teilleistung).

Competence Goal
Qualification objective: Students can describe a given nonlinear dynamic model probabilistically and set up the equations for Bayesian inference. If no analytical solution exists, they can estimate the strength of the nonlinearity and select and implement a suitable practical filter for real-time state estimation.

Learning objective: Students are familiar with dynamic state models and methods for recursively estimating the state. The advantages and disadvantages of the various practical filters can be assessed in a problem-oriented manner.

Content
To deal with complex dynamic systems, as they occur for example in robotics, one typically requires both system models and the temporal evolution of the system states. For both system identification and state reconstruction, however, usually only noisy data are available.

For continuous state spaces, an exact calculation of the probability densities is only possible in a few special cases. In practice, general nonlinear systems are often traced back to these special cases by simplifying assumptions. One extreme is linearization with subsequent application of linear estimation theory. However, this often leads to unsatisfactory results and requires additional heuristic measures. At the other extreme are numerical approximation methods, which only evaluate the desired probability densities at discrete points in the state space. Although the basic principle of these methods is usually quite simple, their practical implementation is often difficult and, in particular for higher-dimensional systems, computationally demanding.

As a middle ground, analytical nonlinear estimation methods would therefore often be desirable. In this lecture the main difficulties in the development of such estimation methods are outlined and corresponding solution modules are presented. Based on these building blocks, some analytical estimation methods are discussed in detail as examples, which are very suitable for practical implementation and offer a good compromise between computing effort and performance. Useful applications of these estimation methods are also discussed. Both known methods and the results of current research are presented.

Additional Information
As a theoretical foundation lecture, "Stochastic Information Processing" is an ideal introduction to the ISAS lectures. Conversely, prior knowledge of "Localization of Mobile Agents" (LMA) [course no. 24613] may nevertheless be helpful depending on the type of learner - more concrete applications are highlighted there. In all of our lectures, the content is always derived and explained in detail from the beginning; it is therefore possible to start with SI or LMA.

Workload
[1.5 h lecture + 1.5 h exercise (3 SWS)] x 15
+ [4.5 h follow-up lecture + 3.5 h preparation exercise] x 15
+ 15 h exam preparation
= 180 h ≙ 6 ECTS

Recommendations
Knowledge of the basics of probability theory is helpful.

M 12.221 Module: Student Innovation Lab [M-ETIT-105073]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Prof. Dr. Werner Nahm
Prof. Dr.-Ing. Eric Sax
Prof. Dr. Wilhelm Stork
Prof. Dr. Orestis Terzidis
Prof. Dr.-Ing. Thomas Zwick

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
15 CP	graded	Each term	2 terms	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-110291	Innovation Lab	**WT+ST**	**9 CP**	**Hohmann, Nahm, Sax, Stork, Zwick**
T-WIWI-102864	Entrepreneurship	**WT+ST**	**3 CP**	**Terzidis**
T-WIWI-110166	SIL Entrepreneurship Project	**WT**	**3 CP**	**Terzidis**

Assessment
This module consists of an 60-minute written exam on the contents of the Entrepreneurship lectures, as well as 5 other types of exams on the contents of the seminar Entrepreneurship and Innovation Lab in the form of term papers and presentations. All exams results are graded. Details are given in the module components and will be given during the course.

Prerequisites
none

Competence Goal
Personal competence

- Reflection faculty:
  The students are able to analyze, evaluate and develop an alternative for action for certain elements of action in social interaction
- Decision-making ability:
  The students are able to prepare a decision template in time and to provide the necessary arguments for alternative decisions and therefore are able to decide in time.
- Interdisciplinary teamwork
  Students are able to detect their limits of competence in one domain and to adjust to a the non-specialist domain. The students are able to detect a lack in competence and to compensate this lack via competences of other team members. The students are able to communicate their domain-specific knowledge and develop a basic understanding of other domains.
- Value-based action:
  The students are able to use selected psychological tools to determine their own values. They are able to match these values with team members and reflect if their offer fits these values.

Social competence

- Ability to cooperate:
  The students are able to analyze and judge their cooperative behavior in a group.
- Communication competence:
  The students are able to present their information in persuasive, focused and target group oriented way.
- Ability to deal with conflicts:
  The students are able to detect conflicts in advance, analyze them and name solution concepts.

Innovation and entrepreneurship competence

- Agile product development:
  The students are able to apply methods of agile product development e.g. Scrum.
- Methodical innovation retrieval:
  The students are able to conduct processes for user- and technology-centered innovation to develop sustainable value propositions for certain target groups (e.g. Design Thinking (DT), Technology Application Selection (TAS)-process).
- Orientation on management of new technology-based firms (NTBF):
  The students are able to name central concepts of intellectual property and legal structures. The students are able to name the most important tasks of entrepreneurial leadership. They are able to name the most common form of business modeling and to setup a business plan. The students know important approaches to establish an organization. The students are able to determine the ownership structure in an investment situation. The students are able to name marketing concepts and setup a business model.
- Generate investment readiness:
  The students are able to setup rudimentary revenue and cost plan. Furthermore, they are able to establish a project plan for a company in order to derive an investment plan. The students are able to present their business proposal to investors and develop empathy for the investors.
- Competence to develop a business model:
  The students are able to apply respective tools for business modeling e.g. Business Model Canvas. The students are able to develop and assess alternative business models.
- Risk handling:
  The students are able to name basic risks w.r.t. requirements, technical limitations and profitability. The students are able to apply methods of customer interaction for evaluation of requirements and willingness to pay. The students are able to setup a rudimentary competitors analyze. The students are able to name and identify risks and present potential reactions.

Systemic technical competence

- Problem solution competence:
  The students are able to analyze, assess and structurally solve a technical problem.
- Agile methodology of system development:
  The students are able to name and apply different system development processes.
- Validation in volatile environment:
  The students are able to conduct technical and economical validation under volatile constraints. For this, they are able to name the constraints and interpret the results of the validation.
- Functional decomposition:
  The students are able to identify, interpret and derive functional requirements from complex customer needs.
- Architecture development:
  The students are able to recognize coherences from the functional requirements and derive a suitable system architecture.

Content
This module strives to combine technical, social and personal competences from the technical and entrepreneurial domain. The objective is to prepare students as best as possible for entrepreneurial activity within or outside of an established organization. Our teaching methods are research-based with a practical orientation.

The lecture Entrepreneurship as the essential component offers the theoretical basis and provides insight in important theoretical concepts and empirical evidence. Currently released case studies and practical experiences of successful founders support the theoretical and empirical content. In order to run a company for the long term additional knowledge is important. That's why the lecture also teaches basic principles for opportunity recognition, business modeling, an introduction to entrepreneurial marketing and leadership. Customer-based design methods from the lean startup approach as well as methods of technology-centered innovation are presented. Future founders have to be able to develop and handle resources such as financial and human capital, infrastructure and intellectual property. Further aspects tackle the establishment of an organization and funding of the own project.

The knowledge taught in the lecture Entrepreneurship will be applied in an application-oriented seminar and the labs. Hence we use an action learning approach to extend the taught knowledge by practical skills and reflection capabilities. In an team of five, the students will experience their way from the ideation process to the final pitch in front of investors.

The students are able to choose between the following options concerning the labs:

- The Automation Innovation Lab offers drones as an innovation platform for cooperative swarm solutions.
- The Industry 4.0 Innovation Lab enables innovation in the context of the next industrial revolution via mobile robot platforms.
- In the Interconnected Intelligent Systems Lab innovations in the context of Assisted Living and Smart Housing are enabled by providing a rich assembly set of mobile robots, actuators and sensors.
- The Computer Vision for Health Lab offers a selection of state-of-the-art imaging devices and powerful computing hardware for innovative image-based applications for medicine and healthcare.

The module also presents methods of agile system development (Scrum) along with associated validation methods as well as methods for functional prototyping. Gate plans are used within the module to determine the progress of the project. Methods for single person work and teamwork are presented and applied. Additionally group-specific knowledge of the different roles of team members, solutions to conflict situations and interdisciplinary teams are presented.

Module Grade Calculation
The module grade consists of the written exam of the Lecture Entrepreneurship (40%), of the submissions and presentation of the Innovation Lab (40%) and of the submissions and presentation of the SIL Entrepreneurship Project (20%).

Additional Information
An application is required to participate in this module. Information about the application: www.kit-student-innovation-lab.de.

Workload
Lecture Entrepreneurship: 32h attendance time, 48h preparation and follow-up time, 10h preparation time for assessment

Seminar Entrepreneurship: 34h attendance time, 3h preparation and follow-up time, 53h preparation time for assessment.

Innovation Lab: 8h attendance time, 213h preparation and follow-up time, 49h preparation time for assessment.

This results in a total of 450 hours and a total of 15 LPs for both semesters (15*30/2 = 225).

Recommendations
It is recommended to attend the lecture Entrepreneurship at the same time as the seminar Entrepreneurship Project and the Innovation Lab in the winter semester.

Teaching and Learning Methods
Related courses:

Lecture Entrepreneurship
Seminar Entrepreneurship Project
Innovation Labs
Please note that the courses must be booked in parallel.

Related exams:

Written exams covering the content of lecture Entrepreneurship
Presentation of the Value Profile (seminar Entrepreneurship)
Submission of the Business Plan (seminar Entrepreneurship)
Submission of a Technical Report with requirements list and system architecture (Innovation Lab)
Submission of the reflection of the Gate Plans (Innovation Lab)
Presentation of the High-fidelity (Innovation Lab)

M 12.222 Module: Superconducting Magnet Technology [M-ETIT-106684]

Coordinators: Prof. Dr. Tabea Arndt
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113440	Superconducting Magnet Technology	**ST**	**4 CP**	**Arndt**

Assessment
The examination takes place in form of an oral exam (abt. 30 minutes).

Two timeslots (weeks) for examination dates will be announced (usually near end of lecture period & end of semester)

Prerequisites
none

Competence Goal

- The students have a solid knowledge of architecture and design aspects of applications in magnets, windings and coils in power engineering.
- For the most important magnet applications the students can apply the state of the art, choose between options and can reflect the main benefits.
- The students have a clear understanding of opportunities, benefits and limitations of superconducting windings and magnets.
- The students are able to perform the required design calculations and to solve fundamental design questions independently.

Content
As the materials become increasingly mature and powerful, using superconductivity in a variety of applications of electrical engineering is of rising interest and benefit, too. This module is focuses on Superconducting Magnet Technology:

Windings, coils and magnets may be used as a device by itself (providing high magnetic fields e.g. in MRI, NMR, accelerators, industry magnets, etc.) or as components for Power Systems.

This section will cover the following aspects:

- Unique selling points of superconducting windings.
- Basic approaches and tools to design superconducting windings.
- Discussion of winding architectures
- Criteria to design the appropriate operating temperatures, materials, conductors, cooling technology for the electromagnetic purpose.
- Limits and opportunities when preparing and operating superconducting windings.
- Measures for safe operation of superconducting magnets.
- High-Field Magnets
- Magnets for Fusion Technology
- 3D topologies (e.g. in dipole magnets or motors/ generators)
- New options potentially offered by widespread use of hydrogen.
- New winding topologies

In the exercises, selected magnets will be designed and calculated analytically and with some computational tools (e.g. dipole magnets and compact, cryogen free HTS-magnets)

The lecturer may change the details of the content without further notice. Materials will be offered on ILIAS.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload

1. attendance in lectures and exercises: 15*3 h = 45 h
2. preparation / follow-up: 15*3 h = 45 h
3. preparation of and attendance in examination: 30 h

A total of 120 h = 4 CR

Recommendations
Having knowledge in “Superconducting Materials” is beneficial, but not mandatory.

M 12.223 Module: Superconducting Materials [M-ETIT-105521]

Coordinators: Dr. Jens Hänisch
Prof. Dr. Bernhard Holzapfel

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	2 terms	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114729	Superconducting Materials Lecture	**WT**	**3 CP**	**Hänisch, Holzapfel**
T-ETIT-114730	Superconducting Materials Lab	**ST**	**3 CP**	**Hänisch, Holzapfel**

Assessment
The success contral takes place in two parts:

1. Part I (lecture): in the form of an oral examination lasting approx. 30 minutes.
2. Part II (lab): in the form of a written protocol (approx. 40 pages).

Prerequisites
none

Competence Goal
The students have a good knowledge and can describe and compare the properties of different superconducting materials including those currently employed in energy and electronic applications (niobium-based superconductors, oxocuprates, MgB2) and also promising recently discovered ones (pnictides), including their synthesis methods.

Students have a thorough understanding of the synthesis variations of superconducting materials in bulk, thin film and wire form as well as the close relationship between microstructural properties of superconductors and their current carrying capabilities. They are able to select the appropriate superconducting materials for the different application scenarios of superconductors.

The students are able to talk about topic-related aspects in English using the technical terminology of the field of study.

Content
This lecture series gives an overview on the basic properties of the known different classes of superconducting materials as well as their synthesis routes in bulk, thin film and wire form. Special emphasis s given to the close interaction of micro- and nanoscale microstructural properties and the superconducting electrical transport properties, which are the key to all large scale applications in power and magnet technology.
The lecture series will cover basic properties of superconductors, superconducting elements, classical metallic superconducting alloys and compounds, high temperature superconductors, Fe-based superconductors and some other “exotic” superconductors, synthesis of superconducting films and wires, superconducting critical currents and pinning in type II superconductors as well an overview on the most prominent applications of superconductors in electronics, medicine and power application.
The obligatory practical work covers a few experiments regarding the synthesis and characterization of superconducting materials.
The lecturer reserves the right to alter the contents of the course without prior notification.
Course material will be available on ILIAS. Up-to-date information will be available via the ITEP- homepage prior to the beginning of the semester.

Module Grade Calculation
The module grade is the average grade of the oral exam and the lab protocol.

Additional Information
WiSe: Superconducting Materials Part I - lecture

SoSe: Superconducting Materials Part II - lab

Workload
A workload of approx. 186h is required for the successful completion of the module. This is composed as follows:

- Attendance time in lectures: 30*2h = 60h
- Preparation and follow-up of lectures: 30*3h = 90h
- Preparation for the exam: 30h

Recommendations
It is recommanded to pass the lecture (part I) before the lab (part II).

M 12.224 Module: Superconducting Nanowire Detectors [M-ETIT-105609]

Coordinators: Dr. Konstantin Ilin
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111236	Superconducting Nanowire Detectors	**ST**	**4 CP**	**Ilin**

Assessment
Oral Exam (approx. 20 min.)

Prerequisites
Module "M-ETIT-102332 - Thin films: technology, physics and applications" + Thin Films: Technology, Physics and Applications II must not be started.

Competence Goal
Students should be able to discuss interplay between growth conditions of thin films, physical and geometrical properties of nanostructure made of these films, and performance and suitable areas of application of detectors of radiation based on interaction of these nanostructures with electromagnetic power. The knowledge obtained by students should provide a theoretical basis for the most important steps in development of thin film nanoelectronic devices.

Content
Students will get practically oriented information about technology of thin films including different methods of deposition of thin films like magnetron sputtering, thermal evaporation, pulsed laser ablation, about basics of vacuum technology, and about mechanisms of growth of thin films of different materials at different conditions.

Patterning methods (photo- and e-beam lithography, reactive ion etching, ion milling, and lift-off techniques) suitable for nanometer scale features of electronic devices will be considered in details.

Experimental methods of characterization of material, geometrical, optical, physical, superconducting, electron and phonon properties of thin films, nanostructures made of these films, and devices based on these nanostructures will be discussed.

Consideration of technology and physics of thin film structures will be done on example of development of three types of fast and sensitive detectors of electro-magnetic radiation for applications in optical and THz spectral ranges: superconducting nanowire single-photon detector, hot-electron bolometer, and YBCO ps-fast detector of synchrotron emission. Dependence of detector's performance on their fabrication condition will be analyzed in frame of physical models which describe response mechanisms of the detectors to absorbed radiation.

Practical actualization of the knowledge is possible in frame of Praktikum Nanoelektronik (LVN 23669).

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
The workload in hours is broken down below:

1. Attendance time in lectures in the winter semester 15*3h = 45h
2. Preparation / follow-up of the same 15*3h = 45h
3. Exam preparation and attendance in the same 30h

Recommendations
Previous participation on Module "Physics, Technology and Applications of thin films " is recommanded.

M 12.225 Module: Superconducting Power Systems [M-ETIT-106683]

Coordinators: Prof. Dr.-Ing. Mathias Noe
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113439	Superconducting Power Systems	**WT**	**4 CP**	**Noe**

Assessment
The examination takes place in form of an oral exam (abt. 45 minutes).

Prerequisites
none

Competence Goal

- The students have a solid knowledge of architecture and design aspects of applications in windings and energy technology devices.
- For the most important power system applications the students can apply the state of the art and can reflect the main benefits.
- The students have a clear understanding of opportunities, benefits and limitations of superconducting components and devices.
- The students are able to perform the required design calculations and to solve fundamental design questions independently.

Content
As the materials become increasingly mature and powerful, using superconductivity in a variety of applications of electrical engineering is of rising interest and benefit, too. This module focuses on Superconducting Power Systems.

It will provide an overview of the state of the art, will give an insight into the basic setup, the design, the characteristic parameters and the specific operation behaviour of the following applications:

· Power Transmission Cables and Lines
· Motors and Generators
· Transformers
· Fault Current Limiters
· Magnetic Energy Storage
· Basics of Cryo Technology

For each application a design example is shown and the focus is given on the conceptual design of each application.

The lecturers may change the details of the content without further announcement.
Materials will be offered on ILIAS.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload

1. attendance in lectures and exercises: 15*3 h = 45 h
2. preparation / follow-up: 15*3 h = 45 h
3. preparation of and attendance in examination: 30 h

A total of 120 h = 4 CR

Recommendations
Having knowledge in „Superconducting Materials“ is beneficial.
Successful participation in „Superconductivity for Engineers“

M 12.226 Module: Superconductivity for Engineers [M-ETIT-105611]

Coordinators: Prof. Dr. Bernhard Holzapfel
Prof. Dr. Sebastian Kempf

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Focus Area)
Field of Specialization / Electrical Power Systems and Electromobility (Fundamentals)
Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111239	Superconductivity for Engineers	**WT**	**6 CP**	**Holzapfel**

Assessment
The assessment of success takes place in the form of a written examination lasting 120min.

Prerequisites
none

Competence Goal
Students know the physical fundamentals of superconductivity and can place various theoretical and practical aspects of superconductivity in the overall context. They understand the principles behind specific applications of superconductivity and are able to communicate with experts in the field.

Content
Superconductivity is one of the most fascinating and astonishing effects in solid state physics. It plays technologically an important role in many modern, scientific, medical and industrial applications. It establishes, for example, the basis of realizing high field electromagnets to be used in magnetic resonance imaging systems in healthcare or for guiding charged particle in modern particle accelerators such as the LHC. Moreover, it allows to build state-of-the-art energy systems as well as sensing devices such as magnetic field sensors or energy-dispersive single particle sensors. In addition, it is conceivable that superconductivity will be utilized in near future for energy and traffic engineering applications, e.g. for dissipationless power transmission over large distances or high-speed trains connecting major cities.

Within this context, this module gives a comprehensive introduction in the basics of superconductivity paving the way for the discussion of state-of-the-art applications of superconductivity. In particular, the module will cover the following topics:

- Historical remarks
- Overview of superconducting materials and applications of superconductivity
- Reminder of normal metals: free electron gas, Drude and Sommerfeld model, electrical and thermal properties, band structure
- Phenomena of superconductivity: zero electrical dc resistance, Meissner Ochsenfeld effect
- Thermodynamics and thermal properties of superconductors
- Phenomenological theories of superconductors: Two-fluid model, London theory, Pippard theory, Ginzburg-Landau theory
- Microscopic theory of conventional superconductors
- Type-I and type-II superconductivity
- Magnetic properties of type-I and type-II superconductors
- Irreversible magnetic properties, Bean model
- AC losses
- Electrical and thermal stabilization
- Energy gap and quasiparticle tunneling
- Unconventional superconductors
- High-frequency electrodynamics of superconductors
- Macroscopic quantum effects
- Overview of applications of superconductivity
- Superconducting magnets, especially for fusion
- Superconducting devices in power engineering
- Thermo-mechanical properties of relevant construction materials for superconducting applications

The tutorial is closely connected to the lecture and deepens important aspects from the field of superconductivity. Using exercises, important theories and effects as well as the realization of applications of superconductivity are discussed.

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
A workload of approx. 180h is required for the successful completion of the module. This is composed as follows:

- Attendance time in lectures and exercises: 15*2*2h = 60h
- Preparation and follow-up of lectures: 22*3 h = 66h
- Preparation and follow-up of tutorials: 8*3h = 24h
- Preparation for the exam: 30h

Recommendations
None

M 12.227 Module: Supplementary Studies on Science, Technology and Society [M-FORUM-106753]

Coordinators: Dr. Christine Mielke
Christine Myglas

Organisation: General Studies. Forum Science and Society (FORUM)

Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
16 CP	graded	Each term	3 terms	German	3	1

Election Notes

Students have to self-record the achievements obtained in the Supplementary Studies on Science, Technology and Society in their study plan. FORUM (formerly ZAK) records the achievements as "non-assigned" under "ÜQ/SQ-Leistungen". Further instructions on self-recording of achievements can be found in the FAQ at https://campus.studium.kit.edu/ and on the FORUM homepage at https://www.forum.kit.edu/english/. The title of the examination and the amount of credits override the modules placeholders.

If you want to use FORUM achievements for both your Interdisciplinary Qualifications and for the Supplementary Studies, please record them in the Interdisciplinary Qualifications first. You can then get in contact with the FORUM study services (stg@forum.kit.edu) to also record them in your Supplementary Studies.

In the Advanced Unit you can choose examinations from three subject areas: "About Knowledge and Science", "Science in Society" and "Science in Social Debates". It is advised to complete courses from each of the three subject areas in the Advanced Unit.

To self-record achievements in the Advanced Unit, you have to select a free placeholder partial examination first. The placeholders' title do *not* affect which achievements the placeholder can be used for!

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-FORUM-113578	Lecture Series Supplementary Studies on Science, Technology and Society - Self Registration	**ST**	**2 CP**	**Mielke, Myglas**
T-FORUM-113579	Basic Seminar Supplementary Studies on Science, Technology and Society - Self Registration	**ST**	**2 CP**	**Mielke, Myglas**
Advanced Unit Supplementary Studies on Science, Technology and Society (Elective: at least 12 credits)				
T-FORUM-113580	Elective Specialization Supplementary Studies on Science, Technology and Society / About Knowledge and Science - Self-Registration	**WT+ST**	**3 CP**	**Mielke, Myglas**
T-FORUM-113581	Elective Specialization Supplementary Studies on Science, Technology and Society / Science in Society - Self-Registration	**WT+ST**	**3 CP**	**Mielke, Myglas**
T-FORUM-113582	Elective Specialization Supplementary Studies on Science, Technology and Society / Science in Public Debates - Self Registration	**WT+ST**	**3 CP**	**Mielke, Myglas**
Mandatory				
T-FORUM-113587	Registration for Certificate Issuance - Supplementary Studies on Science, Technology and Society	**WT+ST**	**0 CP**	**Mielke, Myglas**

Assessment
The monitoring is explained in the respective partial achievement.

They are composed of:
- Protocols
- Reflection reports
- Presentations
- Preparation of a project work
- An individual term paper
- An oral examination
- A written exam

Upon successful completion of the supplementary studies, graduates receive a graded report and a certificate issued by the FORUM.

Prerequisites
The course is offered during the course of study and does not have to be completed within a defined period. Enrollment is required for all assessments of the modules in the supplementary studies.
Participation in the supplementary studies is regulated by § 3 of the statutes. KIT students register for the supplementary studies by selecting this module in the student portal and booking a performance themselves. Registration for courses, assessments, and exams is regulated by § 8 of the statutes and is usually possible shortly before the start of the semester.
The course catalog, module description (module manual), statutes (study regulations), and guidelines for creating the various written performance requirements can be downloaded from the FORUM homepage at **https://www.forum.kit.edu/begleitstudium-wtg.php**.

Registration and exam modalities
PLEASE NOTE:
Registration on the FORUM, i.e. additionally via the module selection in the student portal, enables students to receive up-to-date information about courses or study modalities. In addition, registering on the FORUM ensures that you have proof of the credits you have earned. As it is currently (as of winter semester 24-25) not yet possible to continue additional credits acquired in the Bachelor's programme electronically in the Master's programme, we strongly advise you to digitally secure the credits you have earned by archiving the Bachelor's transcript of records yourself and by registering on FORUM.
In the event that a transcript of records of the Bachelor's certificate is no longer available - we can only assign the achievements of registered students and thus take them into account when issuing the certificate.

Competence Goal
Graduates of the Supplementary Studies on Science, Technology, and Society gain a solid foundation in understanding the interplay between science, the public, business, and politics. They develop practical skills essential for careers in media, political consulting, or research management. The program prepares them to foster innovation, influence social processes, and engage in dialogue with political and societal entities. Participants are introduced to interdisciplinary perspectives, encompassing social sciences and humanities, to enhance their understanding of science, technology, and society. The teaching objectives of this supplementary degree program include equipping participants with both subject-specific knowledge and insights from epistemological, economic, social, cultural, and psychological perspectives on scientific knowledge and its application in various sectors. Students are trained to critically assess and balance the implications of their actions at the intersection of science and society. This training prepares them for roles as students, researchers, future decision-makers, and active members of society.

Through the program, participants learn to contextualize in-depth content within broader frameworks, independently analyze and evaluate selected course materials, and communicate their findings effectively in both written and oral formats. Graduates are adept at analyzing social issues and problem areas, reflecting on them critically from a socially responsible and sustainable standpoint.

Content
The Supplementary Studies on Science, Technology and Society can be started in the 1st semester of the enrolled degree programme and is not limited in time. The wide range of courses offered by FORUM makes it possible to complete the program usually within three semesters. The supplementary studies comprises 16 or more credit points (LP). It consists of **two modules: the Basic Module (4 LP) and the Advanced Module (12 LP).**

The **basic Module** comprises the compulsory courses 'Lecture Series Supplementary Studies on Science, Technology and Society' and a basic seminar with a total of 4 LP.

The **Advanced Module** comprises courses totalling 12 LP in the humanities and social sciences subject areas 'On Knowledge and Science', 'Science in Society' and 'Science in Public Debates'. The allocation of courses to the accompanying study programme can be found on the homepage https://www.forum.kit.edu/wtg-aktuelland in the printed FORUM course catalogue.

The 3 thematic subject areas:

Subject area 1: About Knowledge and Science

This is about the internal perspective of science: students explore the creation of knowledge, distinguishing between scientific and non-scientific statements (e.g., beliefs, pseudo-scientific claims, ideological statements), and examining the prerequisites, goals, and methods of knowledge generation. They investigate how researchers address their own biases, analyze the structure of scientific explanatory and forecasting models in various disciplines, and learn about the mechanisms of scientific quality assurance.

After completing courses in the "Knowledge and Science" area, students can critically reflect on the ideals and realities of contemporary science. They will be able to address questions such as: How robust is scientific knowledge? What are the capabilities and limitations of predictive models? How effective is quality assurance in science, and how can it be improved? What types of questions can science answer, and what questions remain beyond its scope?

Subject area 2: Science in Society

This focuses on the interactions between science and different areas of society, such as how scientific knowledge influences social decision-making and how social demands impact scientific research. Students learn about the specific functional logics of various societal sectors and, based on this understanding, estimate where conflicts of goals and actions might arise in transfer processes—for example, between science and business, science and politics, or science and journalism. Typical questions in this subject area include: How and under what conditions does an innovation emerge from a scientific discovery? How does scientific policy advice work? How do business and politics influence science, and when is this problematic? According to which criteria do journalists incorporate scientific findings into media reporting? Where does hostility towards science originate, and how can social trust in science be strengthened?

After completing courses in the "Sciene in Society" area, students can understand and assess the goals and constraints of actors in different societal sectors. This equips them to adopt various perspectives of communication and action partners in transfer processes and to act competently at various social interfaces with research in their professional lives.

Subject area 3: Science in Public Debates

The courses in this subject area provide insights into current debates on major social issues such as sustainability, digitalization, artificial intelligence, gender equality, social justice, and educational opportunities. Public debates on complex challenges are often polarized, leading to oversimplifications, defamation, or ideological thinking. This can hinder effective social solution-finding processes and alienate people from the political process and from science. Debates about sustainable development are particularly affected, as they involve a wide range of scientific and technological knowledge in both problem diagnosis (e.g., loss of biodiversity, climate change, resource consumption) and solution development (e.g., nature conservation, CCS, circular economy).

By attending courses in "Science in Public Debates," students are trained in an application-oriented way to engage in factual debates—exchanging arguments, addressing their own prejudices, and handling contradictory information. They learn that factual debates can often be conducted more deeply and with more nuance than is often seen in public discourse. This training enables them to handle specific factual issues in their professional lives independently of their own biases and to be open to differentiated, fact-rich arguments.

Supplementary credits:
Additional LP (supplementary work) totalling a maximum of 12 LP can also be acquired from the complementary study programme (see statutes for the WTG complementary study programme § 7). § 4 and § 5 of the statutes remain unaffected by this. These supplementary credits are not included in the overall grade of the accompanying study programme. At the request of the participant, the supplementary work will be included in the certificate of the accompanying study programme and marked as such. Supplementary coursework is listed with the grades provided for in § 9.

Module Grade Calculation
The overall grade of the supplementary course is calculated as a credit-weighted average of the grades that were achieved in the advanced module.

Additional Information
Climate change, biodiversity crisis, antibiotic resistance, artificial intelligence, carbon capture and storage, and gene editing are just a few areas where science and technology can diagnose and address numerous social and global challenges. The extent to which scientific findings are considered in politics and society depends on various factors, such as public understanding and trust, perceived opportunities and risks, and ethical, social, or legal considerations.
To enable students to use their expertise as future decision-makers in solving social and global challenges, we aim to equip them with the skills to navigate the interfaces between science, business, and politics competently and reflectively. In the Supplementary Studies, they acquire foundational knowledge about the interactions between science, technology, and society.

They learn:
- How reliable scientific knowledge is produced,
- how social expectations and demands influence scientific research, and
- how scientific knowledge is adopted, discussed, and utilized by society.

The program integrates essential insights from psychology, philosophy, economics, social sciences, and cultural studies into these topics. After completing the supplementary studies programme, students can place the content of their specialized studies within a broader social context. This prepares them, as future decision-makers, to navigate competently and reflectively at the intersections between science and various sectors of society, such as politics, business, or journalism, and to contribute effectively to innovation processes, public debates, or political decision-making.

Workload
The workload is made up of the number of hours of the individual modules:

- Basic Module approx. 120 hours
- Advanced Module approx. 360 hours
> Total: approx. 480 hours

In the form of supplementary services, up to approximately 360 hours of work can be added.

Recommendations
It is recommended to complete the supplementary study program in three or more semesters, beginning with the lecture series on science, technology, and society in the summer semester. Alternatively, you can start with the basic seminar in the winter semester and then attend the lecture series in the summer semester.

Courses in the Advanced Module can be taken simultaneously. It is also advised to complete courses from each of the three subject areas in the advanced unit.

Teaching and Learning Methods
- Lectures
- Seminars/Project Seminars
- Workshops

M 12.228 Module: System Integration and Communication Structures in Industry 4.0 and IoT [M-ETIT-106026]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-112212	System Integration and Communication Structures in Industry 4.0 and IoT	WT	3 CP	Becker

Assessment
Success control takes place in form of an overall oral examination (approx. 20 minutes). The module grade is the grade of the oral examination.

Prerequisites
none

Competence Goal
Students are familiar with the theoretical and practical aspects of system integration of measuring devices (field devices) and measuring systems in the horizontal and vertical automation and communication structures of Industry 4.0 and IoT.

Content

- Comprehensive overview of automation technology since 1970 to 2020 and its future, summarized in a time scale.
- Contexts regarding system integration of the automation of measurement devices and systems.
- Industry 4.0 and IoT and how they fit into automation technology.
- Offline, online and inline automation in their development based on the technical complexity pyramid mentioned above.
- The time scale of milestones in automation from the beginning of Ethernet and fieldbus system developments, the development of the OSI model and the automation pyramid.
- Presentation and discussion of the 7-layer OSI model and the automation pyramid based on it as the interface basis of the important bus systems in automation with regard to horizontal and vertical communication structures.
- AI and predictive maintenance as pillars of Industry 4.0 and IoT as well as their applications over more than 50 years depending on the available technologies of electronics and software.
- Harmonization efforts and standardization in automation and system technology to contain the rapidly growing fieldbus systems.
- Development of field devices and automation components in the context of the influence of electronics, µ-controller, DSP, FPGA, multi-core processor and chip-on-bond developments as well as software developments with examples from measurement technology.
- The problem of adapting field devices and components to the harsh operating and environmental conditions in factory and process automation, illustrated in a complexity pyramid (EMC, protection class IPxx, explosion protection, CE).
- Due to the complexity of the requirements, quality control in measurement technology is the last step in system integration and communication networking in the automation chain based on analytical measurement technology and physical parameters such as pressure, level, flow and temperature.
- The influence of µ-controllers, DSPs (Digital Signal Processors), FPGAs, ASICs etc. on the transition from mainframes, workstations to PCs and the integration of ever smaller electronics into field devices up to the current state of the art is presented.
- Discussion and comparison of the distribution of tasks between SCADA/HMI-MES and ERP systems. In particular, the hierarchical relationships from the ERP system down to the field devices, the communication levels and their typical network topologies with the different bus systems are discussed.
- Presentation of future technologies in various industries and their innovation in measurement technology with modern communication and automation interfaces. Examples for automotive, solar, semiconductor and environmental technology
- Distribution of bus systems in the markets explained and what you should know about them at a glance.
- In this context, I also show the problems that SMEs have faced and continue to face in automation and how many companies are still trying to catch up in automation technology today. The focus here is also on the adaptation of old devices to modern automation interfaces and the development of new platforms for automation-compatible applications, for some of which solutions are presented.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload

1. Attendance time in lectures: 15*2 h = 30 h
2. Preparation/follow-up of the same: 15*2 h = 30 h
3. Exam preparation and attendance in the same: 30 h

Total: 90 LP = 3 CR

M 12.229 Module: System-on-Chip Lab [M-ETIT-107763]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Prof. Dr. Ivan Peric

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Field of Specialization / Microelectronics, Photonics, and Quantum Technologies (Lab Course)
Field of Specialization / Information and Communication Technology (Lab Course)
Field of Specialization / Automation, Robotics, and Systems Engineering (Lab Course)
Electives (Lab Course)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-115155	System-on-Chip Lab	**WT**	**6 CP**	**Becker, Peric**

Assessment
Success control takes place in the form of other types of examination. The examination consists of:

- Preparation sheets for the introductory lab material (approx. 6 pages each) (7 submissions per group)
- GitLab project involving the implementation and documentation of project work on a course-relevant topic (1 project per group, 6 weeks processing time)
- Presentation on the project work (approx. 15 minutes per group, followed by a 5-minute Q&A session)
- Oral exam on the lab and project content (15 minutes per student)

The overall impression is rated.

Prerequisites
none

Competence Goal
Students can reproduce basic knowledge of digital and analog circuit design and hardware-related software programming. In practice, students are able to apply these methods in the following areas using a current system-on-chip architecture:

- Design of a system architecture for mixed-signal systems
- Simulation of the designed digital and analog circuits
- Debugging the implementations at the simulation and implementation level
- Verification of the overall system developed through test benches

In addition, they can apply the hardware / software code design approach and can evaluate implementation targets based on the given requirements (FPGA and ASIC).

Content
In the System-on-Chip Laboratory, a complete mixed-signal hardware architecture for audio playback based on a System-on-Chip (SoC) is developed.

The system design includes the creation of necessary sub-components, their integration into an overall system, and the simulation and verification of the individual components and the overall system. A prototype is implemented and tested on an FPGA basis. The integration is then prepared for a possible ASIC production. Analog circuits are also considered and designed to build an audio amplifier.

Module Grade Calculation
The module grade results from the combination of the processing of the exercise sheets, the evaluations during the lab and a final presentation including discussion of the results developed in the project. Details will be given during the lecture.

Workload

1. Presence time in laboratory appointments: 15 * 4 = 60 hours
2. Preparation / post-processing: 7 * 6 = 42 hours
3. Project preparation: 3 * 10 = 30 hours
4. Preparation of the final presentation: 30 hours
5. Preparation for the oral exam: 18 hours

Total: 180 h

Recommendations

- Knowledge of Verilog Hardware Description Language, e.g. from Digital Circuit Design
- Knowledge in the design of analog circuits (amplifier circuits, stability considerations), e.g. from the Analog Circuit Design
- Knowledge of VHDL design, e.g. from Hardware Modeling and Simulation
- Knowledge of simulation of digital circuits, e.g. from Hardware Modeling and Simulation
- Knowledge of hardware design processes and algorithms, e.g. from Hardware Synthesis and Optimisation

M 12.230 Module: Systems and Software Engineering [M-ETIT-100537]

Coordinators: Prof. Dr.-Ing. Eric Sax
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Field of Specialization / Automation, Robotics, and Systems Engineering (Fundamentals)
Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100675	Systems and Software Engineering	**WT**	**6 CP**	**Sax**

Assessment
Written exam of 90 minutes.

Students are given the opportunity to earn a grade bonus through separate task assignments. If the grade of the written exam is between 4.0 and 1.3, the bonus improves the grade by a maximum of one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the lecture. Bonus points do not expire and remain valid for exams taken at a later date.

Prerequisites
none

Competence Goal
• Students are able to analyse and explain the functional principles and applications of embedded systems.

• Students are able to evaluate and apply maturity models as well as Software Development Life Cycle models including the waterfall model, V-model, prototyping model, agile models, and DevOps.

• Students are able to apply various creativity techniques to develop innovative solutions to problems. They will be able to derive and analyse requirements.

• Students are familiar with diagram formats software modelling languages; they can evaluate and create these based on problem descriptions of an application area. They will be able to create and evaluate functional, data-oriented, algorithmic, state-oriented, and object-oriented views.

• Students are able to understand and apply various aspects of the realization of embedded systems. They will be able to consider implementation alternatives: hardware, co-design and scheduling aspects.

• Students are familiar with the various testing phases in a project and can explain them. They can assess the reliability of a system and understand the concept of functional safety.

Content
The focus of the course is on processes and methods for the design of systems composed of electrical, electronic and electronically programmable systems that contain software, hardware and mechanical components. The desired competencies of the course include the knowledge and goal-oriented use of modeling techniques, design processes, description and representation tools as well as specification languages that correspond to the current state of the art.

Module Grade Calculation
The grade is determined by the written exam and the bonus points.

Workload

- 15 weeks of 1.5h attendance in lecture and of 1.5h attendance in each exercise = 45 h
- 15 weeks of 2h preparation and follow-up per lecture and at least 3h preparation for each exercise (includes processing of exercise sheets and additional learning material, the processing of tasks for the acquisition of bonus points) = 75h
- Preparation for the exam = 60h

In total: 180h

Recommendations
Knowledge in Digital Technology and Information and Automation Technology (e.g. module M-ETIT-102102 and M-ETIT-106336)

M 12.231 Module: Systems Engineering for Automotive Electronics [M-ETIT-100462]

Coordinators: Prof. Dr.-Ing. Jürgen Bortolazzi
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100677	Systems Engineering for Automotive Electronics	**ST**	**4 CP**	**Bortolazzi**

Assessment
The type and method (written or oral examination) of the assessment will be announced at the beginning of the course.

The examination will take place without any aids.

Prerequisites
none

Competence Goal
Students are familiar with the systematic development process of electrical and electronic systems and architectures in the field of vehicle technology and the automotive industry. They are able to apply the tools supporting systematic development and to describe electrical and electronic architectures based on models. You will be able to analyze and evaluate systems in the domains of functional and physical modeling.

Content
The lecture provides knowledge of methods, techniques and procedures used in the development phases of electrical and electronic systems for vehicles.

Module Grade Calculation
The module grade is the grade of the written or oral examination. Attendance of the laboratory/exercise for the lecture is a prerequisite for admission to the examination.

Additional Information
The type of assessment (written or oral examination) will be announced at the beginning of the course.

The lecture will be examined in writing on the main date; for the follow-up date, the examination may also be oral.

The examination takes place without aids.

Attendance of the laboratory/exercise for the lecture is a prerequisite for admission to the examination.

Workload
1. attendance time in lecture and exercise: 60 h

2. preparation/follow-up of the same: 30 h

3. exam preparation: 30 h

Recommendations
It is recommended to attend the lecture SE (23611)

M 12.232 Module: Team Project: Sensors and Electronics [M-ETIT-105465]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Prof. Dr.-Ing. Thomas Zwick

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	pass/fail	Each term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111007	Team Project: Sensors and Electronics	**WT+ST**	**3 CP**	**Ulusoy, Zwick**

Assessment
Final design report (approx. 8-10 pages)

Prerequisites
An application is required to participate in this module. Information about the application: https://www.ihe.kit.edu/VorlesungenWS_4850.php

Competence Goal

- Students independently expand their knowledge in a selected field of sensors and electronics beyond lecture contents
- Students will go through multiple phases from conceptual design to measurement and thereby gain practical experience
- Students are able to respect practical limits when applying theoretical knowledge
- Students can consider available measurement equipment in their design process
- Students are able to work in a team and organize a project independently and under their own responsibility

Content
Students will develop a system or a component in the field of sensors and electronics. The project's content has to go beyond the content of a single lecture or require the combination of multiple lectures. It has to be connected to general challenges or problems in research. On top of the theoretical aspects students will have to plan and execute their project's realization. Manufacturing tolerances/limits and available measurement equipment have to be respected. This module can be combined with an international student design competition.

Module Grade Calculation
No Grade. Only pass or fail.

Workload
A total of 90 hours (per Student) is estimated for the completion of this module.

1. Planning and conceptual design: 10h
2. Simulation & Design: 50h
3. Assembly, Verification, Measurement: 15h
4. Final report: 15h

Recommendations
Lectures at the Institute for Radio Frequency Engineering and Electronics (IHE)

M 12.233 Module: Technical Acoustic [M-ETIT-101835]

Coordinators: Dr. Nicole Ruiter
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-104579	Technical Acoustic	**WT**	**3 CP**	**Ruiter**

Assessment
Success control is carried out as part of an overall oral examination (approx. 20 minutes) of the selected courses, with which the minimum requirement for CP is met.

Prerequisites
none

Competence Goal
The students understand the basics of acoustics and their technical applications and can understand the basic technical implementation.

Content
The lecture deals with the basics of sound and sound propagation. In addition to sound generation, the measurement and analysis methods for sound, the perception of sound in humans and is also discussed. Selected applications and their technical implementation are presented.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
The workload includes:

Attendance time in lectures (2 h 15 appointments each) = 30 h

Self-study (3 h 15 appointments each) = 45 h

Preparation / post-processing = 20 h

Total effort approx. 95 hours = 3 LP

M 12.234 Module: Technical Optics [M-ETIT-100538]

Coordinators: Prof. Dr. Cornelius Neumann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
5 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100804	Technical Optics	**WT**	**5 CP**	**Neumann**

Assessment
Success is assessed in the form of a written examination lasting 120 minutes. The module grade is the grade of the written examination.

Prerequisites
none

Competence Goal
Students learn the basics of imaging and non-imaging optics, as well as their applications using examples of optical observation and measurement methods, data storage, micro- and nano-optics, as well as the manufacturing methods for optical components. The course allows students to gain an overview of the many possible applications of optical technology.

They are able to apply what they have learned to the design of various optical systems and develop their own concepts.

Based on the examples learned, they are aware of the social and societal influence of new optical technologies and are able to assess the effects of new developments in research and industrial applications.

Content
Technical optics deals with the essential physical principles of optics, as well as a variety of technical applications of optical systems. These range from applications in the automobile, medicine, measurement technology, printing, optical data storage, to micro/nano optics and manufacturing processes for plastic and glass optics.

The following chapters are covered:

- Motivation
- Fundamentals
- Reflection & Refraction
- Absorption
- Mirrors
- Prisms & lenses
- Applications: Prism pole, Fresnel lens, telescope, camera
- Diffraction & Interference
- Application: Microscope
- Paraxial beam matrices
- Application: Focusing of beams
- Application: Distance & angle measurement
- Optics in data storage
- Micro- and nano-optics
- Production of optics

The lecturer reserves the right to deviate from the content given here in the context of the current lecture without special notice.

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
1. attendance time lecture: 15 * 2 h = 30 h
2. preparation and follow-up time lecture: 15 * 2 h = 30 h
3. attendance time exercise: 15 * 2 = 30 h
4. preparation and follow-up time for exercise: 15 * 2 h = 30 h
5. exam preparation and attendance: 30 h

Total: 150 h = 5 LP

Recommendations
Previous attendance of the lecture on lighting technology.

M 12.235 Module: Telematics [M-INFO-107243]

Coordinators: Prof. Dr. Martina Zitterbart
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules) (Usage until 3/31/2027)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-114269	Telematics	**WT**	**6 CP**	**Zitterbart**

Assessment
See partial achievements (Teilleistung)

Prerequisites
See partial achievements (Teilleistung)

Competence Goal
Students

- master protocols, architectures, and methods and algorithms that are used on the Internet for routing and for establishing a reliable end-to-end connection, as well as various media allocation procedures in local networks.
- have an understanding of the systems and the problems that appear in a global, dynamic network as well as the mechanisms used to remedy them.
- are familiar with current developments such as SDN and data center networking.
- know methods to manage and administrate networks.

Students master the basic protocol mechanisms for establishing reliable end-to-end communication. Students have detailed knowledge of the mechanisms used in TCP for congestion and flow control and can discuss the issue of fairness with multiple parallel transport streams. Students can analytically determine the performance of transport protocols and know methods that fulfill special requirements of TCP, such as high data rates and short latencies. Students are familiar with current topics such as problems introduced by utilization of middle boxes in the Internet, the use of TCP in data centers and multipath TCP. Students can use transport protocols in practice.

Students know the functions of routers in the Internet and can reproduce and apply common routing algorithms. Students can reproduce the architecture of a router and know different approaches to buffer placement as well as their advantages and disadvantages.

Students understand the distinction of routing protocols into interior and exterior gateway protocols and have detailed knowledge of the functionality and properties of common protocols such as RIP, OSPF and BGP. The students are familiar with current topics such as SDN.

Students know the function of media allocation and can classify and analytically evaluate media allocation processes. Students have in-depth knowledge of Ethernet and are familiar with various Ethernet forms and their differences, especially current developments such as real-time Ethernet and data center Ethernet. Students can reproduce and apply the spanning tree protocol.
Students can reproduce the technical characteristics of DSL. Students are familiar with the concept of label switching and can compare existing approaches such as MPLS.

Content

- Introduction
- End-to-end data transport
- Routing protocols and architectures
- Media allocation
- Bridges
- Data transmission
- Further selected examples
- Network management

Additional Information
Offered for the last time in WS 26/27!!!!!

Workload
Lecture with 3 SWS plus follow-up/exam preparation, 6 CP.
6 CP corresponds to approx. 180 working hours, of which
approx. 60 hours lecture attendance
approx. 60 hours preparation/follow-up work
approx. 60 hours exam preparation

M 12.236 Module: Test of Embedded Systems in Industrial Contexts [M-ETIT-100546]

Coordinators: Prof. Dr.-Ing. Eric Sax
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Automation, Robotics, and Systems Engineering (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100811	Test of Embedded Systems in Industrial Contexts	**WT**	**4 CP**	**Sax**

Assessment
Success control takes place in form of an overall oral examination (approx. 25 minutes).

Prerequisites
none

Competence Goal
After completing the module, students will be able to group and name the test methods taught. Furthermore, students will be able to select suitable test methodologies based on the theoretical principles for specific applications and test them in various scenarios. To this end, students can use the demonstrated state-of-the-art technologies and have an insight into current tools. The practical content of the lecture can be successfully applied by the students in other contexts, e.g. in standard software development.

Content
The lecture provides knowledge of methods, technologies and procedures used in the testing of software for embedded systems. In the associated practical exercise, exercises are worked on and current test tools are used.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
1. attendance time in lectures + exercise: 60h
2. preparation/post-processing of exercise and lecture = 35h
3. exam preparation and attendance = 20h

Recommendations
Knowledge of the basics of applied computer science, for example attending the practical course in information technology, is helpful.

M 12.237 Module: The Energy Transition of the Electrical Transmission Grid [M-ETIT-105618]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111248	The Energy Transition of the Electrical Transmission grid	**ST**	**3 CP**	**Leibfried**

Assessment
Success control takes place in form of an oral examination of approx. 20 minutes on selected topics from the course.

Prerequisites
none

Competence Goal
Students know the targets for the implementation of the energy transition and the role of the transmission system operators. Students will understand the legal and economic framework within which the electricity transmission system operators work. The relevance in practice of what students have learned in theory so far should be made clear. The dynamic development that is currently taking place in the transmission grids should be presented as a promising field of activity for the future. The responsibilities of the transmission system operators and the regulator are known and can be differentiated. The business model and the value chain of the grid operators are known and the influences on the business model can be understood. The organization and tasks of a transmission system operator are known.

Content

- Introduction:
  Structure of the electrical energy system
  Political and social goals of the energy transition
  Derived goals of the European Union and Germany
  Consequences for the generation portfolio
  German and European planning processes and their scenarios
  Core processes and organization of a transmission system operator

- Effects of the energy transition on grid operators
  Expansion of renewable generation and feed-in priority
  Merit order principle, generation adequacy
  Grid adequacy, demand-oriented grid
  Grid congestion, peak capping and redispatch
  Effects on grid stability
  Cooperation between distribution grid operators and transmission grid operators
  Planning processes and their topics,
  Differentiation between target grid planning and system analyses

- Interconnected operation and energy industry
  System management and grid control in Germany and in the European interconnected grid
  Trading and regulation

- Role and responsibility of transmission system operators
  System management, grid control technology, grid control, dispatch
  Grid operation, maintenance, fault clearance, technology
  Regulation and grid management
  Grid expansion and grid development

- Technology used and innovations
  Line construction
  Protection technology, control technology, communication, IT
  New equipment, STATCOM, HVDC, UPFC, LQR
  Outlook on development topics, reactive and automated Operations management, grid booster

- Trends and future challenges
  Grid adequacy with changing dimensioning
  Use cases
  Ensuring grid stability without large power plants
  Ensuring the power balance with fluctuating generation
  Curative grid operation management and new operating resources
  Storage and changing customer behaviour

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
The course consists of seven block lectures of 3 hours each and an excursion. The dates will be announced on notice boards.

Each credit point corresponds to 30 hours of work by the student. The workload includes

1. attendance time in lecture and excursion (30 h = 1 CP)

2. self-study time for preparation and follow-up of the lecture including exam preparation (60 h = 2 CP)

This results in a total of 90 hours = 3 CP.

Recommendations
Knowledge of the lectures "Elektrische Energienetze" and "Energieübertragung und Netzregelung" is helpful.

M 12.238 Module: Theoretical Foundations of Cryptography [M-INFO-105584]

Coordinators: Prof. Dr. Jörn Müller-Quade
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-111199	Theoretical Foundations of Cryptography	**WT**	**6 CP**	**Müller-Quade**

Assessment
See Partial Achievements (Teilleistung).

Prerequisites
See Partial Achievements (Teilleistung).

Competence Goal
The student
- can motivate and explain the basic concepts of cryptography, point out their differences and relate them to each other.
- is able to compare and evaluate security models and goals.
- knows and understands definitions and constructions, and their relationships and dependencies. For example, properties such as "one-way", "collision-resistant", "pseudo-random", "IND-CPA", "IND-CCA", "EUF-CMA", etc., and candidates, constructions, and schemes with such properties.
- understands elementary proof techniques (such as reductions and hybrid arguments) and can apply them
- can reproduce, check and explain security proofs.
- can construct and evaluate simple new procedures and find possible attacks.
- can prove (simple) secure schemes to be secure using the techniques learned.

Content
The lecture deals with the theoretical foundations of cryptography, with a focus on non-interactive foundations. The lecture consists of three main parts.
1. Aspects of complexity theory
2. Secret-key cryptography
3. Public-Key Cryptography

The usual contents include:
- Asymptotic security, one-way functions, pseudorandomness and indistinguishability
- Secret-key cryptography (encryption, security notions such as IND-CPA, IND-CCA, authenticity, and authenticated encryption)
- Public-key encryption (security notions in this situation, in particular CCA security)
- Signatures (definition and basic constructions)
- Outlooks on more advanced topics (e.g. as part of the exercises)

The lecture is supplemented by an exercise that recapitulates, deepens and applies the material in a new context.

The specific content of the lecture and exercise varies depending on the focus chosen. It serves as a basis for further lectures and seminars, for example on cryptographic protocols (interactive cryptography) and advanced non-interactive cryptography.

Workload
Attendance time in the lecture: 48 h

Preparation and follow-up of the same: 52 h

Exam preparation and attendance in the same: 80 h

M 12.239 Module: Thermal Solar Energy [M-MACH-102388]

Coordinators: apl. Prof. Dr. Ron Dagan
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-105225	Thermal Solar Energy	**WT**	**4 CP**	**Dagan**

Assessment
A performance assessment is obligatory; oral exam about 30 minutes

Prerequisites
none

Competence Goal
Based on the elaboration of the basic physics knowledge of the solar irradiation, heat radiation, optics and thermal-hydraulics, the student will be able to

- select solar thermal components such as mirrors, glasses, selective absorbers and insulation materials and their manufacturing processes and to calculate and assess their performance,
- identify different collector types and to indicate their potential field of application,
- characterize the entire solar thermal collector system with respect to its performance and derive from the collector characteristics its suitability for different types of use,
- embed collectors into a technical overall system for heat (household, process heat, heat storage networks) or electricity generation (power plant), to calculate the system efficiency and independently develop the basics of its optimization.
- identify adequate thermal storage types for the temporal separation of generation and consumption, to dimension them appropriately and to integrate them into a system concept,
- evaluate solar thermal systems in their entirety (capacity, estimation of system dynamics, response behavior, efficiency) and know options for integration into networks (heat, cold, electricity).

Content
Fundamentals of thermal solar energy from solar irradiation (influence of time and place, modifications in the atmosphere) and their implementation in a collector to integration into a technical overall system. In detail:

1. introduction to the energy demand and evaluation of the application potential of solar thermal energy.

2. primary energy source SUN: Sun, solar constant, solar radiation (scattering, absorption in the atmosphere, direct-diffuse radiation, angular influences, radiation balance).

3. solar collectors: basic design of a collector, basics of determining the efficiency, significance of concentration and its limitations, solar thermal collector types (designs, efficiency, system technology).

4. passive mechanisms of solar thermal energy: heat conduction in solids and gases, radiation heat transport in transparent and opaque bodies Design requirements and physical principles of solar thermal glasses, mirrors and selective absorbers. Goal oriented selection of materials and manufacturing processes.

5. momentum and heat transport: basic equations of single- and multi-phase transport, basic ideas of local and system engineering calculation methods, stability limits.

Optional

6. solar thermal low-temperature systems: collector variants, methods for system simulation, planning and dimensioning of systems, system-related system design and stagnation scenarios and their handling.

7. solar thermal high-temperature systems: solar thermal power plants (classification of system components, loss mechanisms, upwind power plants), coupling of collector with energy generation process.

At the end:

8. Thermal energy storage: Explanation of terms (energy contents, storage forms and materials, potentials ...), storage concepts (system structure, design ratio), system integration.

9. Solar air conditioning: Determination of cooling capacity, indoor climate, solar cooling methods and evaluation of air conditioning.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
regular lecture attendance: 30 h
self-study: 60 h (incl. supplementary searches)

exam preparation 30 h

Recommendations
desirable are reliable knowledge in physics in optics and thermodynamics

Basics in heat and mass transfer, material science, energy technology and fluid mechanics

Teaching and Learning Methods
Präsentation complemented by printouts

Literature
supply of lecture material in printed and electronic form

Stieglitz & Heinzel; Thermische Solarenergie -Grundlagen-Technologie- Anwendungen. Springer Vieweg Verlag. 711 Seiten. ISBN 978-3-642-29474-7

M 12.240 Module: Ultrasound Imaging [M-ETIT-100560]

Coordinators: Dr. Nicole Ruiter
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100822	Ultrasound Imaging	**ST**	**3 CP**	**Ruiter**

Assessment
Success control takes place in form of an overall oral examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
Students will master the methods of ultrasound imaging commonly used in medicine today, understand their functional principles and physical principles and be able to comprehend their technical implementation.

Content
Ultrasound applications in medicine: 3D/4D ultrasound, Doppler, Tissue Harmonic Imaging, Compounding, Elastography, Ultrafast US-Imaging, Ultrasound Contrast Agents, Ultrasound Tomography, Ultrasound Therapy. In each case with functional principle, physical principles, technical implementation and medical applications.

- Areas of application of ultrasound in medicine
- Fundamentals and basic illustration
- 2D/3D/4D ultrasound
- Elastography
- (Tissue) Doppler
- Tissue Harmonic Imaging
- Image errors, limitations as an opportunity,
- Compounding
- Ultrasound safety and therapy
- Ultrafast US imaging, SAFT and tomography
- Ultrasound contrast agents

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
1. attendance times in lectures: 30 h
2. preparation/follow-up of the same: 30 h
3. oral examination and attendance at the same: 30 h

M 12.241 Module: Verified Numerical Methods [M-ETIT-104493]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-109184	Verified Numerical Methods	**WT**	**4 CP**	**Hohmann**

Assessment
Success control takes place in form of an oral examination (approx. 20 minutes) on the selected course.

Prerequisites
none

Competence Goal

- Students know the basics of verified numerical methods for the inclusion of solutions of (finite-dimensional) systems of equations and differential equations.
- Students are familiar with all aspects from modeling and the development of verified numerical methods to algorithmic implementation and concrete programming, e.g. in MATLAB/INTLAB.
- Students are proficient in the application of verified numerical methods to practical tasks.

Content

- Interval arithmetic
- Basic concepts of functional analysis
  - Sobolev spaces
  - Embedding and embedding theorems
  - Fixed point formulation
  - Fixed point theorem
- Verified numerical methods for linear systems of equations
- Verified numerical methods for (finite-dimensional) nonlinear equations
- Computer-aided proof methods for differential equations
- Inclusion of eigenvalues

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
The workload includes:

1. Attendance time in lecture/exercise (2+1 SWS: 45h1.75 LP)
2. Preparation/follow-up of lecture/exercise (60h2 CP)
3. Preparation/attendance time for oral examination (7.5h0.25 CP)

Recommendations
Knowledge of the following modules is recommended:

- Mathematik I-III in Bachelor
- M-MATH-100536 - Numerische Methoden
- M-ETIT-104595 - Numerische Methoden für partielle Differentialgleichungen
-

M 12.242 Module: Visual Perception in the Automobile [M-ETIT-100497]

Coordinators: Prof. Dr. Cornelius Neumann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100777	Visual Perception in the Automobile	**ST**	**3 CP**	**Neumann**

Assessment
Success control takes place in the form of an overall oral examination of approx. 20 minutes.

Prerequisites
none

Competence Goal
Students learn about the physiological effects of automotive lighting technology on drivers and other road users. They also gain an insight into the planning and design of test subject studies.

They are able to assess the physiological influences of different technologies on driving safety and to develop and evaluate simple plans for experimental studies.

Students are sensitized to the consequences of faulty developments in the field of automotive lighting and are able to assess and shape these in their later professional life.

Content

- Recapitulation: The human eye
- Mesopic vision
- Perception of signal functions
- Human-machine interaction in display technology
- Vehicle interior
- Perception and glare from headlights
- Advertising

The lecturer reserves the right to deviate from the content given here in the context of the current lecture without special notice.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
The workload includes:

1. attendance time in lectures, exercises: 30 h

2. preparation/follow-up of the same: 30 h

3. exam preparation: 30 h

Recommendations
Previous attendance of the lecture on Lichttechnik.

M 12.243 Module: Water – Energy – Environment Nexus in a Circular Economy: Research Proposal Preparation [M-CIWVT-106680]

Coordinators: Prof. Dr. Andrea Iris Schäfer
Organisation: KIT Department of Chemical and Process Engineering
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
5 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-113433	Water – Energy – Environment Nexus in a Circular Economy: Research Proposal Preparation	ST	5 CP	

Assessment
The Learning control is an examination of another type:
Research proposal of 10 pages and an oral presentation of 10 minutes (individual work). The grade will be a composite of the proposal (submission in week 13 before class) and oral & poster presentation (all day workshop with researcher participation).

Competence Goal
The goal of this course is to get an overview of current challenges in the circular economy focused on the water – energy – environment nexus. Based on individual student interest a topic will be identified and a research plan developed encompassing a thorough background research to establish the state-of-the-art, identification of a specific research problem and research questions suitable to solve this problem. Concepts of novelty and excellence will be explored in an international context. Following the individual topic choice, the research proposal will be developed individually in a tutor group (divided into water, energy, environment) while lectures on required skills will accompany this process. As an outlook beyond this course, criteria to consider when looking for research careers such as applying for funding/scholarships, considering choices in research environment and supervision, performance indicators in research and university rankings will be introduced to enable informed decisions. The proposal will be communicated in writing, as a brief presentation and as a poster, which equips students brilliantly not only for a masters thesis but also a future research publication or a PhD.

Content
In a time of limiting resources, climate change and ever increasing demand for resources the concept of a circular economy is inevitable to create a more sustainable utilization of our key resources, water, energy and 'environment'. Concepts of zero liquid discharge, water reuse, carbon net zero, resource recovery and environmental pollution reduction are all part of this concept where where waste is returned to use. The water – energy – environment nexus is the particular focus of ths course. Global water issues, water and wastewater treatment, desalination, water reuse, micropollutants, decentralized systems, water & sanitation in international development, renewable energies, environmental pollution, climate change, resource recovery – and many more topics will inspire future research.

Module Grade Calculation
The module grade is the grade of the examination of another type.

Workload

- Contact time: lectures and tutorials 60 hrs (4 SWS)
- Group and self study: 50 hrs
- Preparation of assessments and participation at the group presentations (one full day): 30 hrs

M 12.244 Module: Wearable Robotic Technologies [M-INFO-107113]

Coordinators: Prof. Dr.-Ing. Tamim Asfour
Prof. Dr.-Ing. Michael Beigl
Organisation: KIT Department of Informatics
Part of: Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-114145	Wearable Robotic Technologies	**ST**	**4 CP**	**Asfour, Beigl**

Assessment
See partial achievements (Teilleistung)

Prerequisites
See partial achievements (Teilleistung)

Competence Goal
Students can explain the fundamentals, motivation, and applications of wearable robotic technologies including exoskeletons, orthoses, and prostheses. They have comprehensive knowledge of human anatomy, biomechanics, and the neuromuscular system relevant to wearable robot design.

Students understand the core components and challenges associated with the design of wearable robotic technologies, including mechanisms, actuation technologies, sensing approaches, and wearable computing concepts. They can analyze design requirements and challenges including kinematic compatibility and human-robot interfaces for wearable robots.

Students can explain control strategies for prosthetic and exoskeleton systems, understand both physical and cognitive aspects of human-robot interaction, and comprehend concepts such as body schema and symbiotic human-machine interaction in wearable systems.

Content
The lecture provides a comprehensive introduction to wearable robotic technologies. It covers human anatomy and biomechanics as foundations for wearable robot design. Joint biomechanics, locomotion and gait analysis, the neuromuscular system, and models of the human body are discussed to understand how wearable robots can be designed and controlled.

Design principles and mechanisms of wearable robots are addressed. Actuation technologies are covered in depth: requirements analysis based on human motion and user needs, compliant actuation approaches (serial elastic, parallel elastic, variable stiffness actuation), and strategies for increasing efficiency such as energy harvesting, multi-articulation, and underactuation.

The lecture covers wearable computing principles such as body signals and phenomena, design principles for body-worn systems, attachment considerations based on body maps for heat tolerance, weight distribution, and flexibility, and fabric design with textile-embedded electronics.

Exoskeleton systems are presented, covering application areas, design challenges such as redundancy, kinematic compatibility, and misalignments, various types of exoskeletons, and the general design process. Prosthetic systems cover types of hand and lower limb prostheses, hand synergies for control, and mechanical interfaces through socket design.

Human-robot interaction addresses both physical and cognitive interfaces. Control strategies are presented for prosthetic systems (semi-autonomous grasping) and exoskeleton systems (hierarchical control, assistance evaluation). The concept of body schema from both neuroscience and robotics perspectives is discussed, emphasizing symbiotic human-machine interaction in tightly coupled wearable systems. State-of-the-art examples across various application domains illustrate current research and developments.

Workload
Lecture with 2 SWS, 4 ECTS.
4 ECTS corresponds to ca. 120 hours, thereof
ca. 15 * 2h = 30h attendance in the lecture
ca. 15 * 3h = 45h preparation and follow-up of the lecture
ca. 45h preparation for the exam and exam itself

M 12.245 Module: Workshop Finite Element Method in Electromagnetics [M-ETIT-107147]

Coordinators: Prof. Dr. Martin Doppelbauer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Field of Specialization / Electrical Power Systems and Electromobility (Focus Area)
Electives (Elective Modules)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114166	Workshop Finite Element Method in Electromagnetics	**ST**	**3 CP**	**Doppelbauer**

Assessment
Success control takes place in the form of different types of examination consisting of a written assignment in the form of an written report (20-50 pages).

Prerequisites
none

Competence Goal
In this course, students acquire basic knowledge about the application of the finite element method in electromagnetic analysis: mathematical principles, levels of abstraction, model creation and result analysis.

After successfull attendance, students can set up and run simulations using ANSYS Maxwell and interpret the software interface and workflow. They can create a FEM models of rotating electric machines and validate their correctness. They can implement optimization strategies for machine design, considering various parameters such as efficiency, torque, and losses. Finally, students can analyze simulation and optimization results, identify inconsistencies, and critically evaluate the outcomes.

Content

- Introduction to the mathematical basics of the finite element method (FEM) of electromagnetics
- Presentation of the industry-standard software ANSYS Maxwell
- Construction of a model of a permanently excited synchronous machine
- Presentation and implementation of optimization strategies for the design of machines with regard to various parameters
- Introduction to results analysis

The module teaches students

- How to use industry-standard software from the field of electromagnetic FEM
- Solve basic practical tasks in the field of electromagnetic FEM
- Approaches to optimizing various parameters using the example of electrical machines
- Question and evaluate the results of a simulation or optimization

Module Grade Calculation
The module grade is the grade of the written paper.

Workload

1. Attendance time: 20h
2. Preparation and follow-up time: 10h
3. Project work: approx. 60h

   Total approx. 90 h, corresponds to 3 CP

Recommendations
Knowledge from the modules "Elektrische Maschinen und Stromrichter" and "Entwurf elektrischer Maschinen" is desired.

13 Module components

T 13.1 Module component: Adaptive Optics [T-ETIT-107644]

Coordinators: Ph.D. Szymon Gladysz
Prof. Dr. Ulrich Lemmer

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-103802 - Adaptive Optics

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2313724**	Adaptive Optics	**2 SWS**	**Lecture /**	**Gladysz**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Type of Examination: Oral examination

Duration of Examination: approx. 30 Minutes

Modality of Exam: The oral exam will be scheduled during the semester break.

The module grade is the grade of the oral exam.

Prerequisites
None.

Recommendations
Basic knowledge of statistics.

Workload
90 hours

T 13.2 Module component: Advanced Communications Engineering [T-ETIT-113676]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106815 - Advanced Communications Engineering

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2310540**	Advanced Communications Engineering	**3 SWS**	**Lecture /** 🗣	**Jäkel**
WT 26/27	**2310541**	Exercises for 2310540 Advanced Communications Engineering	**1 SWS**	**Practice /** 🗣	**Jäkel**

Legend: 📱 Online, 🎛 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment takes place in the form of a written examination lasting 120 min.

The module grade is the grade of the written exam.

Prerequisites
none

T 13.3 Module component: Analog Circuit Design [T-ETIT-114457]

Coordinators: Prof. Dr. Ivan Peric
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107364 - Analog Circuit Design

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each winter term	2

Courses					
WT 26/27	**2312664**	Analog Circuit Design	**2 SWS**	**Lecture /**	**Peric**
WT 26/27	**2312666**	Practical Exercise for 2312664 Analog Circuit Design	**1 SWS**	**Practice /**	**Peric**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in form of an oral examination (approx. 20 minutes) and a written report with the results of the exercises. Details will be given during the lecture.

Prerequisites
none

T 13.4 Module component: Antennas and Beamforming [T-ETIT-113920]

Coordinators: Prof. Dr.-Ing. Marwan Younis
Prof. Dr.-Ing. Thomas Zwick

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-106956 - Antennas and Beamforming

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	2308465	Antennas and Beamforming	2 SWS	Lecture / 🗣	Younis, Zwick
WT 26/27	2308466	Tutorial for 2308465 Antennas and Beamforming	1 SWS	Practice / 🗣	Younis, Zwick

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes.
The module grade is the grade of the written exam.

Prerequisites
none

Recommendations
Knowledge of the basics of radio frequency technology and some basic knowledge on communication and radar systems is recommended.

T 13.5 Module component: Applied Information Theory [T-ETIT-114412]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107333 - Applied Information Theory

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2310537**	Applied Information Theory	**3 SWS**	**Lecture /** 🗣	**Jäkel**
WT 26/27	**2310539**	Exercises for 2310537 Applied Information Theory	**1 SWS**	**Practice /** 🗣	**Jäkel**

Legend: 💻 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Prerequisites
none

T 13.6 Module component: Authentication and Encryption [T-INFO-110824]

Coordinators: Prof. Dr. Jörn Müller-Quade
Organisation: KIT Department of Informatics
Part of: M-INFO-105338 - Authentication and Encryption

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	**2400045**	**Authentication and Encryption**	**2 SWS**	**Lecture /** 🗣	**Müller-Quade, Raiber, Bayreuther**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of an oral examination, usually lasting 30 minutes in accordance with §4 Para. 2 No. 2 SPO

Prerequisites
Module may not be taken together with the module M-INFO-100743 Digital Signatures

Recommendations
Students should be familiar with the contents of the module "Theoretical Foundations of Cryptography"

T 13.7 Module component: Automotive Engineering I [T-MACH-102203]

Coordinators: Dr.-Ing. Martin Gießler
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-100501 - Automotive Engineering I

Type	Credits	Grading	Term offered	Version
Written examination	8 CP	graded	Each winter term	4

Courses					
WT 26/27	**2113809**	**Automotive Engineering I**	**4 SWS**	**Lecture /** 🗣	**Gießler**

Legend: 🖥 Online, 🏵 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination

Duration: 120 minutes

Auxiliary means: none

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-100092 - Automotive Engineering I (in German) must not have been started.

Additional Information
The course is offered in English.

T 13.8 Module component: Automotive Engineering I (in German) [T-MACH-100092]

Coordinators: Dr.-Ing. Martin Gießler
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-100501 - Automotive Engineering I

Type	Credits	Grading	Term offered	Expansion	Language	Version
Written examination	8 CP	graded	Each winter term	1 semesters		4

Courses					
WT 26/27	**2113805**	**Automotive Engineering I**	**4 SWS**	**Lecture /**	**Gießler**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Written examination

Duration: 120 minutes

Auxiliary means: none

Prerequisites
The module component "T-MACH-102203 - Automotive Engineering I" is not started or finished. The module components "T-MACH-100092 - Grundlagen der Fahrzeugtechnik I" and "T-MACH-102203 - Automotive Engineering I" can not be combined.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-102203 - Automotive Engineering I must not have been started.

Additional Information
The course is offered in German.

Workload
240 hours

T 13.9 Module component: Automotive Engineering II [T-MACH-114435]

Coordinators: Dr.-Ing. Martin Gießler
Organisation: KIT Department of Mechanical Engineering

KIT Department of Business and Economics
Part of: M-MACH-100502 - Automotive Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	2

Courses					
ST 2026	**2114855**	Automotive Engineering II	**2 SWS**	**Lecture /** 🗣	**Gießler**

Legend: 📱 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written Examination

Duration: 90 minutes

Auxiliary means: none

Prerequisites
Only one out of the two moduls "M-MACH-102117 - Grundlagen der Fahrzeugtechnik II"and "M-MACH-114435 - Automotive Engineering II" is allowed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-102117 - Automotive Engineering II (in German) must not have been started.

Additional Information
The course is offered in English.

Workload
120 hours

T 13.10 Module component: Automotive Engineering II (in German) [T-MACH-102117]

Coordinators: Dr.-Ing. Martin Gießler
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-100502 - Automotive Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	2114835	Automotive Engineering II	2 SWS	Lecture / 🗣	Gießler
ST 2026	2114855	Automotive Engineering II	2 SWS	Lecture / 🗣	Gießler

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written Examination

Duration: 90 minutes

Auxiliary means: none

Prerequisites
Only one out of the two moduls "M-MACH-102117 - Grundlagen der Fahrzeugtechnik II"and "M-MACH-114435 - Automotive Engineering II" is allowed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-114435 - Automotive Engineering II must not have been started.

Additional Information
The course is offered in German.

Workload
120 hours

T 13.11 Module component: Automotive Vision [T-MACH-114149]

Coordinators: Dr. Martin Lauer
Prof. Dr.-Ing. Christoph Stiller

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-107148 - Automotive Vision

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	3

Courses					
ST 2026	**2138340**	**Automotive Vision**	**3 SWS**	**Lecture /** 🗣	**Lauer, Bätz**

Legend: 📱 Online, 💮 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Type of Examination: written exam

Duration of Examination: 60 minutes

Prerequisites
none

Additional Information
The course is offered in English

Workload
180 hours

T 13.12 Module component: Autonomous Driving in Practice [T-MACH-115202]

Coordinators: Kevin Simon
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-107806 - Autonomous Driving in Practice

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2113820**	Autonomous Driving in Practice	**3 SWS**	**Lecture / Practice /** 🗣	**Frey, Simon**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The final grade is based on: 2 written exams (duration about 30 minutes) and the final presentation of the practical task (duration 30 minutes max.), including the final report (2-3 pages). Submission of the homework is mandatory.

The overall impression will be evaluated.

Recommendations
Prior experience in autonomous driving, automotive engineering, and machine learning is a plus.

Additional Information
Due to capacity limitations, the lecture is limited to 16 students. If necessary, a selection process will be conducted. Places will be allocated based on the students' academic progress (semester level and subject-specific programming skills).

The course will be conducted in English.

Workload
180 hours

T 13.13 Module component: Basic Seminar Supplementary Studies on Science, Technology and Society - Self Registration [T-FORUM-113579]

Coordinators: Dr. Christine Mielke
Christine Myglas
Organisation: General Studies. Forum Science and Society (FORUM)
Part of: M-FORUM-106753 - Supplementary Studies on Science, Technology and Society

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	2 CP	pass/fail	Each summer term	1 semesters	1

Assessment
Study achievement in the form of a presentation or a term paper or project work in the selected course.

Prerequisites
None

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- FORUM (ehem. ZAK) Begleitstudium

Recommendations
It is recommended that the basic seminar be completed during the same semester as the lecture series "Science in Society". If it is not possible to attend the lecture series and the basic seminar in the same semester, the basic seminar can also be attended in the semesters before the lecture series.
However, attending courses in the advanced unit before attending the basic seminar should be avoided.

T 13.14 Module component: Basics of Converter Control [T-ETIT-100717]

Coordinators: Dr.-Ing. Andreas Liske
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100400 - Basics of Converter Control

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	2

Courses					
ST 2026	**2306330**	Basics of Converter Control	**2 SWS**	**Lecture /**	**Liske**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The success control takes place as part of an online examination in text form with face-to-face supervision (§5, statutes for conducting online examinations at KIT) lasting 60 minutes.

The examination takes place on KIT computers in a pool room.

The module grade is the grade of the online examination.

Prerequisites
none

T 13.15 Module component: Batteries and Fuel Cells Laboratory [T-ETIT-100708]

Coordinators: Dr.-Ing. Andre Weber
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100381 - Batteries and Fuel Cells Laboratory

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2304235**	**Batteries and Fuel Cells Laboratory**	**4 SWS**	**Practical course /** 🗣	**Weber**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of other types of examinations.
The grade is made up of the grades for the colloquium before each attempt (20%), attendance and participation (30%) and the experimental reports (50%).

Prerequisites
none

Recommendations
The contents of the lectures "Batteries and Fuel Cells" and "Battery and Fuel Cell Systems" are assumed to be known. Students who have not (yet) attended these lectures must familiarize themselves with the contents in advance.

T 13.16 Module component: Batteries, Fuel Cells, and Electrolysis [T-ETIT-113986]

Coordinators: Prof. Dr.-Ing. Ulrike Krewer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107005 - Batteries, Fuel Cells, and Electrolysis

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	3

Courses					
WT 26/27	**2304240**	Batteries, Fuel Cells and Electrolysis	**2 SWS**	**Lecture /**	**Krewer**
WT 26/27	**2304241**	Practical Exercise to 2304240 Batteries, Fuel Cells and Electrolysis	**2 SWS**	**Practice /**	**Krewer, Sonder**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in the form of a graded written examination lasting 120 minutes.

Prerequisites
The following module components **must not** have started:

- T-ETIT-100983 - Batterien und Brennstoffzellen
- T-ETIT-114097 - Batterien, Brennstoffzellen und ihre Systeme

The following module components **must** have started:

- T-ETIT-114957 - Batteries, Fuel Cells, and Electrolysis - Group Project

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-ETIT-100983 - Batteries and Fuel Cells must not have been started.
2. The module component T-ETIT-114957 - Batteries, Fuel Cells, and Electrolysis - Group Project must have been started.

Additional Information
For details on content and qualification objectives see "M-ETIT-107005 - Batteries, Fuel Cells, and Electrolysis".

The course is offered in English.

Workload
150 hours

T 13.17 Module component: Batteries, Fuel Cells, and Electrolysis - Group Project [T-ETIT-114957]

Coordinators: Prof. Dr.-Ing. Ulrike Krewer
Organisation: KIT Department of Electrical Engineering and Information Technology
KIT Department of Mechanical Engineering
Part of: M-ETIT-107005 - Batteries, Fuel Cells, and Electrolysis

Type	Credits	Grading	Term offered	Version
Coursework	1 CP	pass/fail	Each winter term	1

Assessment
Success control takes place in the form of an ungraded written technical report (approx. 7-10 pages).

Prerequisites
none

Additional Information
For details on content and qualification objectives see "M-ETIT-107005 - Batteries, Fuel Cells, and Electrolysis".

The course is offered in English.

Workload
30 hours

T 13.18 Module component: Battery and Fuel Cell Systems [T-ETIT-114802]

Coordinators: Dr.-Ing. Andre Weber
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107551 - Battery and Fuel Cell Systems

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2304214**	Battery and Fuel Cell Systems	**2 SWS**	**Lecture /**	**Weber**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in the form of an oral examination lasting approx. 20 minutes.
The module grade is the grade of the oral examination.

Prerequisites
none

T 13.19 Module component: Biologically Inspired Robots [T-MACH-113856]

Coordinators: Prof. Dr.-Ing. Arne Rönnau
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-106903 - Biologically Inspired Robots

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	**2122330**	Biologically Inspired Robots	**2 SWS**	**Lecture /** 🗣	**Rönnau**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of an oral examination (approx. 15-20 minutes)

Prerequisites
none

Recommendations
It is recommended to listen to the course "Robotics I" beforehand .

Additional Information
The course is offered in English.

Workload
90 hours

T 13.20 Module component: BioMEMS - Microsystems Technologies for Life-Sciences and Medicine I [T-MACH-100966]

Coordinators: Prof. Dr. Andreas Guber
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-100489 - BioMEMS - Microsystems Technologies for Life Sciences and Medicine I

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	3

Courses					
WT 26/27	**2141864**	**BioMEMS - Microsystems Technologies for Life-Sciences and Medicine I**	**2 SWS**	**Lecture /** 🗣	**Guber**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam (75 Min.)

Prerequisites
none

Additional Information
The course is offered in German

Workload
120 hours

T 13.21 Module component: BioMEMS - Microsystems Technologies for Life-Sciences and Medicine II [T-MACH-100967]

Coordinators: Prof. Dr. Andreas Guber
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-100490 - BioMEMS - Microsystems Technologies for Life Sciences and Medicine II

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	**2142883**	**BioMEMS - Microsystems Technologies for Life-Sciences and Medicine II**	**2 SWS**	**Lecture / 🗣**	**Guber, Ahrens**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam (75 Min.)

Prerequisites
none

Additional Information
The course is offered in German

Workload
120 hours

T 13.22 Module component: BioMEMS - Microsystems Technologies for Life-Sciences and Medicine III [T-MACH-100968]

Coordinators: Prof. Dr. Andreas Guber
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-100491 - BioMEMS - Microsystems Technologies for Life Sciences and Medicine III

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	**2142879**	**BioMEMS - Microsystems Technologies for Life-Sciences and Medicine III**	**2 SWS**	**Lecture /** 🗣	**Guber, Ahrens**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam (75 Min.)

Prerequisites
none

Additional Information
The course is offered in German

Workload
120 hours

T 13.23 Module component: Business Innovation in Optics and Photonics [T-ETIT-104572]

Coordinators: Prof. Dr. Werner Nahm
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101834 - Business Innovation in Optics and Photonics

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each winter term	2

Courses					
WT 26/27	**2305742**	Business Innovation in Optics and Photonics	**2 SWS**	**Lecture /** 🗣	**Riedel, Nahm**
WT 26/27	**2305743**	Exercise for 2305742 Business Innovation in Optics and Photonics	**1 SWS**	**Practice /** 🗣	**Riedel, Nahm**

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Type of Examination: examination of another type

Duration of Examination: 4 group presentations à 20 minutes (approx.)

Modality of Exam: The exam consists of four group presentations. 2nd day: Technology Presentation. 3rd day: Development plan presentation. 4th day: Business Canvas presentation. Final presentation: Business pitch. The overall impression is rated.

Prerequisites
none

Recommendations
Good knowledge in optics & photonics. Personal motivation and interest for getting deeper into business development aspects, methods and tools. Commitment to active, regular and coninuous participation in the group work.

T 13.24 Module component: Channel Coding: Algebraic Methods for Communications and Storage [T-ETIT-111244]

Coordinators: Prof. Dr.-Ing. Laurent Schmalen
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105616 - Channel Coding: Algebraic Methods for Communications and Storage

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	**2310546**	Channel Coding: Algebraic Methods for Communications and Storage	**2 SWS**	**Lecture /** 🗣	**Fengler, Schmalen**

Legend: 💻 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The exam is held as an oral exam of approx. 20 min.

The module grade of the module corresponds to the grade of the oral exam.

Prerequisites
none

Recommendations
Previous attendance of the lectures "Communication Engineering I" and "Probability Theory" is recommended.

T 13.25 Module component: Channel Coding: Graph-Based Codes [T-ETIT-111245]

Coordinators: Prof. Dr.-Ing. Laurent Schmalen
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105617 - Channel Coding: Graph-Based Codes

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	6 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	**2310520**	Channel Coding: Graph-Based Codes	**3 SWS**	**Lecture /** 🗣	**Schmalen**
WT 26/27	**2310521**	Exercise for 2310520 Channel Coding: Graph-Based Codes	**1 SWS**	**Practice /** 🗣	**Schmalen**

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 30 minutes in which preparatory tasks are solved.

Prerequisites
none

Recommendations
Previous attendance of the lectures "Communication Engineering I" and "Theory of Probability" is recommended. Knowledge from the lectures "Applied Information Theory" and "Verfahren der Kanalcodierung" is helpful.

T 13.26 Module component: Circular Factory [T-MACH-114974]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-107636 - Circular Factory

Type	Credits	Grading	Term offered	Version
Written examination	8 CP	graded	Each summer term	3

Courses					
ST 2026	**2150620**	Circular Factory	**6 SWS**	**Lecture / Practice /**	**Lanza, Baumgärtner**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
written exam, duration 120 minutes

Prerequisites
T-MACH-113983 must not have been started.

Additional Information
The course is offered in English.

Workload
240 hours

T 13.27 Module component: Clinical Translation and Evaluation of Biomedical Technologies [T-ETIT-115194]

Coordinators: Prof. Dr. Werner Nahm
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107798 - Clinical Translation and Evaluation of Biomedical Technologies

Type	Credits	Grading	Version
Written examination	3 CP	graded	1

Courses					
WT 26/27	**2305301**	Clinical Translation and Evaluation of Biomedical Technologies	**2 SWS**	**Lecture /** 🗣	**Nahm**

Legend: 💻 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 90 minutes.

Prerequisites
none

T 13.28 Module component: Coding of Audiovisual Signals [T-ETIT-114615]

Coordinators: Prof. Dr.-Ing. Laurent Schmalen
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107440 - Coding of Audiovisual Signals

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each summer term	1 semesters	2

Courses					
ST 2026	**2310565**	Coding of Audiovisual Signals	**2 SWS**	**Lecture /** 🗣	**Schmalen**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in the form of an oral examination of approx. 20 min.

The module grade is the grade of the oral exam.

Prerequisites
see module

Recommendations
see module

T 13.29 Module component: Communication Networks and Systems Lab [T-ETIT-115146]

Coordinators: Prof. Dr.-Ing. Peter Rost
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107761 - Communication Networks and Systems Lab

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each term	1

Courses					
ST 2026	**2310522**	Mobile Communications Workshop	**2 SWS**	**Practical course /** 🗣	**Rost**
WT 26/27	**2310513**	Communication Networks and Systems Lab	**4 SWS**	**Practical course /** 🗣	**Rost**

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success controll takes place in the form of an oral examination.

The oral exam lasts approx. 25 min.

Prerequisites
"T-ETIT-115147 – Communication Networks and Systems Lab - Reports" must be passed in order to register for this oral examination.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-ETIT-115147 - Communication Networks and Systems Lab - Reports must have been passed.

T 13.30 Module component: Communication Networks and Systems Lab - Reports [T-ETIT-115147]

Coordinators: Prof. Dr.-Ing. Peter Rost
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107761 - Communication Networks and Systems Lab

Type	Credits	Grading	Term offered	Version
Coursework	0 CP	pass/fail	Each term	1

Courses					
ST 2026	**2310522**	Mobile Communications Workshop	**2 SWS**	**Practical course /** 🗣	**Rost**
WT 26/27	**2310513**	Communication Networks and Systems Lab	**4 SWS**	**Practical course /** 🗣	**Rost**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success controll takes place in the form of a coursework (through reports on each experiment).
The reports are graded as "pass" or "fail." One report is approx. 15 pages long.

Prerequisites
none

T 13.31 Module component: Communication Systems and Protocols [T-ETIT-101938]

Coordinators: Dr.-Ing. Jens Becker
Prof. Dr.-Ing. Jürgen Becker

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-100539 - Communication Systems and Protocols

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	1

Courses					
ST 2026	**2311616**	Communication Systems and Protocols	**2 SWS**	**Lecture /** 🗣	**Becker, Becker**
ST 2026	**2311618**	Tutorial for 2311616 Communication Systems and Protocols	**1 SWS**	**Practice /** 🗣	**Stammler**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination consists of a written examination of 120 min.

Prerequisites
none

Recommendations
Knowledge of the basics from the lecture "Digitaltechnik"is helpful.

T 13.32 Module component: Communications Engineering Lab [T-ETIT-114159]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107136 - Communications Engineering Lab

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each term	1

Courses					
ST 2026	**2310517**	Communications Engineering Lab	**4 SWS**	**Practical course /** 🗣	**Schmalen, Jäkel, Edelmann**
WT 26/27	**2310517**	Communications Engineering Lab	**4 SWS**	**Practical course /** 🗣	**Schmalen, Jäkel**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination consists of the participation in the experiments and an oral examination of approx. 30 min. The overall impression is rated.

The module grade results of the participation in the experiments and an oral examination. Details will be given during the lecture.

Prerequisites
none

Recommendations
Previous attendance of the lectures "Signals and Systems" and "Communications Engineering I".

T 13.33 Module component: Components of Power Systems [T-ETIT-113445]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106689 - Components of Power Systems

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2307397**	Components of Power Systems	**2 SWS**	**Lecture /**	**Kaptue Kamga**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The examination takes place in form of an oral examination lasting approx. 20 minutes.
The module grade is the grade of the oral exam.

Prerequisites
none

T 13.34 Module component: Compressed Sensing and Approximate Message Passing: Theory and Applications [T-ETIT-114487]

Coordinators: Dr.-Ing. Alexander Fengler
Prof. Dr.-Ing. Laurent Schmalen

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-107375 - Compressed Sensing and Approximate Message Passing: Theory and Applications

Type	Credits	Grading	Version
Oral examination	3 CP	graded	1

Courses					
WT 26/27	**2310543**	Compressed Sensing and Approximate Message Passing: Theory and Applications	**2 SWS**	**Lecture /** 🗣	**Fengler**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in the form of an oral examination of approx. 30 min.

Prerequisites
see module

Recommendations
see module

T 13.35 Module component: Computational Imaging [T-INFO-112573]

Coordinators: Dr.-Ing. Johannes Meyer
Organisation: KIT Department of Informatics
Part of: M-INFO-106190 - Computational Imaging

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	1

Courses					
WT 26/27	**2400173**	Computational Imaging	**3 SWS**	**Lecture / Practice /**	**Meyer, Beyerer**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The assessment takes the form of a written examination, usually lasting 60 minutes in accordance with Section 4 (2) No. 1 SPO.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take place
- in the form of an oral examination in accordance with Section 4 (2) No. 2 SPO or
- in the form of a written examination in accordance with Section 4 (2) No. 1 SPO.

Prerequisites
None.

T 13.36 Module component: Computational Intelligence in Mechanical Engineering [T-MACH-114110]

Coordinators: Stefan Meisenbacher
apl. Prof. Dr. Ralf Mikut
apl. Prof. Dr. Markus Reischl

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-105296 - Computational Intelligence

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2105016**	Computational Intelligence in Mechanical Engineering	**2 SWS**	**Lecture /**	**Mikut, Reischl**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Written exam (Duration: 1h)

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

T 13.37 Module component: Control of Power-Electronic Systems [T-ETIT-111897]

Coordinators: Dr.-Ing. Andreas Liske
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105915 - Control of Power-Electronic Systems

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	2

Courses					
ST 2026	**2306337**	Control of Power-Electronic Systems	**3 SWS**	**Lecture /** 🗣	**Liske, Sattler, Göhner**
ST 2026	**2306338**	Tutorial for 2306337 Control of Power-Electronic Systems	**1 SWS**	**Practice /** 🛠	**Liske, Göhner**

Legend: 💻 Online, 🛠 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment takes place as part of an online examination in text form with face-to-face supervision (§5, statutes for conducting online examinations at KIT) lasting 60 minutes.

The examination takes place on KIT computers in a pool room.

The module grade is the grade of the online examination.

Prerequisites
none

T 13.38 Module component: Control Technology [T-MACH-105185]

Coordinators: Dr.-Ing. Philipp Gönnheimer
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-105348 - Control Technology

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	2

Courses					
ST 2026	**2150683**	**Control Technology**	**2 SWS**	**Lecture / 🗣**	**Gönnheimer**

Legend: 📱 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written Exam (60 min)

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

T 13.39 Module component: Cryogenic Engineering [T-CIWVT-108915]

Coordinators: Prof. Dr.-Ing. Steffen Grohmann
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-104356 - Cryogenic Engineering

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2250140**	Cryogenic Engineering	**2 SWS**	**Lecture /** 🗣	**Grohmann**
WT 26/27	**2250141**	Cryogenic Engineering - Exercises	**1 SWS**	**Practice /** 🗣	**Grohmann**

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination is an oral examination with a duration of about 30 minutes (section 4 subsection 2 number 2 SPO).

Prerequisites
None

T 13.40 Module component: Cryptographic Protocols [T-INFO-111261]

Coordinators: Prof. Dr. Jörn Müller-Quade
Organisation: KIT Department of Informatics
Part of: M-INFO-105631 - Cryptographic Protocols

Type	Credits	Grading	Term offered	Version
Oral examination	5 CP	graded	Each summer term	2

Courses					
ST 2026	**2400154**	Cryptographic Protocols	**2 SWS**	**Lecture /**	**Mechler**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success is assessed in the form of an oral examination, usually lasting 30 minutes in accordance with §4 Para. 2 No. 2 SPO

Prerequisites
If the module M-INFO-104119 Secure multi-party calculation has already been checked, the module Cryptographic protocols cannot be checked.

Recommendations
Students should be familiar with the contents of the module "Theoretical Foundations of Cryptography".

Additional Information
The event takes place as an "inverted classroom". The material is presented in the form of videos, which are then interactively deepened in the subsequent face-to-face event.

T 13.41 Module component: Cryptographic Protocols II: Universal Composability [T-INFO-114602]

Coordinators: Prof. Dr. Jörn Müller-Quade
Organisation: KIT Department of Informatics
Part of: M-INFO-107435 - Cryptographic Protocols II: Universal Composability

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2400022**	**Cryptographic Protocols II: Universal Composability**	**2 SWS**	**Lecture / 🗣**	**Müller-Quade, Mechler**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) approx. 30 minutes.

Prerequisites
None.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-111584 - Universal Composability in Cryptography must not have been started.

Recommendations
Students should be familiar with the contents of the modules "Theoretische Grundlagen der Kryptographie" (Theoretical Foundations of Cryptography) and „Kryptographische Protokolle“ (Cryptographic Protocols I).

T 13.42 Module component: Current Topics of Solar Energy [T-ETIT-100780]

Coordinators: Prof. Dr.-Ing. Michael Powalla
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100507 - Current Topics of Solar Energy

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	2

Courses					
WT 26/27	**2313748**	Current Topics of Solar Energy	**2 SWS**	**Seminar /**	**Powalla**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success contraol takes placein form of a 30-minute presentation and the preparation of a short written paper. The overall impression is rated.

Prerequisites
none

T 13.43 Module component: Cyber-Physical Modeling [T-ETIT-113908]

Coordinators: Prof. Dr.-Ing. Mike Barth
Prof. Dr.-Ing. Sören Hohmann

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-106953 - Cyber-Physical Modeling

Type	Credits	Grading	Version
Written examination	6 CP	graded	1

Courses					
ST 2026	**2303310**	Cyber Physical Modeling	**3 SWS**	**Lecture /** 🗣	**Hohmann, Barth**
ST 2026	**2303311**	Tutorial to 2303310 Cyber Physical Modeling	**1 SWS**	**Practice /** 🗣	**Hohmann, Barth, Thömmes, Schicketanz**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in the form of a written examination lasting 90 min.
The module grade is the grade of the written exam.

Prerequisites
none

T 13.44 Module component: Deep Learning and Neural Networks [T-INFO-114219]

Coordinators: Prof. Dr. Jan Niehues
Organisation: KIT Department of Informatics
Part of: M-INFO-107197 - Deep Learning and Neural Networks

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2400024**	**Deep Learning and Neural Networks**	**4 SWS**	**Lecture / 🗣**	**Niehues**

Legend: 📱 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Prerequisites
T-INFO-101383 - Neural networks must not be started.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-109124 - Deep Learning and Neural Networks must not have been started.

Recommendations
Prior successful completion of the core module "Cognitive Systems" is recommended.

T 13.45 Module component: Deep Learning for Computer Vision: Advanced Topics [T-INFO-115078]

Coordinators: Prof. Dr.-Ing. Rainer Stiefelhagen
Organisation: KIT Department of Informatics
Part of: M-INFO-107695 - Deep Learning for Computer Vision: Advanced Topics

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	1

Assessment
Success is assessed in the form of a written examination (usually lasting 60 minutes) in accordance with Section 4 (2) No. 1 SPO.

Prerequisites
None.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-111494 - Deep Learning for Computer Vision II: Advanced Topics must not have been started.

Recommendations
Knowledge of deep learning basics is strongly recommended.

T 13.46 Module component: Deep Learning for Computer Vision: Advanced Topics [T-INFO-115257]

Coordinators: Prof. Dr.-Ing. Rainer Stiefelhagen
Organisation: KIT Department of Informatics
Part of: M-INFO-107843 - Deep Learning for Computer Vision: Advanced Topics

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	1

Assessment
Success is assessed in the form of a written examination (lasting 60 minutes) in accordance with § 4 Abs. 2 No. 1 SPO .

Prerequisites
None.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-111494 - Deep Learning for Computer Vision II: Advanced Topics must not have been started.
2. The module component T-INFO-115078 - Deep Learning for Computer Vision: Advanced Topics must not have been started.

Recommendations
Knowledge of deep learning basics is strongly recommended.

T 13.47 Module component: Design and Architectures of Embedded Systems (ESII) [T-INFO-114254]

Coordinators: Prof. Dr.-Ing. Jörg Henkel
Organisation: KIT Department of Informatics
Part of: M-INFO-107230 - Design and Architectures of Embedded Systems (ESII)

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Assessment
The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Prerequisites
None.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-101368 - Design and Architectures of Embedded Systems (ES2) must not have been started.

Recommendations
Knowledge of computer structures is helpful.

T 13.48 Module component: Design of Electrical Machines [T-ETIT-100785]

Coordinators: Prof. Dr. Martin Doppelbauer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100515 - Design of Electrical Machines

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	2

Courses					
WT 26/27	**2306324**	Design of Electrical Machines	**2 SWS**	**Lecture /**	**Doppelbauer**
WT 26/27	**2306325**	Tutorial for 2306324 Design of Electrical Machines	**1 SWS**	**Practice /**	**Doppelbauer, Parche**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in the form of a written examination lasting 120 minutes.

Prerequisites
none

T 13.49 Module component: Detectors for Applications in Space and Astronomy [T-ETIT-100761]

Coordinators: Prof. Theo Scherer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100541 - Detectores for Applications in Space and Astronomy

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2312678**	**Detectores for Applications in Space and Astronomy**	**2 SWS**	**Lecture**	**Scherer**

Assessment
Sucess control takes place in form of an overall oral examination lasting approx. 20 minutes.

Prerequisites
none

Recommendations
Basic knowledge of physics and electrical engineering

T 13.50 Module component: Digital Circuit Design [T-ETIT-114458]

Coordinators: Prof. Dr. Ivan Peric
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107365 - Digital Circuit Design

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each summer term	2

Courses					
ST 2026	**2312683**	Digital Circuit Design	**2 SWS**	**Lecture /**	**Peric**
ST 2026	**2312685**	Practical Excercise to 2312683 Digital Circuit Design	**1 SWS**	**Practice /**	**Peric**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in form of an oral examination (approx. 20 minutes) and a written report with the results of the exercises. Details will be given during the lecture.

Prerequisites
none

T 13.51 Module component: Digital Hardware Design Laboratory [T-ETIT-104570]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-102264 - Digital Hardware Design Laboratory

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2311637**	**Laboratory in Design Automation**	**4 SWS**	**Practical course /** 🗣	**Becker**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment

- Success is assessed in a final oral examination (approx. 30 min) and during the laboratory sessions on the basis of test protocols.
- The oral exams are taken in groups of two students. Each session takes approx. 30 minutes with approx. 15 minutes per student.
- During the laboratory the students have to write seven reports with around 1 page per report.

Prerequisites

The module M-ETIT-102266 Digital Hardware Design Laboratory must not have been started or completed.

Modeled Prerequisites

The following conditions have to be fulfilled:

1. The module M-ETIT-102266 - Digital Hardware Design Laboratory must not have been started.

Recommendations

Previous knowledge of the design and design automation of electronic systems is recommended.

T 13.52 Module component: Digital Hardware Design Laboratory [T-ETIT-104571]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-102266 - Digital Hardware Design Laboratory

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2311645**	**Digital Hardware Design Laboratory**	**4 SWS**	**Practical course /** 🗣	**Becker**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Control of success is carried out in an oral examination of approx. 15 min. as well as exercises in form of laboratory reports and/or oral interrogations during the laboratory. The overall impression is rated.

Prerequisites
none

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-ETIT-102264 - Digital Hardware Design Laboratory must not have been started.

Recommendations
Previous knowledge in design and design automation for electronic systems (e.g. from the lectures SAE, No. 23606, HSO, No. 23619 or HMS, No. 23608) is recommended.

Additional Information
The module ETIT-102264 („Praktikum Entwurf digitaler Systeme“) must not have been started or completed.

T 13.53 Module component: Digital Real Time Simulations for Energy Technologies [T-ETIT-113449]

Coordinators: Prof. Dr.-Ing. Giovanni De Carne
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106690 - Digital Real Time Simulations for Energy Technologies

Type	Credits	Grading	Version
Examination of another type	3 CP	graded	1

Courses					
ST 2026	**2314013**	Digital Real Time Simulations for Energy Technologies	**2 SWS**	**Lecture /** 🗣	**De Carne**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of other types of examination. It consists of an assessment from an exercise on HiL and an oral overall examination (approx. 15 minutes) explaining the exercise results. The overall impression is evaluated.
The module grade results of the assessment of an exercise and the oral exam. Details will be given during the lecture.

Prerequisites
none

T 13.54 Module component: Digital Signal Processing in Optical Communications – with Practical Exercises [T-ETIT-106852]

Coordinators: Prof. Dr.-Ing. Sebastian Randel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-103450 - Digital Signal Processing in Optical Communications – with Practical Exercises

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each summer term	3

Courses					
ST 2026	**2309472**	Digital Signal Processing in Optical Communications	**2 SWS**	**Lecture /** 🗣	**Randel, Bölük, Winter, Xu**
ST 2026	**2309473**	Digital Signal Processing in Optical Communications (Practical Exercises)	**2 SWS**	**Practice /** 🗣	**Randel**

Legend: 💻 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The exercise sheets and the oral questionnaire are used to rate other types of examinations. The overall impression is assessed. Duration about 20 minutes.

Prerequisites
none

Workload
180 hours

T 13.55 Module component: Digital Twin Engineering [T-ETIT-112224]

Coordinators: Prof. Dr.-Ing. Mike Barth
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106040 - Digital Twin Engineering

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each winter term	2

Courses					
WT 26/27	**2301486**	Digital Twin Engineering	**2 SWS**	**Lecture /**	**Barth, Dorn, Knapp**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The examination takes place in form of other types of examination. It consists of a model library developed in the course of a semester-long project in the modeling language Modelica and a presentation of the library lasting 25 minutes. The quality of the model library is evaluated within the framework of the criteria: documentation, formal correctness, functionality, usability, HMI and modeling level of detail. The presentation is evaluated as an additional aspects. The overall impression is evaluated.

The assessment of the developed model library and the presentation of the library will be included in the module grade. More details will be given at the beginning of the course.

Prerequisites
none

T 13.56 Module component: Distributed Discrete Event Systems [T-ETIT-100960]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100361 - Distributed Discrete Event Systems

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	**2302106**	Distributed Discrete Event Systems	**2 SWS**	**Lecture /**	**Heizmann, Baßler**
ST 2026	**2302108**	Practice to Distributed Discrete Event Systems	**1 SWS**	**Practice /**	**Baßler**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Success control takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

Recommendations
Knowledge of probabilities and measurement technology (e.g. from the Bachelor's module "Mess- und Regelungstechnik") and system theory (e.g. from the Bachelor's module "Signale und Systeme") is helpful.

T 13.57 Module component: Educational Development for Student Teachers - Basic Level [T-ETIT-100797]

Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Coursework	2 CP	pass/fail	Each term	1

Assessment
Success monitoring consists of participation in attendance modules (attendance requirement of 80%) and submission of a written reflection portfolio.

Attendance is required both to perform the work as a team on-site and to provide practical techniques and skills that cannot be learned in pure self-study.

Prerequisites
Part-time work as a tutor at KIT during program participation.

T 13.58 Module component: Elective Specialization Supplementary Studies on Science, Technology and Society / About Knowledge and Science - Self-Registration [T-FORUM-113580]

Coordinators: Dr. Christine Mielke
Christine Myglas

Organisation: General Studies. Forum Science and Society (FORUM)

Part of: M-FORUM-106753 - Supplementary Studies on Science, Technology and Society

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Assessment
Another type of examination assessment under § 5, section 3 involves a presentation, term paper, or project work within the chosen course.

Prerequisites
None

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- FORUM (ehem. ZAK) Begleitstudium

Recommendations
The contents of the basic module are helpful. The basic module should be completed or attended in parallel, but not after the advanced module.
The reading recommendations for primary and specialist literature are determined individually by the respective lecturers according to the subject area and course.

Additional Information
This placeholder can be used for any achievement in the Advanced Unit of the Supplementary Studies.

In the Advanced Module, students can choose their own individual focus, e.g. sustainable development, data literacy, etc. The focus should be discussed with the module coordinator at the FORUM.

T 13.59 Module component: Elective Specialization Supplementary Studies on Science, Technology and Society / Science in Public Debates - Self Registration [T-FORUM-113582]

Coordinators: Dr. Christine Mielke
Christine Myglas

Organisation: General Studies. Forum Science and Society (FORUM)

Part of: M-FORUM-106753 - Supplementary Studies on Science, Technology and Society

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Assessment
Another type of examination assessment under § 5, section 3 involves a presentation, term paper, or project work within the chosen course.

Prerequisites
None

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- FORUM (ehem. ZAK) Begleitstudium

Recommendations
The contents of the basic module are helpful. The basic module should be completed or attended in parallel, but not after the advanced module.
The reading recommendations for primary and specialist literature are determined individually by the respective lecturers according to the subject area and course.

Additional Information
This placeholder can be used for any achievement in the Advanced Unit of the Supplementary Studies.

T 13.60 Module component: Elective Specialization Supplementary Studies on Science, Technology and Society / Science in Society - Self-Registration [T-FORUM-113581]

Coordinators: Dr. Christine Mielke
Christine Myglas

Organisation: General Studies. Forum Science and Society (FORUM)

Part of: M-FORUM-106753 - Supplementary Studies on Science, Technology and Society

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Assessment
Another type of examination assessment under § 5, section 3 involves a presentation, term paper, or project work within the chosen course.

Prerequisites
None

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- FORUM (ehem. ZAK) Begleitstudium

Recommendations
The contents of the basic module are helpful. The basic module should be completed or attended in parallel, but not after the advanced module.
The reading recommendations for primary and specialist literature are determined individually by the respective lecturers according to the subject area and course.

Additional Information
This placeholder can be used for any achievement in the Advanced Unit of the Supplementary Studies.

T 13.61 Module component: Electric Drives and Power Electronics Lab [T-ETIT-114162]

Coordinators: Prof. Dr. Martin Doppelbauer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107138 - Electric Drives and Power Electronics Lab

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2306331**	Lab Course Electrical Drives and Power Electronics	**4 SWS**	**Practical course /**	**Brodatzki, Hiller**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment

Success control takes place in the form of other types of examination. It consists of one oral examination per experiment (approx. about 20 min. per person). The overall impression is assessed.

The assessments of the oral examinations are included in the module grade. Further details will be provided at the beginning of the course.

Prerequisites

none

T 13.62 Module component: Electric Drives for E-Mobility [T-ETIT-113936]

Coordinators: Prof. Dr. Martin Doppelbauer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106971 - Electric Drives for E-Mobility

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	2

Courses					
ST 2026	**2306500**	Electric Drives for E-Mobility	**2 SWS**	**Lecture /**	**Doppelbauer, Beck**
ST 2026	**2306501**	Practice to 2306500 Electric Drives for E-Mobility	**1 SWS**	**Practice /**	**Beck**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The success control takes place in the form of a written examination of 120 minutes.

Prerequisites
none

Recommendations
Basic knowledge in the fields of electric machines (types and operation of three-phase machines), electric power systems (complex alternating current theory, three-phase systems, equivalent circuit diagrams) and electronic drives (three-phase bridge circuit, modulation) is strongly recommended (e.g. as given in the BSc-modules "M-ETIT-107133 - Lineare Elektrische Netze", "Elektromagnetische Felder", "M-ETIT-106419 - Elektrische Energietechnik" and "M-ETIT-102124 - Elektrische Maschinen und Stromrichter").

Basic knowledge in the field of hybrid and electric vehicles is helpful (e.g. as given in the BSc-module components “T-ETIT-113612 - Hybrid and Electric Vehicles” or "T-ETIT-100784 – Hybride und elektrische Fahrzeuge").

T 13.63 Module component: Electric Power Generation and Power Grid [T-ETIT-103608]

Coordinators: Dr.-Ing. Bernd Hoferer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101917 - Electric Power Generation and Power Grid

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	2

Courses					
ST 2026	**2307399**	Electric Power Generation and Power Grid	**2 SWS**	**Lecture /** 🗣	**Hoferer**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral examination (approx. 20 minutes)

Prerequisites
none

T 13.64 Module component: Electric Power Transmission & Grid Control [T-ETIT-110883]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105394 - Electric Power Transmission & Grid Control

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each summer term	1 semesters	2

Courses					
ST 2026	**2307376**	Electric Power Transmission & Grid Control	**2 SWS**	**Lecture /** 🗣	**Leibfried**
ST 2026	**2307377**	Tutorial for 2307376 Electric Power Transmission & Grid Control	**2 SWS**	**Practice /** 🗣	**Weber**

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes. The module grade is the grade of the written exam.

Prerequisites
none

T 13.65 Module component: Electrocatalysis [T-ETIT-111831]

Coordinators: Dr. Philipp Röse
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105883 - Electrocatalysis

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	3

Courses					
ST 2026	**2304300**	Electrocatalysis	**3 SWS**	**Lecture /** 🗣	**Röse**
ST 2026	**2304301**	Exercise to 2304300 Electrocatalysis	**1 SWS**	**Practice /** 🗣	**Röse**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

T 13.66 Module component: Electrolysis and Electrosynthesis [T-ETIT-115334]

Coordinators: TT-Prof. Dr. Johanna Schröder
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107914 - Electrolysis and Electrosynthesis

Type	Credits	Grading	Version
Written examination	6 CP	graded	1

Courses					
WT 26/27	**2304361**	Lecture Electrolysis and Electrosynthesis	**2 SWS**	**Lecture /** 🗣	**Schröder**
WT 26/27	**2304362**	Exercise Electrolysis and Electrosynthesis	**2 SWS**	**Practice /** 🗣	**Schröder**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 90 minutes.

Prerequisites
none

T 13.67 Module component: Electrolyzers and their Electrodes [T-ETIT-114993]

Coordinators: TT-Prof. Dr. Johanna Schröder
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107648 - Electrolyzers and their Electrodes

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
ST 2026	**2304360**	Seminar Electrolyzers and their Electrodes	**2 SWS**	**Seminar / 🗣**	**Schröder**
WT 26/27	**2304360**	Seminar Electrolyzers and their Electrodes	**2 SWS**	**Seminar / 🗣**	**Schröder**

Legend: 💻 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success controll takes place in form of an other type of examination and consists of a written report on the scientific topic (max. 20 pages) and a presentation followed by a discussion, 15 minutes each. The overall impression is rated.

Prerequisites
none

T 13.68 Module component: Electromagnetics and Numerical Calculation of Fields [T-ETIT-100640]

Coordinators: Dr. Yongbo Deng
Prof. Dr. Ulrich Lemmer

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-100386 - Electromagnetics and Numerical Calculation of Fields

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2141110**	Electromagnetics and Numerical Calculation of Fields	**2 SWS**	**Lecture /** 🗣	**Rockstuhl**
WT 26/27	**2141111**	Exercise for 2141110 Electromagnetics and Numerical Calculation of Fields	**1 SWS**	**Practice /** 🗣	**Rockstuhl, NN**

Legend: 💻 Online, 💐 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control is carried out in the form of a written test of 120 minutes.

Prerequisites
none

Recommendations
Fundamentals of electromagnetic field theory.

T 13.69 Module component: Electronics and EMC [T-ETIT-100723]

Coordinators: Dr. Martin Sack
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100410 - Electronics and EMC

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2307378**	Electronics and EMC	**2 SWS**	**Lecture /**	**Sack**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success is assessed in an overall oral examination (approx. 20 minutes).

Prerequisites
none

T 13.70 Module component: Energy from Biomass [T-CIWVT-108828]

Coordinators: Dr.-Ing. Siegfried Bajohr
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-104288 - Biomass Based Energy Carriers

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2231510**	Biomass Based Energy Carriers	**2 SWS**	**Lecture /** 🗣	**Bajohr**
WT 26/27	**2231511**	Exercises on 2231510 Biomass Based Energy Carriers	**1 SWS**	**Practice /** 🗣	**Bajohr, und Mitarbeitende**

Legend: 💻 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Learning control is an oral exam lasting approx. 20 minutes.

Prerequisites
None

T 13.71 Module component: Energy Storage and Network Integration [T-ETIT-104644]

Coordinators: Prof. Dr.-Ing. Mathias Noe
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101969 - Energy Storage and Network Integration

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	2

Courses					
WT 26/27	2312687	Energy Storage and Network Integration	2 SWS	Lecture / 🗣	Grilli, De Carne
WT 26/27	2312689	Tutorial for 2312687 Energy Storage and Network Integration	1 SWS	Practice / 🗣	De Carne, Grilli

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in an overall oral examination (30 minutes).

Prerequisites
The course "T-MACH-105952 – Energiespeicher und Netzintegration" must not have started.

Recommendations
Basic knowledge in the fields of Electrical Engineering and Thermodynamics is helpful.

Additional Information
Exam and Lecture will be held in English.

T 13.72 Module component: Engineer's Field of Work [T-MACH-105721]

Coordinators: Prof. Dr. Martin Doppelbauer
Prof. Dr.-Ing. Marcus Geimer

Organisation: KIT Department of Mechanical Engineering

Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Coursework (written)	2 CP	pass/fail	Each summer term	3

Courses					
ST 2026	**2114917**	Engineer's Field of Work	**2 SWS**	**Lecture /** 🗣	**Doppelbauer, Geimer**

Legend: 📱 Online, 💮 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written test

Duration: 60 minutes

Prerequisites
none

Additional Information
The course is offered in German.

Further information can be found on our homepage:
https://www.fast.kit.edu/mobima/13438_15120.php

Workload
60 hours

T 13.73 Module component: Entrepreneurship [T-WIWI-102864]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-ETIT-105073 - Student Innovation Lab

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each term	2

Courses					
ST 2026	2545001	Entrepreneurship	2 SWS	Lecture / 🗣	Terzidis, Dang
WT 26/27	2545001	Entrepreneurship	2 SWS	Lecture / 🏘	Terzidis, Dang

Legend: 💻 Online, 🏘 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes) (following §4(2), 1 of the examination regulation).

Students are offered the opportunity to earn a grade bonus through separate assignments. If the grade of the written exam is between 4.0 and 1.3, the bonus improves the grade by a maximum of one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the lecture.

Prerequisites
None

Recommendations
None

T 13.74 Module component: Fabrication and Characterisation of Optoelectronic Devices [T-ETIT-103613]

Coordinators: Prof. Dr. Ulrich Wilhelm Paetzold
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101919 - Fabrication and Characterisation of Optoelectronic Devices

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2313760**	**Fabrication and Characterization of Optoelectronic Devices**	**2 SWS**	**Lecture /**	**Paetzold**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Type of Examination: written exam
Duration of Examination: 120 Minutes
Modality of Exam: One written exam offered at the end of each semester.

Prerequisites
none

T 13.75 Module component: Field Propagation and Coherence [T-ETIT-100976]

Coordinators: Prof.Dr.Dr.h.c. Wolfgang Freude
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100566 - Field Propagation and Coherence

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2309466**	Field Propagation and Coherence	**2 SWS**	**Lecture /** 🗣	**Freude**
WT 26/27	**2309467**	Tutorial for 2309466 Field Propagation and Coherence	**1 SWS**	**Practice /** 🗣	**Freude, N.N.**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Type of Examination: oral exam

Duration of Examination: approx. 20 minutes

Modality of Exam: Oral examination, usually one examination day per month during the summer and winter terms. An extra questions-and-answers session will be held for preparation if students wish so.

Prerequisites
none

Recommendations
Minimal background required: Calculus, differential equations and Fourier transform theory. Electrodynamics and field calculations or a similar course on electrodynamics or optics is recommended.
Solution of the problems on the exercise sheet, which can be downloaded as homework each week, is highly recommended. Also, active participation in the problem classes and studying in learning groups are strongly advised.

T 13.76 Module component: Foundations of Technology Ethics [T-GEISTSOZ-115075]

Coordinators: Dr. phil. Simon Derpmann
Organisation: KIT Department of Humanities and Social Sciences
Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Coursework	2 CP	pass/fail	Each summer term	1

Courses					
ST 2026	**5000010**	**Introduction to the Ethics of Technology and Responsible Engineering**	**2 SWS**	**Course**	**Derpmann**

Assessment
The course is assed through a written in-person exam lasting 60 minutes.

Prerequisites
T-ETIT-111923 – Technikethik - ARs ReflecTIonis must not have been started.

Additional Information
The course is offered in English.

Learning objectives: Engineering decisions shape lives, institutions, and ecosystems—yet they are rarely value-neutral. This course trains students to identify and critically analyze the ethical dimensions of technological systems, reconstruct and evaluate moral arguments, and navigate value conflicts in engineering practice. Through structured debates, they develop the capacity to justify coherent positions and articulate the responsibilities of engineers, firms, and institutions toward society.

Upon completion students are prepared to critically engage with ethical and political challenges in technology—in professional practice, public discourse, and their own role in shaping a responsible technological future.

Workload
60 hours

T 13.77 Module component: Geodetic Space Techniques for Engineers, Exam [T-BGU-112871]

Coordinators: Prof. Dr.-Ing. Hansjörg Kutterer
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-106347 - Geodetic Space Techniques for Engineers

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	6020161	Geodetic Spatial Methods	2 SWS	Lecture / 🗣	Mayer
WT 26/27	6020162	Geodetic Spatial Methods, Exercises	1 SWS	Practice / 🗣	Mayer

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral examination lasting approximately 20 minutes, in accordance with § 4(2)No.2 of the Study Regulations.

Prerequisites
The module component T-BGU-111169 - Geodetic Space Techniques for Engineers (prerequisite)—must be successfully completed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-BGU-111169 - Geodetic Space Techniques, Prerequisite must have been passed.

Workload
90 hours

T 13.78 Module component: Geodetic Space Techniques, Prerequisite [T-BGU-111169]

Coordinators: Prof. Dr.-Ing. Hansjörg Kutterer
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-106347 - Geodetic Space Techniques for Engineers

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	1 CP	pass/fail	Each winter term	1 semesters	1

Assessment
Completion of four worksheets to accompany the lecture (approximately 8 pages in length). Assessment is based on an accomplishment (§ 4, para. 3 of the Study Regulations).

Prerequisites
none

Workload
30 hours

T 13.79 Module component: Handling Characteristics of Motor Vehicles I [T-MACH-105152]

Coordinators: Dr.-Ing. Fabian Weitz
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-105288 - Handling Characteristics of Motor Vehicles I

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2113807**	Handling Characteristics of Motor Vehicles I	**2 SWS**	**Block /** 🗣	**Weitz**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Verbally

Duration about 30 up to 40 minutes

Auxiliary means: none

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

T 13.80 Module component: Hardware Modeling and Simulation [T-ETIT-100672]

Coordinators: Dr.-Ing. Jens Becker
Prof. Dr.-Ing. Jürgen Becker

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-100449 - Hardware Modeling and Simulation

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	2

Courses					
WT 26/27	**2311608**	Hardware Modeling and Simulation	**2 SWS**	**Lecture /** 🗣	**Becker, Becker**
WT 26/27	**2311610**	Tutorial for 2311608 Hardware Modeling and Simulation	**1 SWS**	**Practice /** 🗣	**Sotiropoulos**

Legend: 📱 Online, 🛜 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Achievement is examined in the form of a written examination lasting 120 minutes.

Prerequisites
none

T 13.81 Module component: Hardware Synthesis and Optimization [T-ETIT-113922]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106963 - Hardware Synthesis and Optimization

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2311619**	Hardware Synthesis and Optimization	**3 SWS**	**Lecture /** 🗣	**Becker**
ST 2026	**2311621**	Tutorial for 2311619 Hardware Synthesis and Optimization	**1 SWS**	**Practice /** 🗣	**Schmidt**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place within the framework of an oral overall examination (approx. 30 minutes).
The module grade is the grade of the oral exam.

Prerequisites
none

T 13.82 Module component: Hardware/Software Co-Design [T-ETIT-114623]

Coordinators: Dr.-Ing. Tanja Harbaum
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107444 - Hardware/Software Co-Design

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2311620**	Hardware/Software Co-Design	**3 SWS**	**Lecture /** 🗣	**Harbaum, Becker**
WT 26/27	**2311623**	Tutorial for 2311620 Hardware/ Software Co-Design	**1 SWS**	**Practice /** 🗣	**Paraskevas**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place within the framework of an oral overall examination (approx. 25 minutes).
The module grade is the grade of the oral exam.

Prerequisites
none

Recommendations
Basic knowledge in the fields of digital technology and information technology is helpful.

T 13.83 Module component: Hardware/Software Co-Design [T-ETIT-100671]

Coordinators: Dr.-Ing. Tanja Harbaum
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100453 - Hardware/Software Co-Design

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2311620**	Hardware/Software Co-Design	**3 SWS**	**Lecture /** 🗣	**Harbaum, Becker**
WT 26/27	**2311623**	Tutorial for 2311620 Hardware/ Software Co-Design	**1 SWS**	**Practice /** 🗣	**Paraskevas**

Legend: 💻 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Achievement will be examined in an oral examination (approx. 20 minutes).

Prerequisites
none

T 13.84 Module component: High-Power Microwave Technology [T-ETIT-114453]

Coordinators: Prof. Dr.-Ing. John Jelonnek
Dr.-Ing. Alexander Marek

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-107363 - High-Power Microwave Technology

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2308435**	**High-Power Microwave Technology**	**2 SWS**	**Lecture /**	**Marek, Jelonnek**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The success control is carried out as part of an oral overall examination (approx. 20 minutes).

Prerequisites
none

T 13.85 Module component: High-Voltage Technology [T-ETIT-110266]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Dr.-Ing. Michael Suriyah

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105060 - High-Voltage Technology

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	**2307360**	High-Voltage Technology	**2 SWS**	**Lecture /** 🗣	**Suriyah**
WT 26/27	**2307362**	Tutorial for 2307362High-Voltage Technology	**2 SWS**	**Practice /** 🗣	**Wöhr, Suriyah**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success conraol takes place in the form of a written examination lasting 120 minutes.

Prerequisites
none

T 13.86 Module component: High-Voltage Test Technique [T-ETIT-101915]

Coordinators: Dr.-Ing. Rainer Badent
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100417 - High-Voltage Test Technique

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2307392**	High-Voltage Test Technique	**2 SWS**	**Lecture /** 🗣	**Badent**
WT 26/27	**2307394**	Tutorial for 2307392 High-Voltage Test Technique	**2 SWS**	**Practice /** 🗣	**Braun**

Legend: 💻 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in an overall oral examination (approx. 20 minutes) on the selected course.

Prerequisites
none

Recommendations
High voltage technology

T 13.87 Module component: Human Brain and Central Nervous System: Anatomy, Information Transfer, Signal Processing, Neurophysiology and Therapy [T-INFO-101262]

Coordinators: Prof. Dr.-Ing. Tamim Asfour
Hon.-Prof. Dr. Uwe Spetzger

Organisation: KIT Department of Informatics

Part of: M-INFO-100725 - Human Brain and Central Nervous System: Anatomy, Information Transfer, Signal Processing, Neurophysiology and Therapy

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each term	2

Courses					
ST 2026	**24678**	Human Brain and Central Nervous System: Anatomy, Information Transfer, Signal Processing, Neurophysiology and Therapy	**2 SWS**	**Lecture /** 🗣	**Spetzger**
WT 26/27	**2424139**	Human Brain and Central Nervous System: Anatomy, Information Transfer, Signal Processing, Neurophysiology and Therapy	**2 SWS**	**Lecture /** 🗣	**Spetzger**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (lasting 45 minutes) in accordance with Section 4 (2) No. 1 SPO.

Prerequisites
None.

Recommendations
Attending the internships and seminars in the field of medical technology at the institute is recommended, as initial practical and theoretical experience in the many different areas is imparted and deepened.

T 13.88 Module component: Industrial Business Administration [T-WIWI-100796]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Coursework (written)	3 CP	pass/fail	Each winter term	1

Courses					
WT 26/27	**2581040**	Industrial Business Administration	**2 SWS**	**Lecture /**	**Fichtner**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The assessment of this course is a ungraded written examination (60 min).

Prerequisites
None

T 13.89 Module component: Industrial Circuitry [T-ETIT-100716]

Coordinators: Dr.-Ing. Andreas Liske
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100399 - Industrial Circuitry

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2306327**	**Industrial Circuitry**	**2 SWS**	**Lecture /**	**Liske**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success is assessed in an overall oral examination (approx. 20 minutes).

Prerequisites
none

T 13.90 Module component: Information Fusion [T-ETIT-106499]

Coordinators: Michael Heizmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-103264 - Information Fusion

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2302139**	Information Fusion	**2 SWS**	**Lecture /** 🗣	**Heizmann**
WT 26/27	**2302141**	Exercize for 2302139 Information Fusion	**1 SWS**	**Practice /** 🗣	**Heizmann, Hoffmann**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination lasting 120 minutes.

Prerequisites
none

Recommendations
Knowledge of the basics of stochastics is helpful.

T 13.91 Module component: Information Systems in Power Engineering Lab [T-ETIT-114183]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107159 - Information Systems in Power Engineering Lab

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2307388**	Information Systems in Power Engineering Lab	**4 SWS**	**Practical course /** 🗣	**Leibfried, und Mitarbeiter**

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Success control takes place in the form of other types of examination and is assessed in form of 3 experiments.

The module grade results of the assessment of the 3 experiments.

- 20 points are awarded for each experiment (max. 10 for preparation and max. 10 for performance).
- This results in a total of 60 points.
- At least 27 points must be achieved in order to pass the module.

Prerequisites
none

T 13.92 Module component: Information Technology in Industrial Automation Systems [T-ETIT-100698]

Coordinators: Dr.-Ing. Peter-Axel Bort
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100367 - Information Technology in Industrial Automation Systems

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2302144**	Information Technology in Industrial Automation Systems	**2 SWS**	**Lecture /** 🗣	**Bort**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in fom of an oral examination (20-25 minutes).

Prerequisites
none

T 13.93 Module component: Innovation Lab [T-ETIT-110291]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Prof. Dr. Werner Nahm
Prof. Dr.-Ing. Eric Sax
Prof. Dr. Wilhelm Stork
Prof. Dr.-Ing. Thomas Zwick

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105073 - Student Innovation Lab

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	9 CP	graded	Each term	2 semesters	1

Courses					
ST 2026	**2303192**	**Innovation Lab**	**2 SWS**	**Project / 🗣**	**Hohmann, Zwick, Sax, Stork, Terzidis**
WT 26/27	**2303192**	**Innovation Lab**	**2 SWS**	**Project / 🗣**	**Hohmann, Zwick, Sax, Stork, Nahm, Schmalen, Rost**

Legend: 💻 Online, 💠 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in the form of other types of examination and includes:

- Submission of a Technical Report (approx. 15-20 pages) with requirements list and system architecture
- Submission of the reflection of the Gantt Plans (approx. 8-12 pages)
- Presentation of the high-fidelity prototype (approx. 15 min.)

Prerequisites
none

T 13.94 Module component: Integrated Intelligent Sensors [T-ETIT-100961]

Coordinators: Prof. Dr. Wilhelm Stork
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100457 - Integrated Intelligent Sensors

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2311630**	Integrated Intelligent Sensors	**2 SWS**	**Lecture /** 🗣	**Stork**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Achievement will be examined in an oral examination (approx. 20 minutes).

Prerequisites
none

T 13.95 Module component: Integrated Photonics [T-ETIT-114418]

Coordinators: Prof. Dr.-Ing. Christian Koos
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107344 - Integrated Photonics

Type	Credits	Grading	Version
Oral examination	6 CP	graded	1

Courses					
WT 26/27	**2309440**	Integrated Photonics	**2 SWS**	**Lecture /** 🗣	**Koos**
WT 26/27	**2309441**	Tutorial to Integrated Photonics	**2 SWS**	**Practice /** 🗣	**Koos, N.N.**

Legend: 💻 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment takes place in the form of an oral examination (approx. 25 minutes); appointments individually on demand.

Prerequisites
none

T 13.96 Module component: Intellectual Property Rights and Strategies in Industrial Companies [T-MACH-105442]

Coordinators: Dipl.-Ing. Frank Zacharias
Organisation: KIT Department of Mechanical Engineering

Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each term	1

Courses					
ST 2026	**2147160**	Patents and Patentstrategies in innovative companies	**2 SWS**	**Lecture /** 🗣	**Zacharias**
WT 26/27	**2147161**	Intellectual Property Rights and Strategies in Industrial Companies	**2 SWS**	**Lecture /** 🗣	**Zacharias**

Legend: 💻 Online, 🌐 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam (ca. 20 min)

Prerequisites
none

Recommendations
None

Additional Information
The course is offered in German.

Workload
120 hours

T 13.97 Module component: Introduction to Automotive and Industrial Lidar Technology [T-ETIT-111011]

Coordinators: Prof. Dr. Wilhelm Stork
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105461 - Introduction to Automotive and Industrial Lidar Technology

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	3 CP	graded	Each winter term	1 semesters	2

Courses					
WT 26/27	**2311604**	Introduction to Automotive and Industrial Lidar Technology	**2 SWS**	**Lecture /** 🗣	**Heußner**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination consists of an oral exam and a short oral presentation. The overall impression is rated.

Recommendations
none

T 13.98 Module component: Introduction to Energy Economics [T-WIWI-102746]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-100498 - Introduction into Energy Economics

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	8

Courses					
ST 2026	**2581010**	Introduction to Energy Economics	**2 SWS**	**Lecture /** 🗣	**Fichtner**
ST 2026	**2581011**	Übungen zu Einführung in die Energiewirtschaft	**2 SWS**	**Practice /** 🗣	**Layer, Fichtner, Raab**

Legend: 💻 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (90 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None.

T 13.99 Module component: Introduction to Microsystem Technology I [T-MACH-114100]

Coordinators: Dr. Vlad Badilita
Prof. Dr. Jan Gerrit Korvink

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-107657 - Introduction to Microsystem Technology I

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2141861**	Introduction to Microsystem Technology I	**2 SWS**	**Lecture /** 🗣	**Korvink, Badilita**

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written examination (60 min)

Prerequisites
T-MACH-114035 and T-MACH-105182 must not have started

Workload
120 hours

T 13.100 Module component: Introduction to the Scientific Method (Seminar, Englisch) [T-ETIT-111317]

Coordinators: Prof. Dr. Werner Nahm
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	1 CP	pass/fail	Each term	1 semesters	1

Courses					
ST 2026	**2305745**	Introduction to the Scientific Method	**1 SWS**	**Seminar /** 🗣	**Nahm**
WT 26/27	**2305746**	Introduction to the Scientific Method	**1 SWS**	**Seminar /** 🗣	**Nahm**

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in the form of a study achievement.

The exam consists of the preparation and the presentation of 2 seminar papers. A seminar paper typically consists of 4-6 presentation slides together with a presentation of 20-30 minutes, including discussion.

Prerequisites
none

Additional Information
Detailled information on contents, competence goals, and work load at:

M-ETIT-105665 – Introduction to the Scientific Method (Seminar)

T 13.101 Module component: Introduction to the Scientific Method (Seminar, German) [T-ETIT-111316]

Coordinators: Prof. Dr. Werner Nahm
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	1 CP	pass/fail	Each term	1 semesters	1

Courses					
ST 2026	**2305744**	Introduction to the Scientific Method (Seminar, German)	**1 SWS**	**Seminar / 🗣**	**Nahm**
WT 26/27	**2305504**	Einführung in die wissenschaftliche Methode	**1 SWS**	**Seminar / 🗣**	**Nahm**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in form of an course work.

The assessment consists of the preparation and presentation of two seminar papers. A seminar paper typically consists of 4–6 presentation slides and a 20–30-minute presentation, including a discussion.

Prerequisites
none

Additional Information
For detailed information on course content, qualification goals, and course load, see:

M-ETIT-105664 – Einführung in die wissenschaftliche Methode (Seminar)

T 13.102 Module component: Introduction to Video Analysis [T-INFO-101273]

Coordinators: Prof. Dr.-Ing. Jürgen Beyerer
Organisation: KIT Department of Informatics
Part of: M-INFO-100736 - Introduction to Video Analysis

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**24684**	Introduction to Video Analysis	**2 SWS**	**Lecture /**	**Arens**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success is assessed in the form of an oral examination (usually lasting 30 minutes) in accordance with Section 4 (2) No. 2 SPO.

Prerequisites
None.

T 13.103 Module component: IT/OT-Security Seminar [T-ETIT-113648]

Coordinators: Prof. Dr.-Ing. Mike Barth
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106789 - IT/OT-Security Seminar

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	**2303201**	IT/OT-Security Seminar	**2 SWS**	**Seminar /**	**Barth, Madsen**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The examination takes place in the form of an oral examinationof approx. 25 min.

The module grade is the grade of the oral exam.

Prerequisites
none

T 13.104 Module component: Lab Course Electrical Power Engineering [T-ETIT-100728]

Coordinators: Dr.-Ing. Rainer Badent
Prof. Dr. Martin Doppelbauer
Prof. Dr.-Ing. Thomas Leibfried

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-100419 - Lab Course Electrical Power Engineering

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each winter term	2

Courses					
WT 26/27	**2307398**	Lab Course Electrical Power Engineering	**4 SWS**	**Practical course /** 🗣	**Hoferer**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Performance is assessed through alternative assessments as part of an ongoing evaluation process. The internship comprises eight experiments, each of which is individually assessed. Each experiment includes a written component with experiment-specific questions (approximately 15 minutes) followed by an oral component (approximately 45 minutes). The overall impression is evaluated.

The final grade is calculated as the arithmetic mean of the eight individual grades.

Prerequisites
none

T 13.105 Module component: Lab Course on Nanoelectronics [T-ETIT-100757]

Coordinators: Prof. Dr. Sebastian Kempf
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100468 - Lab Course on Nanoelectronics

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each term	1

Courses					
ST 2026	**2312669**	Laboratory Nanoelectronics	**4 SWS**	**Practical course /** 🗣	**Kempf, Mitarbeiter*innen**
WT 26/27	**2312669**	Laboratory Nanoelectronics	**4 SWS**	**Practical course /** 🗣	**Kempf, weitere Mitarbeitende**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in form of a written final report (approximately 10–20 pages in length), which should include an introduction to the topic, a description of the experimental procedure, the scientific results, and a discussion of how the results fit into the broader context.

Prerequisites
none

Recommendations
Successful completion of the module “Technology, Physics, and Applications of Thin Films” is recommended.

T 13.106 Module component: Lab course on superconducting materials [T-ETIT-111242]

Coordinators: Prof. Dr. Sebastian Kempf
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105614 - Lab course on superconducting materials

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each term	1

Courses					
ST 2026	**2312695**	Lab Course on Superconducting Materials	**4 SWS**	**Practical course /** 🗣	**Holzapfel, Hänisch**
WT 26/27	**2312650**	Praktikum Supraleitende Materialien	**4 SWS**	**Practical course /** ✕	**Holzapfel**

Legend: 📱 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The control of success takes place in form of the evaluation of a written report (approx. 10-20 pages) which introduces the topic, discusses the execution of the lab course and the scientific results puts the results into the overall context.

Prerequisites
none

T 13.107 Module component: Lab Course on Superconducting Quantum Electronics [T-ETIT-111233]

Coordinators: Prof. Dr. Sebastian Kempf
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105605 - Lab Course on Superconducting Quantum Electronics

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	6 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	**2312675**	Lab Course on Superconducting Quantum Electronics	**4 SWS**	**Practical course /** 🗣	**Kempf, Mitarbeiter*innen**
WT 26/27	**2312715**	Praktikum Supraleitende Quantenelektronik	**4 SWS**	**Practical course /** 🗣	**Kempf, weitere Mitarbeitende**

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in form of a written final report (approximately 10–20 pages in length), which should include an introduction to the topic, a description of the experimental procedure, the scientific results, and a discussion of how the results fit into the broader context.

Prerequisites
none

Recommendations
Successful completion of the modules “Quantum Detectors and Sensors” and “Nano- and Quantum Electronics” is recommended.

T 13.108 Module component: Laboratory FPGA Based Circuit Design [T-ETIT-100759]

Coordinators: Prof. Dr. Sebastian Kempf
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100470 - Laboratory FPGA Based Circuit Design

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each term	1

Courses					
ST 2026	**2312674**	Laboratory FPGA Based Circuit Design	**4 SWS**	**Practical course /** 🗣	**Kempf, Mitarbeiter*innen**
WT 26/27	**2312674**	Laboratory FPGA Based Circuit Design	**4 SWS**	**Practical course /** 🗣	**Kempf, weitere Mitarbeitende**

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in form of another type of examination, consisting of the evaluation of the three projects and a final report (30-40 pages). The overall impression is rated.

Prerequisites
none

T 13.109 Module component: Laboratory in Software Engineering [T-ETIT-100681]

Coordinators: Prof. Dr.-Ing. Eric Sax
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100460 - Laboratory in Software Engineering

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each summer term	2

Courses					
ST 2026	**2311640**	Laboratory in Software Engineering	**4 SWS**	**Practical course /**	**Sax**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success is assessed by means of two oral examinations (5-10 min.) during the laboratory, the submitted software project and a final presentation (5-10 min.) with a question and answer session. The overall impression is assessed.

The assessment of the oral questions, the software project and the final presentation with Q&A session are included in the module grade. Further details will be provided at the beginning of the course.

Prerequisites
none

T 13.110 Module component: Laboratory Lighting Technology [T-ETIT-104726]

Coordinators: Prof. Dr. Cornelius Neumann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-102356 - Laboratory Lighting Technology

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each term	1

Courses					
ST 2026	**2313715**	Laboratory Lighting Technology	**4 SWS**	**Practical course /** 🗣	**Trampert, Neumann**
WT 26/27	**2313715**	Laboratory Lighting Technology	**4 SWS**	**Practical course /** 🗣	**Neumann, Trampert**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control tales place in form of four experiments. The overall impression is rated.

Prerequisites
none

T 13.111 Module component: Laboratory Mechatronics [T-MACH-105370]

Coordinators: Prof. Dr. Veit Hagenmeyer
Prof. Dr.-Ing. Christoph Stiller

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-102699 - Laboratory Mechatronics

Type	Credits	Grading	Term offered	Version
Coursework	4 CP	pass/fail	Each summer term	6

Assessment
The laboratory course is offered exclusively as ungraded course work. The assessment consists of a group colloquium at the beginning of the individual specialization phases (Part 1). During the group phase (Part 2), students must also successfully complete various major assignments and present their results in a final presentation.

Prerequisites
None

Additional Information
The course is offered in German

Workload
120 hours

T 13.112 Module component: Laboratory Nanotechnology [T-ETIT-100765]

Coordinators: Prof. Dr. Ulrich Lemmer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100478 - Laboratory Nanotechnology

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each term	1

Courses					
ST 2026	**2313714**	Laboratory Nanotechnology	**4 SWS**	**Practical course /** 🗣	**Trampert, Lemmer**
WT 26/27	**2313714**	Laboratory Nanotechnology	**4 SWS**	**Practical course /** 🗣	**Lemmer, Trampert**

Legend: 🖳 Online, 🖳🗣 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control tales place in form of four experiments. The overall impression is rated.

Prerequisites
none

Additional Information
The module grade is based on oral examination components and the evaluation of written assignments. Further details will be provided at the beginning of the course.

T 13.113 Module component: Laboratory Solar Energy [T-ETIT-104686]

Coordinators: Dr.-Ing. Klaus Trampert
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-102350 - Laboratory Solar Energy

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each term	1

Courses					
ST 2026	**2313708**	Laboratory Solar Energy	**4 SWS**	**Practical course /** 🗣	**Trampert, Paetzold, Richards**
WT 26/27	**2313716**	Laboratory Solar Energy	**4 SWS**	**Practical course /** 🗣	**Richards, Trampert, Paetzold**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in form of four experiments. The overall impression is rated.

Prerequisites
none

Recommendations
Knowledge of the theoretical foundations of the individual experiments is helpful. It is recommended that you take this module after attending the relevant courses, as knowledge of the theoretical foundations is helpful but not strictly required. If you do not have the foundational knowledge from the corresponding modules, this will require more preparation time for the respective experiment.

Helpful modules: Solid-State Electronics

Additional Information
The module grade is based on oral examination components and the evaluation of written assignments. Further details will be provided at the beginning of the course.

T 13.114 Module component: Laser Metrology [T-ETIT-100643]

Coordinators: Prof. Dr. Marc Eichhorn
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100434 - Laser Metrology

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2303200**	Laser Metrology	**2 SWS**	**Lecture /**	**Eichhorn**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The exam will be taken as an oral examination (about 20 minutes). The individual appointments for examination are offered at two previously determined dates.

Prerequisites
none

T 13.115 Module component: Laser Physics [T-ETIT-100741]

Coordinators: Prof. Dr. Marc Eichhorn
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100435 - Laser Physics

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	2301480	Laserphysics	2 SWS	Lecture / 🗣	Eichhorn
WT 26/27	2301481	Exercise for 2301480 Laserphysics	1 SWS	Practice / 🗣	Eichhorn

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The exam will be taken as an oral examination (about 20 minutes). The individual appointments for examination are offered regularly at two previously determined dates.

Prerequisites
none

T 13.116 Module component: Lecture Series Supplementary Studies on Science, Technology and Society - Self Registration [T-FORUM-113578]

Coordinators: Dr. Christine Mielke
Christine Myglas

Organisation: General Studies. Forum Science and Society (FORUM)

Part of: M-FORUM-106753 - Supplementary Studies on Science, Technology and Society

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	2 CP	pass/fail	Each summer term	1 semesters	1

Courses					
ST 2026	**1130716**	Lecture Series Science in Society	**2 SWS**	**Lecture /** 🗣	**Post, Mielke**

Legend: 🖳 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Active participation, learning protocols, if applicable.

Prerequisites
None

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- FORUM (ehem. ZAK) Begleitstudium

Recommendations
It is recommended that you complete the lecture series "Science in Society" before attending events in the advanced module and in parallel with attending the basic seminar.
If it is not possible to attend the lecture series and the basic seminar in the same semester, the lecture series can also be attended after attending the basic seminar.
However, attending events in the advanced module before attending the lecture series should be avoided.

Additional Information
The basic module consists of the lecture series "Science in Society" and the basic seminar. The lecture series is only offered during the summer semester.
The basic seminar can be attended in the summer or winter semester.

T 13.117 Module component: Liberalised Power Markets [T-WIWI-107043]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-105403 - Liberalised Power Markets

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	4

Courses					
WT 26/27	**2581998**	Liberalised Power Markets	**2 SWS**	**Lecture /** 🗣	**Fichtner**
WT 26/27	**2581999**	Übungen zu Liberalised Power Markets	**2 SWS**	**Practice /** 🗣	**Fichtner**

Legend: 💻 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Recommendations
None

Workload
180 hours

T 13.118 Module component: Lighting Engineering [T-ETIT-100772]

Coordinators: Prof. Dr. Cornelius Neumann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100485 - Lighting Engineering

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2313739**	Lighting Engineering	**2 SWS**	**Lecture /**	**Neumann**
WT 26/27	**2313741**	Lighting Engineering (Tutorial to 2313739)	**1 SWS**	**Practice**	**Neumann**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Success is assessed in the form of an oral examination (approx. 20 minutes) on the selected courses, which together fulfill the minimum LP requirement.

Prerequisites
none

T 13.119 Module component: Localization of Mobile Agents [T-INFO-115014]

Coordinators: Prof. Dr.-Ing. Uwe Hanebeck
Organisation: KIT Department of Humanities and Social Sciences
KIT Department of Informatics
Part of: M-INFO-107663 - Localization of Mobile Agents

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	6 CP	graded	Each summer term	1 semesters	3

Courses					
ST 2026	**24613**	**Localization of Mobile Agents**	**3 SWS**	**Lecture / Practice /**	**Kim, Hanebeck, Frisch**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The assessment is carried out as an oral examination, lasting 15-30 minutes in accordance with Section 4 (2) No. 2 SPO. Participants can contact the chair at any time to arrange an appointment for the oral examination. (§ 4 Abs. 2 Nr. 2 SPO)

Prerequisites
T-INFO-115015 Digital exercise must be successfully completed before exam.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-115015 - Localization of Mobile Agents - Pass must have been started.
2. The module component T-INFO-101377 - Localization of Mobile Agents must not have been started.

Recommendations
Knowledge of a higher-level programming language with mature libraries for scientific and numerical computing (e.g., Julia, Matlab, Python) is an advantage.

T 13.120 Module component: Localization of Mobile Agents - Pass [T-INFO-115015]

Coordinators: Prof. Dr.-Ing. Uwe Hanebeck
Organisation: KIT Department of Informatics
Part of: M-INFO-107663 - Localization of Mobile Agents

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	0 CP	pass/fail	Each summer term	1 semesters	1

Courses					
ST 2026	**24613**	**Localization of Mobile Agents**	**3 SWS**	**Lecture / Practice /**	**Kim, Hanebeck, Frisch**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment is carried out in form of course work (German Studienleistung, § 4 Abs. 3 SPO). Students must complete a digital exercise in Ilias which is ungraded. The digital exercise can be repeated as often as desired.

Prerequisites
None.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-114169 - Localization of Mobile Agents Pass must not have been started.

Recommendations
Knowledge of a higher-level programming language with mature libraries for scientific and numerical computing (e.g., Julia, Matlab, Python) is an advantage.

T 13.121 Module component: Low Power Design [T-INFO-101344]

Coordinators: Prof. Dr.-Ing. Jörg Henkel
Organisation: KIT Department of Informatics
Part of: M-INFO-100807 - Low Power Design

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2424672**	Low Power Design	**2 SWS**	**Lecture /**	**Henkel, Nassar, Khdr**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The assessment is carried out as an oral examination lasting 25-30 minutes, in accordance with Section 4 (2) No. 2 SPO.

Prerequisites
None.

Recommendations
- Basic knowledge from the modules “Design and Architectures of Embedded Systems (ESII)” and “Optimization and Synthesis of Embedded Systems (ESI)” are helpful but not essential for understanding of this lecture.
- The lecture is equally suitable for students from both computer science as well as electrical engineering department.
- The Lab of “Low Power Design and Embedded Systems” enables students to apply some of the theoretical knowledge gained from the lecture in practice.

T 13.122 Module component: Machine Learning 1 - Basic Methods [T-WIWI-106340]

Coordinators: Prof. Dr.-Ing. Johann Marius Zöllner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-105003 - Machine Learning 1

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	4

Courses					
WT 26/27	**2511500**	Machine Learning 1 - Fundamental Methods	**2 SWS**	**Lecture /**	**Zöllner**
WT 26/27	**2511501**	Exercises to Machine Learning 1 - Fundamental Methods	**1 SWS**	**Practice /**	**Zöllner, Polley, Fechner**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success is assessed in the form of a written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.

A grade bonus can be earned by successfully completing exercises. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by up to one grade level (0.3 or 0.4). Details will be announced in the lecture.

Prerequisites
None.

Workload
150 hours

T 13.123 Module component: Machine Learning 2 - Advanced Methods [T-WIWI-106341]

Coordinators: Prof. Dr.-Ing. Johann Marius Zöllner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-105006 - Machine Learning 2

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	5

Courses					
ST 2026	**2511502**	Machine Learning 2 - Advanced Methods	**2 SWS**	**Lecture /**	**Zöllner, Fechner, Polley, Stegmaier**
ST 2026	**2511503**	Exercises for Machine Learning 2 - Advanced Methods	**1 SWS**	**Practice /**	**Zöllner, Fechner, Polley, Stegmaier**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success is assessed in the form of a written examination (70 minutes). The examination is offered every semester and can be repeated at any regular examination date.

Prerequisites
None.

Workload
150 hours

T 13.124 Module component: Machine Learning and Optimization in Communications [T-ETIT-110123]

Coordinators: Prof. Dr.-Ing. Laurent Schmalen
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-104988 - Machine Learning and Optimization in Communications

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4 CP	graded	Each summer term	1 semesters	3

Courses					
ST 2026	**2310560**	Machine Learning and Optimization in Communications	**2 SWS**	**Lecture /** 🗣	**Schmalen**
ST 2026	**2310561**	Practice to 2310560 Machine Learning and Optimization in Communications	**1 SWS**	**Practice /** 🗣	**Schmalen, Karanov**

Legend: 💻 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination of 120 minutes.

Prerequisites
see module

Recommendations
see module

T 13.125 Module component: Machine Learning and Optimization in Energy Systems [T-WIWI-113073]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-106604 - Machine Learning and Optimization in Energy Systems

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	4

Courses					
WT 26/27	**2581050**	**Machine Learning and Optimization in Energy Systems**	**3 SWS**	**Lecture / Practice /** 🗣	**Dengiz, Yilmaz, Kleinebrahm**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The assessment of this course is a written examination (60 min) or an oral exam (30 min) depending on the number of participants.

Workload
120 hours

T 13.126 Module component: Machine Tools and High-Precision Manufacturing Systems [T-MACH-110962]

Coordinators: Prof. Dr.-Ing. Jürgen Fleischer
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-105107 - Machine Tools and Industrial Handling

Type	Credits	Grading	Term offered	Version
Oral examination	8 CP	graded	Each winter term	1

Courses					
WT 26/27	**2149910**	**Machine Tools and High-Precision Manufacturing Systems**	**6 SWS**	**Lecture / Practice /**	**Fleischer**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Oral exam (40 minutes)

Prerequisites
T-MACH-102158 - Machine Tools and Industrial Handling must not be commenced.
T-MACH-109055 - Machine Tools and Industrial Handling must not be commenced.
T-MACH-110963 - Machine Tools and High-Precision Manufacturing Systems must not be commenced.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-110963 - Machine Tools and High-Precision Manufacturing Systems must not have been started.

Additional Information
The course is offered in German.

Workload
240 hours

T 13.127 Module component: Machine Vision [T-MACH-105223]

Coordinators: Dr. Martin Lauer
Prof. Dr.-Ing. Christoph Stiller

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101923 - Machine Vision

Type	Credits	Grading	Term offered	Version
Written examination	8 CP	graded	Each winter term	4

Courses					
WT 26/27	**2137308**	**Machine Vision**	**4 SWS**	**Lecture / Practice /** 🗣	**Lauer, Merkert, Truetsch**

Legend: 📱 Online, 🌐 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Type of Examination: written exam

Duration of Examination: 60 minutes

Prerequisites
None

Additional Information
The course is offered in English

Workload
240 hours

T 13.128 Module component: Master's Project [T-ETIT-115350]

Coordinators: Prof. Dr.-Ing. Mike Barth
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107928 - Master's Project

Type	Credits	Grading	Term offered	Version
Coursework	6 CP	pass/fail	Each term	1

Assessment
Success control takes place in the form of another type of examination. It consists of a short report of the project results, a presentation of the results.

The overall impression is rated.

Prerequisites
none

T 13.129 Module component: Master's Seminar on Scientific Study Design in Biomedical Engineering [T-ETIT-115047]

Coordinators: Dr.-Ing. Axel Loewe
Prof. Dr. Werner Nahm
Prof. Dr.-Ing. Maria Francesca Spadea

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Coursework	1 CP	pass/fail	Each term	1

Courses					
ST 2026	**2305253**	**Seminar über aktuelle Forschungsarbeiten in der Biomedizinischen Technik**	**4 SWS**	**Seminar / 🗣**	**Nahm, Loewe, Spadea**
WT 26/27	**2305253**	**Seminar on Current Research in Biomedical Engineering**	**4 SWS**	**Seminar / 💠**	**Nahm, Loewe, Spadea**

Legend: 💻 Online, 💠 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Course work of another type:

1. Giving a presentation of approx. 25 minutes followed by approx. 15 minutes of discussion.
2. Active participation in the discussion of 10 seminar presentations.

Prerequisites
none

Recommendations
Supplementary seminar recommendations:

- Introduction to the scientific method
- Seminar über ausgewählte Kapitel der Biomedizinischen Technik
- Journal Club

Additional Information
Attendance is compulsory for ten seminar presentations. Compulsory attendance is necessary for the practical instruction and practice of techniques and skills in academic discussion that cannot be acquired through self-study alone.

To facilitate a competent discussion, the seminar focuses on the subject area of biomedical engineering.

Detailled information on contents, competence goals, and work load at:

M-ETIT-107681 – Master's Seminar on Scientific Study Design in Biomedical Engineering

T 13.130 Module component: Master's Thesis [T-ETIT-114213]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107191 - Master's Thesis

Type	Credits	Grading	Term offered	Version
Final Thesis	30 CP	graded	Each term	1

Assessment
The Master's Thesis module has 30 credits. It consists of the Master's Thesis and a presentation. The presentation must be carried out within the processing time in accordance with SPO Section §14(4).

Prerequisites
SPO Section §14 Master's Thesis module (1) A prerequisite for admission to the Master's Thesis module is that the student has successfully completed module examinations totaling 75 credits, including the subject Basics of the specialization according to Section §19(2).

Final Thesis
This module component represents a final thesis. The following periods have been supplied:

Submission deadline	6 months
Maximum extension period	3 months
Correction period	8 weeks

This thesis requires confirmation by the examination office.

T 13.131 Module component: Measurement Technology [T-ETIT-112147]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105982 - Measurement Technology

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	1

Courses					
WT 26/27	**2302118**	Exercise for 2302117 Measurement Technology	**1 SWS**	**Practice /**	**Heizmann, Schmerbeck**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes. The module grade is the grade of the written examination.

Prerequisites
T-ETIT-101937 – Messtechnik (German version) must not have started.

T 13.132 Module component: Mechatronic Measurement Systems Lab [T-ETIT-114341]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107294 - Mechatronic Measurement Systems Lab

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2302123**	Mechatronic Measurement Systems Lab	**4 SWS**	**Practical course /**	**Heizmann, Steffens**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The success control is carried out in the form of a oral examination of approx. 20 min. The module grade is the grade of the oral exam.

Prerequisites
none

T 13.133 Module component: Medical Image Processing for Guidance and Navigation [T-ETIT-113425]

Coordinators: Prof. Dr.-Ing. Maria Francesca Spadea
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106672 - Medical Image Processing for Guidance and Navigation

Type	Credits	Grading	Term offered	Version
Oral examination	9 CP	graded	Each winter term	2

Courses					
WT 26/27	**2305297**	**Medical Image Processing for Guidance and Navigation**	**6 SWS**	**Lecture / Practice /**	**Spadea, Raggio, Riggio, Hopp**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The examination takes place within the framework of an oral overall examination of approx. 30 minutes about the lecture including a presentation and discussion of the project developed during the course. The overall impression is rated.

The module grade is the grade of the oral exam.

A bonus can be earned for submitting homework that will be provided during the lecture time.

The exact criteria for awarding a bonus will be announced at the beginning of the lecture period. If the grade in the oral exam is between 4.0 and 1.3, the bonus improves the grade by 0.3 or 0.4.

Bonus points do not expire and are retained for any examinations taken at a later date.

Prerequisites
none

T 13.134 Module component: Microactuators [T-MACH-101910]

Coordinators: Prof. Dr. Manfred Kohl
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-100487 - Microactuators

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	**2142881**	Microactuators	**2 SWS**	**Lecture /**	**Kohl**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
written exam, 60 min.

Prerequisites
T-MACH-114036 must not be started

Additional Information
The course is offered in German

Workload
120 hours

T 13.135 Module component: Microenergy Technologies [T-MACH-105557]

Coordinators: Dr. Jingyuan Xu
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-102714 - Microenergy Technologies

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	2

Courses					
ST 2026	**2142897**	Microenergy Technologies	**2 SWS**	**Lecture /** 🗣	**Xu**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral examination (approx 30 min.)

Prerequisites
none

Additional Information
The course is offered in English.

Workload
120 hours

T 13.136 Module component: Microsystem Technology [T-ETIT-100752]

Coordinators: Prof. Dr. Wilhelm Stork
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100454 - Microsystem Technology

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2311625**	Microsystem Technology	**2 SWS**	**Lecture /**	**Stork**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Achievement will be examined in an oral examination (approx. 20 minutes).

Prerequisites
none

T 13.137 Module component: Microwave Engineering Lab [T-ETIT-113938]

Coordinators: Prof. Dr.-Ing. Thomas Zwick
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106973 - Microwave Engineering Lab

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each term	1

Courses					
ST 2026	**2308415**	Microwave Engineering Lab	**4 SWS**	**Practical course /** 🗣	**Pauli**
WT 26/27	**2308415**	Microwave Engineering Lab	**4 SWS**	**Practical course /** 🗣	**Pauli**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
In preparation for the laboratory experiments, each laboratory group must work together on a number of tasks as homework before the experiment and submit a single copy to the supervisor immediately before the start of the experiment. The tasks for the experiment itself are completed and recorded during the experiment. The protocol should be handed in to the supervisor immediately after the experiment. Before each experiment is carried out, there is a written or oral examination (approx. 20 minutes, no aids) on the content of the experiment.

The grade for the experiments is made up of the preparation, the protocol and the written or oral assessment of the learning objectives for each experiment. The final grade for the entire laboratory results from the overall impression of the performance. Students who appear unprepared for the respective experiment may not take part in the experiment. The experiment must be repeated at another time.

Prerequisites
none

Recommendations
Knowledge of microwave measurement technology and RF components and systems is helpful.

T 13.138 Module component: Microwave Module Design [T-ETIT-111375]

Coordinators: Dr. Thomas Geist
Prof. Dr.-Ing. Thomas Zwick

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105701 - Microwave Module Design

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	**2308426**	Microwave Module Design	**2 SWS**	**Lecture /**	**Geist**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The success control is carried out as part of an oral overall examination (approx. 20 minutes) of the selected courses, which is in total meet the minimum requirement for LP.

Prerequisites
none

Recommendations
Knowledge of the basics of radio frequency technology and communications technology is helpful.

T 13.139 Module component: Microwaves Measurement Techniques [T-ETIT-114046]

Coordinators: Prof. Dr.-Ing. Thomas Zwick
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107007 - Microwaves Measurement Techniques

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	**2308420**	Microwaves Measurement Techniques	**2 SWS**	**Lecture /** 🗣	**Pauli**
ST 2026	**2308422**	Practical Exercise to 2308420 Microwaves Measurement Techniques	**1 SWS**	**Practice /** 🗣	**Pauli**

Legend: 💻 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Success control takes place in the form of an oral examination (approx. 20 minutes).
The module grade is the grade of the oral exam.

Prerequisites
none

T 13.140 Module component: Mixed-Signal IC Design [T-ETIT-111845]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105893 - Mixed-Signal IC Design

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2308443**	**Mixed-Signal IC Design**	**2 SWS**	**Lecture /**	**Ulusoy**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in form of an oral examination (approx. 30 min.)

Prerequisites
none

T 13.141 Module component: MMIC Design Laboratory [T-ETIT-111006]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105464 - MMIC Design Laboratory

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	6 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	**2308423**	MMIC Design Laboratory	**4 SWS**	**Practical course /**	**Ulusoy**
WT 26/27	**2308438**	MMIC Design Laboratory	**4 SWS**	**Practical course /**	**Ulusoy**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The written report (approx. 30-40 slides) and the oral presentation (approx. 40 min.) are used to mark the course. The overall impression is assessed.

Prerequisites
none

T 13.142 Module component: Mobile Communications [T-ETIT-112127]

Coordinators: Prof. Dr.-Ing. Peter Rost
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105971 - Mobile Communications

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2310523**	Mobile Communications	**2 SWS**	**Lecture /** 🗣	**Rost**
WT 26/27	**2310524**	Exercises for 2310523 Mobile Communications	**1 SWS**	**Practice /** 🗣	**Rost**

Legend: 💻 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Prerequisites
none

T 13.143 Module component: Mobile Communications II [T-ETIT-112679]

Coordinators: Prof. Dr.-Ing. Peter Rost
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106244 - Mobile Communications II

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	**2310514**	Mobile Communications II	**2 SWS**	**Lecture /**	**Rost**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Prerequisites
none

T 13.144 Module component: Modeling Physiological Systems [T-ETIT-113630]

Coordinators: Dr.-Ing. Axel Loewe
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106782 - Modeling Physiological Systems

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2305302**	**Modeling Physiological Systems**	**2 SWS**	**Lecture /**	**Loewe**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The examination takes place in form of a written examination lasting 90 min.

The module grade is the grade of the written exam.

Prerequisites
"T-ETIT-114690 – Modeling Physiological Systems - Workshop" must be passed in order to register for this written examination.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-ETIT-114690 - Modeling Physiological Systems - Workshop must have been passed.

T 13.145 Module component: Modeling Physiological Systems - Workshop [T-ETIT-114690]

Coordinators: Dr.-Ing. Axel Loewe
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106782 - Modeling Physiological Systems

Type	Credits	Grading	Term offered	Version
Coursework	0 CP	pass/fail	Each summer term	1

Courses					
ST 2026	**2305303**	Exercise to 2305302 Modeling Physiological Systems	**2 SWS**	**Practice / 🗣**	**Loewe, Kruthoff**

Legend: 💻 Online, 🌀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in the form of ungraded course works. The workshop tasks (3 exercise sheets) must be submitted.

Prerequisites
none

Additional Information
This success control must be passed in order to register for the written examination.

T 13.146 Module component: Modelling and Simulation of Electrochemical Systems [T-ETIT-100781]

Coordinators: Dr.-Ing. Andre Weber
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100508 - Modelling and Simulation of Electrochemical Systems

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2304217**	Modellbildung elektrochemischer Systeme	**2 SWS**	**Lecture /** 🗣	**Weber**

Legend: 🖳 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Success control takes place in the form of an oral examination lasting approx. 20 minutes.

Prerequisites
none

Recommendations
The contents of the lecture "M-ETIT-107005 - Batteries, Fuel Cells, and Electrolysis" are assumed to be known. Students who have not (yet) heard this lecture are recommended to work through the lecture notes for this lecture in advance.

T 13.147 Module component: Modern Radio Systems Engineering [T-ETIT-100735]

Coordinators: Prof. Dr.-Ing. Thomas Zwick
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100427 - Modern Radio Systems Engineering

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each term	2

Courses					
ST 2026	**2308430**	Modern Radio Systems Engineering	**3 SWS**	**Lecture /** 🗣	**Zwick, Pauli**
ST 2026	**2308431**	Tutorial to 2308430 Modern Radio Systems Engineering	**1 SWS**	**Practice /** 🗣	**Long, Chen**
WT 26/27	**2308430**	Modern Radio Systems Engineering	**3 SWS**	**Lecture /** 🗣	**Zwick, Pauli**
WT 26/27	**2308431**	Tutorial to 2308430 Modern Radio Systems Engineering	**1 SWS**	**Practice /** 🗣	**Chen, Zwick**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in the form of an oral examination of approx. 20 minutes. The module grade is the grade of the oral examination.

Prerequisites
none

T 13.148 Module component: Modern VLSI Technologies [T-ETIT-113864]

Coordinators: Prof. Dr. Jasmin Aghassi-Hagmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106921 - Modern VLSI Technologies

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	2

Courses					
ST 2026	**2308441**	Modern VLSI Technologies	**2 SWS**	**Lecture /** 🗣	**Aghassi-Hagmann, Cadilha Marques**
ST 2026	**2308442**	Tutorial Modern VLSI Technologies	**2 SWS**	**Practice /** 🗣	**Cadilha Marques**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in form of an oral examination with a duration of approx. 20 minutes.
The module grade is the grade of the oral examination.

Prerequisites
"T-ETIT-114743 – Modern VLSI Technologies - Laboratory" must be passed in order to register for this oral examination.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-ETIT-114743 - Modern VLSI Technologies Lab must have been passed.

T 13.149 Module component: Modern VLSI Technologies Lab [T-ETIT-114743]

Coordinators: Prof. Dr. Jasmin Aghassi-Hagmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106921 - Modern VLSI Technologies

Type	Credits	Grading	Term offered	Version
Coursework	0 CP	pass/fail	Each summer term	1

Courses					
ST 2026	**2308442**	**Tutorial Modern VLSI Technologies**	**2 SWS**	**Practice / 🗣**	**Cadilha Marques**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
50% of approx. 10 ungraded lab exercises have to be sucessfully completed.

Prerequisites
none

T 13.150 Module component: Multimodal Artificial Intelligence [T-INFO-115041]

Coordinators: Prof. Dr. Jan Niehues
Prof. Dr.-Ing. Rainer Stiefelhagen
Organisation: KIT Department of Informatics
Part of: M-INFO-107676 - Multimodal Artificial Intelligence

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	2

Courses					
ST 2026	**2400141**	**Multimodal Artificial Intelligence**	**4 SWS**	**Lecture /**	**Niehues, Stiefelhagen**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) approx. 60 minutes.

Prerequisites
None.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-112768 - Advanced Artificial Intelligence must not have been started.
2. The module component T-INFO-114220 - Advanced Artificial Intelligence must not have been started.
3. The module component T-INFO-111491 - Deep Learning for Computer Vision I: Basics must not have been started.
4. The module component T-INFO-111494 - Deep Learning for Computer Vision II: Advanced Topics must not have been started.

T 13.151 Module component: Multivariable Control Systems [T-ETIT-114727]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107515 - Multivariable Control Systems

Type	Credits	Grading	Version
Written examination	6 CP	graded	1

Courses					
WT 26/27	2303193	Multivariable Control Systems	3 SWS	Lecture / 🗣	Kluwe
WT 26/27	2303194	Tutorial to 2303194 "Multivariable Control Systems"	1 SWS	Practice / 🗣	Fehn

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

The module grade is the grade of the written exam.

Prerequisites
none

T 13.152 Module component: Nano- and Quantum Electronics [T-ETIT-111232]

Coordinators: Prof. Dr. Sebastian Kempf
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105604 - Nano- and Quantum Electronics

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	**2312668**	Nano- and Quantum Electronics	**3 SWS**	**Lecture /** 🗣	**Kempf**
ST 2026	**2312670**	Tutorial for 2312668 Nano- and Quantum Electronics	**1 SWS**	**Practice /** 🗣	**Kempf, Mitarbeiter*innen**

Legend: 🖳 Online, 🖳🗣 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success takes place in the form of a written examination lasting 120min. The grade corresponds to the result of the written examination.

Prerequisites
none

Recommendations
Successful completion of the modules "Superconductivity for Engineers" and „Einführung in die Quantentheorie für Elektrotechniker“ is recommended.

T 13.153 Module component: Navigation and Localization Techniques [T-ETIT-111829]

Coordinators: Prof. Dr.-Ing. Thomas Zwick
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105881 - Navigation and Localization Techniques

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2308440**	Navigation and Localization Techniques	**2 SWS**	**Lecture /** 🗣	**Sand**

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The success control is carried out as part of an oral overall examination (approx. 20 minutes) of the selected courses, which in total meet the minimum requirement for LP.

Prerequisites
none

T 13.154 Module component: NMR Methods for Product and Process Analysis [T-CIWVT-111843]

Coordinators: apl. Prof. Dr. Gisela Guthausen
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-105890 - NMR Methods for Product and Process Analysis

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2245130**	**NMR for Engineers**	**2 SWS**	**Lecture /**	**Guthausen**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Learning control is an oral examination with a duration of about 30 minutes.

Prerequisites
None.

T 13.155 Module component: Nonlinear Control Systems [T-ETIT-114989]

Coordinators: Dr.-Ing. Mathias Kluwe
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107644 - Nonlinear Control Systems

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2303320**	Nonlinear Control Systems	**3 SWS**	**Lecture /** 🗣	**Kluwe**
ST 2026	**2303321**	Exercises to 2303320 Nonlinear Control Systems	**1 SWS**	**Practice /** 🗣	**Handwerker**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

T 13.156 Module component: Nonlinear Optics [T-ETIT-101906]

Coordinators: Prof. Dr.-Ing. Christian Koos
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100430 - Nonlinear Optics

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	2

Courses					
ST 2026	**2309468**	Nonlinear Optics	**2 SWS**	**Lecture /** 🗣	**Koos**
ST 2026	**2309469**	Nonlinear Optics (Tutorial)	**2 SWS**	**Practice /** 🗣	**Koos**

Legend: 💻 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Type of Examination: oral exam

Duration of Examination: approx. 30 Minutes

Modality of Exam: The oral exam is offered continuously upon individual appointment.

Prerequisites
none

Recommendations
Solid mathematical and physical background, basic knowledge in optics and photonics.

Additional Information
The module grade is the grade of the oral exam.

There is a bonus system based on the problem sets that are solved during the tutorials: During the term, 3 problem sets will be collected in the tutorial and graded without prior announcement. If for each of these sets more than 70% of the problems have been solved correctly, a bonus of 0.3 grades will be granted on the final mark of the oral exam.

T 13.157 Module component: Numerical Methods - Exam [T-MATH-111700]

Coordinators: apl. Prof. Dr. Peer Kunstmann
TT-Prof. Dr. Xian Liao
Prof. Dr. Wolfgang Reichel

Organisation: KIT Department of Mathematics

Part of: M-MATH-106972 - Numerical Methods with Programming Practice

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	1

Courses					
ST 2026	**0180300**	Numerical Methods (Electrical Engineering, Meteorology, Remote Sensing, Geoinformatics)	**2 SWS**	**Lecture**	**Kunstmann**
ST 2026	**0180400**	Tutorial for 0180300 (Numerical Methods (Electrical Engineering, Meteorology, Remote Sensing, Geoinformatics))	**1 SWS**	**Practice**	**Kunstmann**

Assessment
Success control takes the form of a written examination (120 minutes).

Prerequisites
none

T 13.158 Module component: Numerical Methods - Workshop [T-MATH-113937]

Coordinators: apl. Prof. Dr. Peer Kunstmann
TT-Prof. Dr. Xian Liao
Prof. Dr. Wolfgang Reichel

Organisation: KIT Department of Mathematics

Part of: M-MATH-106972 - Numerical Methods with Programming Practice

Type	Credits	Grading	Term offered	Version
Coursework	1 CP	pass/fail	Each summer term	1

Assessment
Successful participation in the workshop is confirmed by signing the attendance sheet provided at each practice session.

Prerequisites
None

T 13.159 Module component: Numerical Methods for Partial Differential Equations [T-ETIT-104595]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-102311 - Numerical Methods for Partial Differential Equations

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	2

Courses					
ST 2026	**2303600**	Numerical Methods for Partial Differential Equations	**2 SWS**	**Lecture /** 🗣	**Nagato-Plum**
ST 2026	**2303601**	Practice to 2303180 Numerical Methods for Partial Differential Equations	**1 SWS**	**Practice /** 🗣	**Nagato-Plum**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place ni form of an overall oral examination (approx. 20 minutes).

Prerequisites
none

Recommendations
Knowledge of the following modules is recommended:

Higher Mathematics I-III in the Bachelor

"M-MATH-106972 - Numerical Methods with Programming Practice" or "M-MATH-105831 Numerical Methods"

T 13.160 Module component: Optical Design Lab [T-ETIT-100756]

Coordinators: Prof. Dr. Wilhelm Stork
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100464 - Optical Design Lab

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	2

Courses					
ST 2026	**2311647**	Optical Design Lab	**4 SWS**	**Practical course / ×**	**Stork**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The examination consists of an oral exam (approx. 20 min).

Prerequisites
none

Recommendations
Basic knowledge in optics. The participation in the course Optical Engineering is strongly adviced.

T 13.161 Module component: Optical Engineering and Machine Vision [T-ETIT-113941]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106974 - Optical Engineering and Machine Vision

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	2

Courses					
WT 26/27	**2302150**	Optical Engineering and Machine Vision	**3 SWS**	**Lecture /** 🗣	**Heizmann**
WT 26/27	**2302151**	Tutorial to 2302150 Optical Engineering and Machine Vision	**1 SWS**	**Practice /** 🗣	**Heizmann, Leyer**

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 (KSOP/O&P: 90) minutes.
The module grade is the grade of the written examination.

Prerequisites
none

T 13.162 Module component: Optical Networks and Systems [T-ETIT-106506]

Coordinators: Prof. Dr.-Ing. Sebastian Randel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-103270 - Optical Networks and Systems

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	3

Courses					
WT 26/27	**2309470**	Optical Networks and Systems	**3 SWS**	**Lecture /** 🗣	**Randel**
WT 26/27	**2309471**	Tutorial for 2309470 Optical Networks and Systems	**1 SWS**	**Practice /** 🗣	**Randel, N.N.**

Legend: 💻 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Type of Examination: oral exam

Duration of Examination: 20 min (approx.)

Modality of Exam: Oral exams (approx. 20 minutes) are offered throughout the year upon individual appointment.

Prerequisites
none

Recommendations
Interest in communications engineering, networking, and photonics.

T 13.163 Module component: Optical Systems in Medicine and Life Science [T-ETIT-106462]

Coordinators: Prof. Dr. Werner Nahm
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-103252 - Optical Systems in Medicine and Life Science

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	3

Courses					
ST 2026	**2305292**	Optical Systems in Medicine and Life Science	**2 SWS**	**Lecture /**	**Nahm**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Written exam (60 minutes)

Prerequisites
Only one out of the two modules "M-ETIT-100552 - Optische Systeme für Medizintechnik und Life Sciences" and "M-ETIT-103252 - Optical Systems in Medicine and Life Science" is allowed.

Recommendations
Good understanding of optics and optoelectronics.

Additional Information
Language English

T 13.164 Module component: Optical Technologies for Automotive Applications [T-ETIT-100773]

Coordinators: Prof. Dr. Cornelius Neumann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100486 - Optical Technologies for Automotive Applications

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2313740**	Optical Technologies for Automotive Applications	**2 SWS**	**Lecture /** 🗣	**Neumann**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in form ofan overall oral examination (approx. 20 minutes).

Prerequisites
none

Recommendations
Knowledge of the module "M-ETIT-100485 - Lichttechnik"

T 13.165 Module component: Optical Transmitters and Receivers [T-ETIT-100639]

Coordinators: Prof.Dr.Dr.h.c. Wolfgang Freude
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100436 - Optical Transmitters and Receivers

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	2

Courses					
WT 26/27	**2309460**	Optical Transmitters and Receivers	**2 SWS**	**Lecture /** 🗣	**Freude**
WT 26/27	**2309461**	Tutorial for 2309460 Optical Transmitters and Receivers	**2 SWS**	**Practice /** 🗣	**Freude, N.N.**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Access controll takes placein form of an overall oral examination (approx. 20 minutes). The individual dates for the oral examination are offered regularly.

Prerequisites
none

Recommendations
Knowledge of the physics of the pn-transition.

T 13.166 Module component: Optimal Control [T-ETIT-114701]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107497 - Optimal Control

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2303183**	Optimal Control	**2 SWS**	**Lecture /**	**Hohmann**
WT 26/27	**2303185**	Tutorial to 2303183 Optimal Control	**2 SWS**	**Practice /**	**Riffel**
WT 26/27	**2303851**	Accompanying Group Tutorial to 2303183 Optimal Control	**1 SWS**	**Tutorial /**	**Riffel**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of a written exam (120 min) taking place in the recess period.

Prerequisites
none

T 13.167 Module component: Optimization and Synthesis of Embedded Systems (ESI) [T-INFO-114253]

Coordinators: Prof. Dr.-Ing. Jörg Henkel
Organisation: KIT Department of Informatics
Part of: M-INFO-107229 - Optimization and Synthesis of Embedded Systems (ESI)

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Assessment
The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting approx. 20 minutes.

Prerequisites
None.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-101367 - Optimization and Synthesis of Embedded Systems (ES1) must not have been started.

Recommendations
Knowledge of computer structures is helpful.

The prerequisites, if any, are explained in more detail in the module description.

T 13.168 Module component: Optoelectronic Components [T-ETIT-101907]

Coordinators: TT-Prof. Dr. Tobias Huber-Loyola
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100509 - Optoelectronic Components

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	**2309486**	Optoelectronic Components	**2 SWS**	**Lecture /**	**Huber-Loyola, Gablinger, Sauter**
ST 2026	**2309487**	Optoelectronic Components (Tutorial)	**1 SWS**	**Practice /**	**Huber-Loyola**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Type of Examination: oral exam

Duration of Examination: approx. 30 minutes

Modality of Exam: Oral examination, usually one examination day per month during the Summer and Winter terms. An extra questions-and-answers session will be held if students wish so.

Prerequisites
none

Recommendations
Minimal background required: Calculus, differential equations, Fourier transforms and p-n junction physics.

T 13.169 Module component: Optoelectronic Measurement Engineering [T-ETIT-100771]

Coordinators: Dr.-Ing. Klaus Trampert
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100484 - Optoelectronic Measurement Engineering

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2313736**	Optoelectronic Measurement Engineering	**2 SWS**	**Lecture /** 🗣	**Trampert**

Legend: 🖥 Online, 🌀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in form of an overall oral examination (approx. 25 minutes).

Prerequisites
none

Recommendations
Knowledge from Lighting Technology and Technical Optics is an advantage.

T 13.170 Module component: Organic and Flexible Electronics [T-ETIT-114638]

Coordinators: Prof. Dr. Gerardo Hernandez Sosa
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107455 - Organic and Flexible Electronics

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2313768**	Organic and Flexible Electronics	**2 SWS**	**Lecture /**	**Hernandez Sosa**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The control of success takes place within the framework of an oral overall examination (approx. 20 minutes).

Prerequisites
none

Recommendations
Knowledge of semiconductor components

T 13.171 Module component: Pattern Recognition [T-INFO-101362]

Coordinators: Prof. Dr.-Ing. Jürgen Beyerer
Dr. Tim Zander
Organisation: KIT Department of Informatics
Part of: M-INFO-100825 - Pattern Recognition

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	2

Courses					
ST 2026	**24675**	**Pattern Recognition**	**4 SWS**	**Lecture / Practice /**	**Beyerer**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The examination is conducted in the form of a written examination (usually lasting 60 minutes) in accordance with § 4 (2) No. 1 SPO.
Depending on the number of participants, it will be announced six weeks prior to the examination (§ 6 (3) SPO) whether the assessment will take place:

- in the form of an oral examination in accordance with § 4 (2) No. 2 SPO, or
- in the form of a written examination in accordance with § 4 (2) No. 1 SPO.

Prerequisites
None.

Recommendations
Basic knowledge of probability theory, signal processing, and image processing is helpful.

T 13.172 Module component: Photometry and Radiometry [T-ETIT-100789]

Coordinators: Dr.-Ing. Klaus Trampert
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100519 - Photometry and Radiometry

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2313727**	**Photometry and Radiometry**	**2 SWS**	**Lecture /**	**Trampert**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in the form of an overall oral examination (approx. 25 minutes).

Prerequisites
none

Recommendations
Knowledge of the module "M-ETIT-100484 - Optoelektronische Messtechnik" and "M-ETIT-100485 - Lichttechnik" is an advantage.

T 13.173 Module component: Photonic Integrated Circuit Design and Applications [T-ETIT-111896]

Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105914 - Photonic Integrated Circuit Design and Applications

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2309478**	**Photonic Integrated Circuit Design and Applications**	**2 SWS**	**Lecture /**	**Koos, Freude, Randel, Huber-Loyola, Le, Jimenez**
ST 2026	**2309479**	**Photonic Integrated Circuit Design and Applications (Practical Exercise)**	**2 SWS**	**Practical course /**	**Koos, Freude, Randel, Huber-Loyola**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment

- Part 1 – Solutions of problem sets: We will grade your solutions of the various problem sets and design projects. To this end, please upload your solution via the online teaching platform of your respective institution (see above) before the respective deadline. Please merge all pages into a single pdf file, and please use a scanner. Smartphone made snapshots are often illegible, and in this case your solutions cannot not be evaluated. In case there are any technical difficulties with the platforms, you may also submit your solutions by e-mail to picda@ipq.kit.edu before the respective deadline.
- Part 2 - Presentation of one pre-assigned problem set: At the beginning of the term, design projects will be pre-assigned to groups of participants. Each of these groups will explain their approach and results to lecturers and peer students in a short presentation (approx. 15 min), followed by approx. 10 min of public discussion with peer students and professors, and an individual private interview of each group member (approx. 10 min per person).

The overall impression is rated.

Prerequisites
none

T 13.174 Module component: Photonics and Communications Lab [T-ETIT-109173]

Coordinators: Prof. Dr.-Ing. Christian Koos
Prof. Dr.-Ing. Sebastian Randel

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-104485 - Photonics and Communications Lab

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2309490**	**Photonics and Communications Lab**	**4 SWS**	**Practical course /** 🗣	**Koos, Freude, Randel, Huber-Loyola, Kuzmin, Khairy, Reichenbacher, Reza, Schmitz, Wilhelm, Zhang**

Legend: 🖳 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in the form of other types of examination. Success is assessed by solving the tasks relating to the preparation of the experiment (written and oral) and writing an experiment report.

Prerequisites
none

T 13.175 Module component: Photovoltaics [T-ETIT-101939]

Coordinators: Prof. Dr.-Ing. Michael Powalla
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100513 - Photovoltaics

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	2

Courses					
ST 2026	**2313737**	Photovoltaics	**3 SWS**	**Lecture /**	**Lemmer, Powalla**
ST 2026	**2313738**	Tutorial 2313737 Photovoltaik	**1 SWS**	**Practice /**	**Powalla, Lemmer**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Prerequisites
"T-ETIT-100774 - Solar Energy" must not have started.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-ETIT-100774 - Solar Energy must not have been started.

T 13.176 Module component: Physical Foundations of Cryogenics [T-CIWVT-106103]

Coordinators: Prof. Dr.-Ing. Steffen Grohmann
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-103068 - Physical Foundations of Cryogenics

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2250130**	Physical Foundations of Cryogenics	**2 SWS**	**Lecture /** 🗣	**Grohmann**
ST 2026	**2250131**	Exercises on 2250130 Physical Foundations of Cryogenics	**1 SWS**	**Practice /** 🗣	**Grohmann**

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Learning control is an oral examination lasting approx. 30 minutes.

Prerequisites
None

T 13.177 Module component: Physics, Technology and Applications of Thin Films [T-ETIT-111237]

Coordinators: Dr. Konstantin Ilin
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105608 - Physics, Technology and Applications of Thin Films

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	**2312710**	Physics, Technology and Application of Thin Films	**2 SWS**	**Lecture /** 🗣	**Ilin**
WT 26/27	**2312711**	Exercise for 2312710 Physics, Technology and Application of Thin Films	**1 SWS**	**Practice /** 🗣	**Ilin**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place within the framework of an oral overall examination of approx. 20 minutes.

T 13.178 Module component: Power Electronic Systems [T-ETIT-114751]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107533 - Power Electronic Systems

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2306357**	Power Electronic Systems	**3 SWS**	**Lecture /**	**Hiller**
WT 26/27	**2306358**	Practical Exercise to 2306357 Power Electronic Systems	**1 SWS**	**Practice /**	**Hiller, Pfeffer**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Success control takes place in the form of an oral examination lasting approx. 25 minutes.

The module grade is the grade of the oral examination.

Prerequisites
none

T 13.179 Module component: Power Electronics [T-ETIT-109360]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-104567 - Power Electronics

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each summer term	1 semesters	6

Courses					
ST 2026	**2300004**	Ausweich- und Praktikumstermin für ETI-Vorlesungen	**2 SWS**	**Practical course /**	**Hiller, Thönelt**
ST 2026	**2306323**	Power Electronics	**2 SWS**	**Lecture /**	**Hiller**
ST 2026	**2306324**	Tutorial for 2306385 Power Electronics	**2 SWS**	**Practice /**	**Hiller, Thönelt, Becher**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

T 13.180 Module component: Power Electronics for the Energy Transition [T-ETIT-115308]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107888 - Power Electronics for the Energy Transition

Type	Credits	Grading	Version
Oral examination	3 CP	graded	1

Courses					
WT 26/27	**2306360**	Power Electronics for the Energy Transition	**2 SWS**	**Lecture /**	**Hiller**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The success control takes place in form of an oral examination (approx. 25 minutes).

Prerequisites
none

T 13.181 Module component: Power System Protection and Automation [T-ETIT-113164]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106506 - Power System Protection and Automation

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	**2307396**	Power System Protection and Automation	**2 SWS**	**Lecture /** 🗣	**Loitz**

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed as part of an overall oral examination (approx. 20 minutes).

Prerequisites
none

T 13.182 Module component: Power Systems and Economy [T-ETIT-100725]

Coordinators: Dr.-Ing. Bernd Hoferer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100413 - Power Systems and Economy

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2307383**	Power Systems and Economy	**2 SWS**	**Lecture /**	**Weissmüller**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in the form of an overall oral examination (20 minutes)

Prerequisites
none

T 13.183 Module component: Practical Amplifier Design [T-ETIT-114179]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107156 - Practical Amplifier Design

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2308460**	Practical Amplifier Design	**1 SWS**	**Lecture /**	**Ulusoy**
ST 2026	**2308461**	Exercise for 2308460 Practical Amplifier Design	**1 SWS**	**Practice /**	**Ulusoy**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Success control takes place in the form of an oral examination of approx. 30 min.

The module grade is the grade of the oral examination.

Prerequisites
none

T 13.184 Module component: Practical Applications and Modelling of Rarefied Gas Flows [T-CIWVT-115470]

Coordinators: Dr.-Ing. Thomas Giegerich
Organisation: KIT Department of Chemical and Process Engineering
KIT Department of Electrical Engineering and Information Technology
Part of: M-CIWVT-108002 - Practical Applications and Modelling of Rarefied Gas Flows

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2250810**	Practical Applications and Modelling of Rarefied Gas Flows	**2 SWS**	**Lecture /** 🗣	**Giegerich, Tantos**
WT 26/27	**2250811**	Labs on 2250810 Applications and Modelling of Rarefied Gas Flows	**1 SWS**	**Practice /** 🗣	**Tantos, Giegerich**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination is an oral examination with a duration of about 20 minutes (section 4 subsection 2 number 2 SPO).

Prerequisites
None

T 13.185 Module component: Practical Aspects of Electrical Drives [T-ETIT-100711]

Coordinators: Prof. Dr. Martin Doppelbauer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100394 - Practical Aspects of Electrical Drives

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	2

Courses					
WT 26/27	**2306311**	Practical Aspects of Electrical Drives	**2 SWS**	**Lecture /**	**Brodatzki**
WT 26/27	**2306313**	Tutorial to 2306311 Practical Aspects of Electrical Drives	**1 SWS**	**Practice /**	**Schneider**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Prerequisites
none

Additional Information
Shift from SoSe to WiSe, does not take place in WiSe24/25 and SoSe25.

T 13.186 Module component: Practical Course: Smart Energy System [T-INFO-112030]

Coordinators: Dr.-Ing. Richard Jumar
Organisation: KIT Department of Informatics
Part of: M-INFO-105955 - Practical Course: Smart Energy System

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each term	1

Courses					
ST 2026	**2400170**	Laboratory: Smart Energy System Lab	**4 SWS**	**Practical course /** 🗣	**Hagenmeyer, Waczowicz, Jumar, Aghaei Attar**
WT 26/27	**2400230**	Lab Course: Smart Energy System Lab	**4 SWS**	**Practical course /** 🗣	**Hagenmeyer, Jumar, Aghaei Attar**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

An overall grade is awarded which consists of:
40% 10 min presentation (per team) + 5 min of discussion
60% Write up in the style of a research paper of around 5-6 pages (for solo participants) or 7-8 pages (for teams of two)
The students may redraw from the examination before the end of the first laboratory week. The assessment can be repeated once.

Prerequisites
None.

Recommendations
- Knowledge of the fundamentals of energy informatics is a prerequisite.
- Knowledge of the fundamentals of electrical engineering and energy technology is required.
- Knowledge of the basics of mechatronics, data analysis and signal processing is helpful.
- Knowledge of power systems or power electronics is helpful.

T 13.187 Module component: Practical Introduction to Research Software Engineering [T-ETIT-114434]

Coordinators: Dr. Marie Houillon
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Coursework	1 CP	pass/fail	Each term	1

Courses					
ST 2026	**2305180**	Practical Introduction to Research Software Engineering	**1 SWS**	**Practical course /** 🗣	**Houillon, Loewe**
WT 26/27	**2305180**	Practical Introduction to Research Software Engineering	**1 SWS**	**Practical course /** 🗣	**Gerach, Loewe, Houillon**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment takes place in the form of other types of examination, that is an evaluation of a software project.
The module is ungraded. The module is passed with successful assessment of the coursework.

Prerequisites
none

Additional Information
Detailled information on contents, competence goals, and work load at:
M-ETIT-107354 – Practical Introduction to Research Software Engineering

T 13.188 Module component: Practical Machine Learning [T-ETIT-113426]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Prof. Dr.-Ing. Sander Wahls

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-106673 - Practical Machine Learning

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each summer term	2

Courses					
ST 2026	**2302200**	Practical Machine Learning	**4 SWS**	**Lecture / Practice /**	**Gardi**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success is assessed by the submission of the scientific essay (approx. 10 pages) and a 30-minute presentation of the team project.

The module grade results from the team project accompanying the semester and the presentation of the team project. The overall impression is assessed. Further details will be provided at the beginning of the course.

Prerequisites
none

T 13.189 Module component: Practical Tools for Control Engineers [T-ETIT-113628]

Coordinators: Dr.-Ing. Balint Varga
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106780 - Practical Tools for Control Engineers

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	**2303210**	Practical Tools for Control Engineers	**2 SWS**	**Lecture /** 🗣	**Varga**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The examination takes place in form of other types of examination. It consists of an oral overall examination in the amount of 25 minutes and a homework programming task. The examination includes questions from the lecture slides and the presentation of the homework assignment. The homework must be submitted two weeks before of the oral exam. The overall impression is evaluated.

Prerequisites
none

Recommendations
The contents of the modules “Optimization of Dynamic Systems (ODS)” and “Regelung linearer Mehrgrößensysteme (RLM)” are helpful for the lecture.

Workload
120 hours

T 13.190 Module component: Printed and Thin-Film Electronics [T-ETIT-114417]

Coordinators: Prof. Dr. Jasmin Aghassi-Hagmann
Dr. Dr. Michael Hirtz

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-107343 - Printed and Thin-Film Electronics

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2308480**	**Printed and Thin-Film Electronics**	**2 SWS**	**Block / 🗣**	**Aghassi-Hagmann, Hirtz**

Legend: 📱 Online, 🌐 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment takes place in form of an oral examination (approx. 20 minutes).

Prerequisites
none

T 13.191 Module component: Printed Flexible Electronics Lab [T-ETIT-113075]

Coordinators: Prof. Dr. Gerardo Hernandez Sosa
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106464 - Printed Flexible Electronics Lab

Type	Credits	Grading	Term offered	Version
Coursework	6 CP	pass/fail	Each term	2

Courses					
ST 2026	**2313765**	Printed Flexible Electronics Lab	**4 SWS**	**Practical course /** 🗣	**Hernandez Sosa**
WT 26/27	**2313765**	Printed Flexible Electronics Lab	**4 SWS**	**Practical course /** 🗣	**Hernandez Sosa**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination consists of a written journal article and an oral presentation of the student's work, both given in English. The overall impression is rated.

The module is passed with successful assessment of the written paper and the oral presentation. Details will be given during the lecture. The module is ungraded.

Prerequisites
The M-ETIT-100475 Modul: Plastic Electronics / Polymerelektronik should be started.

Additional Information
- The lab is limited to a number of 6 participants due to capacity reasons. If necessary, a selection procedure will be carried out. Places will be allocated taking into account the students' academic progress. Details will be announced on the lecture website.
- The Lab course will take place in the clean room Facilities of InnovationLab in Heidelberg. Speyerer str. 4, 69115 Heidelberg where the research laboratories of Prof. Hernandez-Sosa are located.
- The 4 th experiment will take place at KIT Campus North, Institute of Nanotechnology, in the research unit and laboratories of Prof. Jasmin Aghassi-Hagmann

Attendance of at least 80% is compulsory during the seminar course. Compulsory attendance is necessary for actively contributing to the discussion of the topics presented by all students.

T 13.192 Module component: Process Analysis: Modeling, Data Mining, Machine Learning [T-ETIT-111214]

Coordinators: Dr.-Ing. Christian Borchert
Prof. Dr.-Ing. Michael Heizmann

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105594 - Process Analysis: Modeling, Data Mining, Machine Learning

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each summer term	1 semesters	2

Courses					
ST 2026	**2302145**	**Process Analysis: Modeling, Data Mining, Machine Learning**	**2 SWS**	**Lecture / 🗣**	**Borchert**

Legend: 💻 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in the form of an oral examination lasting approx. 30 minutes.

Recommendations
Basics in: Mathematics, differential equations, linear algebra, statistics, basic knowledge of Matlab

T 13.193 Module component: Project Management in the Development of Products for Safety-Critical Applications [T-ETIT-109148]

Coordinators: Dr.-Ing. Manfred Nolle
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-104475 - Project Management in the Development of Products for Safety-Critical Applications

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	3

Courses					
WT 26/27	**2311641**	Project Management in the Development of Products for Safety-Critical Applications	**2 SWS**	**/** 🗣	**Nolle**
WT 26/27	**2311643**	Tutorial for 2311641 Project Management in the Development of Products for Safety-Critical Applications	**1 SWS**	**Practice /** 🗣	**Nolle**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination lasting 90 minutes. The module grade is the grade of the written examination.

Prerequisites
none

Recommendations
Basic knowledge of hardware and software design is helpful.

T 13.194 Module component: Pulsed Power Technology and Applications (Lecture) [T-ETIT-111215]

Coordinators: Prof. Dr. Georg Müller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105595 - Pulsed Power Technology and Applications (Lecture)

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	**2307395**	Pulsed Power Technology and Applications	**2 SWS**	**Lecture /** 🗣	**Müller**

Legend: 📱 Online, 🌀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place within the scope of an overall oral examination (approx 20 minutes).

Prerequisites
none

Additional Information
Following the lecture period the oral exam is offered by appointment.

T 13.195 Module component: Pulsed Power Technology and Applications (Tutorial) [T-ETIT-111216]

Coordinators: Prof. Dr. Georg Müller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105596 - Pulsed Power Technology and Applications (Tutorial)

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	5 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	**2307358**	Pulsed Power Technology and Applications	**4 SWS**	**Practice / 🗣**	**Müller**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of other types of examination. It consists of oral examination and discussioning the amount of 30 min and a written report about the results of the experiments conducted (one report per group) The overall impression is evaluated. Details will be given during the lecture.

Prerequisites
none

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-ETIT-111215 - Pulsed Power Technology and Applications (Lecture) must have been passed.

T 13.196 Module component: Quantum Detectors and Sensors [T-ETIT-111234]

Coordinators: Prof. Dr. Sebastian Kempf
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105606 - Quantum Detectors and Sensors

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	**2312706**	Quantum Detectors and Sensors	**3 SWS**	**Lecture /** 🗣	**Kempf**
WT 26/27	**2312707**	Exercise for 2312706 Quantum Detectors and Sensors	**1 SWS**	**Practice /** 🗣	**Zeuner, Kempf**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success takes place in the form of a written examination lasting 120min. The grade corresponds to the result of the written examination.

Prerequisites
None

Recommendations
Successful completion of the module "Superconductivity for Engineers" is recommended.

T 13.197 Module component: Quantum Engineering [T-ETIT-113909]

Coordinators: Prof. Dr. Sebastian Kempf
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106954 - Quantum Engineering

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2312100**	Quantum Engineering	**3 SWS**	**Lecture /** 🗣	**Kempf**
ST 2026	**2312101**	Tutorial to 2312100 Quantum Engineering	**1 SWS**	**Practice /** 🗣	**Kempf, Mitarbeiter*innen**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

The module grade is the grade of the written exam.

Prerequisites
none

T 13.198 Module component: Quantum Machine Learning [T-ETIT-111838]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Prof. Dr.-Ing. Sander Wahls

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105889 - Quantum Machine Learning

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	2302126	Quantum Machine Learning	2 SWS	Lecture /	Gardi

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in form of an oral examination lasting approx. 20 minutes.

Prerequisites
none

T 13.199 Module component: Radar Systems Engineering [T-ETIT-100729]

Coordinators: Prof. Dr.-Ing. Marwan Younis
Prof. Dr.-Ing. Thomas Zwick

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-100420 - Radar Systems Engineering

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	2

Courses					
WT 26/27	**2308454**	Radar Systems Engineering	**3 SWS**	**Lecture /** 🗣	**Zwick, Younis**
WT 26/27	**2308455**	Exercise to 2308454 Radar Systems Engineering	**1 SWS**	**Practice /** 💠	**Zwick, Younis**

Legend: 📱 Online, 💠 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in form of a written overall examination (120 minutes) of the selected course, which in total meets the minimum requirement for LP.

Prerequisites
none

T 13.200 Module component: Radio Frequency Integrated Circuits and Systems [T-ETIT-110358]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105123 - Radio Frequency Integrated Circuits and Systems

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	2

Courses					
ST 2026	**2308419**	Radio Frequency Integrated Circuits and Systems	**2 SWS**	**Lecture /** 🗣	**Ulusoy**
ST 2026	**2308421**	Workshop for 2308421 Radio Frequency Integrated Circuits and Systems	**2 SWS**	**Practice /** 🗣	**Ulusoy, Tsai**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success criteria will be determined by an oral examination (approx. 20-30 min.)

Recommendations
The lecture materials to „Grundlagen der Hochfrequenztechnik“ and „Halbleiterbauelemente“ are recommended.

T 13.201 Module component: Radio Receivers [T-ETIT-106431]

Coordinators: Prof. Dr. Friedrich Jondral
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-103241 - Radio Receivers

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2310531**	Radio Receivers	**2 SWS**	**Lecture /** 🗣	**Jondral**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in the form of an oral examination lasting approx. 20 minutes.

The module grade is the grade of the oral examination.

Prerequisites
none

Recommendations
Knowledge of basic engineering mathematics including integral transformations and probability theory as well as basic knowledge of communications engineering.

T 13.202 Module component: Radio-Frequency Electronics [T-ETIT-113910]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106955 - Radio-Frequency Electronics

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	2308503	Radio-Frequency Electronics	3 SWS	Lecture / 🗣	Ulusoy, Pauli
WT 26/27	2308504	Exercise for 2308503 Radio-Frequency Electronics	1 SWS	Practice / 🏵	Kuo

Legend: 💻 Online, 🏵 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

The module grade is the grade of the written examination.

Prerequisites
none

T 13.203 Module component: Real Time Control of Electrical Drives [T-ETIT-111898]

Coordinators: Dr.-Ing. Andreas Liske
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105916 - Real Time Control of Electrical Drives

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	2306353	Real Time Control of Electrical Drives	3 SWS	Lecture /	Liske
WT 26/27	2306354	Tutorial for 2306353 Real Time Control of Electrical Drives	1 SWS	Practice /	Liske

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in an overall oral examination (approx. 30 minutes).

Prerequisites
none

T 13.204 Module component: Refrigeration B - Foundations of Industrial Gas Processing [T-CIWVT-108914]

Coordinators: Prof. Dr.-Ing. Steffen Grohmann
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-104354 - Refrigeration B - Foundations of Industrial Gas Processing

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2250120**	Refrigeration B	**2 SWS**	**Lecture /** 🗣	**Grohmann**
ST 2026	**2250121**	Exercises on 2250120 Refrigeration B	**1 SWS**	**Practice /** 🗣	**Grohmann, und Mitarbeitende**

Legend: 📱 Online, 🎛 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination is an oral examination with a duration of about 30 minutes (section 4 subsection 2 number 2 SPO).

Prerequisites
None

T 13.205 Module component: Registration for Certificate Issuance - Supplementary Studies on Science, Technology and Society [T-FORUM-113587]

Coordinators: Dr. Christine Mielke
Christine Myglas

Organisation: General Studies. Forum Science and Society (FORUM)

Part of: M-FORUM-106753 - Supplementary Studies on Science, Technology and Society

Type	Credits	Grading	Term offered	Version
Coursework	0 CP	pass/fail	Each term	1

Prerequisites
In order to register, it is mandatory that the basic module and the advanced module have been completed and that the grades for the partial performances in the advanced module are available.

Registration as a partial achievement means the issue of a certificate.

T 13.206 Module component: Renewable Energy-Resources, Technologies and Economics [T-WIWI-100806]

Coordinators: Prof. Dr. Patrick Jochem
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-100500 - Renewable Energy-Resources, Technologies and Economics

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	9

Courses					
WT 26/27	**2581012**	Renewable Energy – Resources, Technologies and Economics	**2 SWS**	**Lecture /** 🗣	**Jochem**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes, in English, answers are possible in German or English) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None.

T 13.207 Module component: Research Seminar Cooperative and Haptic Human-Machine Interactions [T-ETIT-114891]

Coordinators: Dr.-Ing. Balint Varga
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107596 - Research Seminar Cooperative and Haptic Human-Machine Interactions

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2303330**	Research Seminar Cooperative and Haptic Human-Machine Interactions	**3 SWS**	**Seminar / 🗣**	**Varga**

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in the form of another type of examination. It consists of an the preparation of an research paper in IEEE-conference style (4-6 IEEE pages) and an oral presentation (approx. 15 min). The overall impression is rated.

Prerequisites
none

T 13.208 Module component: Robotic Intelligence: Navigation and Mobile Manipulation [T-MACH-114952]

Coordinators: Prof. Dr.-Ing. Arne Rönnau
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-107621 - Robotic Intelligence: Navigation and Mobile Manipulation

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	**2121333**	Robotic Intelligence: Navigation and Mobile Manipulation	**2 SWS**	**Lecture /** 🗣	**Rönnau**
ST 2026	**2121334**	Robotic Intelligence: Navigation and Mobile Manipulation (Exercise)	**1 SWS**	**Practice /** 🗣	**Rönnau**

Legend: 📱 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam: duration 60 minutes. If the written exam is passed, a grade bonus can be achieved by successfully completing the exercises.

Prerequisites
The partial performance "T-MACH-114034 - Robotic Intelligence for Mobile Systems" must not have been started.

Additional Information
The course is offered in English.

Workload
180 hours

T 13.209 Module component: Robotic Intelligence: Perception and Scene Understanding [T-MACH-114813]

Coordinators: Prof. Dr.-Ing. Arne Rönnau
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-107558 - Robotic Intelligence: Perception and Scene Understanding

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each term	1 semesters	1

Courses					
WT 26/27	**2121331**	Robotic Intelligence: Perception and Scene Understanding	**2 SWS**	**Lecture /** 🗣	**Rönnau**
WT 26/27	**2121332**	Robotic Intelligence: Perception and Scene Understanding (Exercise)	**1 SWS**	**Practice /** 🗣	**Rönnau**

Legend: 💻 Online, 🌐 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam: duration 60 minutes. A grade bonus can be achieved by successfully completing the exercises.

Prerequisites
"T-MACH-114034 - Robotic Intelligence for Mobile Systems" must not have started.

Additional Information
The course is offered in English.

Workload
180 hours

T 13.210 Module component: Robotic Winding Technology for Superconducting Wires Lab [T-ETIT-114158]

Coordinators: Prof. Dr. Tabea Arndt
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107135 - Robotic Winding Technology for Superconducting Wires Lab

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	6 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	**2314012**	Robotic Winding Technology for Superconducting Wires Lab	**4 SWS**	**Practical course /** 🗣	**Arndt**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success is assessed by evaluating a written final report (approx. 10-20 pages in length), in which an introduction to the topic, the execution of the experiment, the scientific results and a classification of the results in the overall context should be summarized.

Prerequisites
none

T 13.211 Module component: Robotics II - Humanoid Robotics [T-INFO-114152]

Coordinators: Prof. Dr.-Ing. Tamim Asfour
Organisation: KIT Department of Informatics
Part of: M-INFO-107123 - Robotics II - Humanoid Robotics

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2400074**	Robotics II: Humanoid Robotics	**2 SWS**	**Lecture /** 🗣	**Asfour**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Prerequisites

- M-INFO-100816 - Robotics II - Learning and planning robots Module must not have been started.
- T-INFO-101391 - Anthropomatics: Humanoid RoboticsPartial work must not have been started.

Recommendations
Having visited the lecture on Robotics I – Introduction to Robotics is recommended

T 13.212 Module component: Robotics III - Sensors and Perception in Robotics [T-INFO-114155]

Coordinators: Prof. Dr.-Ing. Tamim Asfour
Organisation: KIT Department of Informatics
Part of: M-INFO-107130 - Robotics III - Sensors and Perception in Robotics

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2400067**	**Robotics III: Sensors and Perception in Robotics**	**2 SWS**	**Lecture / 🗣**	**Asfour, Triebel**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Prerequisites
none.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-109931 - Robotics III - Sensors and Perception in Robotics must not have been started.

Recommendations
Having visited the lectures on Robotics I – Introduction to Robotics is recommended.

T 13.213 Module component: Robotics, Automation, and Systems Control Lab [T-ETIT-114740]

Coordinators: Prof. Dr.-Ing. Mike Barth
Prof. Dr.-Ing. Sören Hohmann

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-107524 - Robotics, Automation, and Systems Control Lab

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each summer term	2

Courses					
ST 2026	**2303195**	**Robotics, Automation, and Systems Control Lab**	**4 SWS**	**Practical course /** 🗣	**Breisacher, Wagemann, Witucki**

Legend: 🖥 Online, 🌍 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment

The examination takes place in the form of other types of examination. It consists of pitches after every thematic block, a final presentation and an oral examination in the amount of 20 minutes. The overall impression is evaluated.

The module grade results of the assessment of the pitches, presentation and the oral exam. Details will be given during the lecture at the beginning of the semester.

Prerequisites

none

Recommendations

Knowledge in automation, control and robotics in general. Recommended, but not mandatory, lectures are:

- Practical Tools for Control Engineers, PTCE – M-ETIT-106780
- Cyber Physical Modeling – M-ETIT-106953
- Seminar Industrial Process and Plant Engineering – M-ETIT-106970
- Optimal Control – M-ETIT-107497
- Multivariable Control Systems – M-ETIT-107515
- Nonlinear Control Systems - in preparation

Additional Information

Due to capacity reasons there are only 25 places available. If necessary, a selection procedure will be carried out. Selection criteria are the progress in studies and in the recommended lectures. The application window is from March 1 for March 31 in Campus Plus (https://plus.campus.kit.edu/). For further information and current news, regularly visit the courses website (https://www.irs.kit.edu/Lectures_5404.php).

T 13.214 Module component: Safe Human-Robot-Collaboration [T-INFO-109911]

Coordinators: Dr.-Ing. Johannes Kurth
Organisation: KIT Department of Informatics
Part of: M-INFO-104877 - Safe Human-Robot-Collaboration

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2400236**	Safe Human-Robot-Collaboration	**2 SWS**	**Block /** 🗣	**Kurth**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 13.215 Module component: Satellite Communications [T-ETIT-110672]

Coordinators: Prof. Dr.-Ing. Laurent Schmalen
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105272 - Satellite Communications

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each summer term	2 semesters	1

Courses					
ST 2026	**2310566**	**Satellite Communications**	**2 SWS**	**Lecture /**	**Lázaro Blasco, Clazzer**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The assessment will be carried out as part of an overall oral exam (about 20 minutes).

Prerequisites
see module

Recommendations
see module

T 13.216 Module component: Scanning Probe Lithography for Biomedical Applications [T-ETIT-115009]

Coordinators: Dr. Dr. Michael Hirtz
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107661 - Scanning Probe Lithography for Biomedical Applications

Type	Credits	Grading	Version
Oral examination	3 CP	graded	1

Courses					
ST 2026	**2308481**	Scanning Probe Lithography for Biomedical Applications	**2 SWS**	**Block /**	**Aghassi-Hagmann, Hirtz**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The examination takes place within the framework of an oral overall examination (approx. 20 minutes).

Prerequisites
none

T 13.217 Module component: Scientific Computing for Engineers [T-MACH-100532]

Coordinators: Prof. Dr. Peter Gumbsch
Dr. Daniel Weygand

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-105369 - Scientific Computing for Engineers

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	3

Courses					
WT 26/27	**2181738**	Scientific computing for Engineers	**2 SWS**	**Lecture /** 🗣	**Weygand, Gumbsch**
WT 26/27	**2181739**	Exercises for Scientific Computing for Engineers	**2 SWS**	**Practice /** 🗣	**Weygand**

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam (90 minutes)

Prerequisites
The module components can not be combined with "Application of advanced programming languages in mechanical engineering" (T-MACH-105390).

Additional Information
The course is offered in German.

Workload
120 hours

T 13.218 Module component: Seamless Engineering [T-MACH-111401]

Coordinators: Prof. Dr.-Ing. Kai Furmans
Prof. Dr.-Ing. Eric Sax

Organisation: KIT Department of Electrical Engineering and Information Technology

KIT Department of Mechanical Engineering

Part of: M-MACH-105725 - Seamless Engineering

Type	Credits	Grading	Term offered	Version
Examination of another type	9 CP	graded	Each winter term	2

Courses					
WT 26/27	**2117072**	**Seamless Engineering - Logistics Robotics Workshop**	**2 SWS**	**Lecture / Practice /** 🗣	**Furmans, Sax**

Legend: 📱 Online, 🛠 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Competence certificate in the form of an examination of another type.

The overall grade is composed as follows:

- assessment of an examination as individual performance as the conclusion of the lecture block.
- assessment of colloquia as individual performance on defined milestones during the project work

The overall impression is evaluated. Failing the final examination or the colloquia does not result in failing the course.

Prerequisites
None

Recommendations
None

Additional Information
The course consists of two components. Theoretical knowledge and basics about structured system design are taught in lecture and exercise. In parallel, a practical part takes place throughout the semester. In this, the students design and implement a mechatronic system in small groups using industry-related hardware and software to deal with a given task in the logistics environment.

The course is offered in English.

Workload
270 hours

T 13.219 Module component: Self-booking-HOC-SPZ-FORUM-STK-graded [T-ETIT-112621]

Organisation:
KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each term	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a graded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum", the "Studienkolleg" or the Center for Applied Cultural Studies and Studium Generale.

Title and credits of the achievement are adopted.

T 13.220 Module component: Self-booking-HOC-SPZ-FORUM-STK-graded [T-ETIT-112619]

Organisation:
KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each term	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a graded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum", the "Studienkolleg" or the Center for Applied Cultural Studies and Studium Generale.

Title and credits of the achievement are adopted.

T 13.221 Module component: Self-booking-HOC-SPZ-FORUM-STK-graded [T-ETIT-112620]

Organisation:
KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	6 CP	graded	Each term	1 semesters	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a graded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum", the "Studienkolleg" or the Center for Applied Cultural Studies and Studium Generale.

Title and credits of the achievement are adopted.

T 13.222 Module component: Self-booking-HOC-SPZ-FORUM-STK-ungraded [T-ETIT-112618]

Organisation:
KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	6 CP	pass/fail	Each term	1 semesters	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a graded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum", the "Studienkolleg" or the Center for Applied Cultural Studies and Studium Generale.

Title and credits of the achievement are adopted.

T 13.223 Module component: Self-booking-HOC-SPZ-FORUM-STK-ungraded [T-ETIT-112616]

Organisation:
KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Coursework	6 CP	pass/fail	Each term	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a graded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum", the "Studienkolleg" or the Center for Applied Cultural Studies and Studium Generale.

Title and credits of the achievement are adopted.

T 13.224 Module component: Self-booking-HOC-SPZ-FORUM-STK-ungraded [T-ETIT-112617]

Organisation:
KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	6 CP	pass/fail	Each term	1 semesters	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a graded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum", the "Studienkolleg" or the Center for Applied Cultural Studies and Studium Generale.

Title and credits of the achievement are adopted.

T 13.225 Module component: Semiconductor Quantum Dots for Quantum Photonics [T-ETIT-115306]

Coordinators: TT-Prof. Dr. Tobias Huber-Loyola
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107886 - Semiconductor Quantum Dots for Quantum Photonics

Type	Credits	Grading	Version
Oral examination	6 CP	graded	1

Courses					
WT 26/27	**2309600**	Lecture Semiconductor Quantum Dots for Quantum Photonics	**3 SWS**	**Lecture /** 🗣	**Huber-Loyola, Barth**
WT 26/27	**2309601**	Exercise Semiconductor Quantum Dots for Quantum Photonics	**1 SWS**	**Practice /** 🗣	**Huber-Loyola**

Legend: 💻 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place within the framework of an oral overall examination (approx. 30 minutes).

Prerequisites
none

T 13.226 Module component: Seminar Advanced Concepts for Flexible and Soft Optoelectronic Devices and Sensors [T-ETIT-113427]

Coordinators: Prof. Dr. Gerardo Hernandez Sosa
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106674 - Seminar Advanced Concepts for Flexible and Soft Optoelectronic Devices and Sensors

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2313766**	Seminar Advanced Concepts for Flexible and Soft Optoelectronic Devices and Sensors	**2 SWS**	**Seminar / 🗣**	**Hernandez Sosa**

Legend: 💻 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination consists of a written journal article and an oral presentation of the student's work, both given in English. The overall impression is rated.

The module grade results of the assessment of the written paper and the oral presentation. Details will be given during the lecture.

Prerequisites
none

T 13.227 Module component: Seminar Battery II [T-ETIT-110801]

Coordinators: Dr.-Ing. Andre Weber
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105321 - Seminar Battery II

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	3 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	**2304226**	Seminar Batteries	**2 SWS**	**Seminar / 🗣**	**Weber**
WT 26/27	**2304226**	Seminar Batteries	**2 SWS**	**Seminar / 🗣**	**Weber**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in form of an examination of another type. It consists of a written paper (20-40 pages) and an oral presentation (approx. 20 min.) of the students work. The overall impression is rated.

Prerequisites
none

T 13.228 Module component: Seminar Creating a Patent Specification [T-ETIT-100754]

Coordinators: Prof. Dr. Wilhelm Stork
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Coursework	3 CP	pass/fail	Each summer term	1

Courses					
ST 2026	**2311633**	**Seminar Creating a Patent Specification**	**2 SWS**	**Seminar / 🗣**	**Schmid, Stork**

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success controll take place in form of a submission of a written paper of a fictional patent specification. The seminar is ungraded and students are considered to have passed upon successful evaluation of their written paper.

Prerequisites
none

Recommendations
Students are expected to have a technical understanding equivalent to approximately the fifth semester.

Additional Information

- The seminar has a limited number of participants
- The selection process begins after the first lecture
- Places will be allocated based on academic progress and degree program. Students majoring in Electrical Engineering and Information Technology, as well as those in master's programs, will be given priority for admission.

For detailed information on course content, qualification goals, and workload, see:
M-ETIT-100458 – Seminar Wir machen ein Patent

T 13.229 Module component: Seminar Electrocatalysis [T-ETIT-111256]

Coordinators: Prof. Dr.-Ing. Ulrike Krewer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105629 - Seminar Electrocatalysis

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	3 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	**2304302**	Seminar Electrocatalysis	**2 SWS**	**Seminar / 🗣**	**Röse**
WT 26/27	**2304238**	Seminar Electrocatalysis	**2 SWS**	**Seminar / 🗣**	**Röse**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The performance assessment is an "examination of another type". It consists of a written elaboration of the scientific topic (max. 25 pages) and a presentation followed by a discussion of 15 minutes each. The overall impression is assessed.

Prerequisites
none

T 13.230 Module component: Seminar Embedded Systems [T-ETIT-115156]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Prof. Dr.-Ing. Eric Sax
Prof. Dr. Wilhelm Stork

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-107764 - Seminar Embedded Systems

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each term	1

Courses					
ST 2026	**2311627**	Seminar Embedded Systems	**2 SWS**	**Seminar /**	**Becker, Sax, Stork**
WT 26/27	**2311627**	Seminar Embedded Systems	**2 SWS**	**Seminar /**	**Becker, Sax, Stork**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in the form of other types of examination. The examination consists of:

- Written report (approx. 6 pages)
- 2 peer reviews (approx. 1 page each)
- Presentation (approx. 15 minutes) followed by discussion (approx. 10 minutes)

The overall impression is rated.

Prerequisites
none

T 13.231 Module component: Seminar Fuel Cell II [T-ETIT-110799]

Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105322 - Seminar Fuel Cell II

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	3 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	**2304227**	Seminar Fuel Cells	**2 SWS**	**Seminar / 🗣**	**Weber**
WT 26/27	**2304227**	Seminar Fuel Cells	**2 SWS**	**Seminar / 🗣**	**Weber**

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in form of an examination of another type. It consists of a written paper (20-40 pages) and an oral presentation (approx. 20 min.) of the students work. The overall impression is rated.

Prerequisites
none

T 13.232 Module component: Seminar Industrial Process and Plant Engineering [T-ETIT-113932]

Coordinators: Prof. Dr.-Ing. Mike Barth
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106970 - Seminar Industrial Process and Plant Engineering

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each summer term	1

Courses					
ST 2026	**2300009**	Seminar Industrial Process and Plant Engineering	**3 SWS**	**Seminar / 🗣**	**Barth, Jilg**
ST 2026	**2303650**	Seminar Industrial Process and Plant Engineering	**3 SWS**	**Seminar / 🗣**	**Barth, Jilg**

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination will be the seminar presentation at the end of the semester. The criteria are:

- Live presentation of the created CAD and simulation models
- Poster design and usage within the presentation
- Answering the questions from the examiners
- Structure of the talk

Prerequisites
none

T 13.233 Module component: Seminar New Power Electronic Systems and Technologies [T-ETIT-114862]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107588 - Seminar New Power Electronic Systems and Technologies

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each term	1

Courses					
ST 2026	**2306317**	Seminar New Power Electronic Systems and Technologies	**3 SWS**	**Seminar / 🗣**	**Hiller**
WT 26/27	**2306317**	Seminar New Power Electronic Systems and Technologies	**3 SWS**	**Seminar / 🗣**	**Hiller**

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in the form of an examination of another type. It consists of a 15-minute final presentation followed by a discussion and a 2- to 3-page written summary paper. The overall impression is evaluated.

The following aspects are assessed:

Presentation:

- Quality of the presentation (form and content)
- Presentation style (structure, style, content)
- Behavior during the discussion

Paper with a summary of the main content:

- Format, spelling, linguistic style (academic/factual)
- Content, (graphical) presentation of the researched findings
- Relevance and quality of the references used, citation style

Prerequisites
none

Additional Information
The number of participants in the seminar may be limited for capacity reasons and depends on the number of seminar topics offered.

Selection may be based on degree program, academic progress, waiting time, and topic assignment.

Presentations and summaries may be given/written in English or German.

T 13.234 Module component: Seminar Novel Concepts for Solar Energy Harvesting [T-ETIT-108344]

Coordinators: Prof. Dr. Bryce Sydney Richards
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-103447 - Seminar Novel Concepts for Solar Energy Harvesting

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each summer term	2

Courses					
ST 2026	**2313761**	Seminar Novel Concepts for Solar Energy Harvesting	**2 SWS**	**Seminar / 🗣**	**Paetzold**

Legend: 🖥 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination consists of two components: a written journal article and an oral presentation on a research topic selected by the student under the lecturer's guidance. Topics are agreed upon during the first weeks of the semester. Both the lecture and the presentation are conducted in English. The preparation of the manuscript is completed as homework.

The journal article must be at least 3 pages long and adhere to the template provided by the lecturer. The oral presentation should last approximately 25 minutes, followed by a 20-minute discussion.

The overall impression is rated.

Prerequisites
none

Recommendations
Good knowledge of semiconductor components/optoelectronics is desirable.

T 13.235 Module component: Seminar on Applied Superconductivity [T-ETIT-111243]

Coordinators: Prof. Dr. Tabea Arndt
Prof. Dr. Bernhard Holzapfel
Prof. Dr. Sebastian Kempf
Prof. Dr.-Ing. Mathias Noe

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105615 - Seminar on Applied Superconductivity

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	3 CP	graded	Each summer term	1 semesters	2

Courses					
ST 2026	**2310542**	Seminar on Applied Superconductivity	**2 SWS**	**Seminar / 🗣**	**Kempf, Arndt, Holzapfel, Noe**

Legend: 💻 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Elaboration of a scientific topic and presentation of a talk on the topic within the seminar of about 30min.

Prerequisites
none

T 13.236 Module component: Seminar Project Management for Engineers [T-ETIT-108820]

Coordinators: Prof. Dr.-Ing. Mathias Noe
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Coursework (oral)	3 CP	pass/fail	Each summer term	2

Courses					
ST 2026	**2312684**	Project Management for Engineers	**2 SWS**	**Seminar / 🗣**	**Noe**

Legend: 💻 Online, 🌐 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success controll takes place in form of an oral comprehensive exam (approx. 30 minutes).

Prerequisites
none

Additional Information
For detailed information on course content, qualification goals, and workload, see:
M-ETIT-104285 – Seminar Projektmanagement für Ingenieure

T 13.237 Module component: Seminar Radiation Protection in Medicine [T-ETIT-115253]

Coordinators: PD Dr. Bastian Breustedt
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107839 - Seminar Radiation Protection in Medicine

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2305290**	Seminar Radiation Protection in Medicine	**2 SWS**	**Seminar / 🗣**	**Breustedt**

Legend: 💻 Online, 🌐 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination consists of a written handout (approx. 2-3 pages) and an oral presentation (approx. 25-35 min.) of the students work. Considered will be the quality of the contents, the style of the presentation, the quality and content of the handout. The overall impression is rated.

Prerequisites
none

T 13.238 Module component: Seminar Selected Topics in Communications [T-ETIT-100962]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100441 - Seminar Selected Topics in Communication

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each term	1

Courses					
ST 2026	**2310512**	Seminar Selected Topics in Communications	**3 SWS**	**Seminar /**	**Schmalen, Jäkel**
WT 26/27	**2310512**	Seminar Selected Topics in Communications	**3 SWS**	**Seminar /**	**Jäkel**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The success controll consists of a written paper (approx. 5 pages in IEEE format) and a presentation (approx. 25 minutes). The overall impression is rated.

Prerequisites
none

T 13.239 Module component: Seminar: Digital Accessibility and Assistive Technologies [T-INFO-111832]

Coordinators: Prof. Dr.-Ing. Rainer Stiefelhagen
Organisation: KIT Department of Informatics
Part of: M-INFO-105884 - Seminar: Digital Accessibility and Assistive Technologies

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2400129**	Seminar Digital Accessibility and Assisitive Technologies	**2 SWS**	**Seminar / 🗣**	**Anken, Stiefelhagen**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-104742 - Seminar Accessibility - Assistive Technologies for Visually Impaired Persons must not have been started.

T 13.240 Module component: Sensors [T-ETIT-101911]

Coordinators: TT-Prof. Dr. Johanna Schröder
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100378 - Sensors

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	2

Courses					
ST 2026	**2304231**	Sensors	**2 SWS**	**Lecture /**	**Schröder**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success is assessed in the form of a written examination lasting 2 hours.

Prerequisites
none

Recommendations
Basic knowledge of materials science (e.g. lecture "Passive components") is helpful.

T 13.241 Module component: Signal Processing for Communication Receivers [T-ETIT-115288]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
KIT Department of Mechanical Engineering
Part of: M-ETIT-107871 - Signal Processing for Communication Receivers

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	1

Assessment
Exam will be held as 25 mins oral. Beforehand, students are elaborating small problems within a 15 mins. period of preparation.

Prerequisites
none

T 13.242 Module component: Signal Processing in Communications [T-ETIT-100747]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100443 - Signal Processing in Communications

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	3

Courses					
ST 2026	**2310534**	Signal Processing in Communications	**3 SWS**	**Lecture /**	**Jäkel**
ST 2026	**2310535**	Practice to 2310534 Signal Processing in Communications	**1 SWS**	**Practice /**	**Jäkel**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in the form of a written examination lasting 120 min.

The module grade is the grade of the written exam.

Prerequisites
none

T 13.243 Module component: Signal Processing Lab [T-ETIT-113369]

Coordinators: Prof. Dr.-Ing. Sander Wahls
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106633 - Signal Processing Lab

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2302134**	Signal Processing Lab	**4 SWS**	**Practical course /**	**Wahls, van Wijk**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success is assessed in the form of a written examination lasting 120 minutes.

Prerequisites
none

Recommendations
Knowledge of the contents of the modules "Signals and Systems", "Measurement Technology" and "Methods of Signal Processing" is strongly recommended.

Additional Information
A prerequisite for admission to the examination is the submission of protocols of all experiments. The quality of the protocols will be assessed; they must be acceptable for admission to the examination.

Compulsory attendance is necessary both for carrying out the work in the team on site and for the practical teaching of skills and hardware experience that cannot be learned in pure self-study.

T 13.244 Module component: Signal Processing Methods [T-ETIT-113837]

Coordinators: Prof. Dr.-Ing. Sander Wahls
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106899 - Signal Processing Methods

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2302113**	Signal Processing Methods	**2 SWS**	**Lecture /** 🗣	**Wahls**
WT 26/27	**2302115**	Tutorial to 2302113 Signal Processing Methods	**2 SWS**	**Practice /** 🗣	**Wahls, Al-Hammadi**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam, 120 minutes.

Prerequisites
none

Recommendations
Familiarity with signals and systems (in particular, Fourier transforms) and probability theory at the Bachelor level is assumed.

T 13.245 Module component: Signal Processing with Nonlinear Fourier Transforms and Koopman Operators [T-ETIT-113428]

Coordinators: Prof. Dr.-Ing. Sander Wahls
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106675 - Signal Processing with Nonlinear Fourier Transforms and Koopman Operators

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	**2302135**	Signal Processing with Nonlinear Fourier Transforms and Koopman Operators	**2 SWS**	**Lecture /** 🗣	**Wahls**
ST 2026	**2302136**	Practice to 2302135 Signal Processing with Nonlinear Fourier Transforms and Koopman Operators	**2 SWS**	**Practice /** 🗣	**Wahls, Liang**

Legend: 🖥 Online, 💐 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination in this module consists of programming assessments and a graded written examination of 120 minutes.

The programming assignments are either pass or fail. They must be passed during the lecture period for admission to the written examination.

The module grade is the grade of the written exam.

Prerequisites
none

T 13.246 Module component: SIL Entrepreneurship Project [T-WIWI-110166]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-ETIT-105073 - Student Innovation Lab

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	1

Assessment
Success is assessed in the form of an alternative exam assessment. The grade results from the evaluation of two seminar papers (each approx. 20 pages). Further details will be provided at the beginning of the course. The points system for the assessment of the two seminar papers is determined by the lecturer of the course. It will be announced at the beginning of the course.

Prerequisites
None

Recommendations
None

Workload
90 hours

T 13.247 Module component: Single-Crystal Growth – Crystal Growth Methods and Applications of Crystals for Electronic and Optical Components [T-ETIT-113293]

Coordinators: Prof. Dr. Marc Eichhorn
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106597 - Single-Crystal Growth – Crystal Growth Methods and Applications of Crystals for Electronic and Optical Components

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	**2303182**	Single-Crystal Growth – Crystal Growth Methods and Applications of Crystals for Electronic and Optical Components	**2 SWS**	**Lecture / ×**	**Ackermann**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in form of an overall oral examination (approx. 20 minutes).
The module grade is the grade of the oral examination.

Prerequisites
none

T 13.248 Module component: Single-Photon Detectors [T-ETIT-108390]

Coordinators: Dr. Konstantin Ilin
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101971 - Single-Photon Detectors

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2312680**	Single-Photon Detectors	**2 SWS**	**Lecture /** 🗣	**Ilin**
WT 26/27	**2312694**	Tutorial for 2312680 Single-Photon Detectors	**1 SWS**	**Practice /** 🗣	**Ilin**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success controll takes place in form of an overall oral examination lasting approx. 20 minutes.

Prerequisites
none

T 13.249 Module component: Software Engineering [T-ETIT-108347]

Coordinators: Dr. Clemens Reichmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100450 - Software Engineering

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	4

Courses					
ST 2026	**2311611**	Software Engineering	**2 SWS**	**Lecture /** 🗣	**Reichmann**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in the form of an oral examination lasting approx. 25 minutes.

The module grade is the grade of the oral examination.

Prerequisites
none

T 13.250 Module component: Solar Energy [T-ETIT-100774]

Coordinators: Prof. Dr. Ulrich Wilhelm Paetzold
Prof. Dr. Bryce Sydney Richards

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-100524 - Solar Energy

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	2313745	Solar Energy	3 SWS	Lecture / 🗣	Richards, Paetzold
WT 26/27	2313750	Tutorial 2313745 Solar Energy	1 SWS	Practice / 🗣	Richards, Paetzold

Legend: 💻 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment

Type of Examination: written exam

Duration of Examination: 120 Minutes

Modality of Exam: One written exam at the end of each semester.

Prerequisites

- Knowledge of optoelectronics is a prerequisite, e.g. M-ETIT-107146 – Bauelemente der Opto- und Nanoelektronik.
- Students are not allowed to take „T-ETIT-101939 - Photovoltaik" in addition to this one.

Modeled Prerequisites

The following conditions have to be fulfilled:

1. The module component T-ETIT-101939 - Photovoltaics must not have been started.

T 13.251 Module component: Solar Thermal Energy Systems [T-MACH-106493]

Coordinators: apl. Prof. Dr. Ron Dagan
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101924 - Solar Thermal Energy Systems

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	6

Courses					
WT 26/27	**2189400**	Solar Thermal Energy Systems	**2 SWS**	**Lecture /** 🗣	**Dagan**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam of about 30 minutes

Prerequisites
none

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-105225 - Thermal Solar Energy must not have been started.

Recommendations
Literature

1. "Solar Engineering of Thermal Processes", 4th Edition, J. Duffie &W. Beckman. Published by Wiley & Sons
2. “Heat Transfer”, 10th Edition, J. P. Holman Mc. Graw Hill publisher
3. “Fundamentals of classical Thermodynamics”, G. Van Wylen & R. E. Sonntag. Published by Wiley &Sons

Workload
120 hours

T 13.252 Module component: Space-Born Microwave Radiometry-Advanced Methods and Applications [T-ETIT-100810]

Coordinators: Prof. Dr.-Ing. Thomas Zwick
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100545 - Space-Born Microwave Radiometry - Advanced Methods and Applications

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2308448**	Space-borne Microwave Radiometry - Advanced Methods and Applications	**2 SWS**	**Lecture /** 🗣	**Jirousek**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The success controll takes place in form of an oral exam (approx. 20 minutes).

Prerequisites
none

Recommendations
Knowledge of the fundamentals of high-frequency technology is helpful.

T 13.253 Module component: Spaceborne Radar Remote Sensing - Exam [T-ETIT-112857]

Coordinators: Prof. Dr.-Ing. Alberto Moreira
Dr. Pau Prats

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-103042 - Spaceborne Radar Remote Sensing

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	**2308428**	Spaceborne Radar Remote Sensing	**2 SWS**	**Lecture /** 🗣	**Prats-Iraola, Moreira**
ST 2026	**2308429**	Tutorial Spaceborne Radar Remote Sensing	**1 SWS**	**Tutorial /** 🗣	**Younis**

Legend: 🖥 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment takes place in the form of a written examination lasting 120 min.

Prerequisites
"T-ETIT-106056 - Spaceborne Radar Remote Sensing" is not allowed to be started or to be completed.

Recommendations
Signal processing and radar fundamentals.

Additional Information
Further information can be found at the internet page of the IHE (https://s.kit.edu/ihe-srrs).

T 13.254 Module component: Spaceborne Radar Remote Sensing - Workshop [T-ETIT-112858]

Coordinators: Prof. Dr.-Ing. Marwan Younis
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-103042 - Spaceborne Radar Remote Sensing

Type	Credits	Grading	Term offered	Version
Examination of another type	2 CP	graded	Each summer term	1

Courses					
ST 2026	**2308427**	Spaceborne Radar Remote Sensing	**1 SWS**	/ 🗣	**Younis, Prats-Iraola**

Legend: 💻 Online, 🌀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment takes place in the form of reports (other types of examination). Those reports have to be submitted as part of the SAR computer workshop (approx. a total of five workshops). Details will be given during the lecture.

Prerequisites
"T-ETIT-106056 - Spaceborne Radar Remote Sensing" is not allowed to be started or to be completed.

Recommendations
Signal processing and radar fundamentals.

Additional Information
Further information can be found at the internet page of the IHE (https://s.kit.edu/ihe-srrs).

T 13.255 Module component: Stochastic Information Processing [T-INFO-115261]

Coordinators: Prof. Dr.-Ing. Uwe Hanebeck
Organisation: KIT Department of Informatics
Part of: M-INFO-107847 - Stochastic Information Processing

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	1

Assessment
Success is assessed in the form of an oral examination, usually lasting 15 - 25 minutes in accordance with Section 4 (2) No. 2 SPO.

Prerequisites
None.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-101366 - Stochastic Information Processing must not have been started.

Recommendations
Knowledge of the basics of probability theory is helpful.

Additional Information
As a theoretical foundation lecture, "Stochastic Information Processing" is an ideal introduction to the ISAS lectures. Conversely, prior knowledge of "Localization of Mobile Agents" (LMA) [course no. 24613] may nevertheless be helpful depending on the type of learner - more concrete applications are highlighted there. In all of our lectures, the content is always derived and explained in detail from the beginning; it is therefore possible to start with SI or LMA.

T 13.256 Module component: Strategy Derivation for Engineers [T-ETIT-111369]

Coordinators: Prof. Dr. Tabea Arndt
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105803 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Coursework (oral)	3 CP	pass/fail	Each winter term	2

Courses					
WT 26/27	**2314010**	Strategy Derivation for Engineers	**2 SWS**	**Seminar /**	**Arndt**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success is assessed in an overall oral examination (approx. 30 minutes).
The course is ungraded. The module is passed with successful assessment of the coursework.

Prerequisites
none

Additional Information
Regular attendance at the course dates is a prerequisite for admission to the examination.
The examination is expected to take place at Campus North.
Detailed information on content, qualification objectives and workload can be found at:
M-ETIT-105697 – Seminar Strategieableitung für Ingenieure

T 13.257 Module component: Superconducting Magnet Technology [T-ETIT-113440]

Coordinators: Prof. Dr. Tabea Arndt
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106684 - Superconducting Magnet Technology

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	**2312698**	Superconducting Magnet Technology	**3 SWS**	**Lecture / Practice /**	**Arndt**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The examination takes place in form of an oral exam (abt. 30 minutes).
Two timeslots (weeks) for examination dates will be announced (usually near end of lecture period & end of semester).
The module grade is the grade of the oral exam.

Prerequisites
none

T 13.258 Module component: Superconducting Materials Lab [T-ETIT-114730]

Coordinators: Dr. Jens Hänisch
Prof. Dr. Bernhard Holzapfel

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105521 - Superconducting Materials

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2312696**	Superconducting Materials Part II	**2 SWS**	**Practical course /** 🗣	**Holzapfel, Hänisch**

Legend: 📱 Online, 🏵 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in the form of other types of examination. It consists of a written protocol (approx. 40 pages).

Prerequisites
none

Recommendations
It is recommanded to pass the lecture (part I) before the lab (part II).

T 13.259 Module component: Superconducting Materials Lecture [T-ETIT-114729]

Coordinators: Dr. Jens Hänisch
Prof. Dr. Bernhard Holzapfel

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105521 - Superconducting Materials

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2312717**	Superconducting Materials Part I	**2 SWS**	**Lecture /** 🗣	**Holzapfel, Hänisch**

Legend: 📱 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The Success control takes place in the form of an oral examination lasting approx. 30 minutes.

Prerequisites
none

Recommendations
It is recommanded to pass the lecture (part I) before the lab (part II).

T 13.260 Module component: Superconducting Nanowire Detectors [T-ETIT-111236]

Coordinators: Dr. Konstantin Ilin
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105609 - Superconducting Nanowire Detectors

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each summer term	1 semesters	2

Courses					
ST 2026	**2312671**	Superconducting Nanowire Detectors	**2 SWS**	**Lecture /** 🗣	**Ilin**
ST 2026	**2312673**	Practice to 2312671 Superconducting Nanowire Detectors	**1 SWS**	**Practice /** 🗣	**Ilin**

Legend: 💻 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral Exam (approx. 20 min.)

Prerequisites
none

T 13.261 Module component: Superconducting Power Systems [T-ETIT-113439]

Coordinators: Prof. Dr.-Ing. Mathias Noe
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106683 - Superconducting Power Systems

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	3

Courses					
WT 26/27	**2314011**	Superconducting Power Systems	**3 SWS**	**Lecture / Practice /**	**Noe**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The examination takes place in form of an oral exam (abt. 45 minutes).
The module grade is the grade of the oral exam.

Prerequisites
none

T 13.262 Module component: Superconductivity for Engineers [T-ETIT-111239]

Coordinators: Prof. Dr. Bernhard Holzapfel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105611 - Superconductivity for Engineers

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each winter term	1 semesters	4

Courses					
WT 26/27	**2312708**	Superconductivity for Engineers	**3 SWS**	**Lecture /** 🗣	**Kempf, Hänisch**
WT 26/27	**2312709**	Exercise for 2312708 Superconductivity for Engineers	**1 SWS**	**Practice /** 🗣	**Hänisch, Arldt**

Legend: 💻 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success takes place in the form of a written examination lasting 120min.
The module grade is the grade of the written examination.

Prerequisites
none

T 13.263 Module component: System Integration and Communication Structures in Industry 4.0 and IoT [T-ETIT-112212]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106026 - System Integration and Communication Structures in Industry 4.0 and IoT

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each winter term	1 semesters	1

Courses					
WT 26/27	**2311614**	System Integration and Communication Structures in Industry 4.0 and IoT	**2 SWS**	**Lecture /** 🗣	**Babel**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in form of an overall oral examination (approx. 20 minutes). The module grade is the grade of the oral examination.

Prerequisites
none

T 13.264 Module component: System-on-Chip Lab [T-ETIT-115155]

Coordinators: Prof. Dr.-Ing. Jürgen Becker
Prof. Dr. Ivan Peric

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-107763 - System-on-Chip Lab

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2311612**	System-on-Chip Lab	**4 SWS**	**Practical course /** 🗣	**Becker, Peric**

Legend: 📱 Online, 💮 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in the form of other types of examination. The examination consists of:

- Preparation sheets for the introductory lab material (approx. 6 pages each) (7 submissions per group)
- GitLab project involving the implementation and documentation of project work on a course-relevant topic (1 project per group, 6 weeks processing time)
- Presentation on the project work (approx. 15 minutes per group, followed by a 5-minute Q&A session)
- Oral exam on the lab and project content (15 minutes per student)

The overall impression is rated.

Prerequisites
none

T 13.265 Module component: Systems and Software Engineering [T-ETIT-100675]

Coordinators: Prof. Dr.-Ing. Eric Sax
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100537 - Systems and Software Engineering

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	3

Courses					
WT 26/27	**2311605**	Systems and Software Engineering	**2 SWS**	**Lecture /** 🗣	**Sax**
WT 26/27	**2311607**	Tutoral for 2311605 Systems and Software Engineering	**2 SWS**	**Practice /** 💠	**Stroop, Hagel, Gerdes**

Legend: 📱 Online, 💠 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam of 90 minutes.

Students are given the opportunity to earn a grade bonus through separate task assignments. If the grade of the written exam is between 4.0 and 1.3, the bonus improves the grade by a maximum of one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the lecture. Bonus points do not expire and remain valid for exams taken at a later date:

The grade is determined by the written exam and the bonus points.

Prerequisites
none

T 13.266 Module component: Systems Engineering for Automotive Electronics [T-ETIT-100677]

Coordinators: Prof. Dr.-Ing. Jürgen Bortolazzi
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100462 - Systems Engineering for Automotive Electronics

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	**2311642**	Systems Engineering for Automotive Electronics	**2 SWS**	**Lecture /**	**Bortolazzi**
ST 2026	**2311644**	Tutorial for 2311642 Systems Engineering for Automotive Electronics	**1 SWS**	**Practice /**	**Beck**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Prerequisites
none

T 13.267 Module component: Team Project: Sensors and Electronics [T-ETIT-111007]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Prof. Dr.-Ing. Thomas Zwick

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105465 - Team Project: Sensors and Electronics

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	3 CP	pass/fail	Each term	1 semesters	2

Courses					
WT 26/27	**2308425**	Team Project: Sensors and Electronics	**2 SWS**	**Project /**	**Ulusoy, Zwick**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Final design report (approx. 8-10 pages)

Prerequisites
none

T 13.268 Module component: Technical Acoustic [T-ETIT-104579]

Coordinators: Dr. Nicole Ruiter
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101835 - Technical Acoustic

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	**2305296**	**Technical Acoustic**	**2 SWS**	**Lecture /**	**Ruiter**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control is carried out as part of an overall oral examination (approx. 20 minutes) of the selected courses, with which the minimum requirement for CP is met.

Prerequisites
none

Additional Information
There is a course with the same name and module component at the KIT Department of Mechanical Engineering (LV 2158107, T-MACH-111382, every summer semester). Please check which course (and associated module component) is offered in your degree program.

T 13.269 Module component: Technical Optics [T-ETIT-100804]

Coordinators: Prof. Dr. Cornelius Neumann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100538 - Technical Optics

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	1

Courses					
WT 26/27	**2313720**	Technical Optics	**2 SWS**	**Lecture /** 🗣	**Neumann**
WT 26/27	**2313722**	Technical Optics (Tutorial to 2313720)	**1 SWS**	**Practice /** 🗣	**Neumann**

Legend: 💻 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination lasting 120 minutes. The module grade is the grade of the written examination.

Prerequisites
none

Recommendations
Previous attendance of the lecture on lighting technology.

T 13.270 Module component: Telematics [T-INFO-114269]

Coordinators: Prof. Dr. Martina Zitterbart
Organisation: KIT Department of Informatics
Part of: M-INFO-107243 - Telematics

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2400007**	Telematics	**4 SWS**	**Lecture /** 🗣	**Zitterbart, Helmig, Mahrt, Seehofer**

Legend: 📱 Online, 🎲 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 90 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take place

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO
- or in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO

Prerequisites
None.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-101338 - Telematics must not have been started.

Recommendations

- Contents of the lecture ***Introduction to computer networks*** or comparable lectures are a prerequisite.
- Attendance of the module-accompanying ***basic practical course Protocol Engineering*** is recommended.

Additional Information
Offered for the last time in WS 26/27!!!!!

T 13.271 Module component: Test of Embedded Systems in Industrial Contexts [T-ETIT-100811]

Coordinators: Prof. Dr.-Ing. Eric Sax
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100546 - Test of Embedded Systems in Industrial Contexts

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2311648**	Test of Embedded Systems in Industrial Contexts	**2 SWS**	**/** 🗣	**Schmerler**
WT 26/27	**2311649**	Test of Embedded Systems in Industrial Contexts (Tutorial to 2311649)	**1 SWS**	**Practice /** 🗣	**Reis**

Legend: 💻 Online, 🖧 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in form of an overall oral examination (approx. 25 minutes).

Prerequisites
none

Recommendations
Knowledge of the basics of applied computer science, for example attending the practical course in information technology, is helpful.

T 13.272 Module component: The Energy Transition of the Electrical Transmission grid [T-ETIT-111248]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105618 - The Energy Transition of the Electrical Transmission Grid

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	**2307357**	The Energy Transition of the Electrical Transmission Grid	**2 SWS**	**Lecture /** 🗣	**Jesberger**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in form of an oral examination of approx. 20 minutes on selected topics from the course.

Prerequisites
none

T 13.273 Module component: Theoretical Foundations of Cryptography [T-INFO-111199]

Coordinators: Prof. Dr. Jörn Müller-Quade
Organisation: KIT Department of Informatics
Part of: M-INFO-105584 - Theoretical Foundations of Cryptography

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	1

Courses					
WT 26/27	**2400237**	Theoretical Foundations of Cryptography	**3 SWS**	**Lecture /** 🗣	**Müller-Quade, Berger, Klooß**

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment takes the form of a written examination (lasting 120 minutes) in accordance with § 4 Para. 2 No. 1 SPO.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take the form of an oral examination of approx.

- in the form of an oral examination of approx. 40 minutes in accordance with § 4 Para. 2 No. 2 SPO or
- in the form of a writtenexamination in accordance with § 4 Para. 2 No. 1 SPO

takes place.

Prerequisites
If the module M-INFO-100836 Selected Chapters of Cryptography has already been examined, the module Theoretical Foundations of Cryptography cannot be examined.

T 13.274 Module component: Thermal Solar Energy [T-MACH-105225]

Coordinators: apl. Prof. Dr. Ron Dagan
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-102388 - Thermal Solar Energy

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	2

Courses					
WT 26/27	**2189400**	Solar Thermal Energy Systems	**2 SWS**	**Lecture /**	**Dagan**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Oral examination of about 30 minutes

Prerequisites
none

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-106493 - Solar Thermal Energy Systems must not have been started.

Workload
120 hours

T 13.275 Module component: Ultrasound Imaging [T-ETIT-100822]

Coordinators: Dr. Nicole Ruiter
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100560 - Ultrasound Imaging

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2305295**	Ultrasound Imaging	**2 SWS**	**Lecture /** 🗣	**Ruiter**

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in form of an overall oral examination (approx. 20 minutes).

Prerequisites
none

T 13.276 Module component: Verified Numerical Methods [T-ETIT-109184]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-104493 - Verified Numerical Methods

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	**2303001**	Verified numerical methods	**2 SWS**	**Lecture /**	**Nagato-Plum**
WT 26/27	**2303002**	Verified Numerical Methods (Tutorial to 2303001)	**1 SWS**	**Practice /**	**Nagato-Plum**

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success control takes place in form of an oral examination (approx. 20 minutes).

Recommendations
Knowledge of the following modules is recommended:

- Mathematik I-III in Bachelor
- M-MATH-100536 - Numerische Methoden
- M-ETIT-104595 - Numerische Methoden für partielle Differentialgleichungen

T 13.277 Module component: Visual Perception in the Automobile [T-ETIT-100777]

Coordinators: Prof. Dr. Cornelius Neumann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100497 - Visual Perception in the Automobile

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2313717**	**Visual Perception in the Automobile**	**2 SWS**	**Lecture /** 🗣	**Neumann**

Legend: 💻 Online, 🌸 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in the form of an overall oral examination of approx. 20 minutes.

Prerequisites
none

T 13.278 Module component: Water – Energy – Environment Nexus in a Circular Economy: Research Proposal Preparation [T-CIWVT-113433]

Organisation: KIT Department of Chemical and Process Engineering

Part of: M-CIWVT-106680 - Water – Energy – Environment Nexus in a Circular Economy: Research Proposal Preparation

Type	Credits	Grading	Term offered	Version
Examination of another type	5 CP	graded	Each summer term	1

Courses					
ST 2026	**2233130**	Circular Economy Water Energy Environment: Research Proposal Preparation	**4 SWS**	**Lecture /** 🗣	**Schäfer**

Legend: 💻 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The Learning control is an examination of another type:
Research proposal of 10 pages and an oral presentation of 10 minutes (individual work). The grade will be a composite of the proposal (submission in week 13 before class) and oral & poster presentation (all day workshop with researcher participation).

Prerequisites
None

T 13.279 Module component: Wearable Robotic Technologies [T-INFO-114145]

Coordinators: Prof. Dr.-Ing. Tamim Asfour
Prof. Dr.-Ing. Michael Beigl
Organisation: KIT Department of Informatics
Part of: M-INFO-107113 - Wearable Robotic Technologies

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	**2400062**	Wearable Robotic Technologies	**2 SWS**	**Lecture /** 🗣	**Asfour, Beigl**
ST 2026	**5016643**	BUT - Attractive Robot Technologies		**Lecture /** 🗣	**Asfour**

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-106557 - Wearable Robotic Technologies must not have been started.

T 13.280 Module component: Workshop Finite Element Method in Electromagnetics [T-ETIT-114166]

Coordinators: Prof. Dr. Martin Doppelbauer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107147 - Workshop Finite Element Method in Electromagnetics

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each summer term	1

Courses					
ST 2026	**2306333**	**Workshop Finite Element Method in Electromagnetics**	**2 SWS**	**Seminar /**	**Brodatzki, Gjeset**

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Success control takes place in the form of different types of examination consisting of a written assignment in the form of an written report (20-50 pages).

The module grade is the grade of the written paper.

Prerequisites
none

Recommendations
Knowledge from the modules "Elektrische Maschinen und Stromrichter" and "Entwurf elektrischer Maschinen" is desired.

14 Appendix

14.1 Definition

MHB, PDF-Version: https://s.kit.edu/mhb-etit-msc25